

Applying the Ortony, Clore and Collins's Model with the Mechanics, Dynamics, and Emotions Gamification Framework:

An Exploratory User Experience Study Based in a Community Context

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Abstract

Recent technological advancements have made the use of non-intrusive emotion measuring techniques accessible. Simultaneously, discoveries in the field of neuroscience shows that emotions and cognition are virtually inextricable. Cooperating with the student platform Studydrive to increase the overall activity and reduce unintended behavior on the site, this paper explored the potential of gamification - the use of game elements in a non-game context - in the context of an online community. Through the application of the OCC-framework the onboarding process was gamified and compared to the non-gamified alternative within an A/B testing scenario. Having analyzed the users' affective states with an artificial intelligence-based emotion-measurement tool, leads to the conclusion that a considerate gamification of the onboarding process of an online community will elicit to a more divergent valanced reaction and consequently make the process more welcoming to the users. Before implementing, the emotional effect of gamification on the regular activity on the platform, as well the precise impact on cognition should be researched. Here, as this study discovers, the personality differences between introverts and extroverts need to be examined.

Keywords: User Experience, Gamification, Emotion Measurement, Affective Computing, Prototyping, Online Community.

List of Abbreviations

AI	Artificial Intelligence
AIC	Akaike's Information Criterion
ANS	Autonomic Nervous System
BIC	Bayesian Information Criterion
DGD	Demographic Game Design Model
E-I	Extraversion-Introversion
EM	Expectation-Maximization
J-P	Judgement-Perception
Max	Maximum Value
MBTI	Myer-Briggs Type Indicator
Min	Minimum Value
MLE	Maximum Likelihood Estimations
MVP	Minimum Viable Product
PFC	Prefrontal Cortex
RQ	Research Question
S-N	Sensation-Intuition
SUS	System Usability Scale
T-F	Thinking-Feeling
UX	User Experience
XP	Experience Points

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1 Introduction

We jump out of our seats out of joy when our favorite soccer team scores the decisive goal against their rival, full of hope play round after round of the captivating online game we have been on for hours and disappointedly sink in our chairs as we lose a card game. Whether you are watching a soccer game, playing a video game, playing cards, or are involved in any other type of game, emotions play a key factor in the creation of an engaging and memorable experience (Mullins & Sabherwal, 2020).

Until recent years, in their attempt to create intense gaming experiences to satisfy the growing demand, game designers were deepening their understanding of the interplay between the game´s mechanics, its dynamics with the player and her emotional response, resulting in an elaborate spectrum of observations, questions, guidelines for realization and more. Soon, these findings were transferred to serious activities. It notably gained far reaching attention through unique use cases of successful implementation and has already proven substantial in areas of Human Computer Interaction, such as personal fitness, learning and work processes. The implementation of gamification, commonly defined as the application of game design principles in non-gaming contexts, can showcase enhancements in multiple industries and consequently attracts the attention of scientists and researchers.

Frameworks and in-depth analyses combining advances in neurosciences, psychology, and technology are being released frequently (Robson et al., 2015). Nevertheless, gamification in a community context has to this date not experienced comparable scientific attention.

1.1 Practical relevance

Online communities are built on active content creators and social interaction, basically user participation and contribution. Focusing on the emotional responses of users, this study examines if and how an online community can overall benefit from the implementation of game elements.

Primarily, the initiator of this paper, Studydrive benefits from this research. Recently, in their aim to decrease internal efforts, increase overall activity and reduce trolling in the online community, they investigated the option to gamify their platform. Since their gamification efforts are currently on hold, I will provide them with a tailored and concrete guidance on what considerations need to be taken into account and how to proceed adequately.

Additionally, this paper will support further operators of online communities on their decision whether to consider gamifying their platform or not, by providing a user-centric infrastructure including possible advantages and drawbacks of the implementation.

Correspondingly, Studydrive users and those of further platforms can, in the long term, profit from more quality content and a more personalized and encouraging platform experience.

1.2 Academic Relevance

It has been thoroughly proven that fulfilling human desires, for reward, self-expression, achievement, competition, and status, and therefore gamification appeals to mobile users and consequently, gamification has grown worldwide from 4.91 billion U.S dollars in 2016 to 11.94 billion U.S. dollars in 2021 and been researched exhaustively (Business Wire, 2017; Weiser et al., 2015; Tondello et al., 2016). It is particularly efficacious and widespread in the corporate context, in knowledge acquisition or in personal behavioral influence (Korn & Schmidt, 2015).

In opposition to the just mentioned areas, academic papers on gamification in a community context appear to be rather scarce.

Previous work has shown that gamification in a community context can increase the overall amount and intensity of emotions elicited when implemented carefully (Papanaoum, 2019). Besides challenging this assumption for the target group of students, this paper intends to determine the effect of personality on the emotions elicited and uses emotion AI in order to allow an unbiased analysis of results.

1.3 Positioning

This will be achieved by using a combination of AI-based emotion measurement and a subsequent self-report on the onboarding process of the student platform Studydrive. The latter will be A/B tested, to directly compare the various elicited emotions and their intensity. All candidates have submitted an extensive personality test and are students that are not particularly familiar with Studydrive.

The two low fidelity prototypes are based on the current onboarding process. With the first version, representing the gamified version of the platform, popular game elements have been added in accordance with current scientific research. In contrast, for the second version, elements that could be identified as game elements have been removed from the present platform, to clearly differentiate the two. Both user flows have been created in Balsamiq Wireframes, a rapid low-fidelity UI wireframing tool (Balsamiq Wireframes, 2021). This was done to answer the following research questions (RQ):

RQ1. Does the gamified onboarding process elicit more and stronger emotions than the non-gamified version?

RQ2. Do personality trait specific affective tendencies occur in the context of an online platform?

To address these questions, the paper first gives an in-depth review of the theory of gamification. The basic idea of the latter including an overview over its initial development and the influence of personality traits are presented. Then, with a focus on emotions, its relation to cognition is introduced. In the following segment, Studydrive is introduced and their work towards an introduction of a gamified version is given in detail, in order to understand the motivation of this paper. Lastly, relevant theory and methods on platform onboarding, user experience and digital prototyping are written about, to support the empirical approach.

2 Conceptual Framework

The following chapter provides an overview over the concept of Gamification, current findings in the neurosciences that enable through an implementation of frameworks and the integration of personality specific aspects, the augmentation of cognitive capacities of users. Furthermore, previous attempts of Studydrive to gamify the platform will be outlined and recommendations resulting from their research are presented. Lastly, the term User Experience and frameworks and methods to emphasize the user-centered approach in an efficient way are described. All these will conduct the creation of the constructs.

2.1 Gamification

The concept of gamification is not completely new. Its origins are found in marketing efforts, including points cards and rewards memberships, in educational structures, such as scholastic levels and grades and workplace performance, introducing competitive games or a sense of childhood play to labor operations (Seaborn & Fels, 2015). In the early to mid-20th century for example, the Soviet Union initiated extensive experiments to explore “socialist competition” aiming to replace capitalist competition with a more engaging and humane approach through the power of games (Nelson, 2012). Starting in 2008, a time in which technology got cheaper, personal data tracking became more common, and a growing videogame industry commissioned extensive research, gamification (re-)emerged by transferring assessments from the game industry to other fields (Karimi & Nickpayam, 2017).

2.1.1 Gamification Theory

The term gamification, which was institutionalized in 2010, has – to this day - no agreed-upon standard definition. Based on current findings, it can be described as “the application of lessons from the gaming domain to change behaviors in non-game situations” (Robson et al., 2015). These transferred lessons are implemented through so-called *affordances*, which are aspects of the system that contribute to a playful experience – representing elements, such as points, leaderboards, levels, badges, and challenges (Mullins & Sabherwal, 2020). When introducing these affordances, one must resist the compulsion to design the game around what features are available, rather than thinking first about the user and her behavior that is to be stimulated. Gamification, at its core should “engage the users on an emotional level and motivate them to achieve their goals”, but factors such as the scope and area of the application and the demographic group and personality types of the target group also have an influence on the outcome of well-planned implementation (Dale, 2014).

Research has shown that individuals of various nature are impacted differently by gamification, because they react differently to given affordances (Hamari et al., 2014). Examples of these divergent preferences have been studied by researchers such as Jia at al. in 2016, who, amongst other findings, demonstrated that extroverts tend to be motivated by points, levels and leaderboards and less emotional stable people are negatively impacted by multiple motivational affordances. When executed correctly, benefits arising from gamification can be as wide-ranging as the possible applications and commonly include increased motivation, stronger emotional experience and more focused attention and engagement (Ghaban & Hendley, 2019; Koivisto & Hamari, 2014).

While generally having a positive impact, shortcomings must be considered (Seaborn & Fels, 2015). In concrete terms, as much as 80% of gamification attempts are said to fail due to inappropriate implementation and consequently, a not adequately researched, standard approach is to be avoided (Gartner, 2012). Negative consequences can show through increased effects of competition and other effects detracting the user from the experience (Jia, 2016). A system that intends to improve through gamification should incorporate several aspects.

Firstly, it is crucial that the end-product creates an immersive experience. While the individual target outcomes may vary from one application to another, it is in all cases of gamification essential to its success that the user perceives the activity as actually playing a game. This is necessary to harvest the benefit of increased attention and engagement (Dale, 2014).

Motivation is a crucial requirement to initiate change in behavior. Accordingly, when the target of a rework includes changing the behavior of users, the system should be grounded on theories of motivation and integrate the user's needs and goals (Fogg, 2009; Eyal, 2014). Human needs are made up of psychological needs and social needs. Psychological needs comprise the desire for *autonomy*, so the desire to make self-determined choices, the desire for *competence*, the desire to become better at something, which can ideally keep the user in a state of "flow" and *relatedness*, the need for engaging in relationships with others. To satisfy the latter, recognition, acceptance and being valued must be signaled clearly. Social needs on the other hand, can be acquired through learning processes and include the longing for achievement, affiliation, and intimacy, as well as leadership and followership. Inversely correlated with motivation, the user's ability to execute a given task must have reached a certain level so that behavior change can be initiated. If triggers anticipate and induce behavior, they can both increase the intrinsic

motivation, by prompting on goals, beliefs, and the self, satisfy extrinsic motivation or increase the ability, through guidance and instruction.

Overall, gamification has established itself as a central element in certain areas. Most notable in the realm of education, many digital solutions, such as *Duolingo* or *Kahoot!* implementing game elements including leaderboards, experience points (XP) and levels (Evgeniy & Danil, 2020). While 25-30% of all studies released on gamification applied to educational measures, only few studies concerning gamification in a community context could be identified and only one would examine the emotions elicited by the technique (Robson et al., 2015; Papanaoum, 2019).

As described in the previous section, the paper by Papnaoum concluded that gamification would both more and more intense emotions and that those candidates that experienced less positive emotions in the gamified platform still tended to prefer it over the more positive and non-gamified version.

2.1.2 Personality Traits and the Myer-Briggs Type Indicator

Personality is one of the human aspects playing a central role in a person's approach to life, influencing the manner of thinking, feeling, and behavior, among other factors. Unsurprisingly, the idea of personality traits, which are defined as patterns of thinking, feeling, or behavior that tend to be consistent over time and across relevant situations (Soto, 2018), is dating back to the earliest writings. It has been known to and described by renowned writers such as Aristotle (384-322 BC). He, for example, saw vanity, modesty and cowardice as significant determinants of moral and immoral behavior and a student of his, Theophrastus (371-287 BC) even presented thirty different personality types in a book which's name can be translated as

‘Traits’. In order to develop a science of traits, it was and still is fundamental to measure and classify the latter (Matthews et al., 2003).

One ground block of this has been laid by the Swiss psychoanalyst Carl Jung in the 20th century whose findings would later lead to the development of the currently prevalent Myers-Briggs personality test. Initially highly influenced by Sigmund Freud, he eventually chose his own approach to the topic, in which he drew inspiration from empirical observations, art and even traditional tales. Originally aspiring to display the complexity of human personality, he detected that the anticipatedly random variation in human behavior, is indeed quite orderly and consistent. The basic differences he discovered were two personality attitudes and four functions. The former is namely extroversion, the attitude to be focused on the outside world, and introversion, where persons would focus more on the inner world. The four functions, defined as modes of orientation of the human brain are thinking, sensation, intuition and feeling. Already back then he concluded that these functions would not be exclusive, but rather on a scale from predominant to inferior (Jung, 1939; Jung, 2014).

The Myer-Briggs Type Indicator (MBTI), which is a self-report inventory, was developed to estimate the variables in Jung’s personality typology and its’ current version has been published back in 1962. It is important to mention upfront that it aims to describe how persons are likely to behave, as opposed to discern their cognitive functions. The test consists of the following four scales:

- The Extraversion-Introversion (E-I) scale was designed to measure the person’s orientation to life. Persons that rate high on extraversion are primarily oriented to the outer world and introverted types more inwards. The first have the tendency to get

caught up with whatever is happening around them, while the latter tend to detach themselves from the surrounding world.

- The Sensation-Intuition (S-N) index describes the person's preferred way of perceiving things. Sensing ones focus on their perceptions received through the senses and notice details and practical aspects of situations, while intuitive types perceive things more vaguely and prefer dealing with abstractions, implied meanings, and hidden possibilities.
- The Thinking-Feeling (T-F) scale measures the person's preferred way of making decisions. Those on the thinking end, rely on logical structures to put clarifying order into situations and are skilled at objectively organizing material. Their ability to impersonally judge whether something is true or false is well-developed. On the contrary, the feeling types are rather skilled at understanding other people's feelings and analyzing subjective concepts.
- Lastly, the Judgement-Perception (J-P) spectrum, aims at measuring the person's way of dealing with the outer world. Those falling into the judging type, are organized and systematic. Their surrounding is planned, as they aspire to regulate life and control it. Perceptive ones on the other hand, are more curious and open-minded, approach life more spontaneous and attempt understanding it and adapting to it. This last scale is not found in Jung's theory and relates to at least one of the other dimensions (Carlyn, 1977).

The MBTI has been deemed reliable and accurate over time and accordingly, many applications have been found for the model that it is the largest test-base of all typological

studies. In an educational context, the MBTI model is being used to reliably predict the learning style of students. As digital learning solutions gain an increasing amount of importance, MBTI supports optimizing them (Fatahi et al., 2018). Personality majorly influences a person's behavior at work and job satisfaction, as the attitude towards people and issues and their decisions and responses are highly dependent on their personality (Jaqua & Jaqua, 2021). Based on this, many companies introduce undertaking personality tests to their employees and virtually all of the Fortune 1000 companies in the USA use the MBTI model for employee relation studies and team building (Bateman, 2004). In alignment with the point above, the Myer-Briggs index is also widely used in career building. Research has shown that there is a continuous correlation between the personality type and future work prospects (Garden, 1997).

Even when one looks at the gaming industry, personal adaption via the MBTI model finds use. It arises through using personality information to adapt aspects of gameplay in order to create a more tailored and enjoyable experience for the player. While it is common practice to simply use the MBTI model, there have even been approaches translating the results of the index into models, specific to the game experience (Dias & Martinho, 2011; Felicia & Pitt, 2009). The Demographic Game Design Model (DGD), for example uses the results from MBTI and provides a specific player type for each 4-letter code elaborating on the style of play in regard to the aspects progress, story and social, best fitted to the latter. Linked to categorial names such as *Casual Wanderer* or *Hardcore Conqueror*, personality type specifics are highlighted and related individual preferences as to Game dynamics are proposed in this instance (Bateman, 2004).

In relation to gamification, personality has also proven to be a decisive factor. While certain users find some affordances to be motivating, others perceive them as distracting. For example, Ghaban & Hendley have shown that certain personality types are affected negatively

by gamification, as they tend to describe game elements as *silly* and *toy-like* (2019). Consequently, it is essential for a successful implementation of game elements to understand the relationship between personality and affordances. Butler has laid out corresponding gameful mechanics that speak to each personality trait's individual attitude, as laid out in Appendix A (2014).

2.1.3 Emotion and Cognition

According to Rüdiger Schnell, "today's historical research into emotions involves two basic and contrary positions: according to the one, human feelings have remained the same for millennia (only the means of expressing them having changed); and according to the other, each emotion has its own history determined by general historical changes". This arises from another division, namely *nature* versus *culture*. Starting with the Enlightenment, nature was representing four properties. The *state of nature*, which is a state in which humans were in the period before the existence of society, a 'primitive' state, describing alien and non-European people, an equation to the human body and its internal and less mutable aspects, such as instincts and lastly a fusion with the environment in general. Culture on the other hand has risen from man-made rationality (Plamper, 2015).

Lorraine Daston, an American science historian and former director at the Max Planck Institute for the History of Science in Berlin, perceives the contrast of nature to culture as a foundation to universalism to social constructivism (Dath, 2021). Universalism, stating that some ideas are applicable universally and social constructivism, asserting that human development is socially situated, and knowledge is established through the interaction with others are opposing

socio-philosophical concepts. While current scientists are aiming to move beyond this dichotomy, no definite and generally agreed postulation has been detected.

However, this does in no way devalue the importance of emotions in our everyday life and the urgency to apply findings on the topic (Plamper, 2015). Plamper, in his scientific work, has amongst others benefited from discussions and comments by Ray Dolan, who in 2002, stated the following.

> The importance of emotion to the variety of human experience is evident in that what we notice and remember is not the mundane but events that evoke feelings of joy, sorrow, pleasure, and pain. Emotion provides the principal currency in human relationships as well as the motivational force for what is best and worst in human behavior. Emotion exerts a powerful influence on reason and, in ways neither understood nor systematically researched, contributes to the fixation of belief (Dolan, 2002).

In alignment with the quote of Dolan, recent advances in cognitive neuroscience provide the view that cognition, referring to the ‘mental activities pertaining to acquisition and application of knowledge, including processes such as attention, learning, language processing, problem-solving, and memory does not exist without emotions (Mullins & Sabherwal, 2020).

Humans’ cognitive activity is motivated from lower, evolutionarily older brain regions by basic emotional and motivational drives that explore circumstantial matters for survival while facilitating secondary processes of learning and memory. Consequently, in order to understand how emotions affect cognition, an understanding of evolutionarily nested hierarchies of the central nervous system’s emotional-affective processes is necessary, as well as an understanding

of the large-scale network responsible for cognitive activity which includes the midbrain's periaqueductal gray and the ventral tegmental area, the basal ganglia, which comprises the amygdala and lastly the diencephalon with the medial temporal lobe (Tyng et al., 2017).

The primary activity of the prefrontal cortex (PFC), located in the foremost anterior region of the frontal lobe, is envisaged as the creation of thoughts and actions in accordance with internal goals (Miller et al., 2002). Upon that it is associated with higher-order cognitive functions like prediction and planning and acts as a control center for selective attention, semantic processing and problem-solving (Barbey et al., 2009; Squire et al., 2013).

With *valence* representing a bipolar dimension that defines stimuli in terms of pleasure and displeasure and *arousal* the intensity of these stimuli, scientific evidence indicates that the PFC responds to emotional valence of non-arousing stimuli. Arousal on the other hand is said to activate the amygdala, which is a brain region critical to regulating flow and integrating information between brain regions in cognitive-emotional interactions (Tyng et al., 2017). It is linked to the processing of fear and proposed to modulate sensory processing through focusing attention on potential threats. Furthermore, it is said to aid the encoding, consolidation, and subjective recollection of memories linked to emotional stimuli (Phelps, 2006). Memory, illustrating a mentally stored representation of a particular incident is closely related to learning, typically occurring as a result of exertion over a period of time (Kazdin, 2000). Because of their importance to survival in an evolutionary perspective, both the consolidation process through which memories become stable over time and associative learning are enhanced by emotional arousal (Phelps, 2006). Attentional focus, a resource that appears to be scarce in the current time, can be captured effectively through stimuli that evoke emotional responses as well. Even

subliminal emotional stimuli are presumed to have the same physiological responses as conscious ones.

According to the *appetitive/aversive model* presented by Lang, Bradley, and Cuthbert in 1997, the combinations of valence and arousal can activate two motivational brain systems. The appetitive system, eliciting basic reactions of approach and the aversive system, producing those of withdrawal. Researchers have then discovered that both positive valence and low arousal generate responses of approach and negative valence, or high arousal elicit withdrawal – a finding termed positivity offset (Robinson, 1998; Cacippo, 2004; Norris et al., 2010).

Nevertheless, certain borders of emotions exceeding limitations need to be considered. While the negative emotional state, stress, can facilitate learning and memory if occurring in the mild or acute intensity, excess or chronic stress will impair learning and memory performance (Tyng et al., 2017).

Nowadays, changes in emotional states cannot merely be measured subjectively by the person communicating feelings and experiences. Behavioral investigations of facial expressions, gesture changes and vocal expressions and the assessment of physiological responses such as responses of the autonomic nervous system (ANS) including the heart rate, skin conductance and blood volume pulses can analyze affective states in an implicit and objective way (Tyng et al., 2017).

2.1.4 Gamification and Emotions

Drawing on the findings of the interrelation between emotion and cognition and the differences in perception by persons of various personality types, gamification itself has the capability to create an immersive and engaging experience that through the precise use of

affordances is activating cognitive states as intended and additionally supports the formation of habits by initiating positive and negative emotions. In concrete terms, gamification is said to influence the episodic memory, associations with a specific time or place, through the creation of an immersive experience. During a state of total immersion, the deepest level of flow, greater negative emotions such as anxiety are experienced, which increase the level of engagement.

Negative emotions triggered by game elements, including anxiety, frustration, anger, and sadness, can increase attention and decision making of users, while the "pretend context of video games may be […] safe enough to practice controlling, or modulating, negative emotions […]" (Granic et al., 2014, p. 72). Additionally, high levels of arousal and positive valence can lead players of video games to a flow state and elicit an overall positive effect on cognition (Gabana et al., 2017). Gameful experiences are also able to promote the acquisition of knowledge through the creation of selected emotions in the user. Beneficial factors can include attributes of the individual, the task, or the technology. It is important though that those emotions promoting effective learning in one context can even be detrimental in another (Mullins & Sabherwal, 2018).

This complex outset has made the use of frameworks popular, two of which will be described, and their linkage will be outlined in the following section.

2.1.4.1 The MDE Framework of Gamification Principles

An adaption of the MDA-Framework, which was formally developed for game design in 2004, has established itself for many gamification professionals nowadays. The MDE-Framework introduces three principles – mechanics, dynamics, and emotions – to illustrate how gamified experiences can be created and outlines the interdependent relationship of the three. It

represents the only gamification framework explicitly integrating emotions as a key factor (Robson et al., 2015).

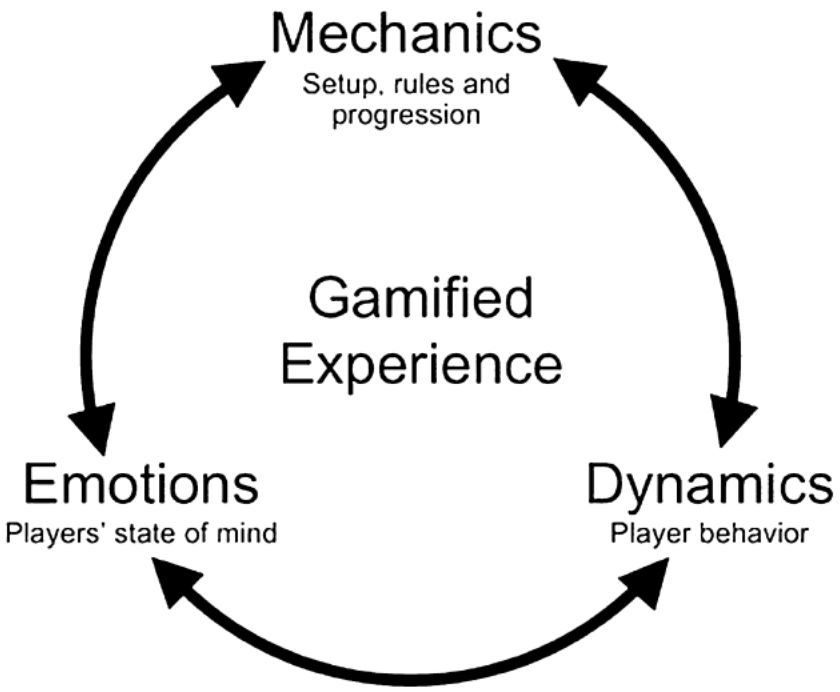


Figure 1. MDE framework of gamification principles

The first principle, *mechanics*, stands for design decisions to specify goals, rules, setting, context, types of interactions and boundaries of a situation. They form the structure that the gamified experience exists in and are known before the experience starts and remain constant over time and from one user to another. Mechanics fall into three categories – setup mechanics, rule mechanics, and progression mechanics. Setup mechanics shape the environment of the experience, including the setting, what objects are needed, and how the objects are to be distributed among players. Rule mechanics shape the concept or goal of the gamified experience to be pursued by prescribing the actions that are permissible and the constraints that limit those actions, and progression mechanics describe different types of instruments that designers embed to affect the experience while it happens. The latter make up the most important principle for gamification by reinforcing desired behavior and providing feedback signaling a player's success

through achievement rewards. Mechanics influence both the dynamics and emotions through an interdependent relationship.

The second principle, the *dynamics*, denotes the type of behavior emerging as users partake in the experience. They are produced by how players follow the mechanics chosen by designers and describe in-game behaviors and strategic actions and interactions that emerge during play. Accordingly, the designer only has an indirect impact on dynamics, as it is directly influenced by the players' characteristics including their personality. They are difficult to predict and might be the basis of unintended consequences arising through gamification.

With the MDE framework highlighting the importance of emotional experiences, the third and last principle is called *emotions*. These emotions are understood as mental affective states and reactions evoked in the experience and are the result of the relationship between how mechanics are interpreted by the player and what dynamics are generated. While a game designers' lever is its mechanics, emotions are the key factor for the players (Lazzaro, 2004).

MDE suggests creating a gamified experience by first focusing on mechanics, then dynamics and only then on players' emotions. Nevertheless, it suggests that emotions are "more important than the rules that make them possible" (Robson et al., 2015).

Initially proposed in 1988 by Ortony, Clore & Collings in their book "The Cognitive Structure of Emotions", the psychological model offers a clear structure of the eliciting conditions of emotions and the variables affecting their magnitude. Similarly, to the MDE framework, it is prominent in the computer science field and is being used to build systems dealing with emotions and incorporating them in fabricated characters. The revisited model includes 31 emotion types. A list of specifications to these types and how they are invoked can be found in the Appendices B and C.

As introduced by Mullins & Saberwal, the obstruction of eliciting intended emotions through the designers' lever, the game mechanics, can be bridged by integrating the reworked

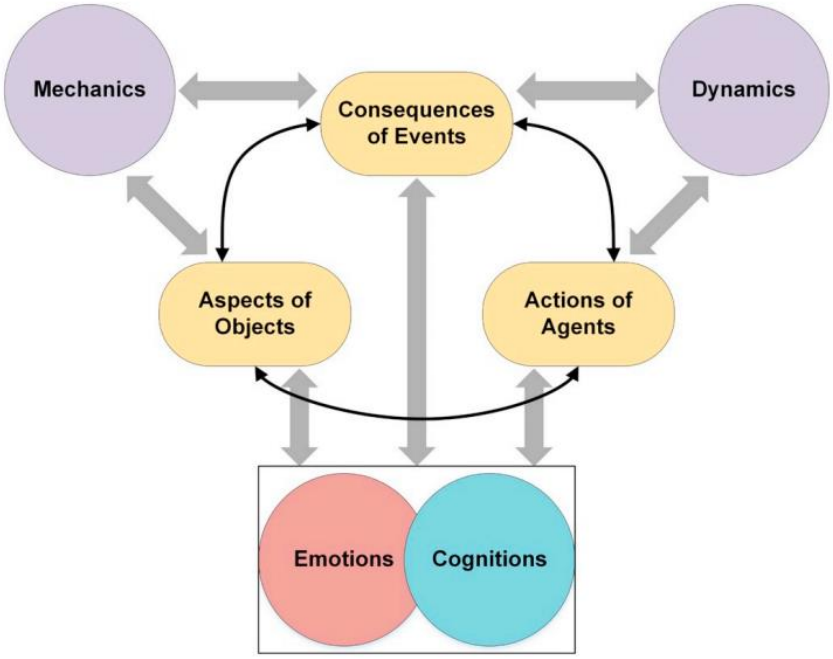


Figure 2. Cognitive-emotional view of gamification

OCC model into the MDE framework and creating a cognitive-emotional view of gamification. According to this framework, these *mechanics* have a direct influence on the *aspects of objects* and the *consequences of events*. In practice, a designer decides on the effect of certain events, like leveling up, unlocking a badge, or completing a challenge. In addition, the aspects of included objects, like experience points needed to level up, the nature of existing badges and requirements and rewards of challenges, are to be pre-defined. How a player, titled agent in this model, interacts with these mechanics is represented by *actions of agents* and in direct correlation with aspects of objects and the consequences of events. Differently to the MDE-framework, the emotions and cognitions are not directly linked to mechanics or dynamics, but rather emerge from these three elements.

2.2 Studydrive

Studydrive, was founded by Philipp Mackeprang and Sven Gasper in 2013 and acquired by Stepstone, a daughter company of Axel Springer, in 2019. Their website and mobile application allow students to share documents, answer questions about their studies or follow campus news. The latter can connect to fellow students in subject groups and use Studydrive as a learning platform, community, and career portal. Currently, the platform counts over 1.7 million active users and close to 700,000 uploaded documents. The service is entirely free for its users, since Studydrive's income is generated through selling targeted job offers to third parties. These companies, looking to fill vacancies, can get in touch with Studydrive users, for example through job advertisements, mailings, or ads.

2.2.1 Gamification Research

Since the platform exclusively provides the infrastructure and is dependent on user created content, activating their community is one of Studydrive's key challenges. Currently, a system including the fictional currency *Karma Points*, which do not have any in-game value and, so-called *course experts*, exceptionally active and productive students that are directly employed at Studydrive and receive money for activity on the platform, is set up to augment motivation and overall activity. Besides an unsolved bias, through which students at larger universities and study programs benefit in relation to those of smaller ones, the current system appears inefficient, costly, and laborious for Studydrive and yet leaves room for unsubstantial posts, comments, and uploads on the platform. Based on this, the growth department explored the idea of gamifying their platform. Initial hypotheses were formulated as follows:

Studydrive - RQ1. Does the gamification system increase user engagement and retention for users?

Studydrive - RQ2. Do our XP-collection, leveling, and related features make sense for the users, and do they motivate the users?

Along these also emerged the doubt that introducing the gamified version might overwhelm new users and impede the onboarding process. Accordingly, a gamified minimum viable product (MVP) was created in a cross-departmental approach that would exclude the Karma-Points-system and included challenges, experience points, levels, badges, monetary rewards. This MVP was tested qualitatively and quantitatively concerning the initial hypotheses.

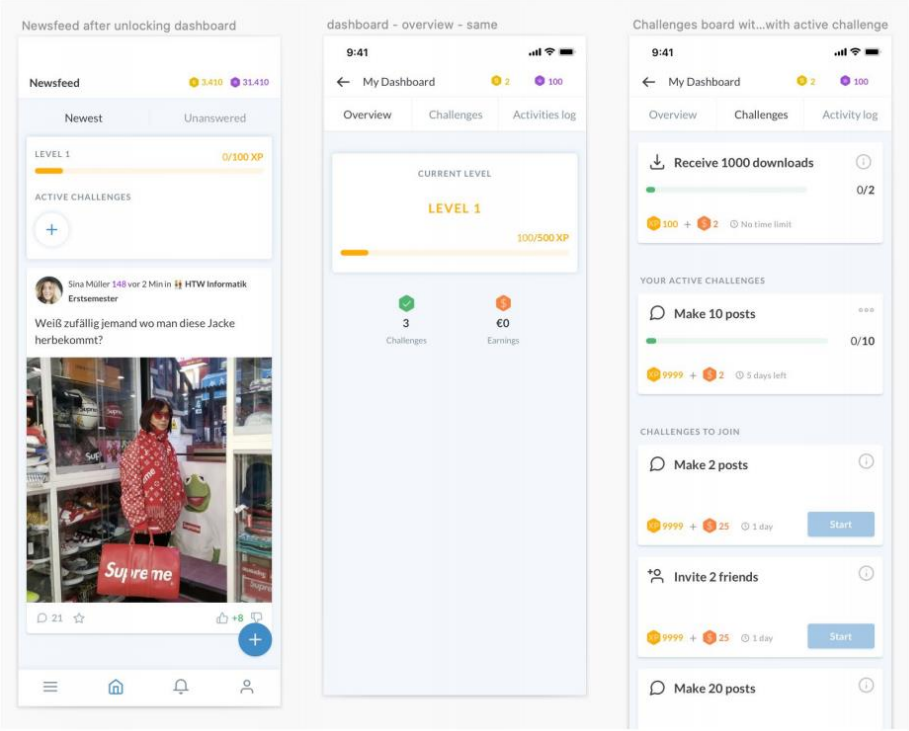


Figure 3. Screenshots of Studydrive's MVP

The first step of the research consisted of interviewing ten students. Five had been unfamiliar with the website – the other five were active users. There was also an equal distribution between small and big universities. The candidates were interviewed in 30-minute-long pre-release sessions concerning the following topics:

- The course expert earning system
- Trolling & spamming
- Karma
- Feeling of unfairness
- Motivational constraints
- Game experience and behavior.

Building on information gathered from these interviews, testing of the MVP over a two-week period was initiated. This test run was carried out with 100 participating course experts. Feedback on the experience of using the prototype was collected through a set of qualitative research methodologies. In *daily check-ins*, the students' first impression, user behavior, motivations, technical issues, and opinions were queried. The user tracking tool *Smartlook* was utilized to evaluate the intuitiveness of the interface, the gamification system itself, as well as user behavior. A subsequent survey aimed to evaluate the trial quantitatively by asking questions on the frequency of use, number of challenges started, amongst others. Lastly, post-release interviews were set up with the same ten users that had participated in the pre-release session, to evaluate the systems perceived usability, taking up again questions from previous research steps.

This extensive research did deliver interesting insights. Most testers experienced the gamified version as intuitive and easy to understand and 43% of them considered XP collection

as motivating. However, testers questioned the incentive or reason behind the system, raising concerns such as “What’s its point?”. With only some users believing that this new system could increase earnings for students from smaller classes and activate inactive universities, many even expressed concern over the content quality on the platform. Due to a lack of motivation, many used the gamification features to a lesser extent in the second week. Essentially, the conclusive results would turn out to be ambiguous and were neither able to prove or disprove the initial hypotheses. The following statement summarizes the outcome adequately. “From the feedback, we weren’t sure if this was what the users actually wanted.” (Studydrive GmbH, 2020).

Accordingly, a second research iteration, with a peripherally adapted prototype was initiated. Despite the intention of yielding more precise results, this rounds approach was similar to the first one, including the same research methods, with only the testing period enlarged by two weeks to a total of four weeks. No pre-release interviews took place. In contrary to the prospect, no clear answer to the initial hypotheses could be submitted once again. On the other hand, explicit recommendations – which can be found in Appendix D – were expressed on aspects of a gamified alternative version of the platform and a new research question was raised:

Studydrive - RQ3. What is the impact of the gamification system on newly registered users?

Based on the still existing ambiguity on whether gamifying Studydrive would be overall beneficial, I decided to call the whole approach into question on a more primal level, including selected recommendations from the second research and aspiring to find an answer to the onboarding research question (Studydrive - RQ3). Founded on the neuroscientific cognitive-

emotional relations and the assumption that eliciting basic reactions of approach is elementarily essential to onboarding, and increased cognition in the user would aid the generation of higher quality content, I will create a prototype and analyze it implicitly in a non-invasive way. Before the latter is outlined, an overview of platform onboarding and user experience is given.

2.3 Platform Onboarding

Onboarding is essential for any online platform and marks the beginning of a relationship between the platform and its users. A carefully planned overall strategy should be in place, so the experience can support new users' engagement. This process consists of a sequence of steps that introduce users to a new product or service and can be designed in multiple ways, with different duration and the amount of element or types of flows. Platforms that entirely depend on users' contributions can, above all, benefit from onboarding design, as it includes initial interests and motivations, addresses instruction and guidance.

Across different types of platforms, and different industries, a typical onboarding approach consists of five elements. The first phase is called the *purpose statement* and communicates its relevance, main benefits and clarifies what can be expected. Following this step and usually before having access to the pages main content and before being able to participate in the platform, comes the *registration*. To be able to make contributions and clarify community rules, users then go through the *informational scaffolding*, offering guidance and instructing on how to use the platform. A user then contributing herself is called the *conversion*. The last onboarding step is the *re-engagement*. It is focused on retaining the users through external or internal actions (Cascaes Cardoso, 2017).

Research of onboarding processes has shown that setting goals to users is relevant. Confronting the user with specific goals lead to higher contribution than do-your-best goals. Furthermore, a u-shaped relationship between the size of the goal and the corresponding contribution was detected. When the goal size goes beyond a certain limit, it demotivates rather than motivates users. Another point that can be mentioned is that group goals tend to be more effective than individual goals and people contribute more when they believe that their own contribution will be unique (Ling et al., 2005). These findings align with the direction of gamified onboarding.

Research in the gamification of employee onboarding, which consists of comparable steps to platform onboarding, has for example shown that an apt implementation can help increasing employee engagement and learning effectiveness (Depura & Garg, 2012). While differences between the two scenarios cannot be neglected, it is not unreasonable to conclude that gamifying the onboarding process of an online platform might, despite not being researched extensively, yield similarly positive results as employee onboarding.

2.4 User Experience

Although the gamification of Studydrive and the onboarding process of communities in general is the focus of this paper, it is important to understand how its implementation can be structured. To give some context, first the area of User Experience (UX), its lifecycle process, and its integration into the Design Thinking concept will be provided.

Before UX can be addressed, the underlying concept of usability in interaction design needs to be explained. It concerns the effectiveness, efficiency, and the general satisfaction of the user with the product (Department of Health and Human Services, 2013). Several methods have

established to measure usability, including the System Usability Scale (SUS). Originally created by John Brooke in 1986, it consists of a questionnaire with ten items, each offering five response options ranging from Strongly agree to Strongly disagree and facilitates the evaluation of different products and services, such as hardware, software, websites, and mobile applications (Department of Health and Human Services, 2013).

Originally popularized by Don Norman, UX is primarily used to represent a commitment to and a focus on higher quality human-computer interaction. Its prevalence in the scientific and corporate is similar but not as widespread as the related term usability. Whereas the latter focuses on the pragmatic side of the user-product relationship, such as users' tasks and their performance, UX balances these pragmatic aspects with other non-task related aspects such as challenge, stimulation, or self-expression. Another point that can be mentioned is that while usability takes an *objective* approach, basing recommendations on observation of the user, UX lays focus on the *subjective*. The focal point of this approach is the user herself and shows interest in the way, she experiences and judges the product, including her thoughts, feelings, and perceptions that result from the interaction (Hassenzahl et al., 2006; Tullis and Albert, 2013).

It is understandable that in a Human-Computer Interaction context, Usability is more widespread, since a computer at a basic level, serves as a tool for humans to solve a specific task or problem (Diaper & Sanger, 2006). Nevertheless, there are major differences between simply doing a job and enjoy doing it. As laid out in section 2.1.3, an increase in emotional experience also increases memory, attention and decision making. Many methods to improve user experience and by improving the hedonic quality also inherently improves the pragmatic quality. Improving the subjective experience consequently also improves the objective performance.

Overall, these activities can be categorized in four elemental groups that are namely *Understand Needs, Design Solutions, Prototype Candidates and Evaluate UX*. Hartson and Pyla have deducted these activities to create a life cycle process called the Wheel (2018). Being a wheel, it goes around in circles, bringing one closer to the goal with each rotation. They elaborate the activities as follows:

- *Understand Needs*, to understand users, work practice, usage, the subject-matter domain, and, ultimately, needs for the design.
- *Design Solutions*, to create designs as solutions.
- *Prototype Candidates* (of promising solutions) to realize and envision promising design candidates.
- *Evaluate UX*, to verify and refine designs with respect to the user experience they afford.

The primary objective of the lifecycle process is to make the transition to production, but to do this successfully, it must undergo decisions as per the following points.

1. When to leave an activity
2. Where to go after any given activity
3. When to revisit a previous process activity
4. When to stop making transitions and proceed to production

So called transition criteria at the end of each process activity shaping the answers to these questions. Since there is no given formula determining the transition criteria, they are generally based on whether goals and objectives of the current iteration of the activity have been

met. Consequently, it is up to the team, to articulate the goals for each process activity and decide when they are met.

Problems that do not have a definite solution and cannot be solved using linear problem-solving techniques, require continuous reformulation. Design thinking is a method for solving the latter. It is based on abductive reasoning and "aims at iteratively designing and validating solutions using a forward-looking approach". Most importantly, it lays upmost importance on the customer, end user or consumer of the product or service. Originally emerging from architectural and industrial design, it is now being implemented across all industries, with varying guidelines depending on its context. Similarly, to the problems it attempts to solve, the process is dynamic and can be complicated, messy, and nuanced. Nevertheless, an underlying process made up of five building blocks guides this process.

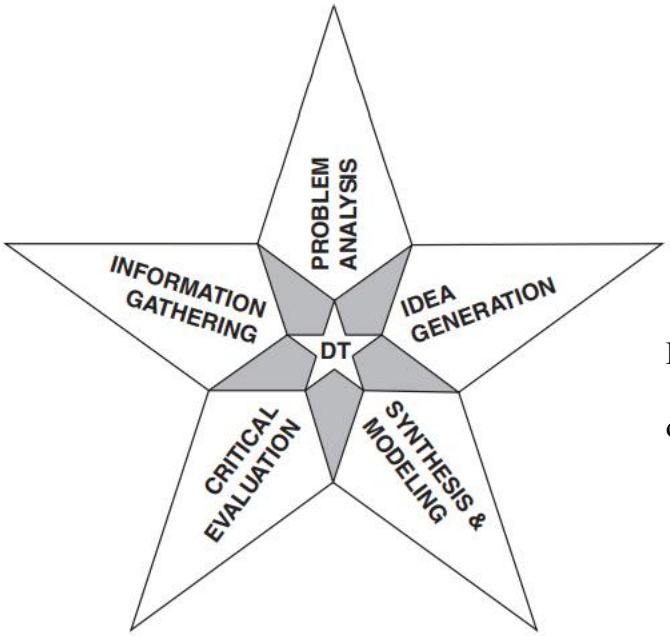


Figure 4. The fundamental building blocks of design thinking that form a "loop".

It starts with *Information Gathering*. In this stage, the context and stakeholders need to be studied, to build a deep understanding of the relevant issues, conflicts, and constraints. Gaining this insight might lead to new ideas. In the second stage, namely *Problem Analysis and*

Definition, extensive analysis would lead to the identification of the most salient problem. *Idea Generation* has the goal of generating many ideas based on the Problem Analysis and Definition. The creation of innovative concepts or outlines of ideas is the goal here. The fourth step goes by the name of *Synthesis through Modeling*, and in this stage, one is to take the best ideas and elaborate them in higher resolution and detail through prototypes, models, or draft solutions. The predominant target of this stage is learning and discovery, so success or failure should not influence one's decision making. Lastly, the *Critical Evaluation* takes place. It is the crucial step of testing the previously created model and validating or nullifying concepts. Possible solutions are to pass through this loop as many times as necessary to the existing problem. It is important to mention nevertheless, that accurately following these given steps might be oversimplifying the actual process. Consequently, the building blocks are dynamics and can be modified, cherry-picked, and prioritized, while embracing the user-driven process (Pressman, 2019).

2.4.1 User Experience Research Methods

For both the proper implementation of the Wheel and Design Thinking in the context of UX, evaluating this UX is crucial to the user driven/centric approach. UX methods, the "way one can carry out the whole part of a given lifecycle activity or sub activity", are used to carry out the latter and various methods have been established for it. In their paper from 2010, Vermeeren et al. already reported a total of 96 established techniques being used to measure and analyze different UX aspects. The most popular methods include self-developed questionnaires, semi-structured interviews, activity logging, standardized questionnaire, and think aloud feedback (Pettersson et al., 2018).

Depending on the intended goal, a different method is recommended. Particularly in early stages within the process, evaluation focuses on qualitative data when it comes to assessing UX issues. Where later, it is recommended to establish metrics and targets, foregoing them in beginning is deemed reasonable. Altogether, the trend in the UX field moves away from a quantitative focus and towards rapid qualitative evaluation (Hartson & Pyla, 2018).

2.4.2 Digital Prototyping

As presented in the sections above, the user centric approach of UX and Design Thinking requires a procedure to test, evaluate and then adapt current concepts.

In the field of manufacturing conventional prototyping techniques can take years to finish. Therefore, rapid prototyping has been developed and is being used to create a part of a product or a 3D model of the product to test it before manufacturing it in larger quantities. Correspondingly, in the realm of digital products, prototypes can be used to test concepts, test the logic, conduct usability tests, or even sell the idea to clients in a cost-effective way. While distinct approaches vary in their form, fidelity, interactivity and lifecycle, all digital prototyping can be considered rapid (La Riva, 2020).

Generally, digital prototypes are divided into three categories based on their fidelity, which describes how similar the prototype is to the final product. Most assumptions should be tested with lower fidelity prototypes and a progress towards a higher fidelity prototype is recommended when specific details or specific user flows are to be tested.

Low-fidelity prototypes keep content and visuals to a minimum and do not resemble the final product. Used to test core concepts, explore different ideas, and detect problems, they come

with the downside that the simplicity might lead to uncertainty during testing and are limited in their interactivity. Examples include paper prototypes, storyboards, or wireframes.

Mid-fidelity prototypes come with limited functionality, clickable areas, basic visual design, and a correct content description. They are generally used to test broad functional features and are suited for the validation of the interaction concept. Examples are clickable prototypes or coded prototypes.

High-fidelity prototypes on the other hand are more detailed and realistic prototypes that both look and work like the final product and include all visual components, interactive elements, and content. Consequently, they are the most time-consuming option and require specialist knowledge to create. The typical example would be a fully coded product (Stevens, 2021; Škerget, 2021).

3 Methods

The following section describes the process followed to set up the prototype testing in alignment with the theoretical background and the findings from Studydrive. It also lays out the measures used to analyze the test results.

3.1 Setup

3.1.1 Prototype

Before the prototype for this test could be created, multiple prospects had to be addressed. These included where its position on the fidelity spectrum would be, accordingly which tool was to be used to create it, whether it should be for mobile or desktop, which section of the platform's user flow was to be tested and which method was to be implemented for the latter.

As presented in section 2.4.2, a low-fidelity prototype is recommended to test concepts. Considering that the concept of a gamified platform with the use case of Studydrive is being tested, a low-fidelity prototype was to be created. Because of its strong record of 13 years, and its intuitive interface, the lo-fi prototyping tool, Balsamiq was chosen. Stylistically resembling paper sketches of wireframes, it allows the creation of clickable areas and interconnectivity between multiple wireframes (Balsamiq, 2021). In alignment with the preceding research from Studydrive, within which prototypes and user tests were employed, the mobile application was chosen over the desktop version. Similarly, the user flow was to take place in the onboarding segment. As stated in 2.2.1, Studydrive was concerned that gamification might overwhelm new users. Additionally, as can be seen in 2.3, the first impression of an application plays a decisive

role. To test the concept an A/B test was selected as the UX research method. Both versions can be found in the attachment (Attachment 1 and 2) and will be described below.



Figure 5.

Non-Gamified Prototype: Newsfeed including the first steps.

The first version of the test was heavily influenced by Studydrive's current user flow. Extracting program elements, such as karma points, that could be considered game elements and do not have a direct effect on the usability of the platform, this version's focus was laid on its functionality. In accordance with previously mentioned onboarding stages, it included the user registration, introduction steps and finally a list of tasks that the candidates would be required to carry out.

Composited by the emotional-cognitive view on gamification, the second iteration was built on top of the first one, was enhanced by adding game elements and includes the recommendations outlined in Appendix D. Herewith, a *level system* was introduced, allowing users to collect experience points through being active on the platform, namely posting content, raising questions, or completing challenges and leveling up in the process. In addition to these XP, users would also be rewarded with so-called credits, a platform specific currency, which can be exchanged for real prices. The reward gained for active participation on the platform, or finishing a challenge was set to correlate with the required effort. Eventually reaching level ten

would allow users to earn real Euros on top of XP, a goal introduced at the start of the user test, which could not be obtained within the limitations of the prototype. A *challenge*, providing a clear description of tasks and portraying the rewards, could be unlocked in an iterative way by finishing the previous challenge. Respectively, variable rewards were woven into this mechanic, as the user never knew which challenge would be awaiting her next. Upon reaching certain milestones, users would unlock *badges* that can be inspected on her user profile page.

Figure 6.

Gamified Prototype: Newsfeed including the first steps.



3.1.2 User Selection

Having completed the prototypes, potential candidates were screened out and surveyed. According to several studies, numerous influences including personality traits, intellectual ability and sex are to be regulated, as their effect might influence results on top of a range of induvial differences (Montag & Panksepp, 2017; Brackett et al., 2004; Cahill, 2003). Consequently, a balance between personality traits and sex was attained in user selection. On top of this, to ensure a consistency with the student platform Studydrive and obtain a sample of homogeneous intellectual ability, candidates would have to be university students and no frequent Studydrive users. To account for a well spread-out selection of personality traits and sex, candidates provided personal data, including course of study, age, and MBTI personality type. A total of 26

candidates participated in this pre-screener, out of which four had to be excluded due to insufficient information. 15 of the latter were male and seven females. Ranging from 19 to 30 years, the average age of all participants was 24. Regarding the study-programs, the sample appeared very diverse, including both students from bachelor's and master's, ranging from Architecture and Social Work to International Business and Informatics. While the majority of candidates would select Studydrive as a student platform they have heard of, only one stated to be familiar with the platform and its functions.

Out of the 22 possible candidates a mixed set of 12, based on gender and their personality type were selected and divided into two analogical groups. The two primary characteristics for the selection were an equal distribution between male and female candidates, as well as an equal proportion of the MBTI scale. The selected cohort, including the induvial age, study program, MBTI type and gender are presented in the figure below, which also depicts the distribution between users that would interact with the gamified prototype (Group A) and the non-gamified version (Group B).

Table 1. Main characteristics of the selected candidates

Group	Age	MBTI Type	Gender
A	25	ENFJ	Male
A	25	INFJ	Female
A	25	ISFJ	Female
A	21	ESTP	Female
A	22	INTJ	Male
A	23	INTJ	Male
B	21	ENFP	Female
B	25	INTJ	Male
B	19	ENTP	Male
B	24	INFP	Male
B	23	INFJ	Female
B	24	ISTP	Male

3.1.3 Testing Process

After being selected, the approved candidates entered the second stage of analysis. As can be seen in the overview in Figure 7, the candidates first went through a short pre-test and then proceeded with the user test, solving all provided tasks in the prototype. Followed by a quantitative survey.

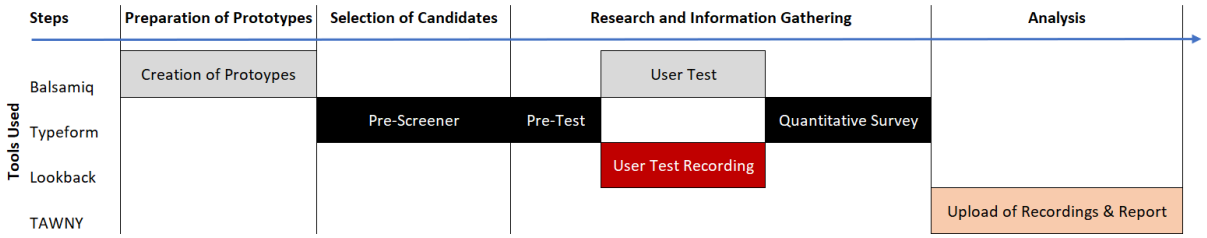


Figure 7. Overview depicting which tools were used for which step of the research

3.2 Measurement and Data Collection

In order to measure the nuanced emotional responses that are elemental to analyze the effectiveness of game elements, the candidates' screen and camera were recorded simultaneously through Lookback. This allowed not only for further in-depth analysis on my side, but also enabled the test to take place in a realistic environment in which a student platform would be accessed, such as the universities' library, or at their desk at home. As proposed by Tyng et al. (2017), the video material was then uploaded to the Emotions Analytics Platform TAWNY, which "analyze[s] human affective states based on state-of-the-art AI algorithms" (TAWNY, 2015).

Having analyzed the recordings, the platform provides multiple tables, representing the affective states during the sessions in numerical values. These states are for one thing, measured and appraised according to the two variables *Valence* and *Arousal*. Valence in this scenario

represents the degree of whether an emotion is positive or negative from 1, positive, to -1 negative. It is important to add that in the version of TAWNY used for this test, a valence level of 0 is to be interpreted in relation to the given data and does not signify an affective state of merely neutral or nonexistent valence (Burgmayr, personal communication, 2021). Arousal on the other hand, explains on a scale from -1 to +1 how intense the emotion is.

Besides, more specific emotions are measured and designated into four categories.

The first, *Sympathy/Happiness*, indicates that the stimulus elicited a state of happiness in the user. Relevant expressions that lead to a designation to this category usually contain a smile. The second category, *Surprise,* can be both positive and negative and usually displays that something happened that was not expected by the candidate. Raising eyebrows and lowering the jaw are indicators for this classification.

The third type is *Focus/Anger* is related to the person getting confused or angry. It can have multiple causes, such as the person not liking or understanding the stimulus, or the task being more demanding. Relevant expressions contain frowning and optionally pressed lips or clenched teeth.

Lastly, *Indifference/Sadness*, is to be interpreted as expressions of indifference, disappointment or even sadness. Notably, testing sessions often illustrate this category when nothing in particular happens, and relevant expressions usually include lowered lip corners. In contrast to the Valence-Arousal scale, the values of the specific emotions, starting at 0 and reaching their maximum at 1.0, do not represent the strength of the given emotional state, but rather how sure the model is (TAWNY, 2015).

After the content is uploaded to TAWNY, the site automatically produces a report that on one side provides already plotted analyses and an export of the raw data. In addition to this qualitative collection of data, the students answered the ten questions of the quantitative SUS.

3.3 Analysis

In this section, to find out whether both Studydrive and further online platform should pursue the implementation of game elements in a structured way, the results of the data analysis will be presented. According to Runkler (2020), data analytics can be separated into four phases. At this point of the paper, the first phase, the *preparation*, which includes data collection and planning, has already been achieved. The last phase, the *postprocessing*, comprising the interpretation, documentation and evaluation will likewise be addressed in another section, namely 3.4. This section will shortly cover the *preprocessing* phase and then present the results of the *analysis*. To answer the first research question, whether the gamified version elicits stronger and more different emotions, the first two variables of implicit UX are considered. As described above in 3.2, valence refers to how negative or positive an emotion is, and therefore explains the variety of elicited emotions. Arousal on the other hand presents the intensity of the emotion felt, so it can be applied to identify the potency of emotions elicited. For basic data arrangement Excel and for more complex computations Statgraphics Centurion 19 and the programing language R were integrated (Statgraphics, n.d.).

3.3.1 Preprocessing of the Data

Preprocessing the given data is necessary because it might contain errors and noise. Two different types of errors, stochastic and deterministic errors, are distinguished. *Stochastic errors*

include measurement errors or random fluctuations that produce noise in the data. While low noise often has little impact on the results, factors influencing the latter need to be addressed. In this case, the video recordings were realized in a non-laboratory setting, therefore, independent distractions and abnormalities in the video recordings are considered. Consequently, the data was alleviated by specific arousal and valence levels that appeared once or twice in the recording throughout all candidates of one group. Additionally, TAWNY would leave cells representing seconds for which no value could be calculated at all empty. These were also eliminated from the table before the analysis. *Deterministic errors* appear when the modeled phenomenon is too complex for the current representation (Yaser et al., 2012). Affective states of humans can be considered a complex phenomenon, consequently an appropriate distribution for the given data will be explored. Additionally, both data sets will be tested for the number of components most appropriate for their distribution. As for the RQ1 both valence, determining how positive or negative an emotion contextualizes and arousal, the intensity of the emotion, are relevant, they will be tested in the upcoming section. When it comes to the explicit usability survey, preprocessing is not essential and both stochastic and deterministic errors can be ignored in this context.

Valence. In a first step, the histograms of both testing groups (Appendix E; Appendix F) were observed. As the shapes of both distributions appear to include two peaks, the models will be tested for their parameter values. Consequently, a selection of likelihood methods is to be applied (Cramer, 1989). To estimate the number of clusters, a selection of the most widely used information criteria, namely the Akaike's Information Criterion (AIC) and the Bayesian Information Criterion (BIC) have been chosen to determine the most suitable representation

(Henson et al., 2007; Akogul & Erisoglu, 2017). For both versions of the prototype, the gamified as well as the control group, two components represent the parameters in the most compelling way (Appendix G).

As described in 3.2, the valence parameter cannot be interpreted absolutely. So, incited by the relativity of its scale, the data of each candidate must be individually standardized, to allow for an unbiased comparison amongst each other. After having been depleted of errors the figures were split up by candidate and consequently standardized via the "STANDARDIZE" function in Excel, representing the z-transformation (Lakshmanan, 2019).

Arousal. As can be seen in Appendices H and I, the emotional intensity data gathered through the variable Arousal, exhibits a multi-modal distribution. Consequently, in accordance with the Valence data, the distributions' modalities were tested via the AIC and the BIC method (Appendix J). Similarly to the first iteration, both tests deemed a bimodal distribution as most likely. In contrast to the relatively generated valence data, arousal is presented in absolute terms, deeming further revision, such as standardizing the information unnecessary. Therefore, no further adaptation needed to be done as part of the pre-processing.

3.3.2 Analysis

Valence Span. Drawing on the standardized data sets of valanced reactions, the dispersity of the sets can be analyzed. For this the valence span, determined by the maximum minus the minimum valence level, was introduced. To allow for proper comparability and analysis the standardized values of the individual candidates' data sets has been used as a source. To receive a more precise and clear answer, the mean of the two groups candidates' minimum and respective maximum

values are first added together, providing the mean minimum and maximum level of the gamified prototype (Min = -2.1598; Max = 4.7916) and the same variables for the non-gamified version (Min = -1.8359; Max = 3.7887). This aggregated mean minimum is then subtracted from the corresponding maximum, presenting a single span value of each group's valence figures. Entailing a difference of 23.59%, the gamified prototype elicited with the value of 6.9513, a larger span than the non-gamified protype which elicited merely 5.6246.

Arousal. In contrast to valence, the arousal figures are delivered in absolute terms and therefore do not require standardization before allowing a viable comparison. To answer the question which prototype overall more intense emotions, the search for the most representative value is relevant. According to Schremmer (2017), both the median and the mean could be options. With the mean being the most popular measurement of central tendency and its downside, extreme outliers, having been cleared from the data set, as depicted in section 3.3.1, it was chosen as the measurement tool. Nevertheless, as can be seen in Appendix K, both the mean and the median both present much higher values for the non-gamified prototype. Taking the mean value, it elicits emotions that are stronger by 96.6931%. Considering that the arousal scale reaches from 0 to 1, both values (Mean of Gamified = 0.0275; Mean of Non-Gamified = 0.0540) are to be categorized as low arousal regardless.

Relative Comparison. Although the isolated figures of the bare valence and arousal figures can provide relevant insights regarding the effects of gamification on the affective states, it is important to recognize the interdependence of the two figures. Valence provides, in an isolated form, the positivity or negativity of emotions elicited. Each second of the recordings receives the

same weight, disregarding the strength of these emotions. Similarly, the arousal solely provides the emotions intensity, disregarding whether it was negative, positive, or neutral. Consequently, a combination of the two is to be considered and examined. For this, the standardized valence figures are matched to the respective arousal values. Similarly, to the isolated figures, both the AIC and the BIC test suggest a bimodal distribution (Appendix L).

A visual representation of the combined distributions has been computed via the RndEM procedure in Statgraphics 19. It randomly selects starting points from data and runs one E-step to obtain the log likelihood. The starting point with the highest log likelihood is then picked as the best choice for Maximum Likelihood Estimations (MLE) in further expectation-maximization (EM) iterations (Maitra, 2009). The resulting frequency histograms can be found in the Appendices M and N and are supplanted by the density fitted distributions below to allow a convenient comparison. Despite the dissimilar numerical ranges for both valence and arousal, obvious differences in density and therefore the frequency of the combined distributions can be pointed out.

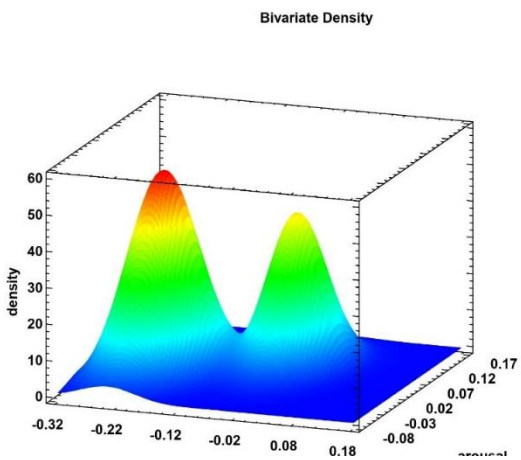


Figure 9. Density fitted distribution depicting arousal and valence of the gamified prototype

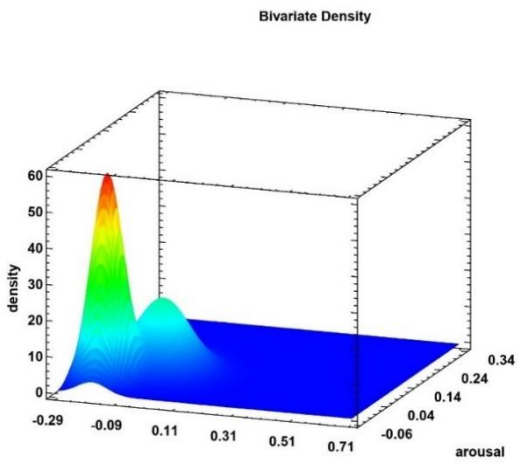


Figure 8. Density fitted distribution depicting arousal and valence of the non-gamified prototype

First, the fact that the isolated valence distribution shows bimodality appears to correlate to the bimodality of the arousal distribution, as two clear pinnacles can be detected in both figures. Corresponding to this relation, the gamified distribution shows a more balanced distribution between cluster of the relatively less positive and less strong emotions and the cluster of relatively more positive and stronger emotions. Overall, the gamified representation appears more dispersed and less unilateral. Nevertheless, the non-gamified graph clearly sits further on the positive side of the arousal scale. It can be concluded that the findings from isolating the isolated metrics have been affirmed by the combined multivariable representation.

Personality differences. Based on the divergent perceptions of affordances from the various personality categories, outlined in section 2.1.2, the variables valence, arousal, sympathy-happiness, focus-anger, surprise, and indifference-sadness were compared by personality trait. The intention was to figure out whether the perceptions applied to gamified online platforms in the same way to other applications and if personality trait specific tendencies occur in this context.

The MBTI four-letter code for each person can vary between two letters per position, so a total of four contrasts were established. The first compared all candidates that fell into the extroversion side of the spectrum to introverts. The second one matched sensing candidates with intuitive ones. The third similarly contrasted feeling individuals against their sensing counterparts and lastly the judging students were opposed to those showing perceiving behavior. The preprocessed data was consequently filtered by personality trait and affective variable for each step, after which the mean of these variables was compared. After statistically verifying the

findings via a t-test, it becomes apparent that the largest difference in responses emerges on the extroversion-introversion scale and valence was the most effected affective figure. The exact figures and respective p-values can be observed in the Appendix O.

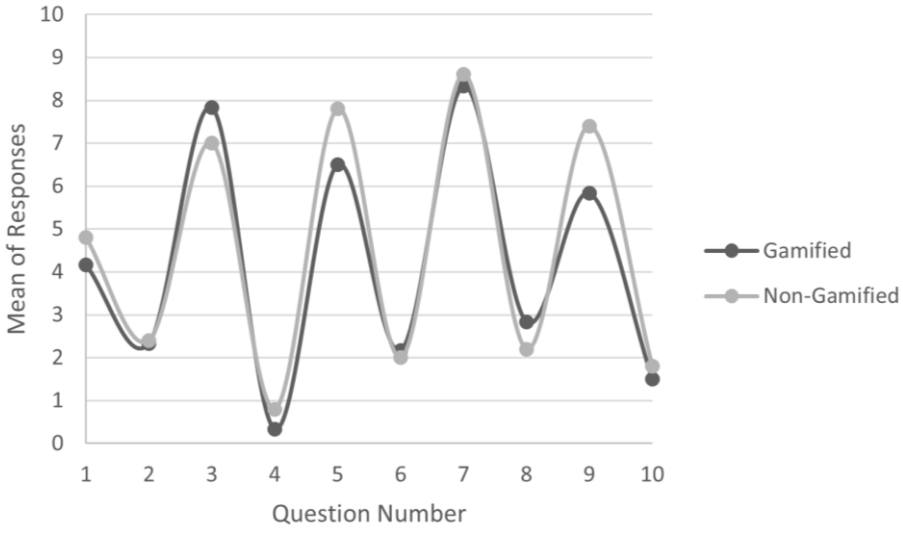
Explicit UX. Consequent to the testing of the prototype, candidates had been confronted with questions determining the prototypes' position on the System Usability Scale (SUS). Examining the arithmetic mean of each group's responses to the question leads to the assumption that no major differences can be detected between the two. To corroborate this a hypothesis test was performed. The two independent responses to each of the ten questions were evaluated via a t-test.

(H0) There is no significant difference in responses to all SUS questions.

(H1) There is a significant difference in responses to all SUS questions.

As can be seen in the Appendix P, all of the questions portrait a p-value of over 0,05, which validifies the null hypothesis and proves the assumption that there is no significant difference between the responses.

Figure 10. SUS - Responses by Prototype



Note. Own illustration based on Appendix P.

3.4 Results

The implicit user experience metrics valence and arousal reveals novel understandings about the gamification of an online community. Supported by the findings from a mixed distribution of the two, analyzing the individual values lead to the following conclusions.

First, the data suggests that, in contrast to previous research on emotions elicited from gamification in a community context, the data from this study disproves that gamification overall produces stronger emotions. The significant difference in average arousal between the data sets even shows that gamification might lead to less intense emotions altogether. With the mean of the gamified prototype equaling 0.0275 and the mean of the non-gamified version lying at 0.0540, both elicited low arousal values, considering the potential range between 0 to 1. Nevertheless, it was shown that gamifying the platform produces a wider range of different affective reactions, which is statistically concluded from the larger valence span and displays visually in the samples' density graphs. The explicit metric on the other hand does not yield varying results for the two groups.

4 Discussion and Conclusion

The following section discusses the findings from section 3.3 and discusses them based on the information obtained throughout this paper. Lastly, the implications on theory and practice are outlined.

4.1 Discussion

Analyzing the emotions elicited through gamification in an implicit and non-invasive way is especially important in the scarcely researched context of an online community. Interestingly, certain scientifically prevalent assumptions could be confirmed, while others were negated. Firstly, it was authenticated that gamifying a community platform does even when tested implicitly, elicit a wider range of emotions. This finding especially supports the assumption that gamification can increase cognition of the user, namely having a positive influence on the attention, decision making, the acquisition of knowledge and more. The overall less strong emotional reaction elicited by the gamified version in comparison to the control group can be determined subordinate, as both arousal values appear very low. Applying the discovery of the positivity offset, allows the conclusion that the gamified version, having more positive valence than the non-gamified version, elicits a more welcoming reaction in users during the onboarding process. Other interesting findings are the correlations between personality and affective response to the stimuli. Based on the comparison of the four personality categories, a relative and an absolute comparison shows that the extroversion-introversion scale delivers the largest differences between valence, arousal and residual emotions. Furthermore, the effect has shown by far strongest in the valence category.

Nevertheless, certain limitations might have influenced the results of this study. Following the recommendations made by UX theory and Design Thinking, a low fidelity prototype had been used for the testing. As the prototyping is to take place before the launch of a product, real conditions of usage are not considered (Pressman, 2019). A fully functional prototype on the other hand could have delivered improved results by increasing the likelihood of creating an immersive and engaging experience for the users and pertain to psychological and social needs more directly than the limited version of it. A limited possibility of implementable mechanics also restricted the programs, such as social stimuli and situations of decision making, lead to only capturing a snapshot of emotions that could have been expressed (Pessoa, 2013; Phelps, 2006). Most importantly, the sample size of merely 12 candidates (n=12) might have led to certain statistical biases. While the results' meaning was approved through statistical hypothesis testing, a definite probability of uncertainty is undoubtedly an elemental part of the outcome. Moreover, the choice of merely consulting a behavioral personality test was limiting the propositions. While including the benefits of the latter, a cognitive personality overview could have provided a completely new set of insights to the study. A last elemental limitation is that the positive effect of the varied emotional experience is based on findings derived in different contexts. While it can be assumed that the same processes will unfold in the context of an online platform, it would be interesting to explore them in the particular instance.

4.2 Implications for Practice

For Studydrive, a clear recommendation for the implementation of game elements to the platform can be made based on the findings of this study. The increased valence span confirms the assumption that it will enhance the overall cognition of users by activating their prefrontal

cortex. Furthermore, applying the positivity-offset, onboarding new students to the platform can be composed in a more welcoming way through gamification. Nonetheless, this does not mean that it should be implemented instantly. On one side, Studydrive needs to call the downsides of imprecisely composed game elements into attention. As much as 80% of gamification attempts are said to fail, and there is no reason to believe that Studydrive with its 1.7 million active users would be an exception. On the other side through the limitations of this general study it can merely be verified that further research on this topic would be rewarding. Before the student platform decides to invest further resources into this matter, it should be aware of the fact that negative emotions, such as stress, frustration and anxiety are a fundamental part of effective gamification. If this does not interfere with the values and vision of the site, the following steps should be taken to enable a compelling implementation.

Future work might continue to use the integration of the OCC model with the MDE-framework to calibrate the platform's mechanics attentively. The impact of the latter on cognition should be measured in targeted research and the same should be done to make sure misuse of these mechanics, such as posting spam messages to the platform, are prevented. This could be achieved through a peak analysis of affective states in combination with the measurements of cognitive performance. Within this whole process, the significance of personality traits should not be underestimated and at least differences between extroverts and introverts are to be considered.

The general finding that gamifying a platform has the potential to yield an improved affective state in the user can be transferred to other applications as well. Nevertheless, the individuality of each platform needs to be accounted for. For example, the limitation of the users, all currently studying and made up of a small selection of ages, needs to be mentioned. Since

effects of emotion on cognition vary in different populations, it is advised that future research of platform operators verifies the results with their own user base (Bennion et al., 2013).

4.3 Implications for Theory

Based on the discussed findings and limitations, the study suggests further research to be done on the implementation of gamification to online communities. In particular, the discovery of the positivity offset motivates more detailed examination of a gamified onboarding process. Most importantly though, the effect on cognition by the promising emotional responses yielded within this study should be examined. Lastly, when working with affective data, such as valence or arousal, scholars should consider testing their data set on bimodality, as this might unveil a more accurate interpretation.

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Appendix

Appendix A: Suggestion of game-related tendencies and mechanics

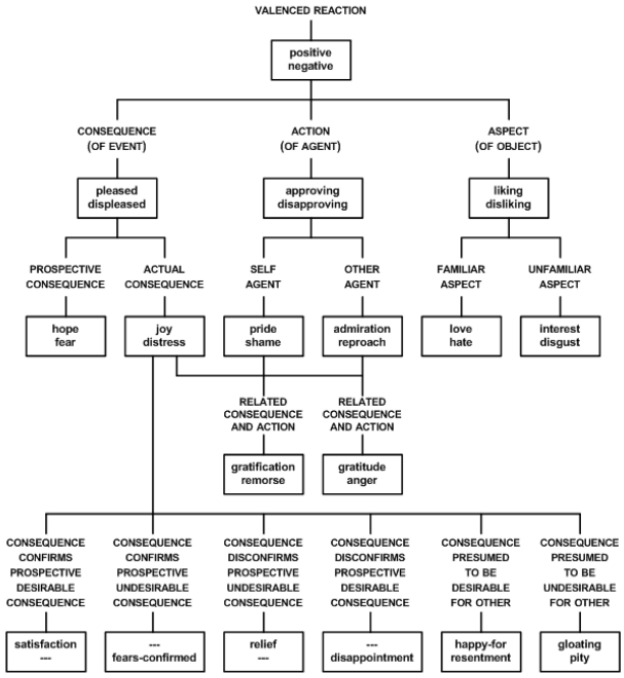
PERSONALITY	TENDENCIES	CORRESPONDING MECHANICS
Extroverts	• Talk more than listen • Think out loud • Act, then think • Like to be around others • Prefer to do many things simultaneously • Be attuned to their external environment • Learn best through doing or discussing	• Voice or face-to-face interaction • Multiplayer (competitive or cooperative) • Opportunities to multitask (arcadestyle engagement) • Avatar customization • Quick/intuitive play
Introverts	• Listen more than talk • Think inside their head • Think, then act • Feel comfortable being alone • Prefer to focus on one thing at a time • Not take action • Be attuned to their inner world • Prefer to communicate in writing	• Text chat, or no interaction • Single-player • Clearly identified goal to pursue • Story arcs/lore • Deep rule sets with optimization opportunities
Sensors	• Focus on details and specifics • Admire practical solutions • Notice details and remember facts • Live in the here-and-now • Trust actual experiences • Like step-by-step instructions • Work at a steady pace	• Few but meaningful choices • Resource management • Real-time challenges • Repeatable sequences of events • Gradually “grindable” progression
Intuitives	• Focus on the big picture and possibilities • Admire creative ideas • Notice things that are new or different • Trust their instincts • Prefer to learn new skills • Like to figure things out for themselves • Work in bursts of energy	• Open-ended or branching decisions • Plot-twists, exploration • The context must make sense • Enable progression along multiple paths, and horizontal progression • Infrequent but long sessions should be rewarding

PERSONALITY	TENDENCIES	CORRESPONDING MECHANICS
Thinkers	• Make decisions based on logic and evidence • Be direct • Appear to be reserved • Be convinced by rational arguments • Value honesty and fairness • Take few things personally • Be motivated by achievement	• Puzzles to solve • Sports, contests, etc. • Awards/badges, public ranks/titles • Leveling up, objective chains
Feelers	• Make decisions based on their values and feelings • Appear to be friendly • Be convinced by how they feel • Be tactful • Value harmony and compassion • Take things personally • Compliment others	• Situations to deal with/figure out • Caretaking/support roles • Negotiation, political activities
Judgers	• Make decisions easily • Pay attention to time • Finish projects • Work first and play later • See the need for rules • Make a plan and stick to it • Find comfort in schedules	• Numerous decisions to make • Timers and deadlines • Collections, progress tracking • Deep, unintuitive rule sets • Min/maxing opportunities • Long, difficult progression
Perceivers	• Have difficulty making decisions • Be less aware of time • Start projects • Play first, work later • Keep their options open • Question the need for many rules • Keep plans flexible • Be spontaneous	• Sandbox style play • No restrictions/penalties • Variety of activities • Simple rules (if any) • Unexpected events, “Easter eggs” • The mechanics themselves should create enjoyment, not as a means to an end

Appendix B: Emotion type specifications

positive and *negative* are valenced reactions (to "something")
pleased is being *positive* about a consequence (of an event)
displeased is being *negative* about a consequence (of an event)
hope is being *pleased* about a prospective consequence (of an event)
fear is being *displeased* about a prospective consequence (of an event)
joy is being *pleased* about an actual consequence (of an event)
distress is being *displeased* about an actual consequence (of an event)
satisfaction is *joy* about the confirmation of a prospective desirable consequence
fears-confirmed is *distress* about the confirmation of a prospective undesirable consequence
relief is *joy* about the disconfirmation of a prospective undesirable consequence
disappointment is *distress* about the disconfirmation of a prospective desirable consequence
happy-for is *joy* about a consequence (of an event) presumed to be desirable for someone else
resentment is *distress* about a consequence (of an event) presumed to be desirable for someone else
gloating is *joy* about a consequence (of an event) presumed to be undesirable for someone else
pity is *distress* about a consequence (of an event) presumed to be undesirable for someone else
approving is being *positive* about an action (of an agent)
disapproving is being *negative* about an action (of an agent)
pride is *approving* of one's own action
shame is *disapproving* of one's own action
admiration is *approving* of someone else's action
reproach is *disapproving* of someone else's action
gratification is *pride* about an action and *joy* about a related consequence
remorse is *shame* about an action and *distress* about a related consequence
gratitude is *admiration* about an action and *joy* about a related consequence
anger is *reproach* about an action and *distress* about a related consequence
liking is being *positive* about an aspect (of an object)
disliking is being *negative* about an aspect (of an object)
love is *liking* a familiar aspect (of an object)
hate is *disliking* a familiar aspect (of an object)
interest is *liking* an unfamiliar aspect (of an object)
disgust is *disliking* an unfamiliar aspect (of an object)

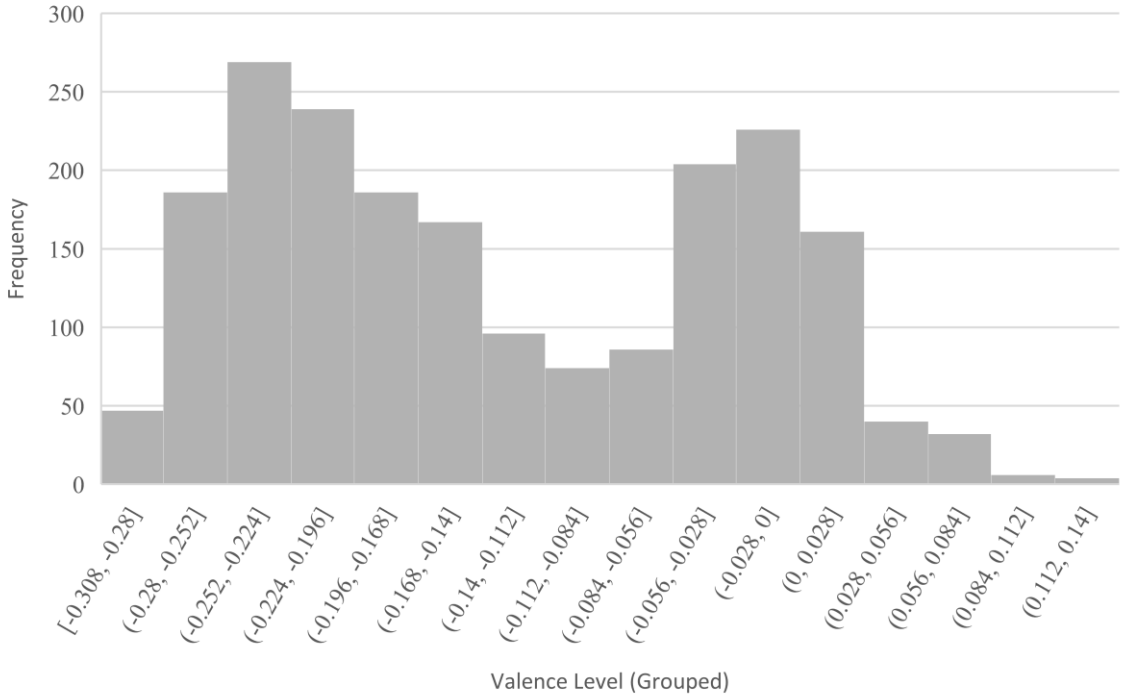
Appendix C: Disambiguated, inheritance-based hierarchy of emotions of the OCC model.



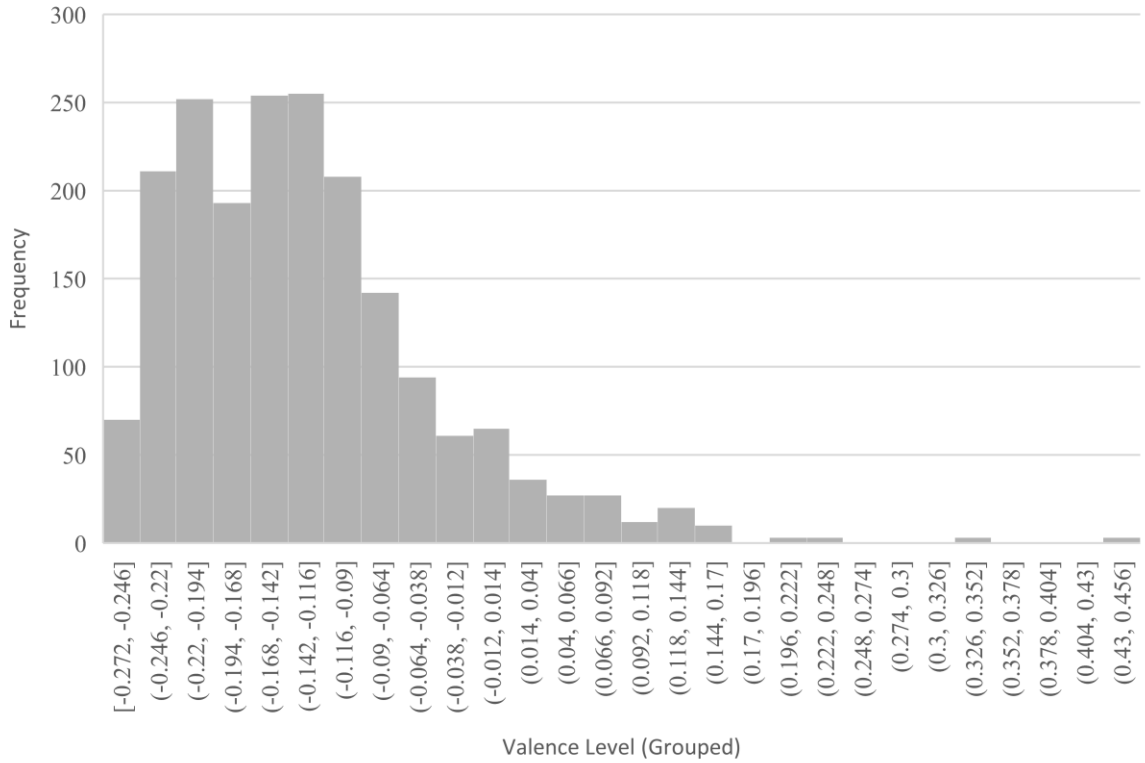
Appendix D: List of Studydrive's recommendations for further gamification research.

Recommendations based on findings from second research approach
Explore options for more types of challenges.
Make less-favorite challenges more attractive in terms of rewards.
Explore design options to reduce number of steps to activate the challenge.
Explore options for pre-joined challenges
Explore conceptual options in cross linking the rewards, e.g. having badges for reaching certain levels.
Reconsider time limits for each challenge.
Explore design options to display the challenges.
Consider adding a text explanation [to challenges].
Explore options to adjust or add surprise and uncontrollable elements such as the weekly challenges, extra XP when levelling up, etc.
Consider increasing the rewards [for flashcard related challenges].
Consider auto-saving function on of flashcards on the web.
Explore design options to encourage users to mark an answer "best answer".
Explore with the devs for notification options, while considering it's just an MVP.

Appendix E: Histogram of grouped valence levels obtained from the gamified prototype.



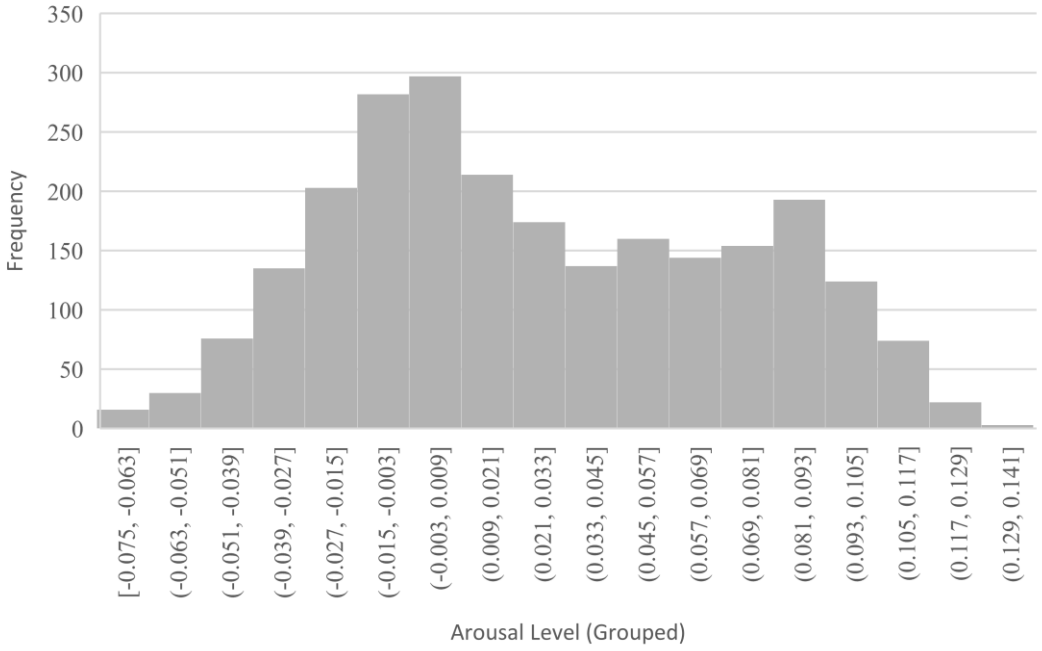
Appendix F: Histogram of grouped valence levels obtained from the non-gamified prototype.



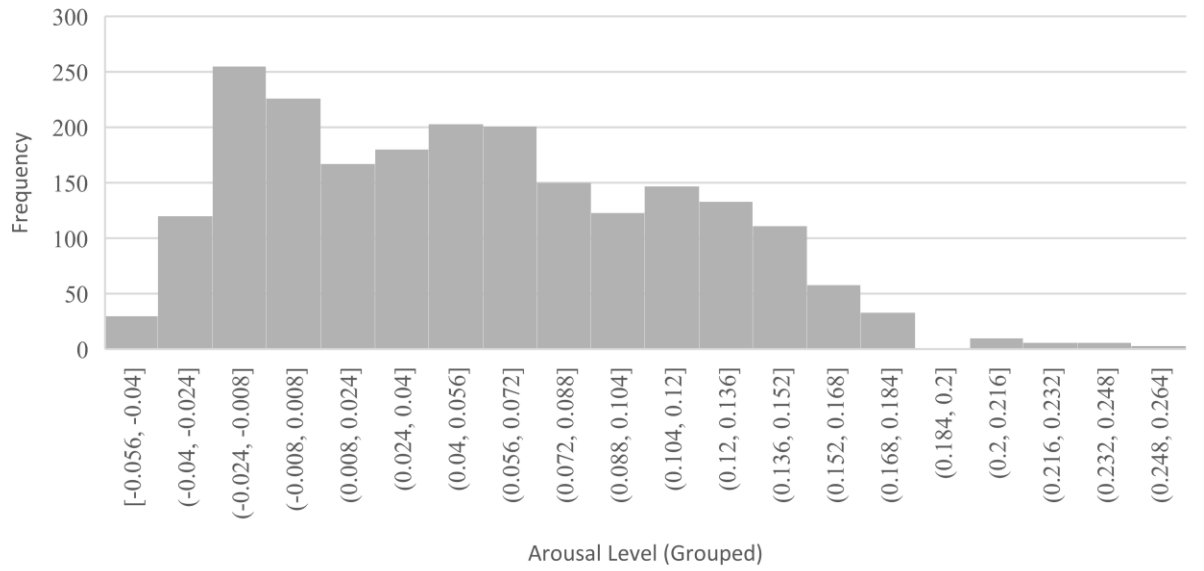
Appendix G: Quality test of models – Valence.

	Gamified		*Non-Gamified*	
Number of components	***AIC***	***BIC***	***AIC***	***BIC***
1	-3472.34	-3461.11	-3772.98	-3761.83
2	-4207.02	-4178.96	-4166.87	-4138.99

Appendix H: Histogram of grouped arousal levels obtained from the gamified prototype.



Appendix I: Histogram of grouped arousal levels obtained from the non-gamified prototype.



Appendix J: Quality test of models – Arousal.

	Gamified		***Non-Gamified***	
Number of components	***AIC***	***BIC***	***AIC***	***BIC***
1	-8160.25	-8148.65	-6030.82	-6019.46
2	- 8554.16	-8525.16	-6351.73	-6323.33

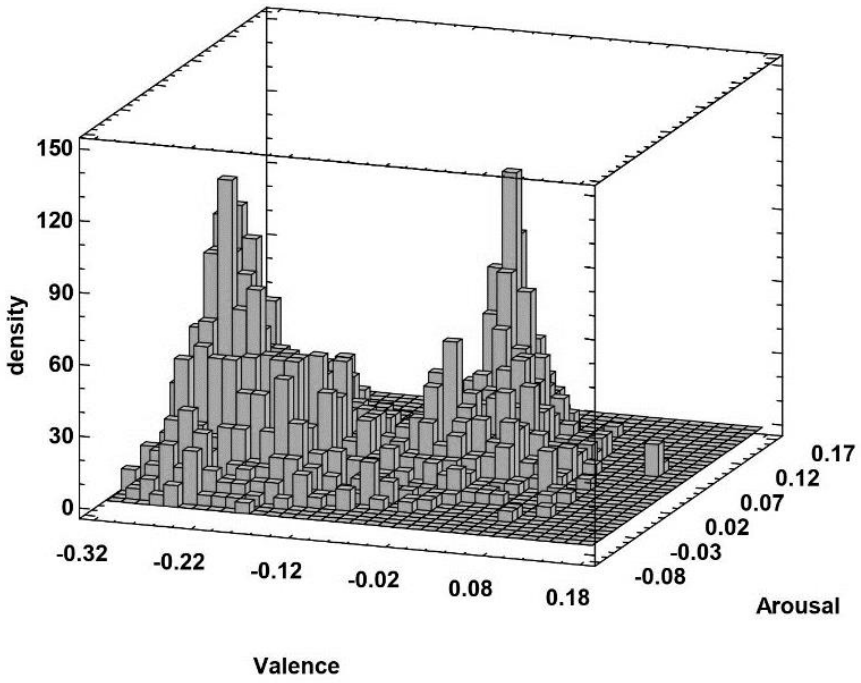
Appendix K: Mean and Median of the respective arousal values.

Prototype	***Mean***	***Median***
Gamified	0.027474159	0.0195
Non-Gamified	0.054039778	0.048

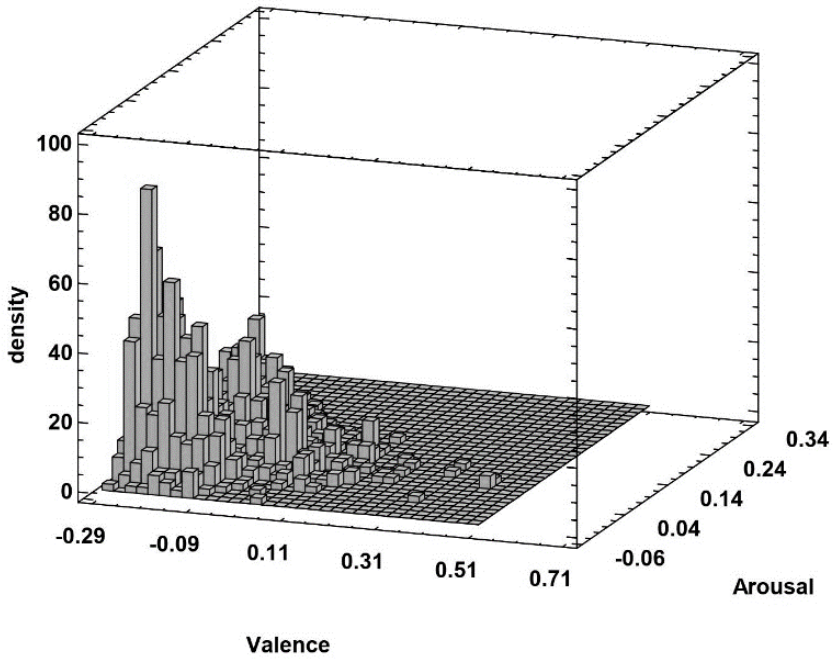
Appendix L: Quality test of models – Arousal and Valence combined.

Number of components	***Log likelihood***	***AIC***	***BIC***
Gamified			
1	636.665	-1263.33	-1234.96
2	944.055	-1866.11	-1803.7
Non-Gamified			
1	-72.6144	155.229	182.759
2	217.948	-413.896	-353.329

Appendix M: Histogram of grouped arousal and valence levels obtained from the gamified prototype.



Appendix N: Histogram of grouped arousal and valence levels obtained from the non-gamified prototype.



Appendix O: Mean of personality-specific affective variables, including their p-values.

Personality	Valence	Arousal	Sympathy-Happiness	Focus-Anger	Surprise	Indifference-Sadness
Extrovert	0.29876252	0.02855535	0.03130914	0.02184016	0.01424593	0.22169256
Introvert	0.13593244	0.04030706	0.02173515	0.03962084	0.00731701	0.14560681
P-Value (E-I)	4.48E-06	9.83E-18	1.2846E-05	0.01013558	0.00418884	1.62E-09
	0.13593244	0.04030706	0.02173515	0.03962084	0.00731701	0.14560681
Intuition						
Sensing	0.05509398	0.035907	0.02244834	0.05320827	0.00904691	0.1566514
P-Value (N-S)	1.13E-09	3.15E-07	0.4832068	6.64E-23	5.09E-10	0.53867922
	0.13593244	0.04030706	0.02173515	0.03962084	0.00731701	0.14560681
Feeling						
Thinking	0.18860935	0.04283135	0.02150706	0.04540924	0.00796983	0.13952004
P-Value (F-T)	1.07E-57	2.14E-66	0.72859265	5.74E-28	4.25E-15	0.84820079
	0.11481549	0.02899801	0.02299623	0.02313711	0.00810134	0.15688284
Judging						
Perceiving	0.23085198	0.04576068	0.02624773	0.04971885	0.00966283	0.17461596
P-Value (J-P)	5.99E-09	5.28E-68	0.24083513	1.673E-06	1.04E-08	7.77E-16

Appendix P: Statistical significance of the divergence of responses to the SUS questionnaire

Question	Question Number	P-Value
I think that I would like to use this system more frequently.	1	0.5169
I found the system unnecessarily complex.	2	0.952
I thought the system was easy to use.	3	0.5019
I think that I would need the support of a technical person to be able to use this system.	4	0.579
I found the various functions in this system were well integrated.	5	0.1866
I thought there was too much inconsistency in this system.	6	0.8666
I would imagine that most people would learn to use this system very quickly.	7	0.7772
I found the system very cumbersome to use.	8	0.6384
I felt very confident using the system.	9	0.3349
I needed to learn a lot of things before I could get going with this system.	10	0.8107

Attachment

Attachment 1: Gamified Prototype




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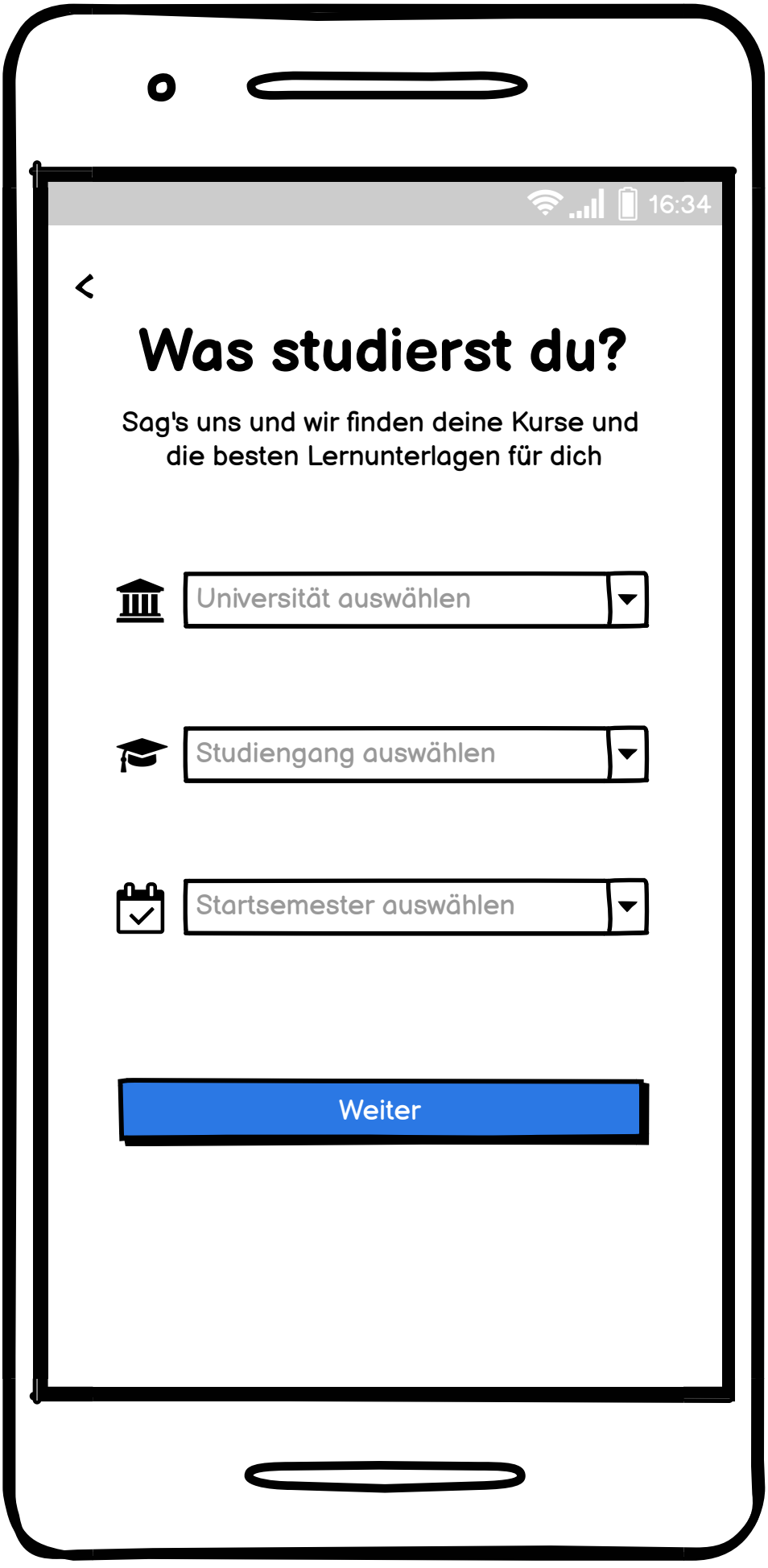

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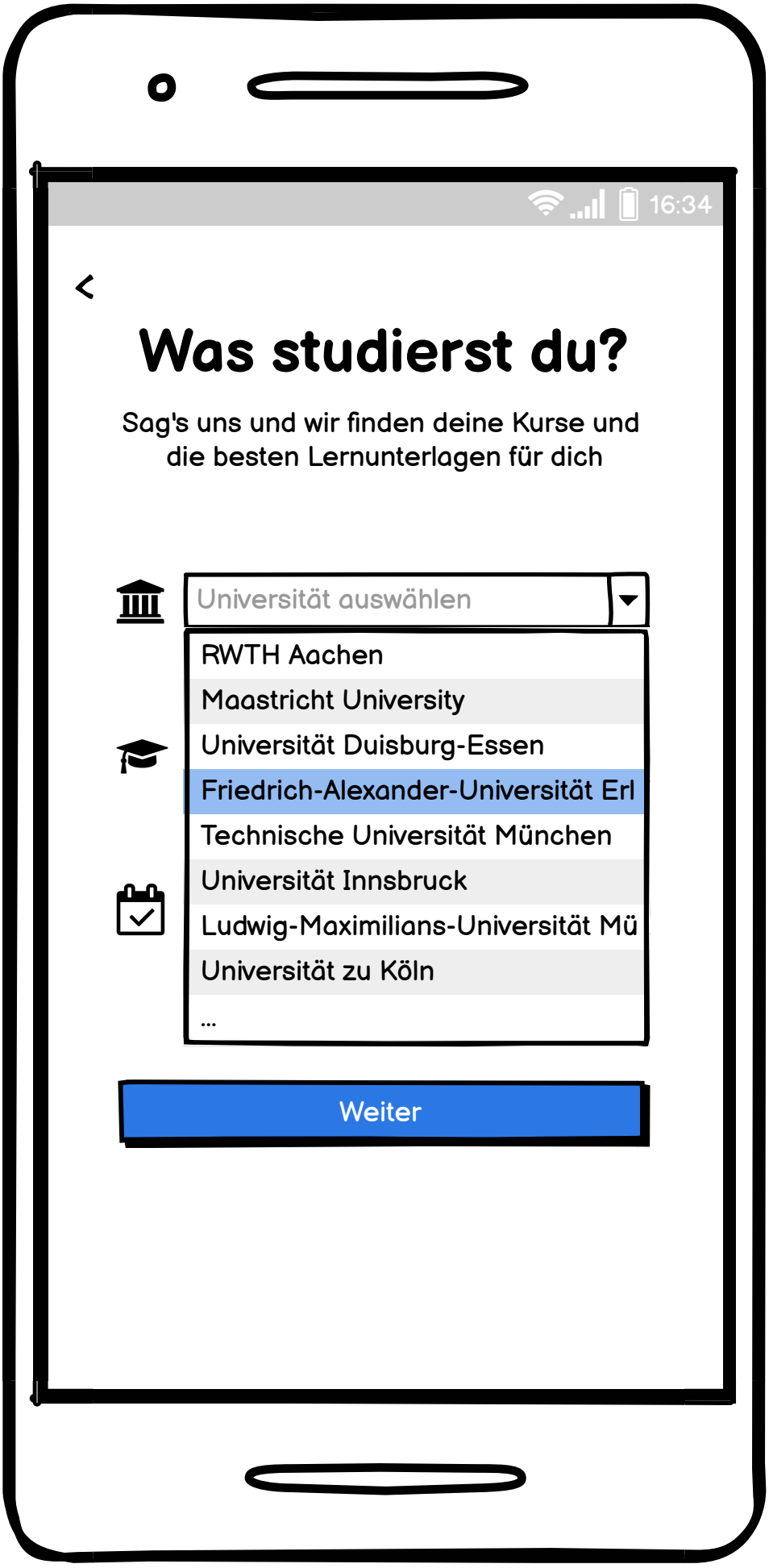

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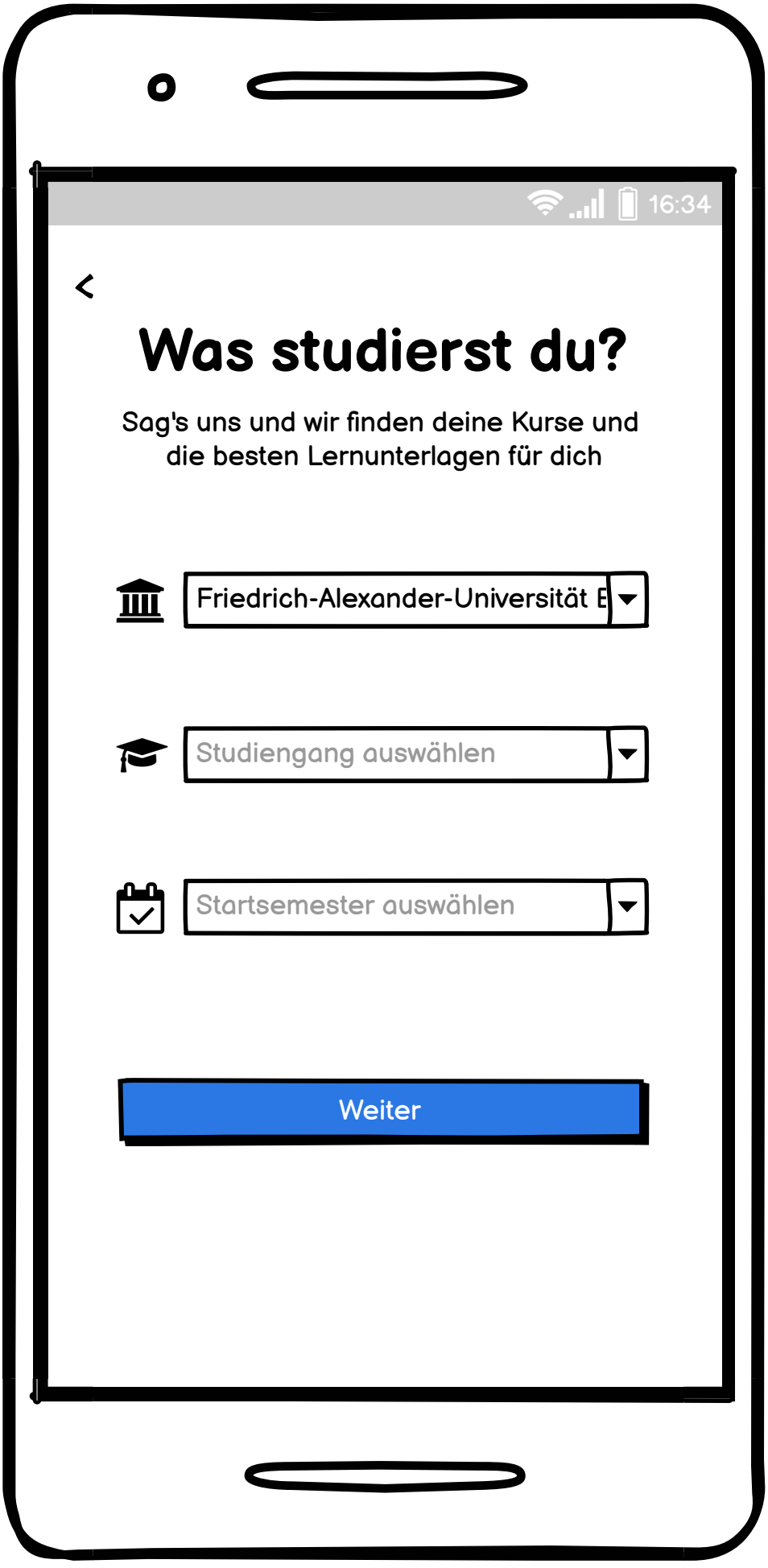

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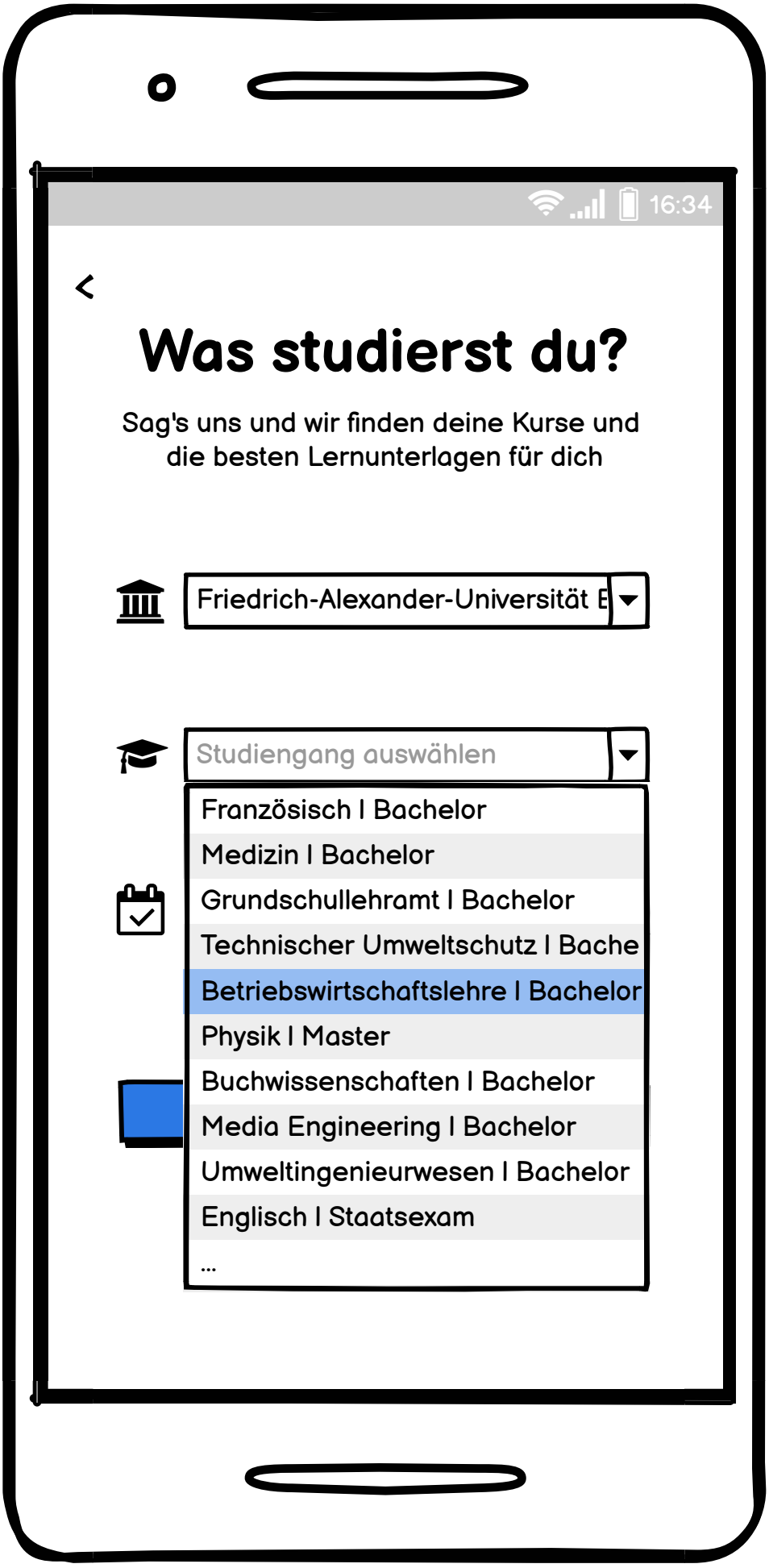

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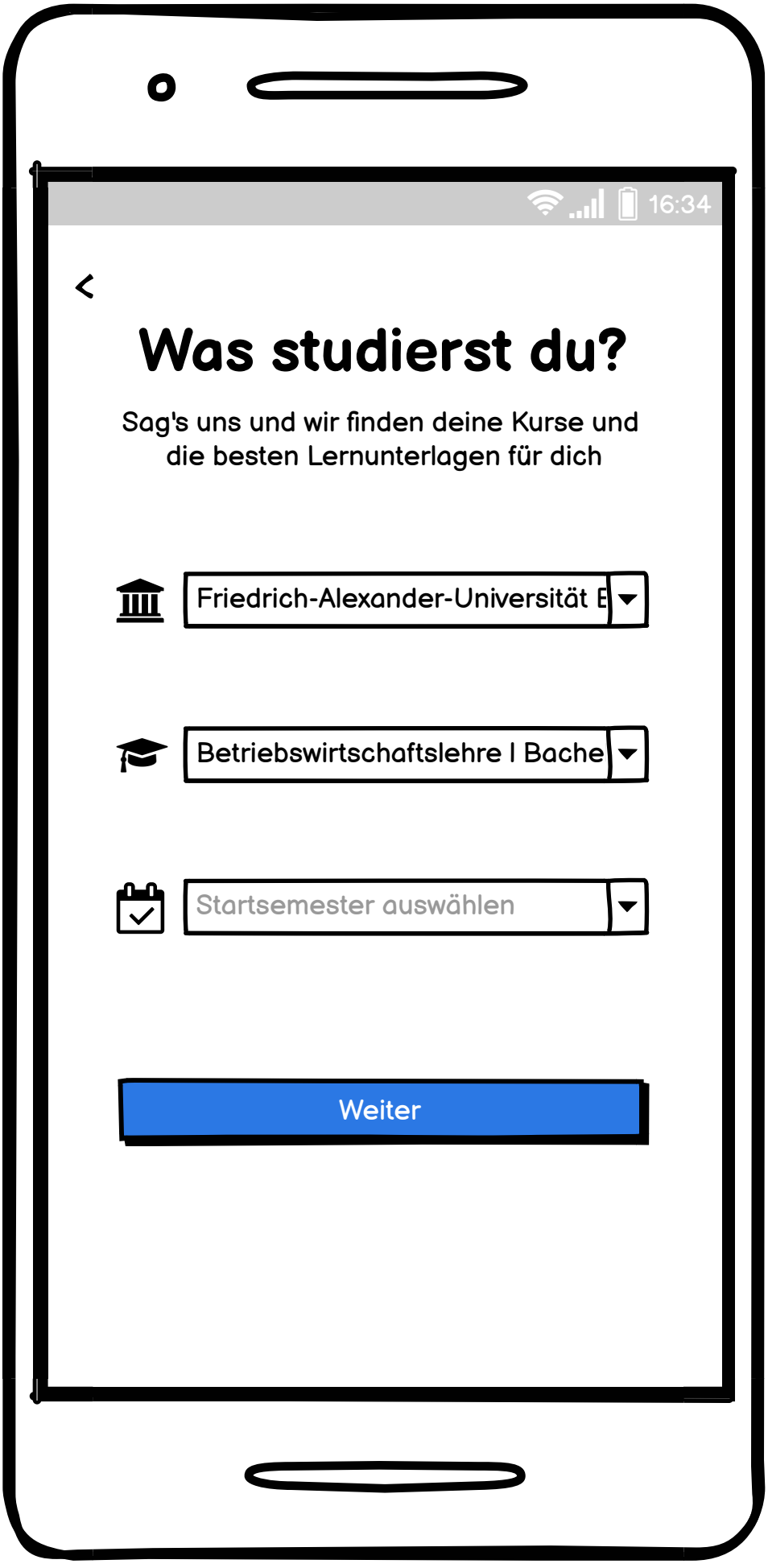

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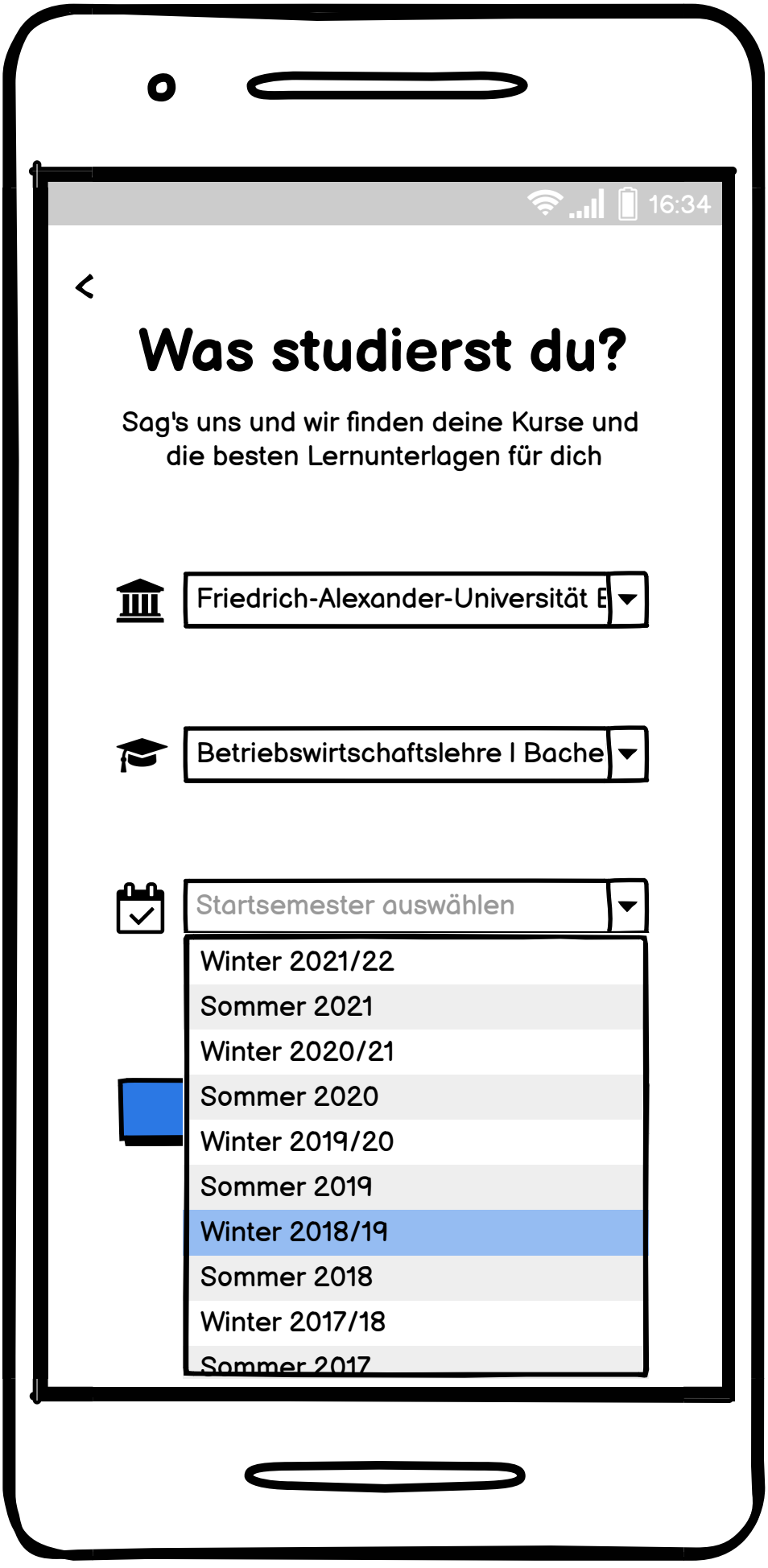

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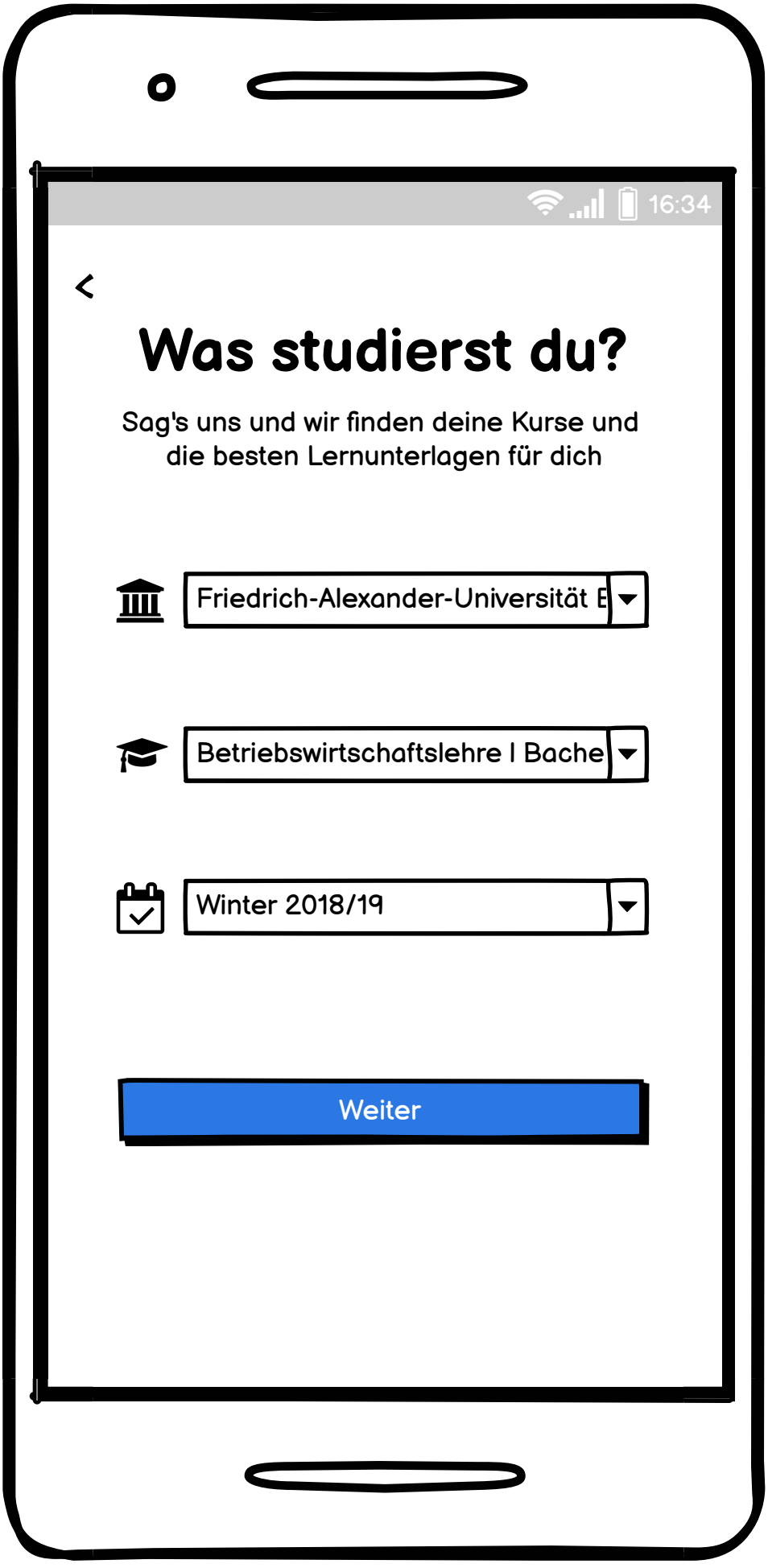

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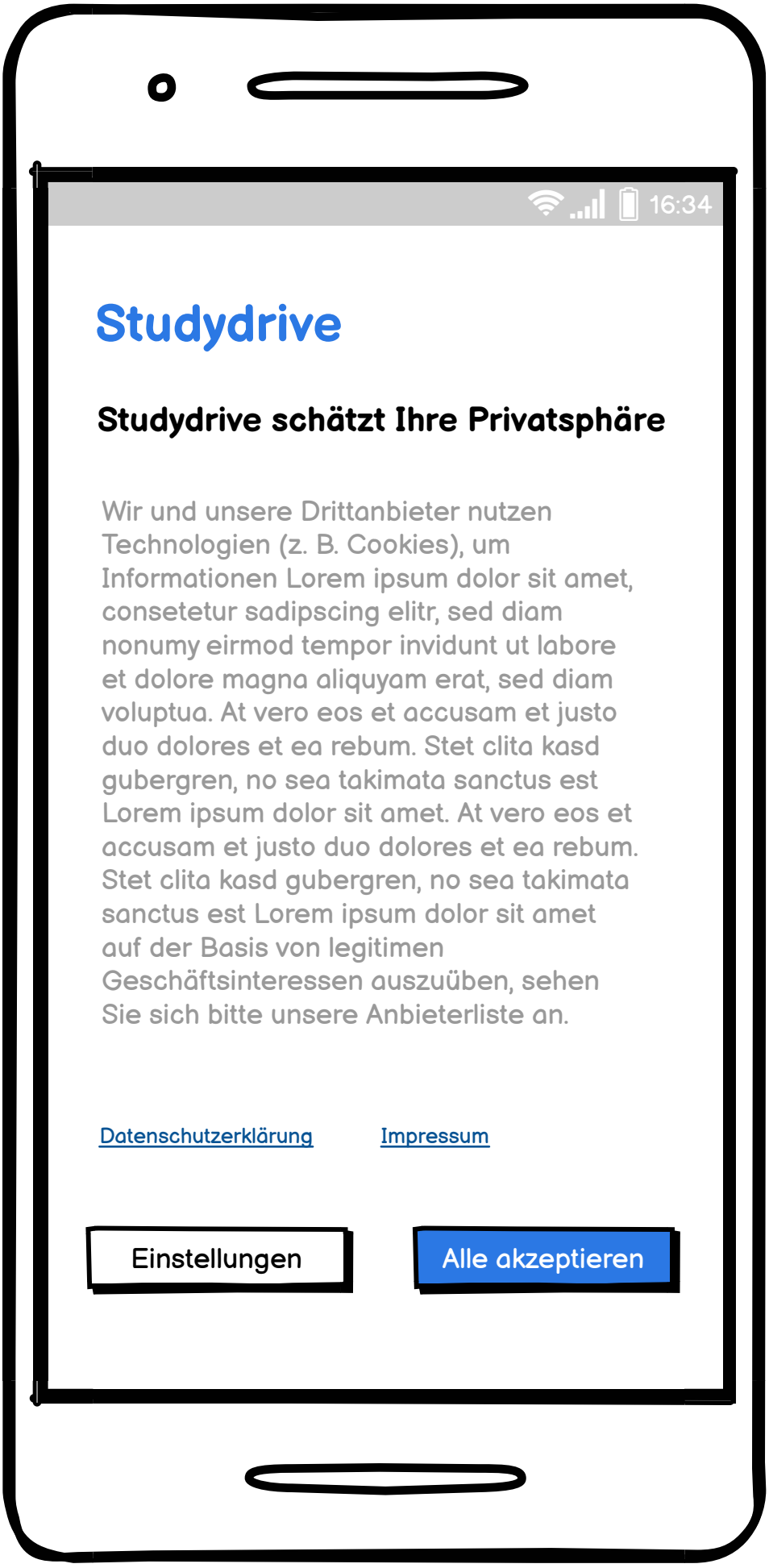

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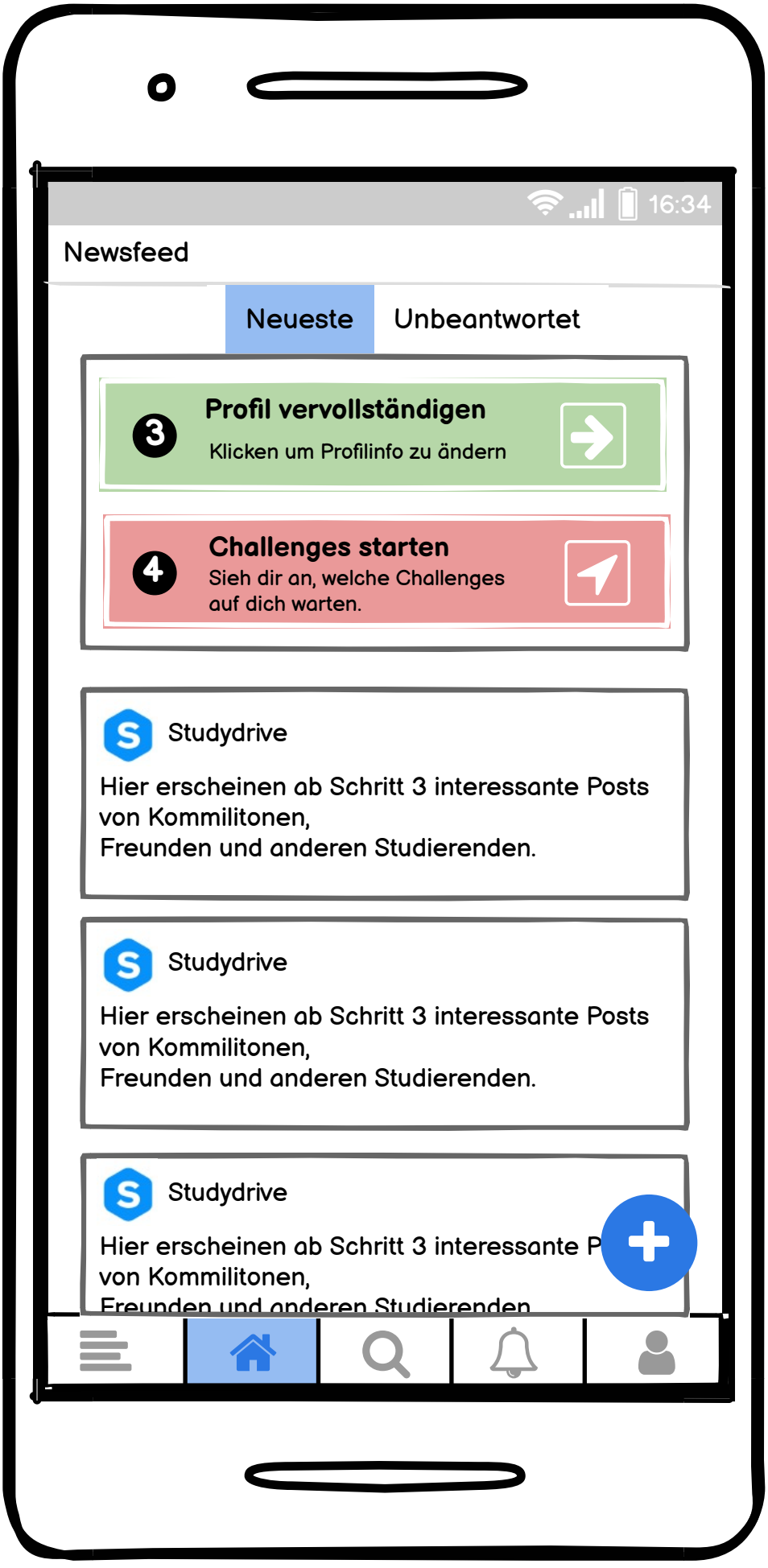

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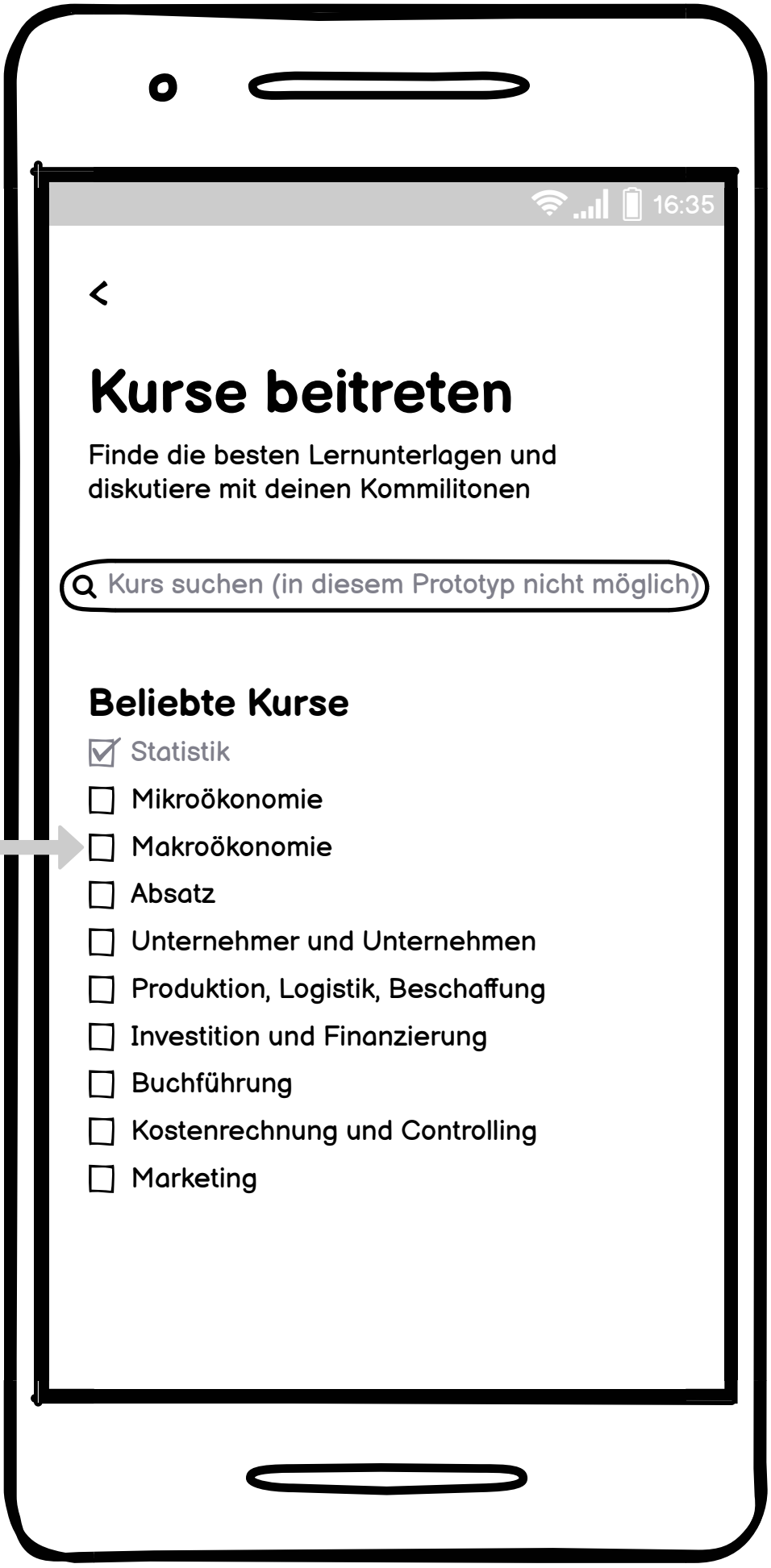


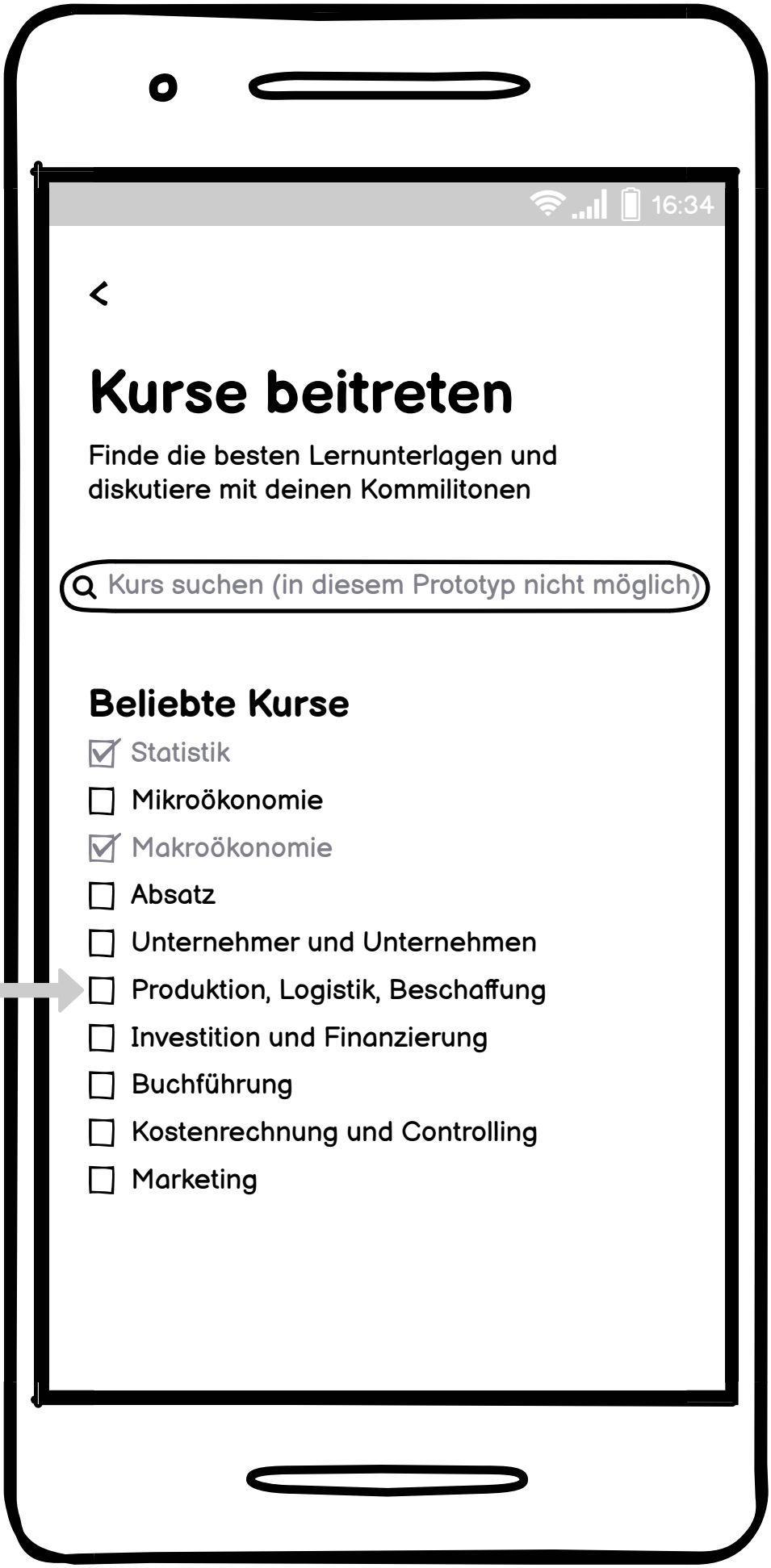
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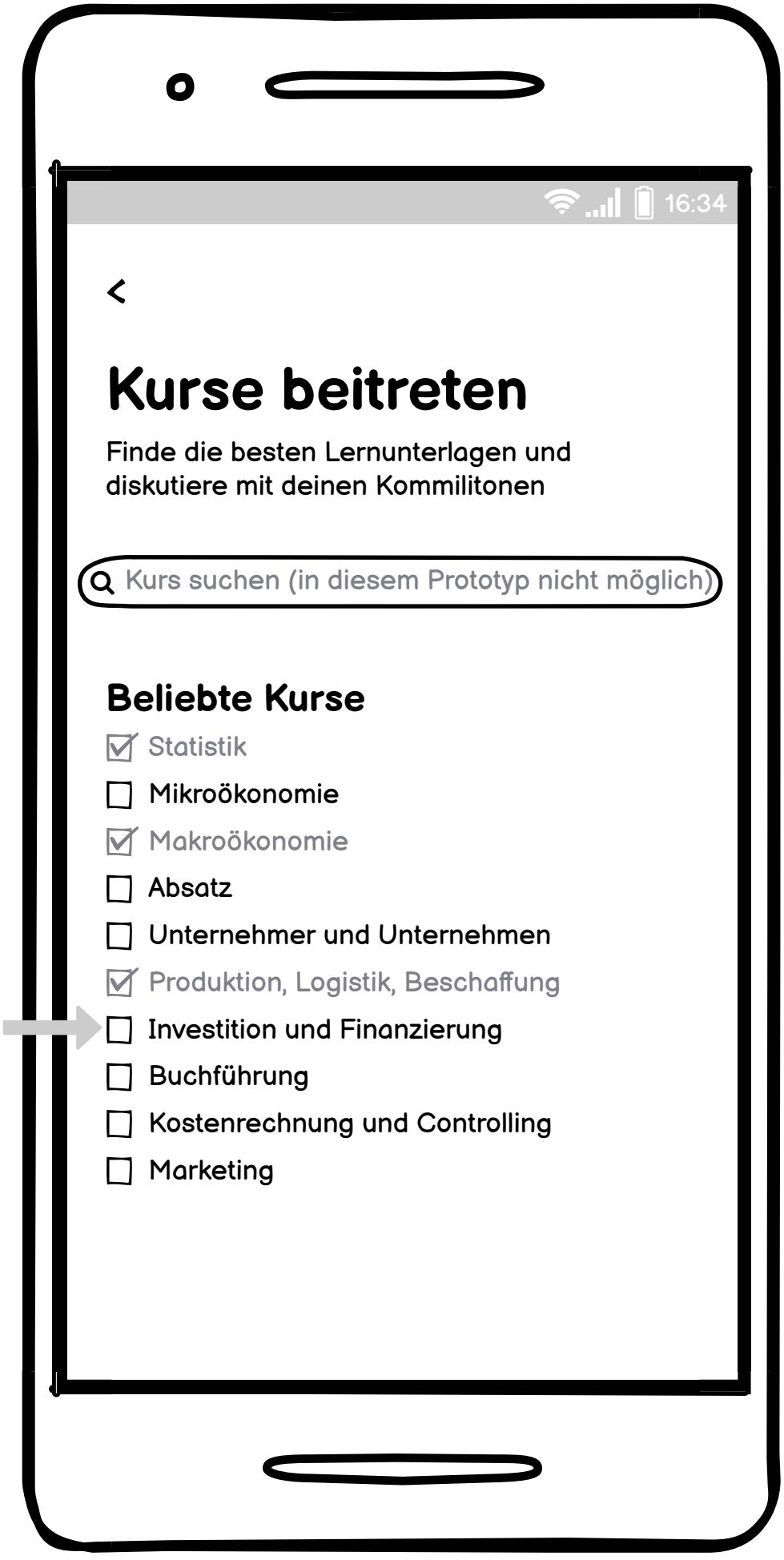


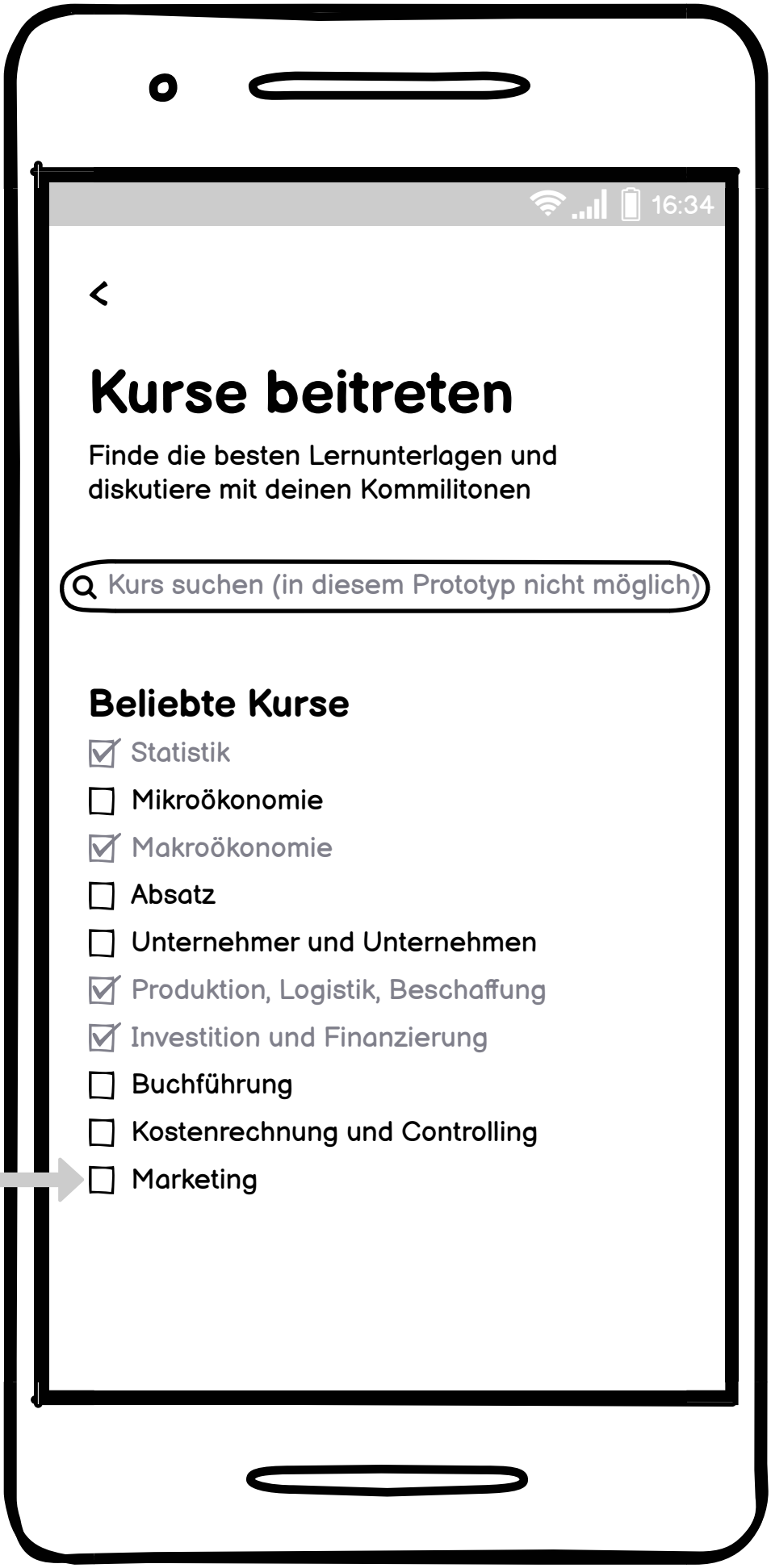
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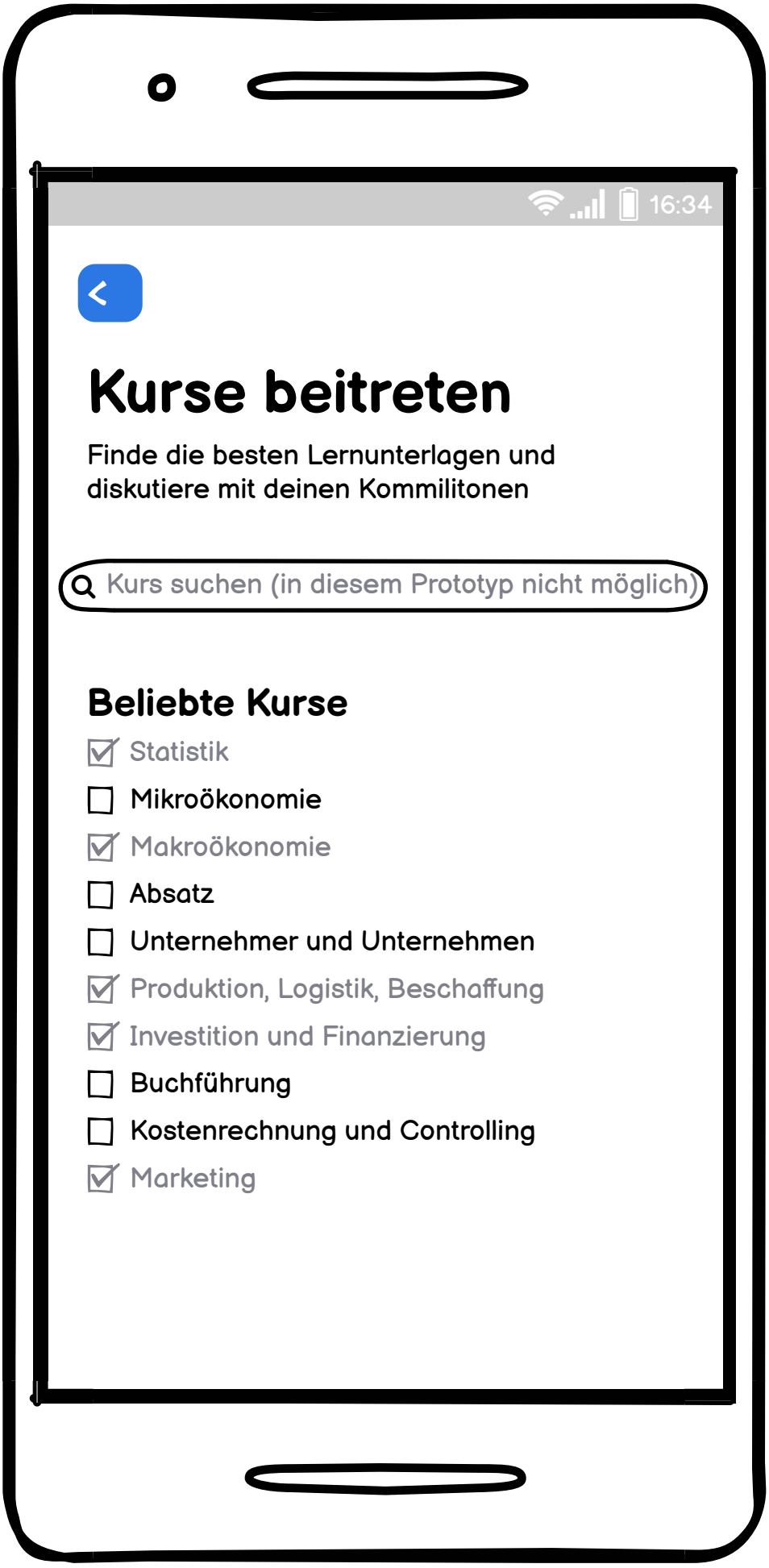

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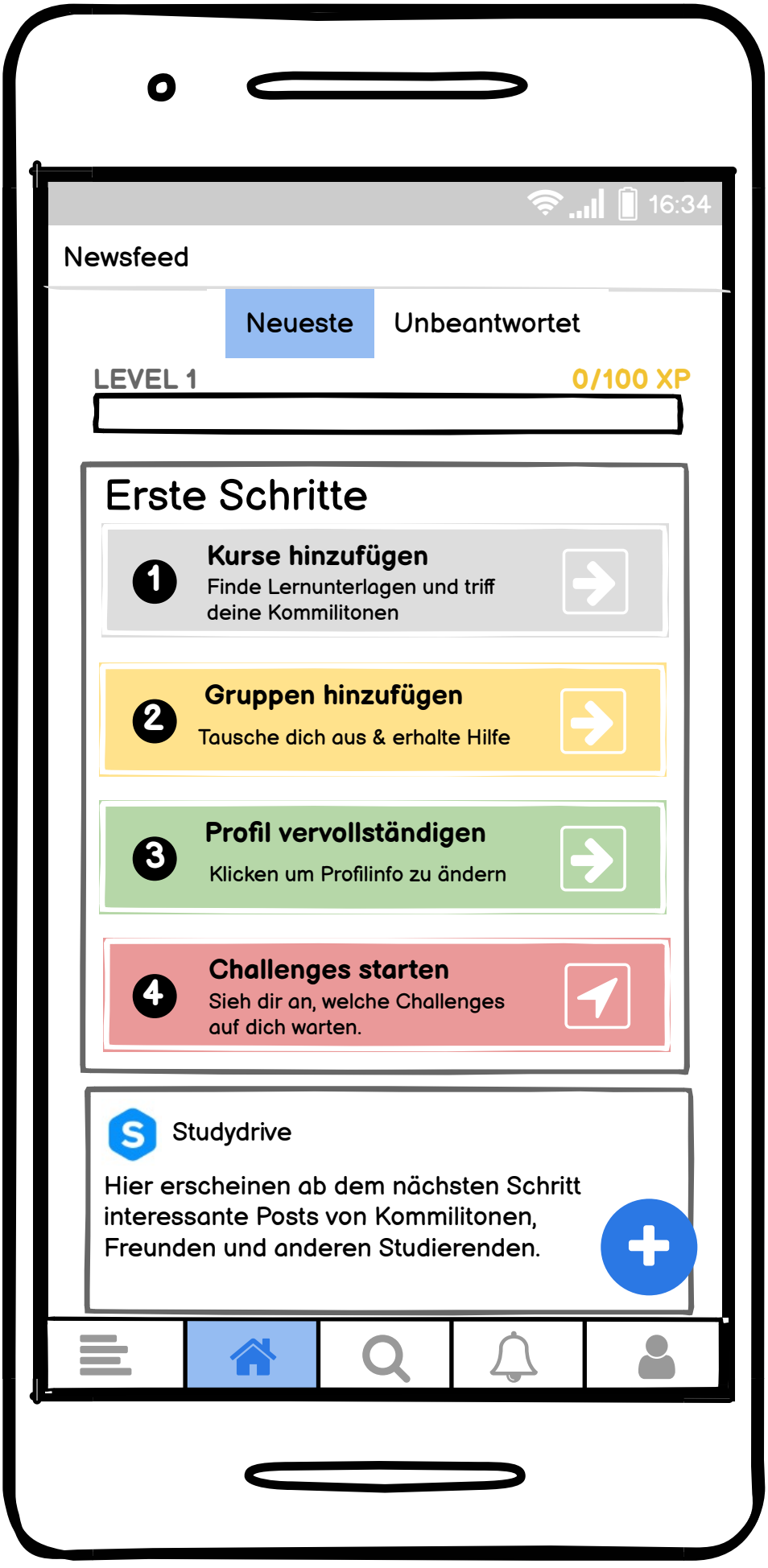

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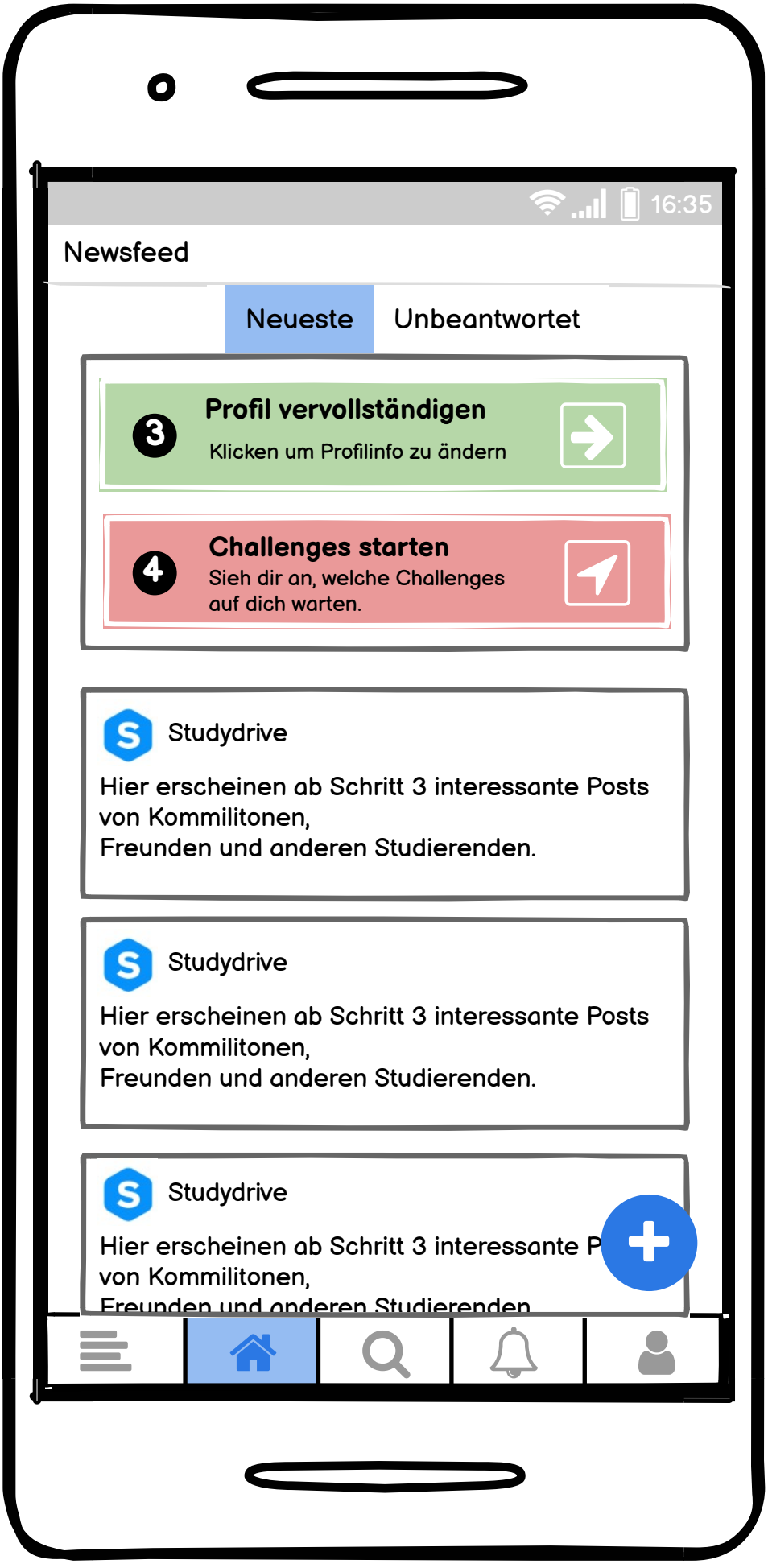

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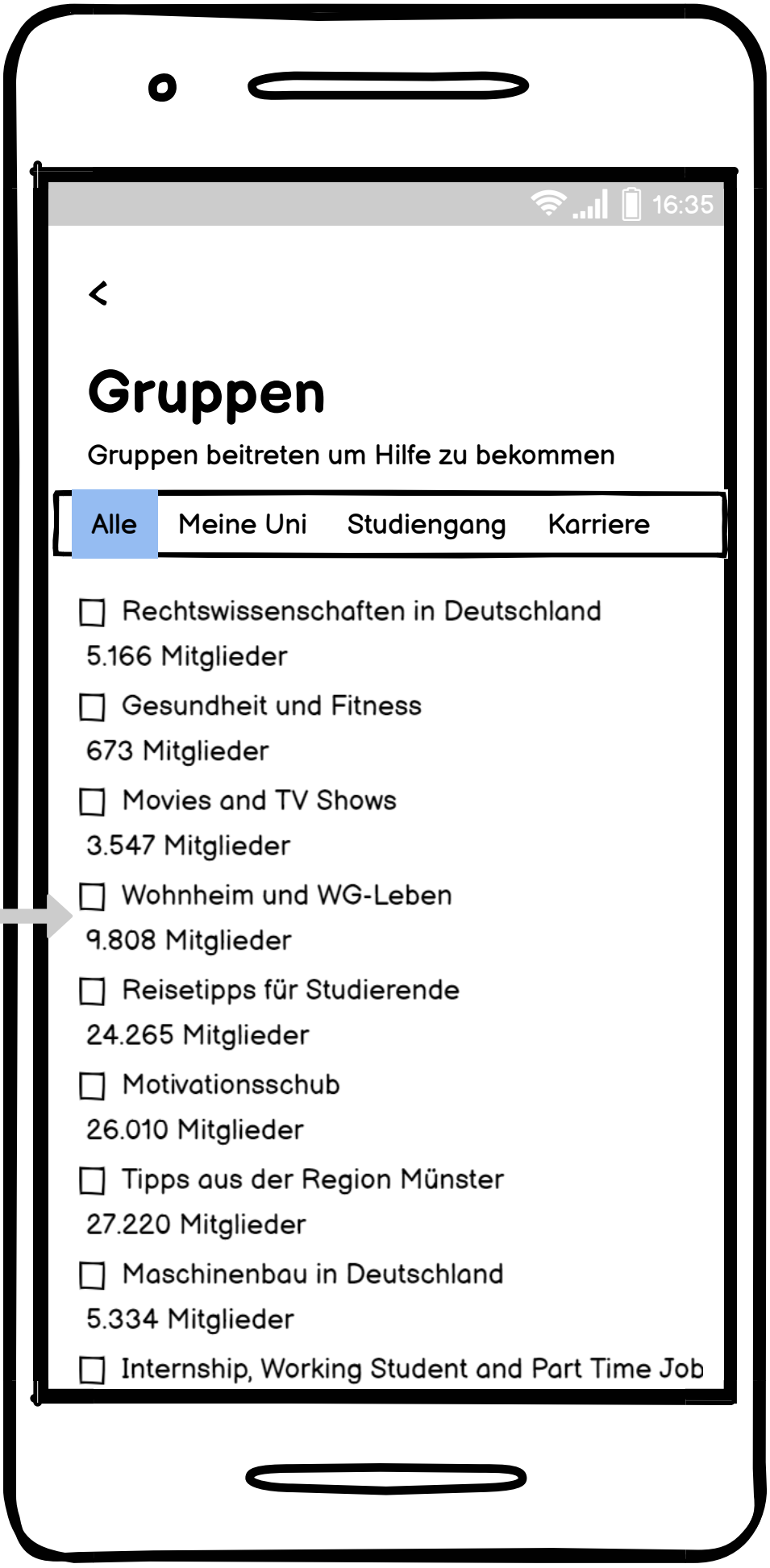

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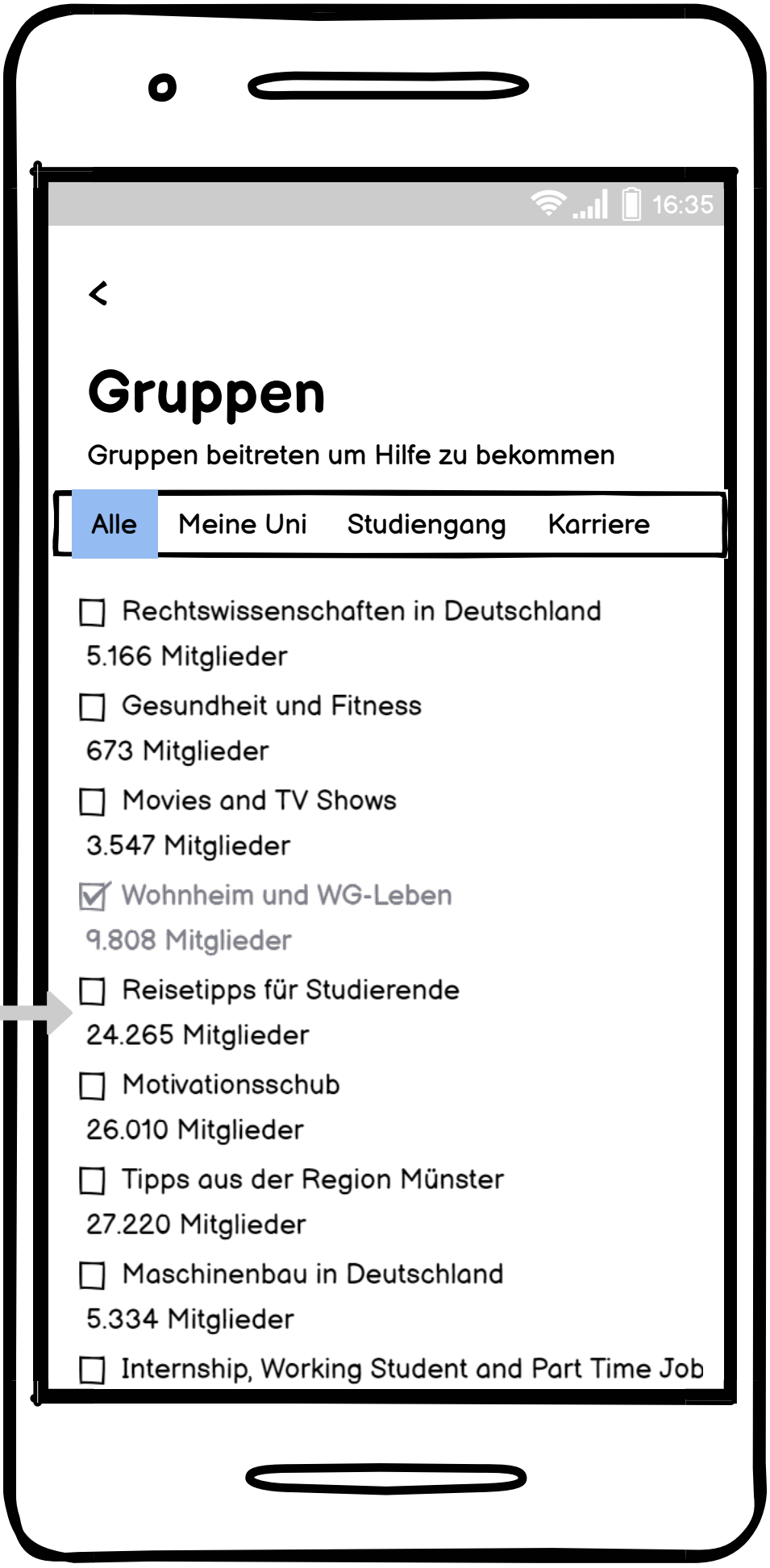


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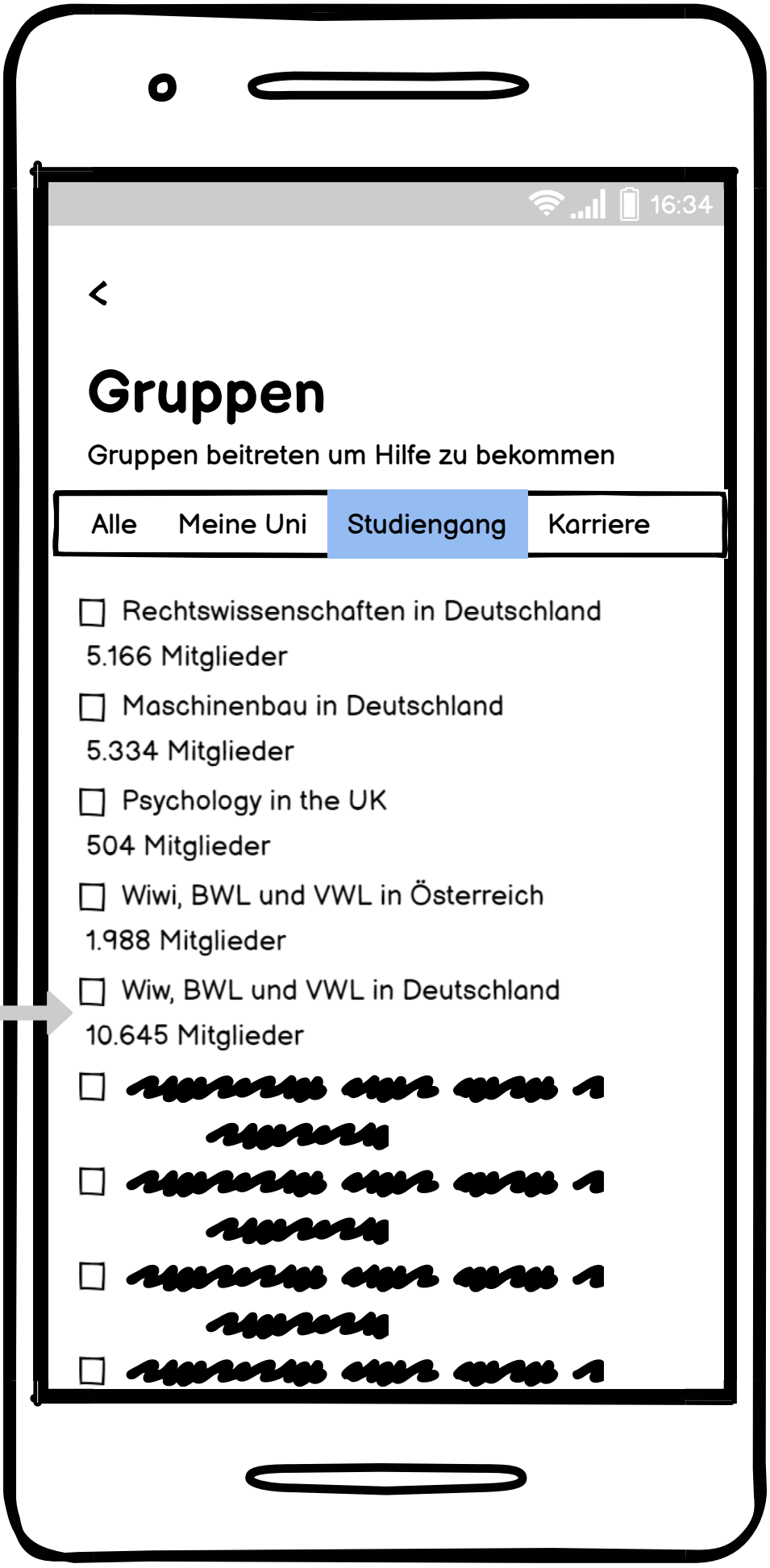
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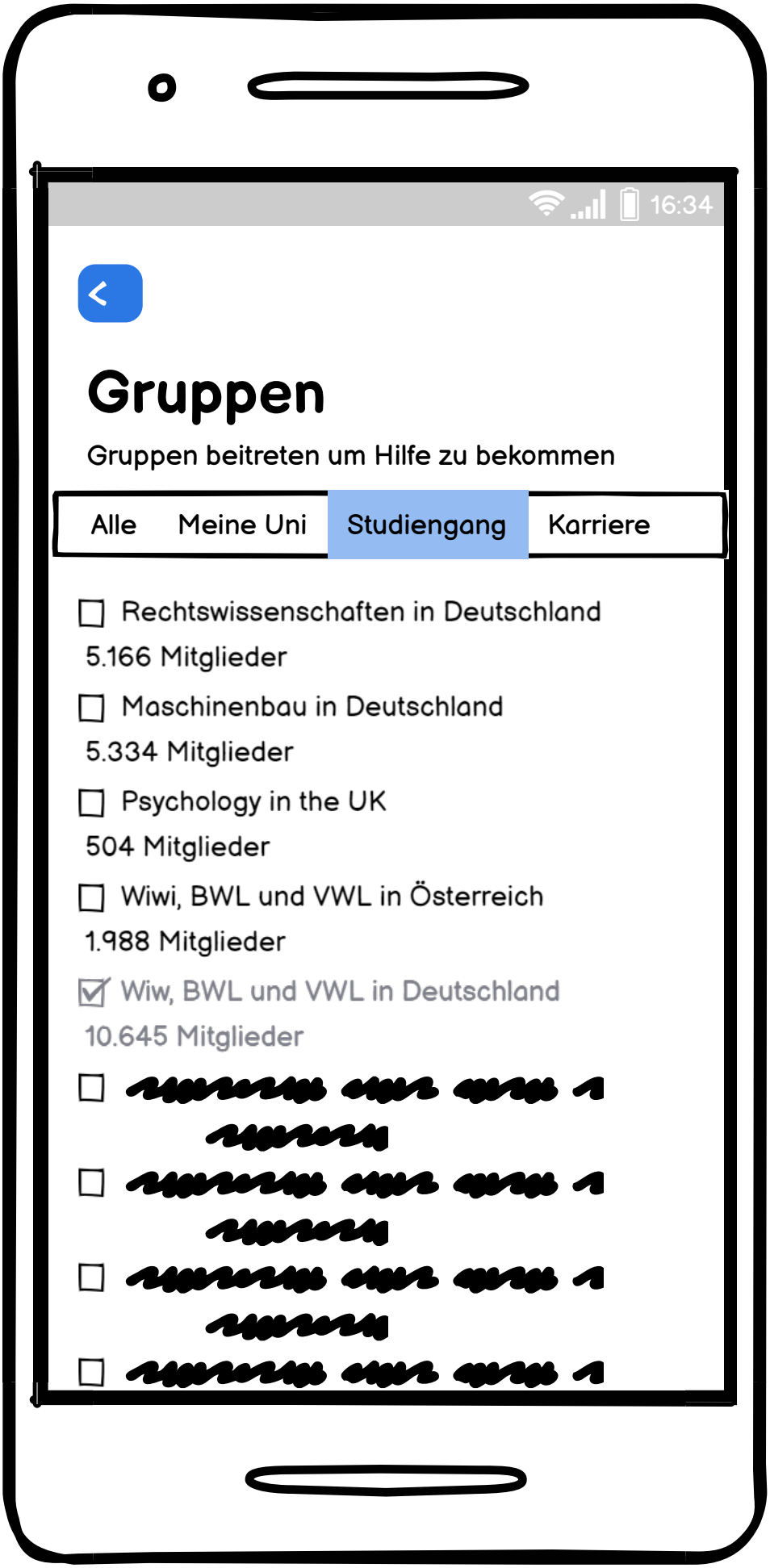

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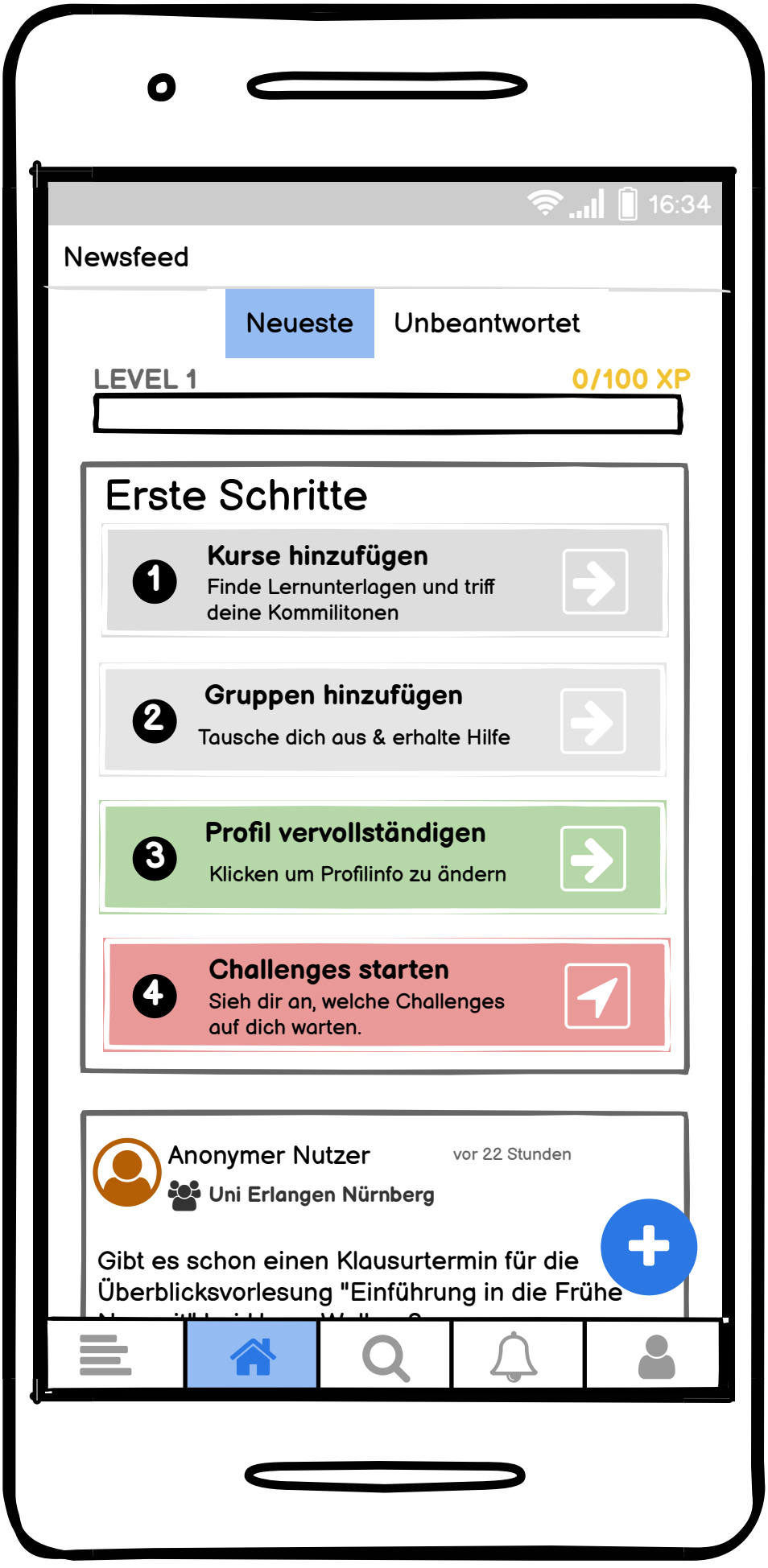

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504 Mitglieder
Wiwi, BWL und VWL in Österreich
1.988 Mitglieder
Wiw, BWL und VWL in Deutschland
10.645 Mitglieder



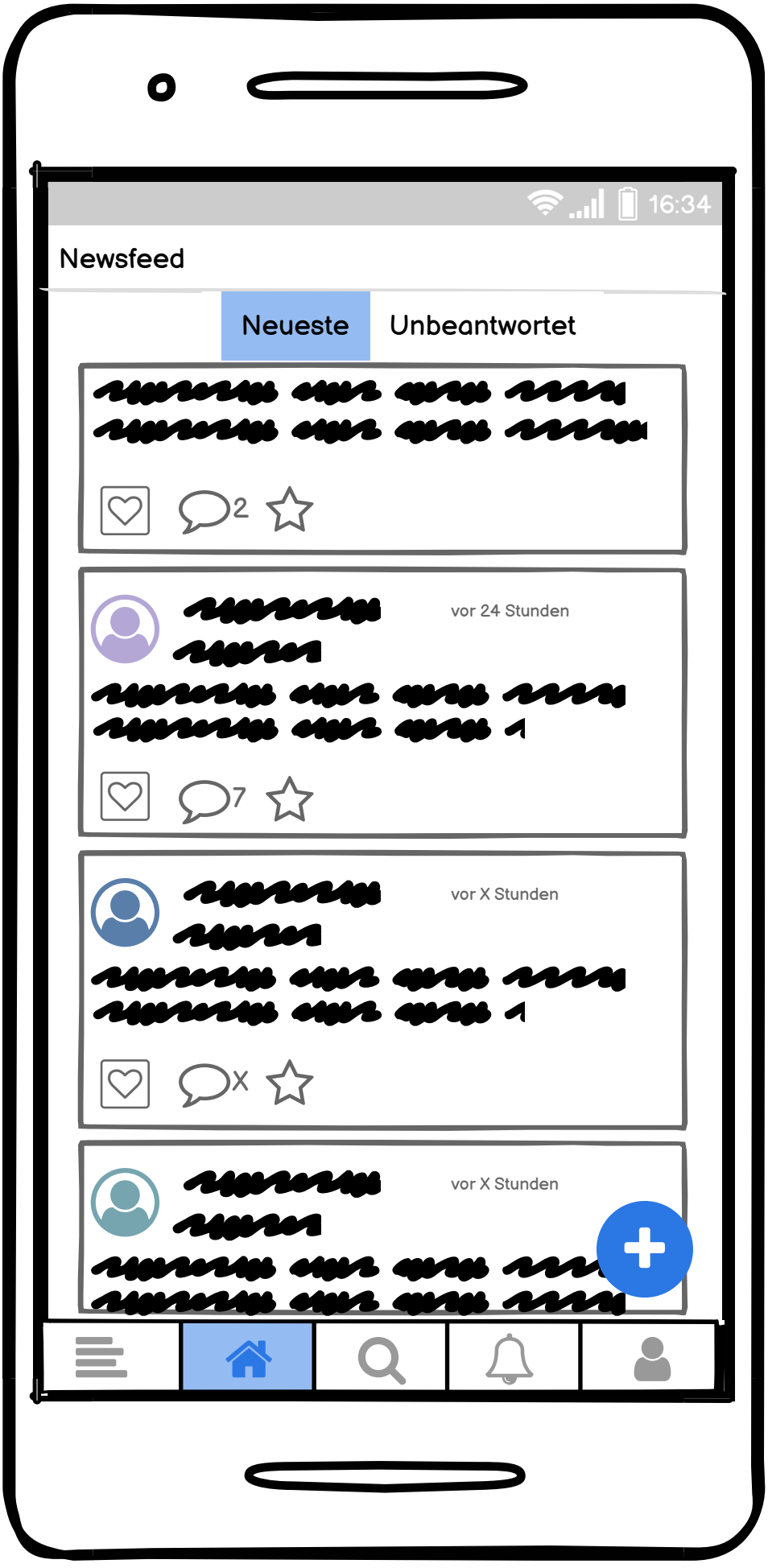
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16:34

Profil Bearbeiten

Friedrich-Alexander-Universität Erlangen

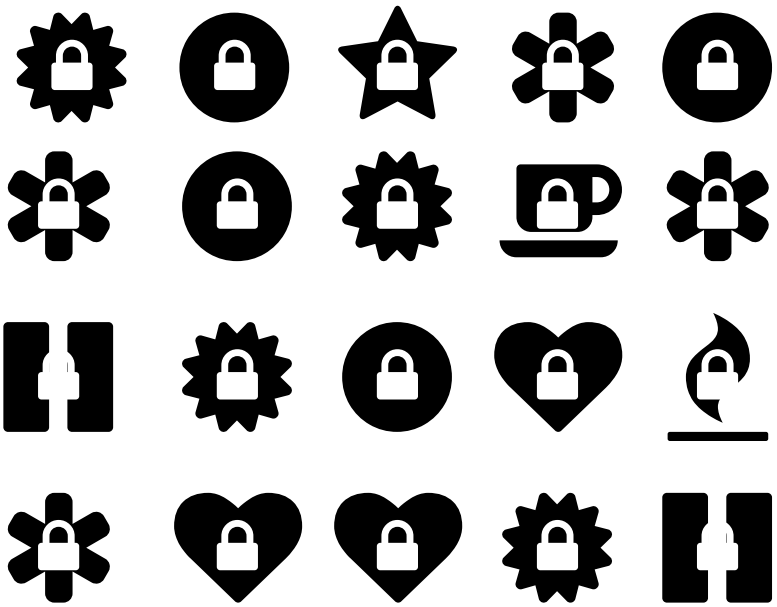
Betriebswirtschaftslehre >

AKTUELLES LEVEL

LEVEL 1

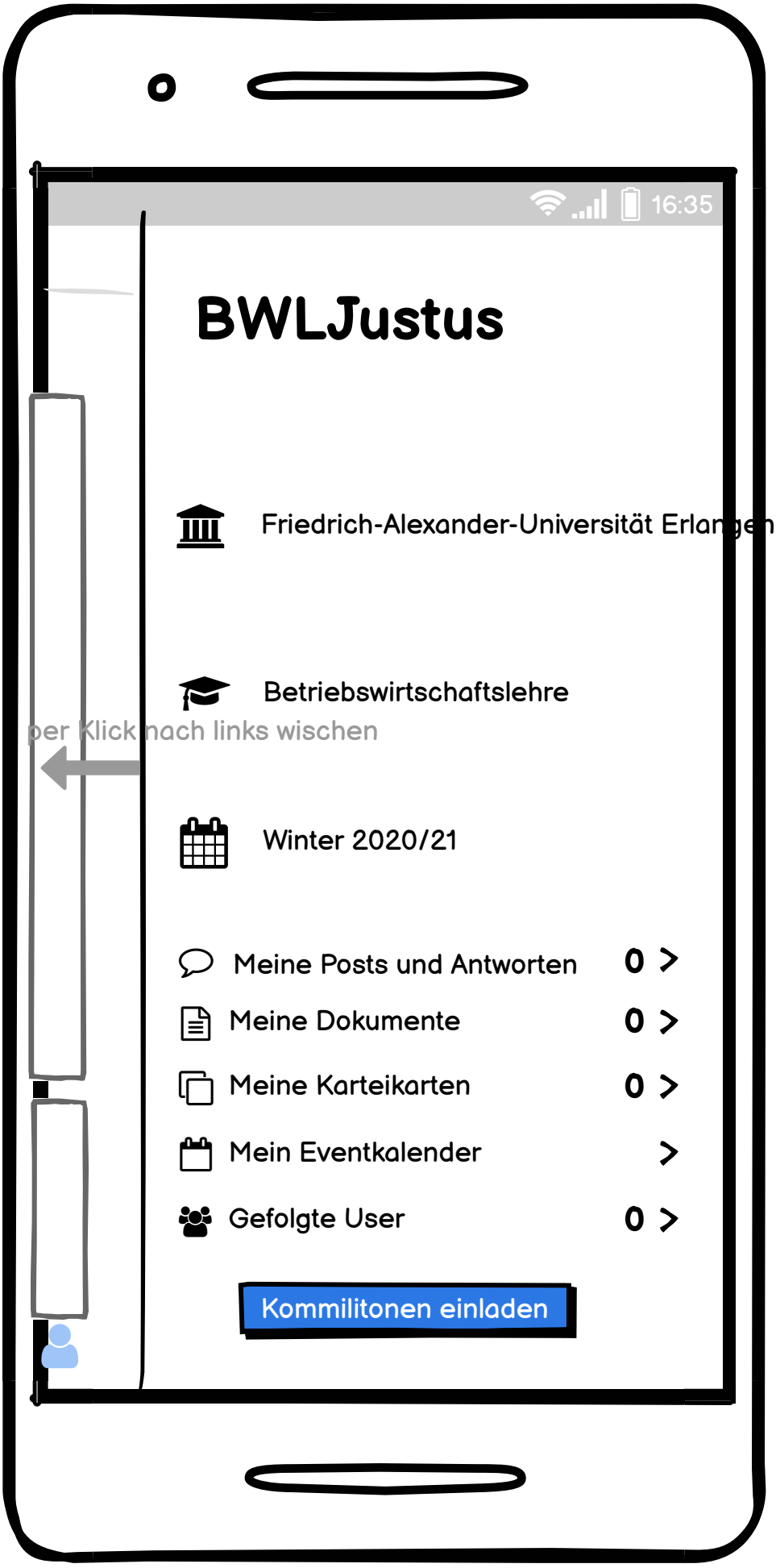
0/100 XP

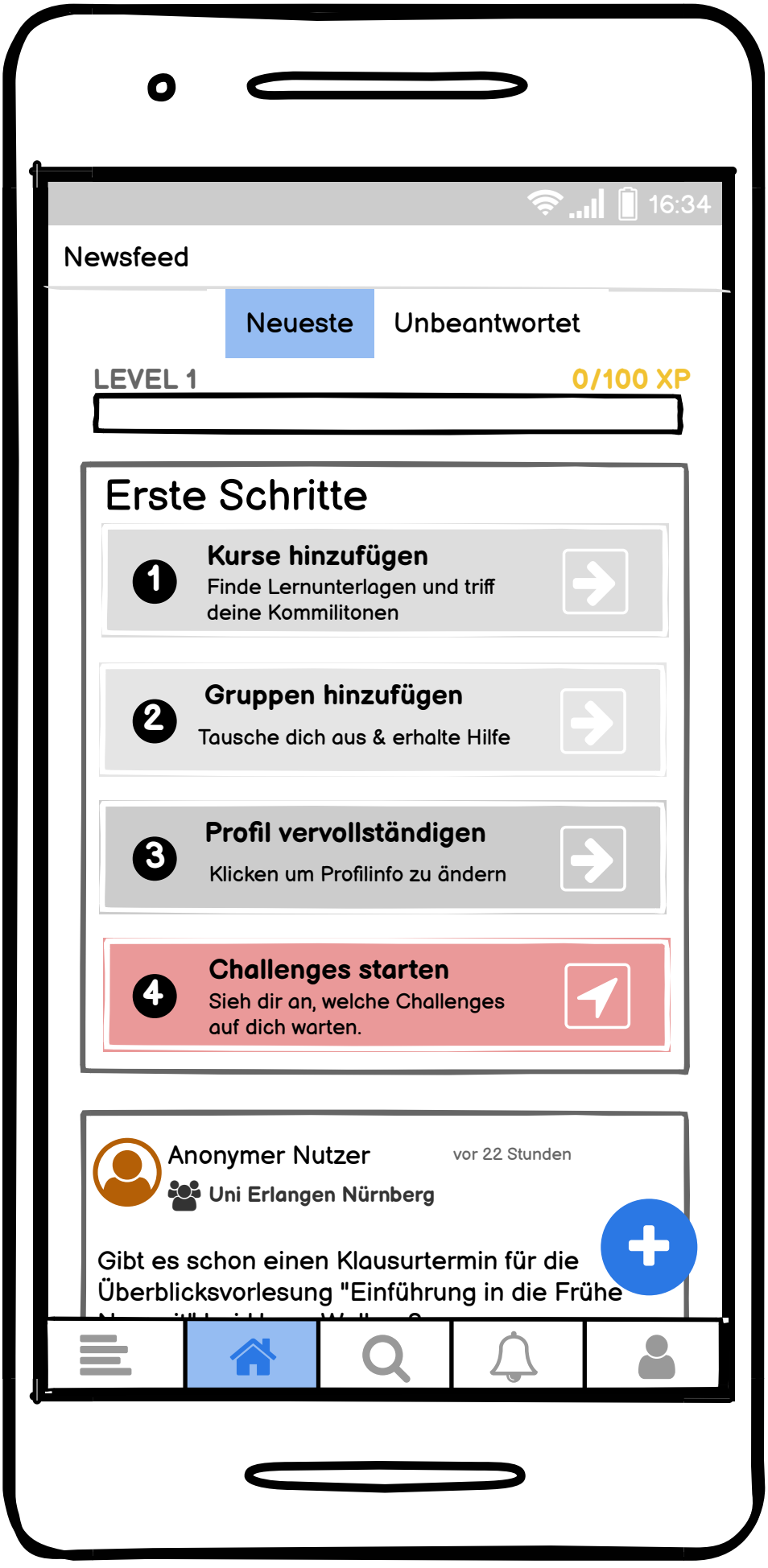
ERFOLGE





use arrows to scroll


16:35
BWLJustus
Friedrich-Alexander-Universität Erlangen
Betriebswirtschaftslehre
per Klick nach links wischen
Winter 2020/21
Meine Posts und Antworten 0 >
Meine Dokumente 0 >
Meine Karteikarten 0 >
Mein Eventkalender >
Gefolgte User 0 >
Kommilitonen einladen



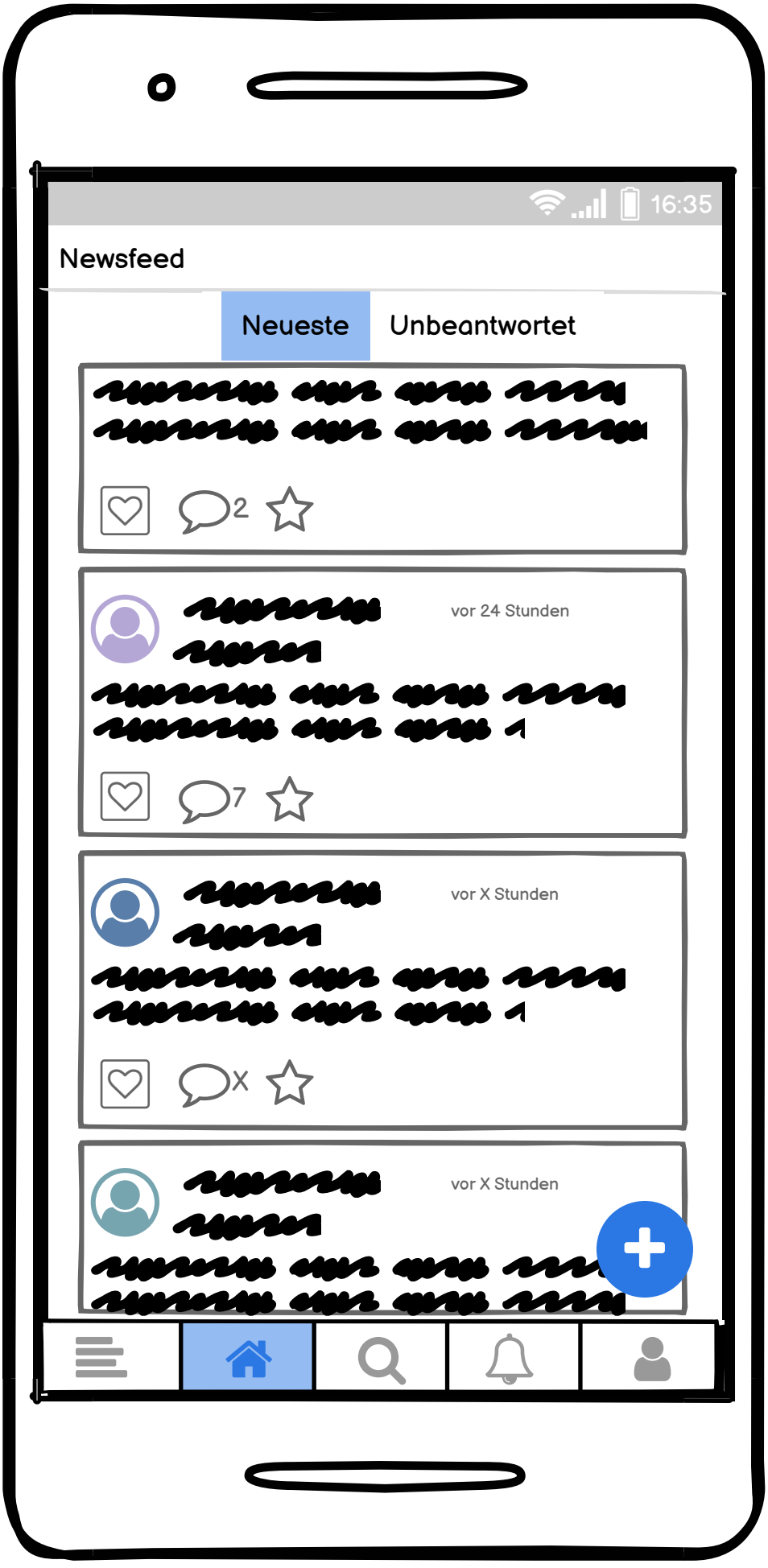
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use arrows to scroll



use arrows to scroll



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16:34

< Das sind Challenges

Durch erledigen von Challenges erhältst du **XP** und **Credits** .

Wenn du auf Studydrive aktiv bist, erhältst du auch außerhalb von Challenges XP und Credits.

Durch das Sammeln von XP, kannst du höhere Level erreichen. Dies schaltet viele Funktionen frei, wie z.B. anpassbares Profilbild, hervorgehobene Fragen, coole Badges.

Ab Level 10 hast du den Kursexperten-Status erreicht. Als Kursexperte erhältst du neben XP und Credits auch echtes Geld für das Erledigen von Challenges. Aktuelle Kursexperten verdienen so bis zu 450€ jeden Monat!

Zeig' mir meine Challenges

Diese Prämien kannst du gewinnen



Spende 5 Euro
450 Credits



10 Kondome
730 Credits



Bierpong Set
810 Credits

use arrows to scroll

16:34

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Zeig' mir meine Challenges

Diese Prämien kannst du gewinnen



Spende 5 Euro
450 Credits



10 Kondome
730 Credits



Bierpong Set
810 Credits



Studydrive T-Shirt
950 Credits



Powerbank
1200 Credits



Studydrive Flasche
1400 Credits



30€ IKEA Gutsch.
2000 Credits

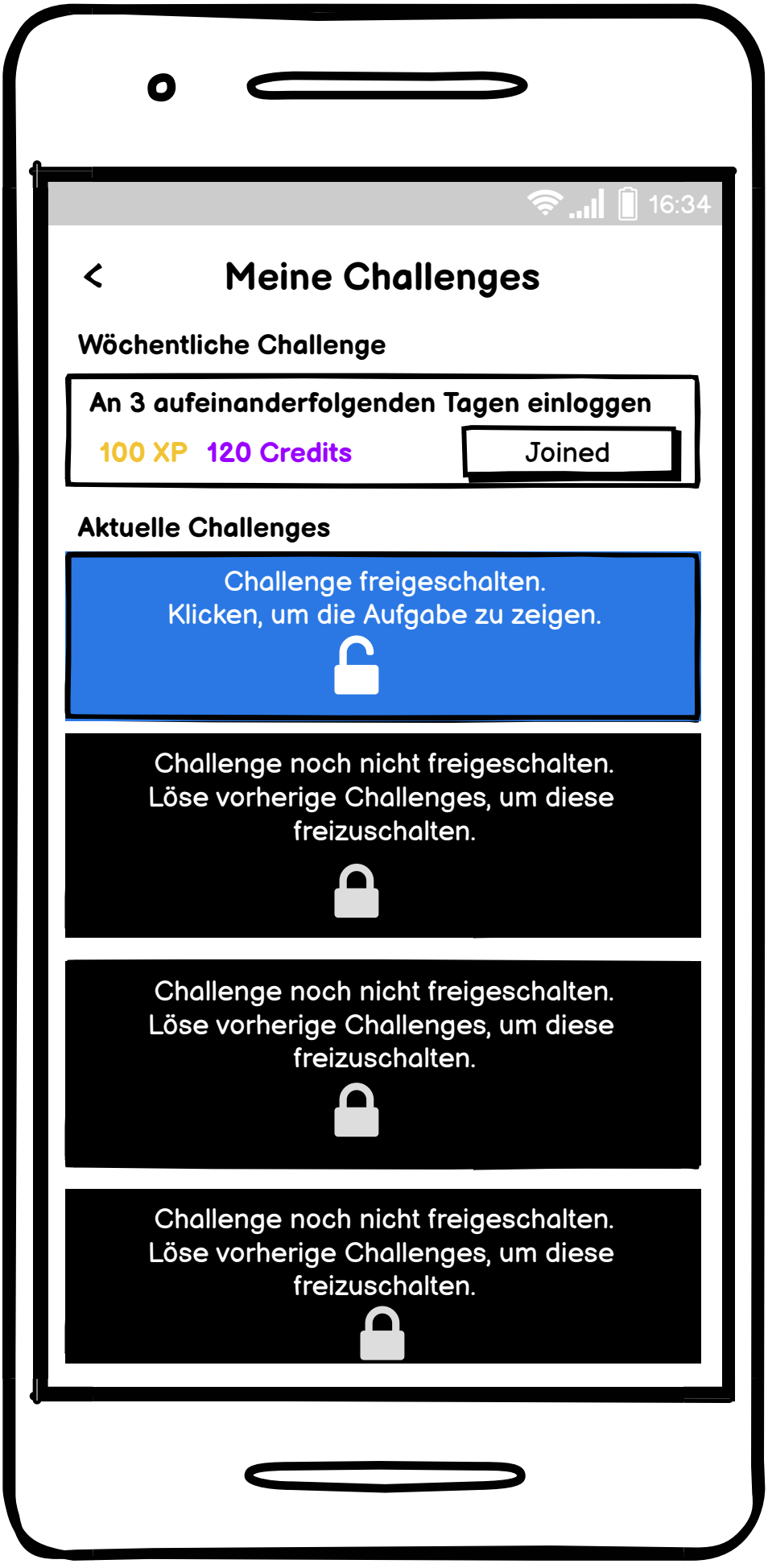


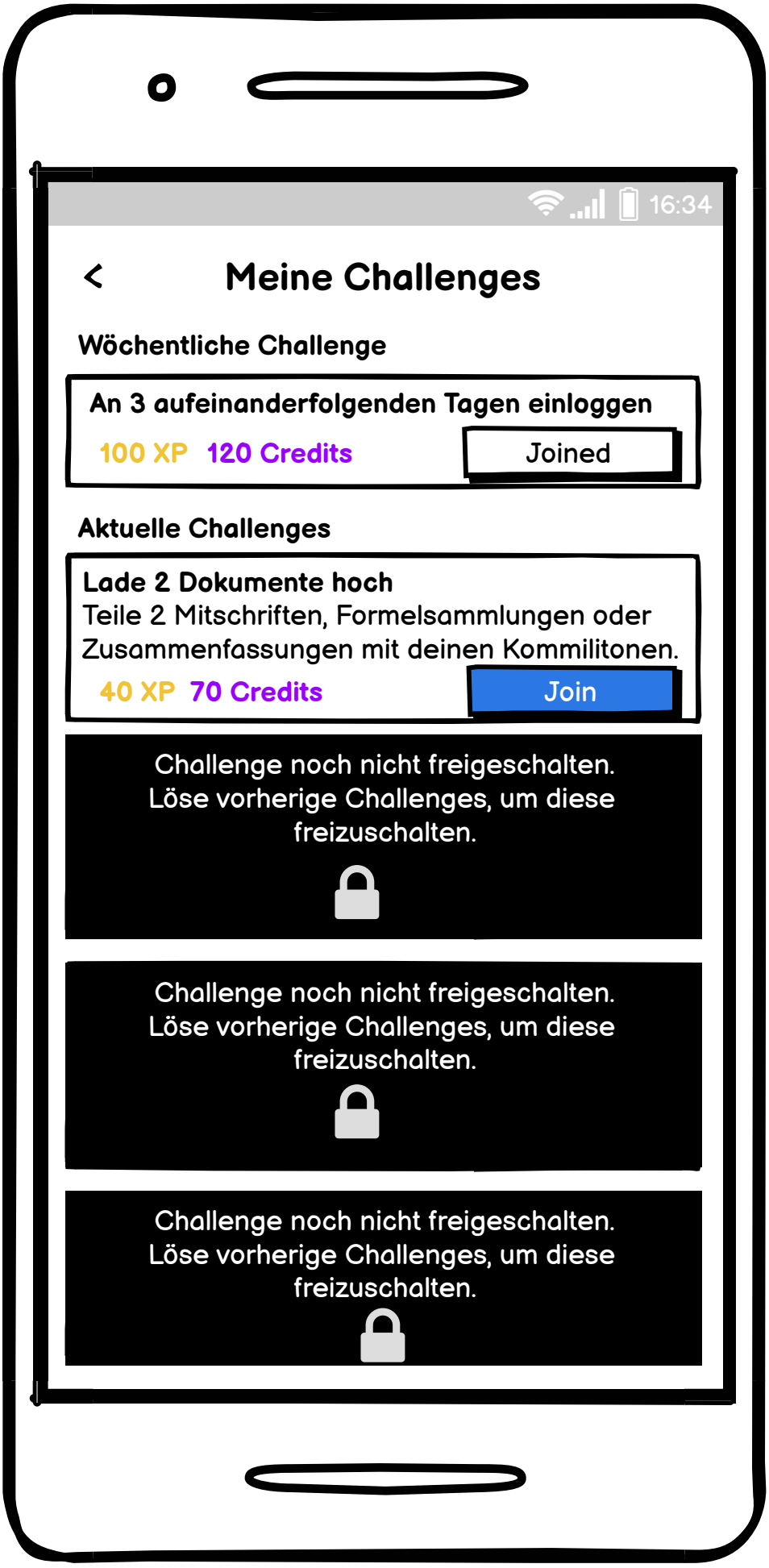
Lautsprecher
2200 Credits

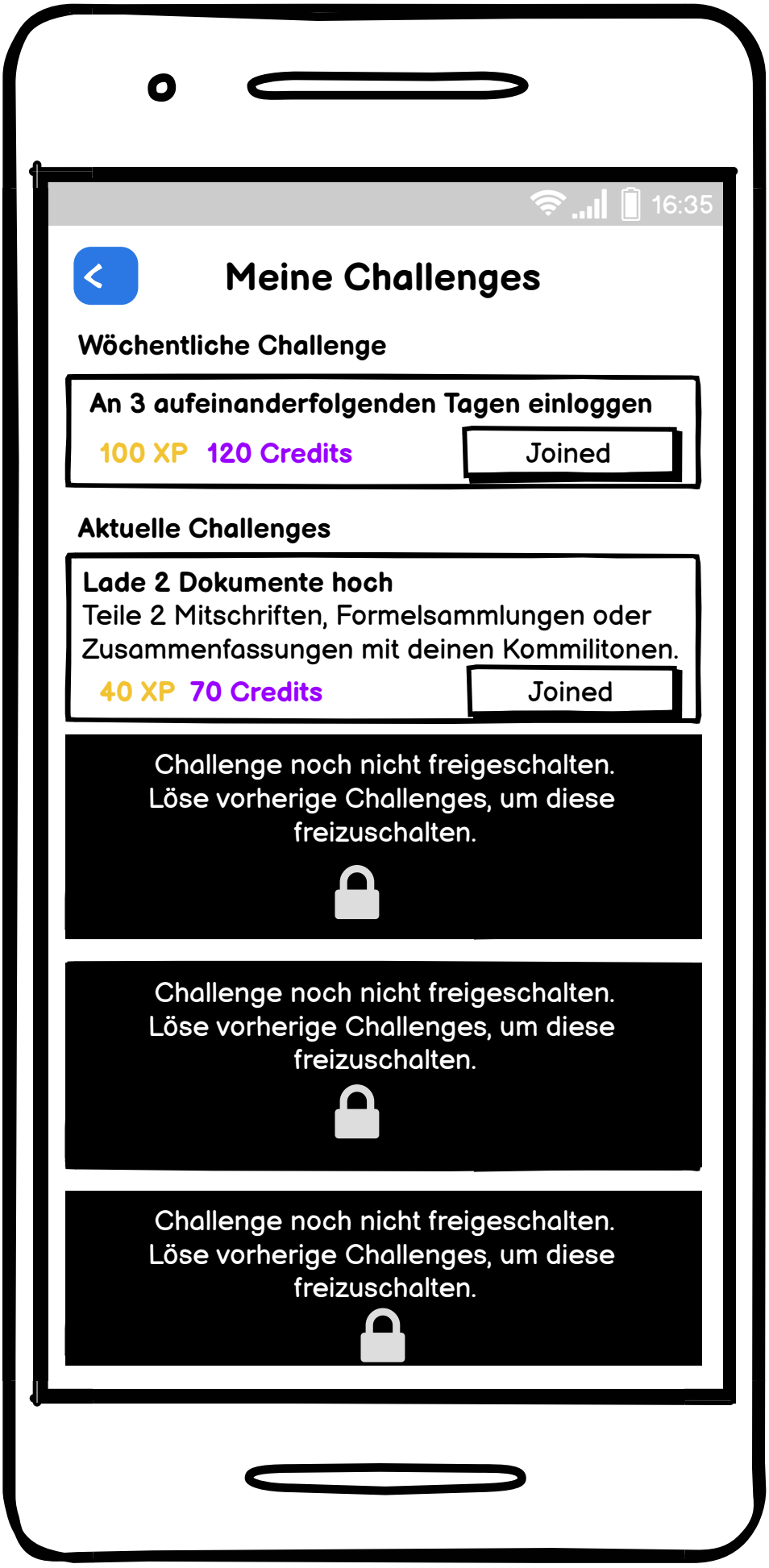


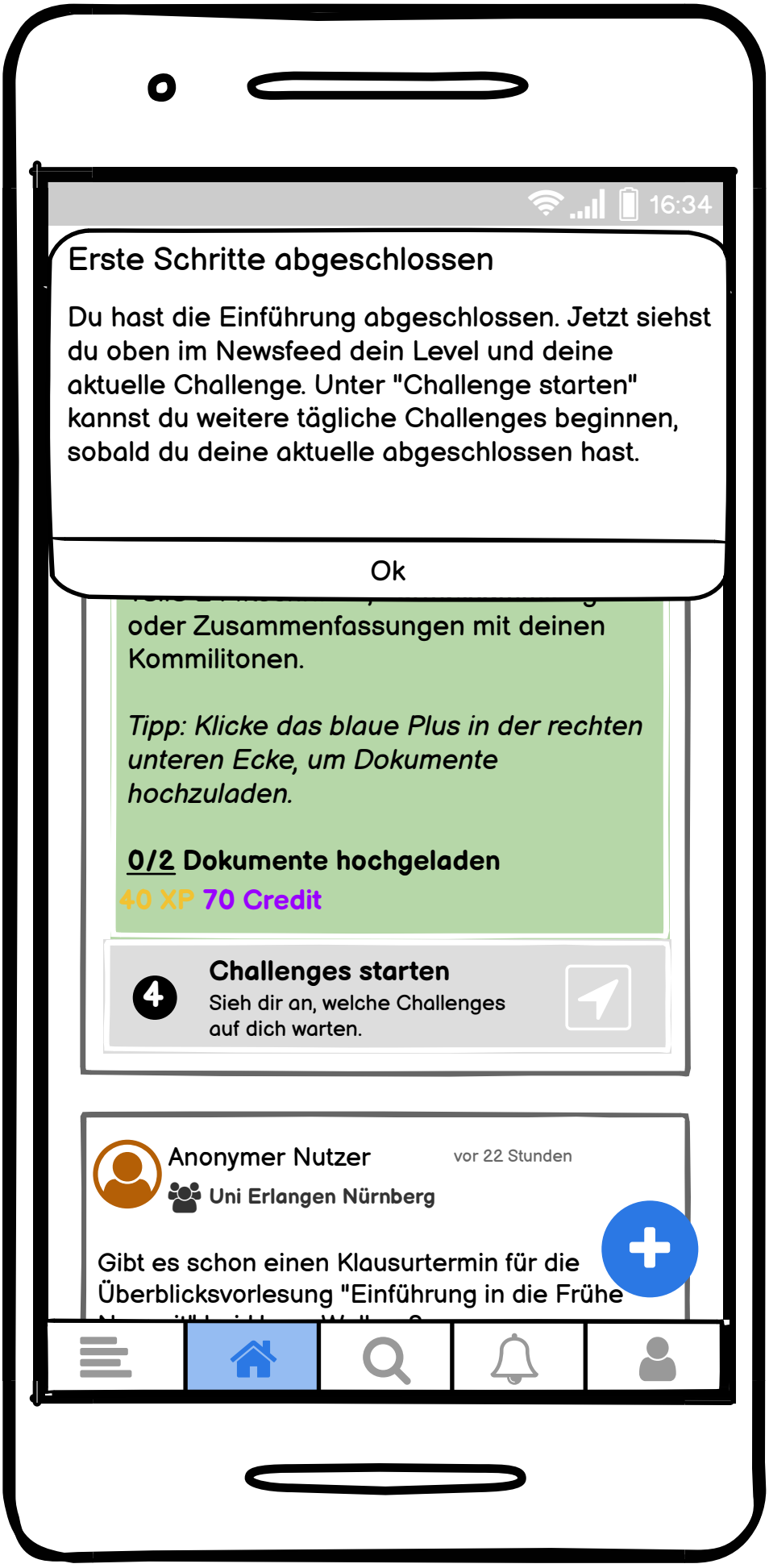
50€ Amazon Gut.
3800 Credits

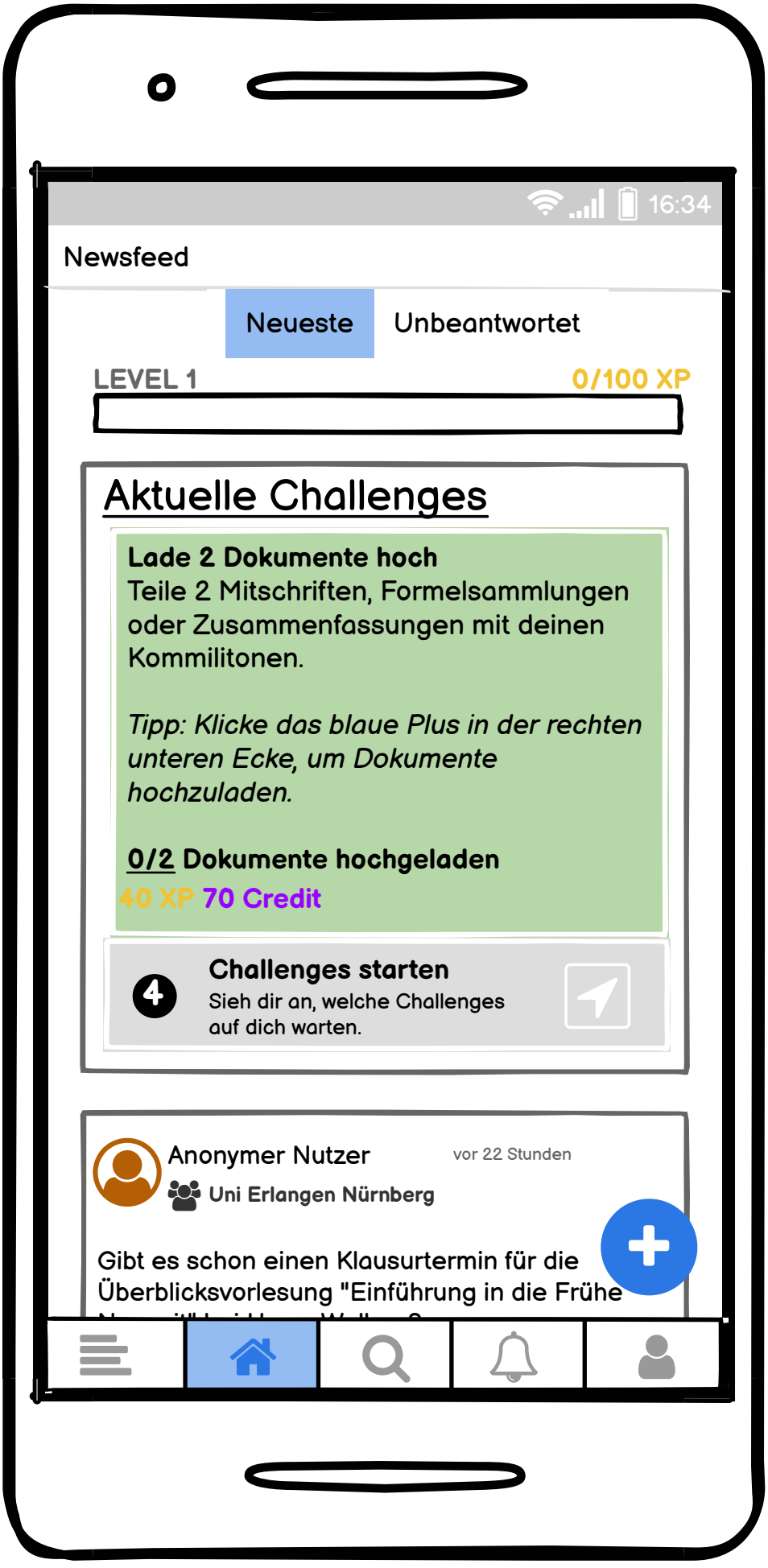
use arrows to scroll


16:34
Meine Challenges
Wöchentliche Challenge
An 3 aufeinanderfolgenden Tagen einloggen
100 XP
120 Credits
Joined
Aktuelle Challenges
Challenge freigeschalten.
Klicken, um die Aufgabe zu zeigen.
Challenge noch nicht freigeschalten.
Löse vorherige Challenges, um diese freizuschalten.
Challenge noch nicht freigeschalten.
Löse vorherige Challenges, um diese freizuschalten.
Challenge noch nicht freigeschalten.
Löse vorherige Challenges, um diese freizuschalten.


16:34
Meine Challenges
Wöchentliche Challenge
An 3 aufeinanderfolgenden Tagen einloggen
100 XP 120 Credits
Joined
Aktuelle Challenges
Lade 2 Dokumente hoch
Teile 2 Mitschriften, Formelsammlungen oder Zusammenfassungen mit deinen Kommilitonen.
40 XP 70 Credits
Join
Challenge noch nicht freigeschalten. Löse vorherige Challenges, um diese freizuschalten.
Challenge noch nicht freigeschalten. Löse vorherige Challenges, um diese freizuschalten.
Challenge noch nicht freigeschalten. Löse vorherige Challenges, um diese freizuschalten.


16:35
Meine Challenges
Wöchentliche Challenge
An 3 aufeinanderfolgenden Tagen einloggen
100 XP 120 Credits
Joined
Aktuelle Challenges
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Joined
Challenge noch nicht freigeschalten. Löse vorherige Challenges, um diese freizuschalten.
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Challenge noch nicht freigeschalten. Löse vorherige Challenges, um diese freizuschalten.


16:34
Erste Schritte abgeschlossen
Du hast die Einführung abgeschlossen. Jetzt siehst du oben im Newsfeed dein Level und deine aktuelle Challenge. Unter "Challenge starten" kannst du weitere tägliche Challenges beginnen, sobald du deine aktuelle abgeschlossen hast.
Ok
oder Zusammenfassungen mit deinen Kommilitonen.
Tipp: Klicke das blaue Plus in der rechten unteren Ecke, um Dokumente hochzuladen.
0/2 Dokumente hochgeladen
40 XP 70 Credit
4
Challenges starten
Sieh dir an, welche Challenges auf dich warten.
Anonymer Nutzer
vor 22 Stunden
Uni Erlangen Nürnberg
Gibt es schon einen Klausurtermin für die Überblicksvorlesung "Einführung in die Frühe



use arrows to scroll



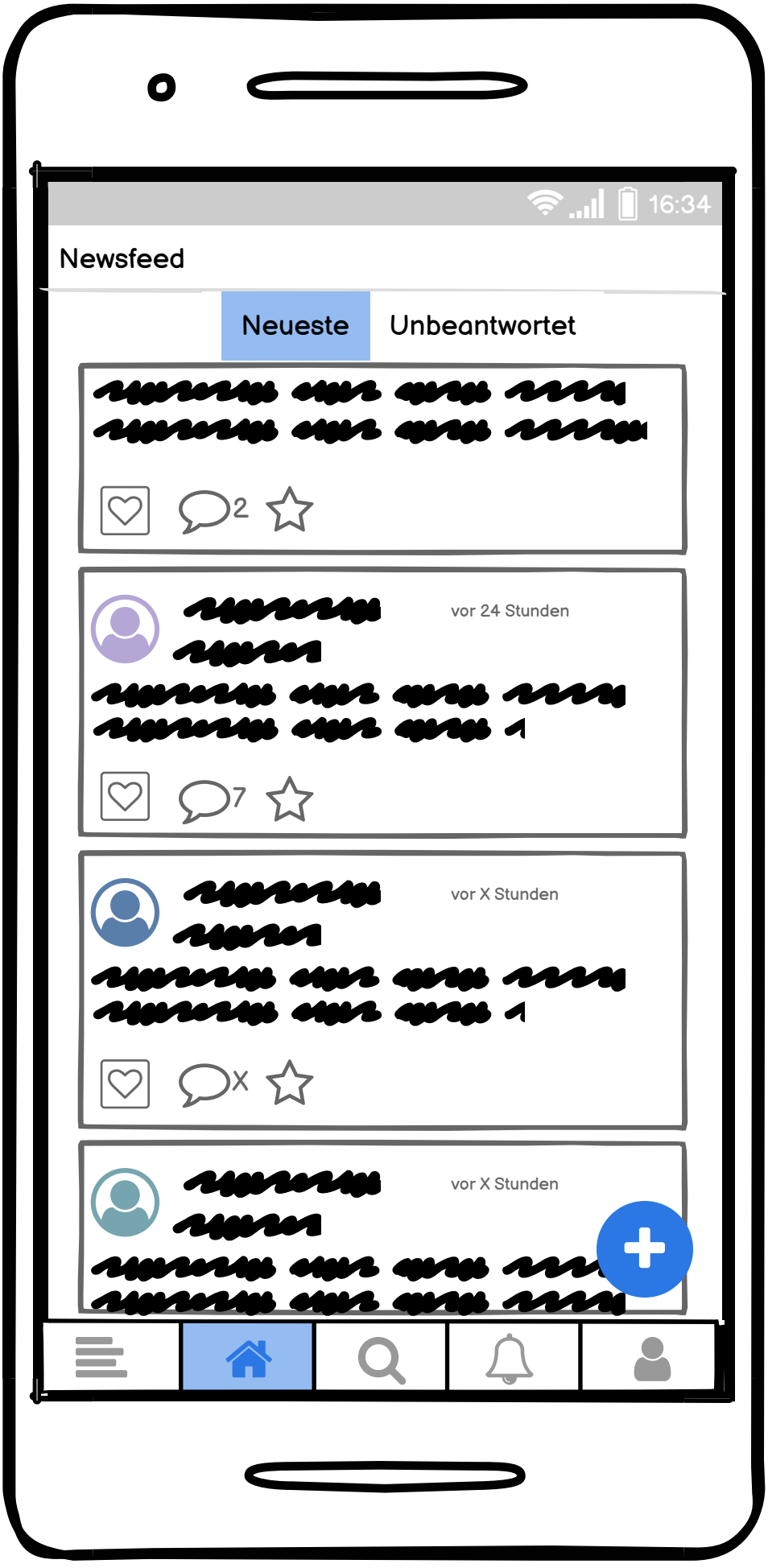
use arrows to scroll


16:35
Newsfeed
Neueste
Unbeantwortet
0/2 Dokumente hochgeladen
40 X 70 Credit
4
Challenges starten
Sieh dir an, welche Challenges auf dich warten.
Anonymer Nutzer
vor 22 Stunden
Uni Erlangen Nürnberg
Gibt es schon einen Klausurtermin für die Überblicksvorlesung "Einführung in die Frühe Neuzeit" bei Herrn Wellner?
2
Frage stellen
Dokument hochladen
Karteikarten erstellen
5
Anonyme Rakete
vor 23 Stunden



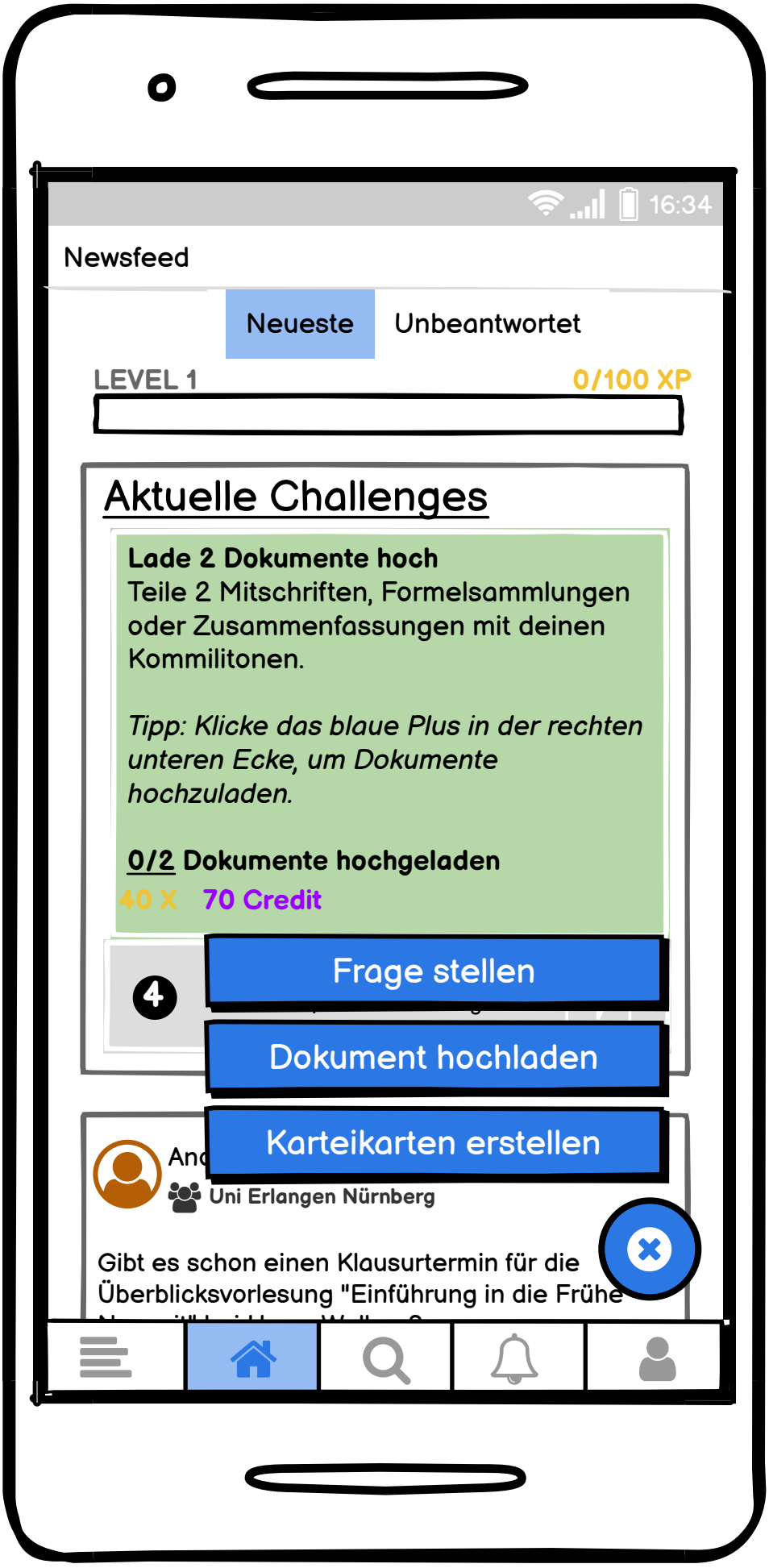
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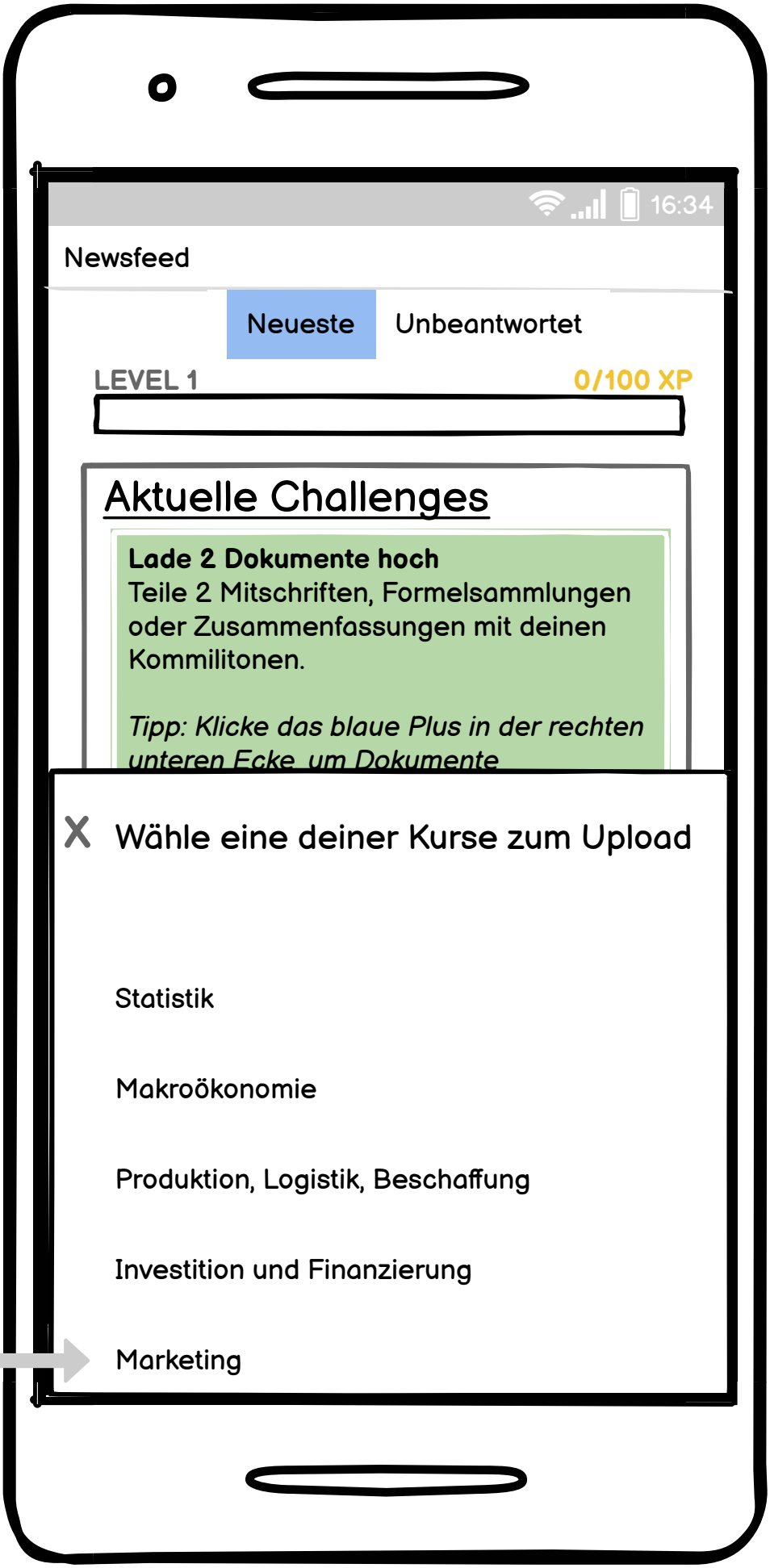

16:34
Newsfeed
Neueste
Unbeantwortet
2
Anonyme Nutzerin
vor 23 Stunden
Marketing
5
Anonyme Rakete
vor 23 Stunden
Frage stellen
Dokument hochladen
Karteikarten erstellen
7



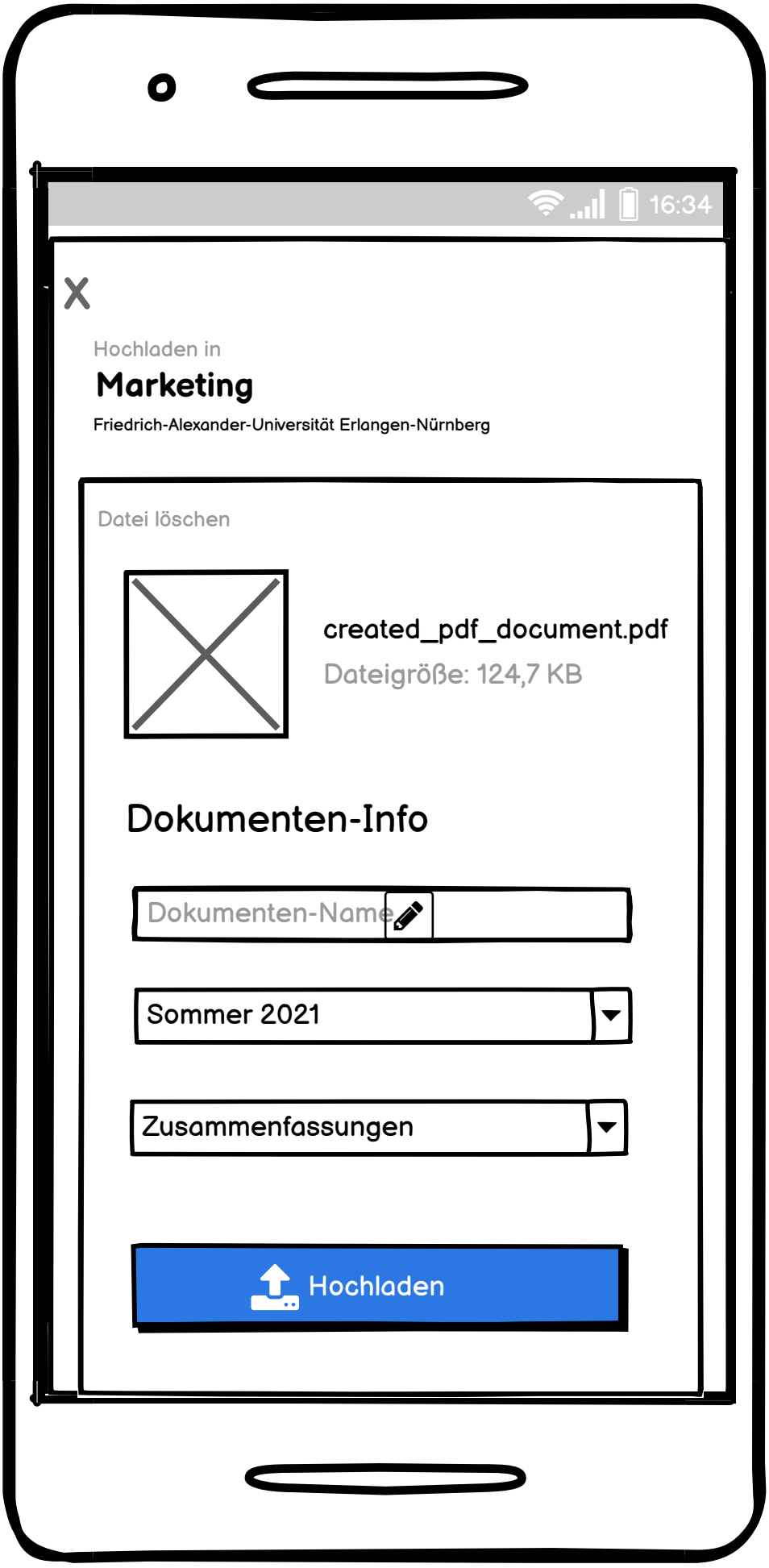
use arrows to scroll


16:34
Newsfeed
Neueste
Unbeantwortet
2
vor 24 Stunden
7
vor X Stunden
Frage stellen
Dokument hochladen
Karteikarten erstellen


16:34
Newsfeed
Neueste
Unbeantwortet
LEVEL 1
0/100 XP
Aktuelle Challenges
Lade 2 Dokumente hoch
Teile 2 Mitschriften, Formelsammlungen oder Zusammenfassungen mit deinen Kommilitonen.
Tipp: Klicke das blaue Plus in der rechten unteren Ecke, um Dokumente hochzuladen.
0/2 Dokumente hochgeladen
40 X 70 Credit
4
Frage stellen
Dokument hochladen
Karteikarten erstellen
Uni Erlangen Nürnberg
Gibt es schon einen Klausurtermin für die Überblicksvorlesung "Einführung in die Frühe


16:34
Newsfeed
Neueste
Unbeantwortet
LEVEL 1
0/100 XP
Aktuelle Challenges
Lade 2 Dokumente hoch
Teile 2 Mitschriften, Formelsammlungen oder Zusammenfassungen mit deinen Kommilitonen.
Tipp: Klicke das blaue Plus in der rechten unteren Ecke, um Dokumente
X
Wähle eine deiner Kurse zum Upload
Statistik
Makroökonomie
Produktion, Logistik, Beschaffung
Investition und Finanzierung
Marketing


16:34
X
Hochladen in
Marketing
Friedrich-Alexander-Universität Erlangen-Nürnberg
Datei hochladen
von deinem Handy, Drive oder Dropbox
Was solltest du teilen?
Deine Notizen, Zusammenfassungen, Lösungen, etc.
Getippte oder handgeschriebene Dateien
Aktuelle & immer noch relevante Dokumente
Bevorzugt diese Formate: .pdf, .doc, .docx, .jped, .xls
Was solltest du nicht teilen?
Ganze Bücher oder Artikel
Einzelne Fotos von mehrseitigen Dokumenten
Vermeide .zip & .rar hochzuladen
Alles was nicht studienrelevant ist oder urheberrechtlich geschützt ist


16:34
Hochladen in
Marketing
Friedrich-Alexander-Universität Erlangen-Nürnberg
Datei löschen
created_pdf_document.pdf
Dateigröße: 124,7 KB
Dokumenten-Info
Dokumenten-Name
Sommer 2021
Zusammenfassungen
Hochladen

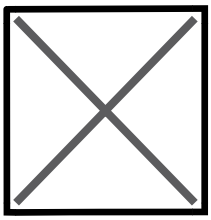
16:34

Hochladen in

Marketing

Friedrich-Alexander-Universität Erlangen-Nürnberg

Datei löschen



created_pdf_document.pdf

Dateigröße: 124,7 KB

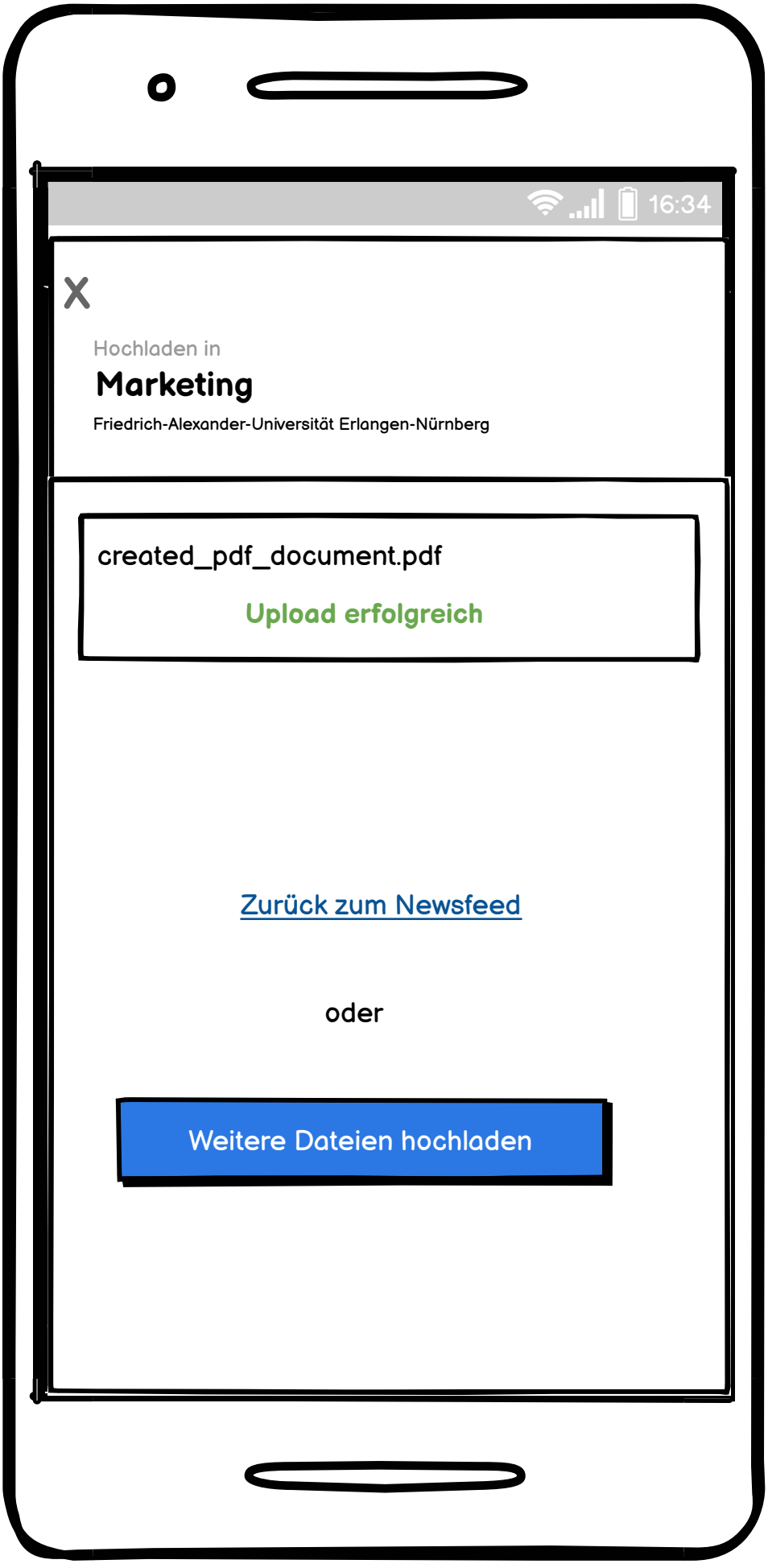
Dokumenten-Info

Meine Marketing Zusammenfassung

Sommer 2021

Zusammenfassungen

Hochladen


16:34
Hochladen in
Marketing
Friedrich-Alexander-Universität Erlangen-Nürnberg
created_pdf_document.pdf
Upload erfolgreich
Zurück zum Newsfeed
oder
Weitere Dateien hochladen

16:34

Neues Badge!

Glückwunsch du bist nun offiziell ein STUDYDRIVE UPLOADER!



Ok

Lade 2 Dokumente hoch
Teile 2 Mitschriften, Formelsammlungen oder Zusammenfassungen mit deinen Kommilitonen.

Tipp: Klicke das blaue Plus in der rechten unteren Ecke, um Dokumente hochzuladen.

1/2 Dokumente hochgeladen
40 X **70 Credit**

4 **Challenges starten**
Sieh dir an, welche Challenges auf dich warten.

Anonymer Nutzer vor 22 Stunden
Uni Erlangen Nürnberg

Gibt es schon einen Klausurtermin für die Überblicksvorlesung "Einführung in die Frühe

16:34

Newsfeed

Neueste | Unbeantwortet

LEVEL 1 0/100 XP

Aktuelle Challenges

Lade 2 Dokumente hoch
Teile 2 Mitschriften, Formelsammlungen oder Zusammenfassungen mit deinen Kommilitonen.

Tipp: Klicke das blaue Plus in der rechten unteren Ecke, um Dokumente hochzuladen.

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Anonymer Nutzer vor 22 Stunden
Uni Erlangen Nürnberg

Gibt es schon einen Klausurtermin für die Überblicksvorlesung "Einführung in die Frühe

use arrows to scroll



use arrows to scroll

16:34

Profil Bearbeiten

Friedrich-Alexander-Universität Erlangen

Betriebswirtschaftslehre >

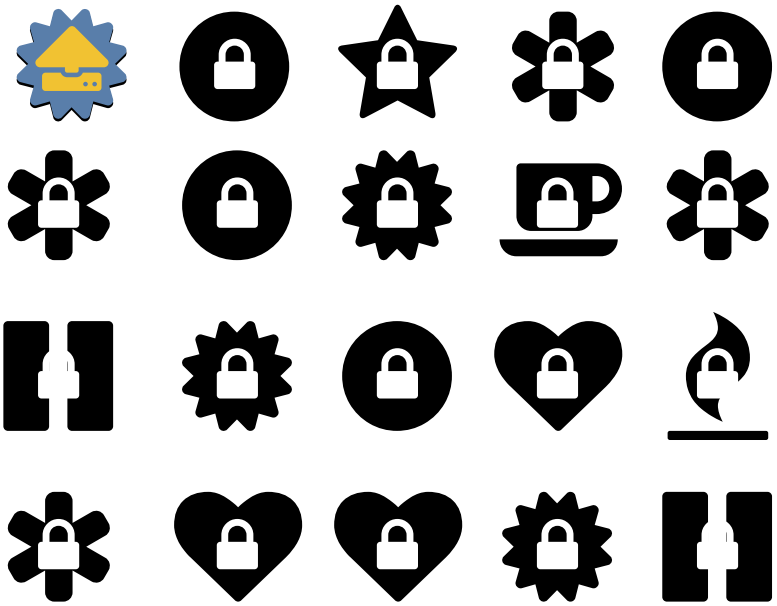
AKTUELLES LEVEL

LEVEL 1

0/100 XP

ERFOLGE

UPLOADER





use arrows to scroll



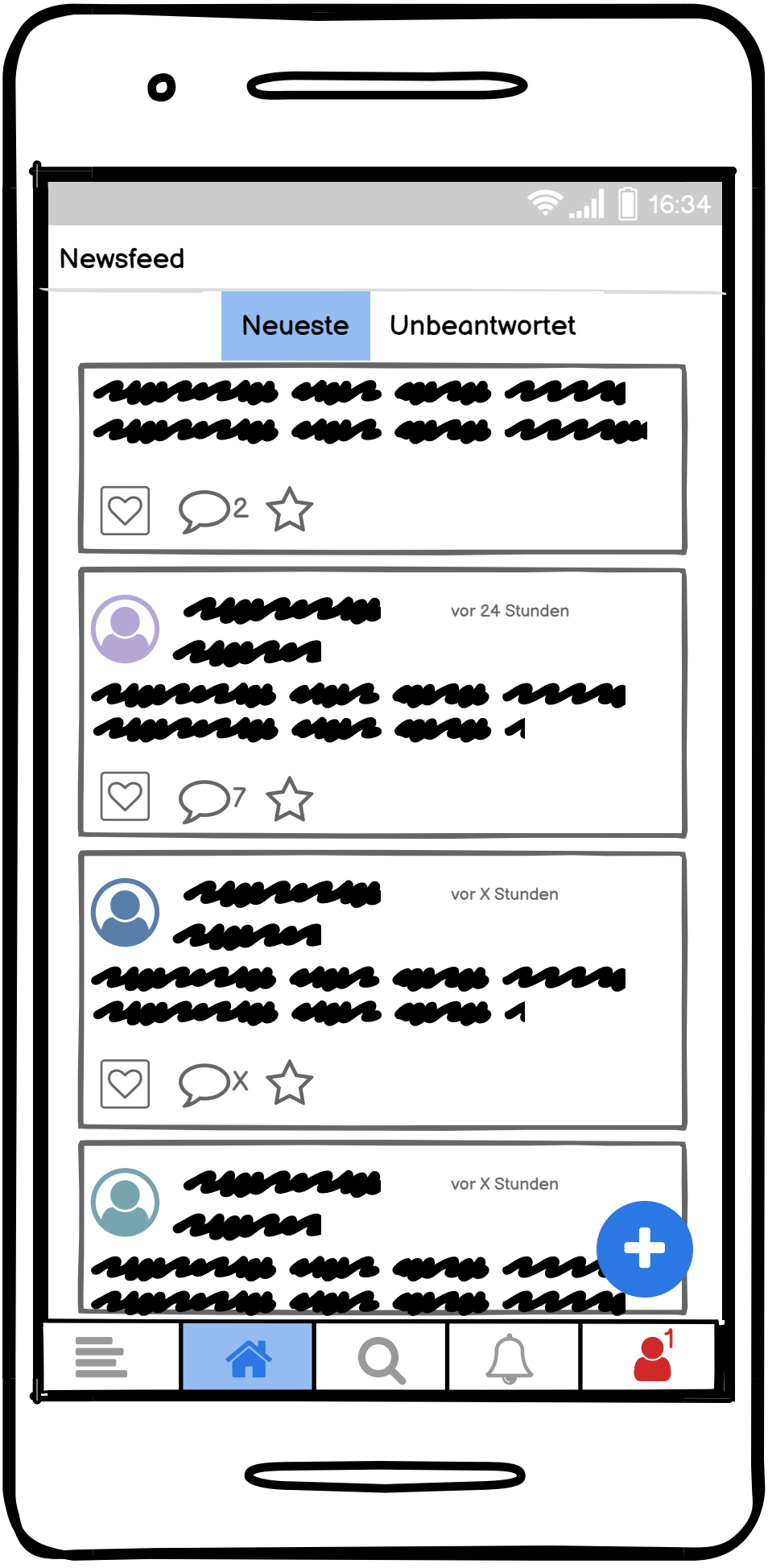
use arrows to scroll


16:34
Newsfeed
Neueste
Unbeantwortet
1/2 Dokumente hochgeladen
40 X 70 Credit
4
Challenges starten
Sieh dir an, welche Challenges auf dich warten.
Anonymer Nutzer
vor 22 Stunden
Uni Erlangen Nürnberg
Gibt es schon einen Klausurtermin für die Überblicksvorlesung "Einführung in die Frühe Neuzeit" bei Herrn Wellner?
2
Frage stellen
Dokument hochladen
Karteikarten erstellen
5
Anonyme Rakete
vor 23 Stunden
1



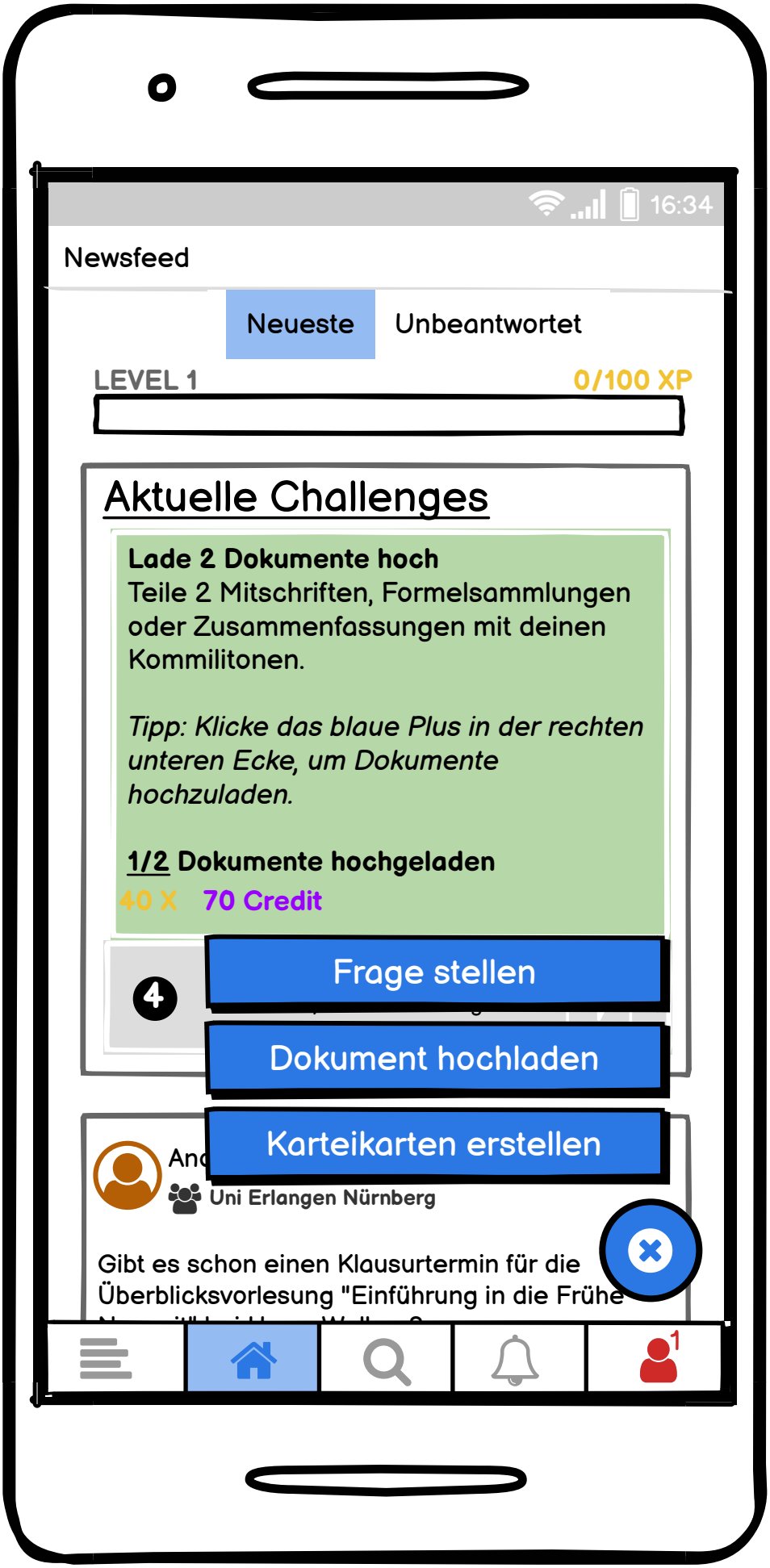
use arrows to scroll


16:34
Newsfeed
Neueste
Unbeantwortet
2
Anonyme Nutzerin
vor 23 Stunden
Marketing
5
Anonyme Rakete
vor 23 Stunden
Frage stellen
Dokument hochladen
Karteikarten erstellen
7
1



use arrows to scroll


16:34
Newsfeed
Neueste
Unbeantwortet
2
vor 24 Stunden
7
vor X Stunden
Frage stellen
Dokument hochladen
Karteikarten erstellen
1


16:34
Newsfeed
Neueste
Unbeantwortet
LEVEL 1
0/100 XP
Aktuelle Challenges
Lade 2 Dokumente hoch
Teile 2 Mitschriften, Formelsammlungen oder Zusammenfassungen mit deinen Kommilitonen.
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Uni Erlangen Nürnberg
Gibt es schon einen Klausurtermin für die Überblicksvorlesung "Einführung in die Frühe
1

16:34

Newsfeed

Neueste | Unbeantwortet

LEVEL 1 0/100 XP

Aktuelle Challenges

Lade 2 Dokumente hoch
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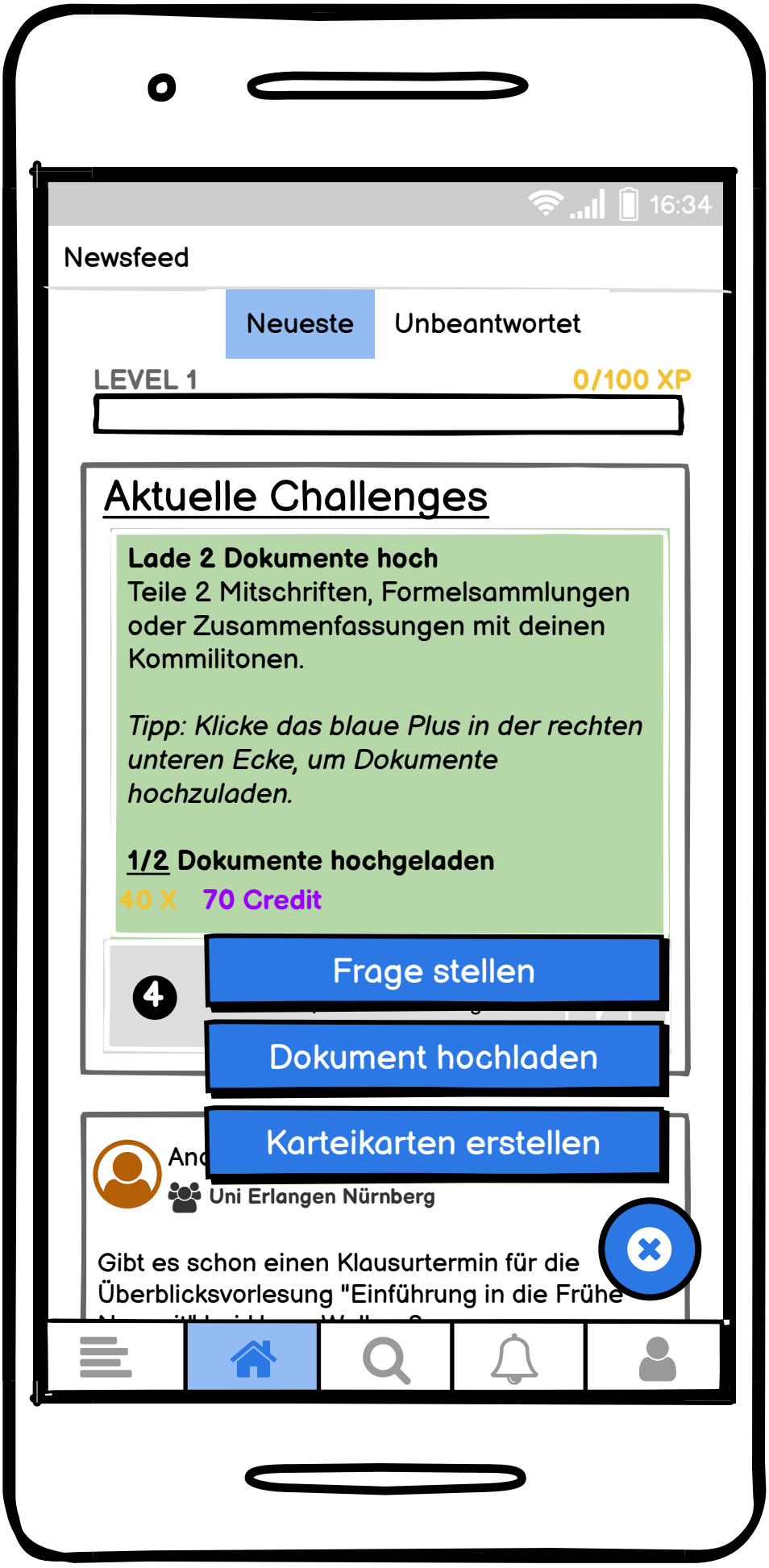
1/2 Dokumente hochgeladen
40 X 70 Credit

4 **Challenges starten**
Sieh dir an, welche Challenges auf dich warten.

Anonymer Nutzer vor 22 Stunden
Uni Erlangen Nürnberg

Gibt es schon einen Klausurtermin für die Überblicksvorlesung "Einführung in die Frühe

use arrows to scroll


16:34
Newsfeed
Neueste
Unbeantwortet
LEVEL 1
0/100 XP
Aktuelle Challenges
Lade 2 Dokumente hoch
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Dokument hochladen
Karteikarten erstellen
Uni Erlangen Nürnberg
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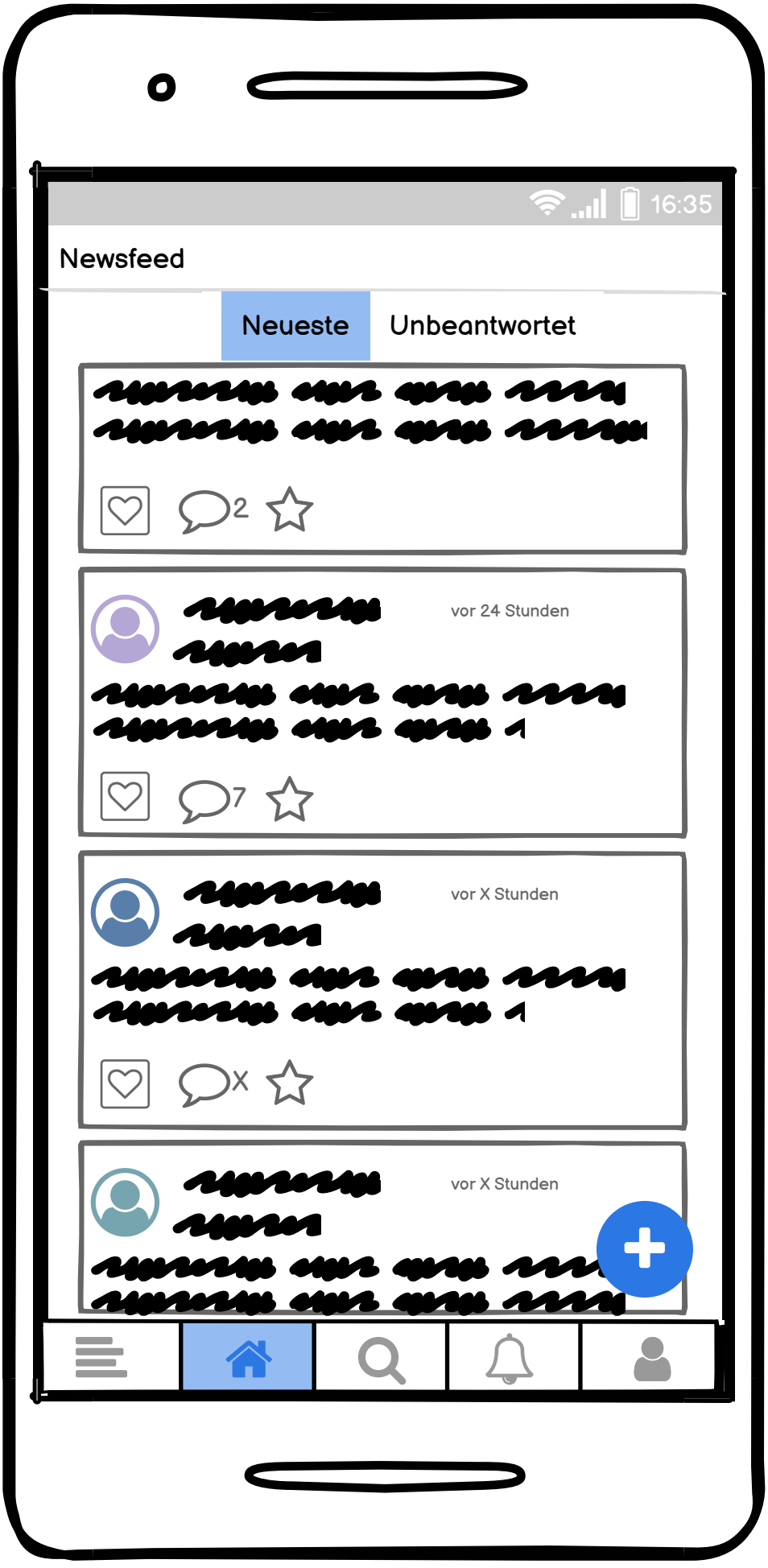
use arrows to scroll


16:34
Newsfeed
Neueste
Unbeantwortet
1/2 Dokumente hochgeladen
40 X 70 Credit
4
Challenges starten
Sieh dir an, welche Challenges auf dich warten.
Anonymer Nutzer
vor 22 Stunden
Uni Erlangen Nürnberg
Gibt es schon einen Klausurtermin für die Überblicksvorlesung "Einführung in die Frühe Neuzeit" bei Herrn Wellner?
2
Frage stellen
Dokument hochladen
Karteikarten erstellen
5
Anonyme Rakete
vor 23 Stunden



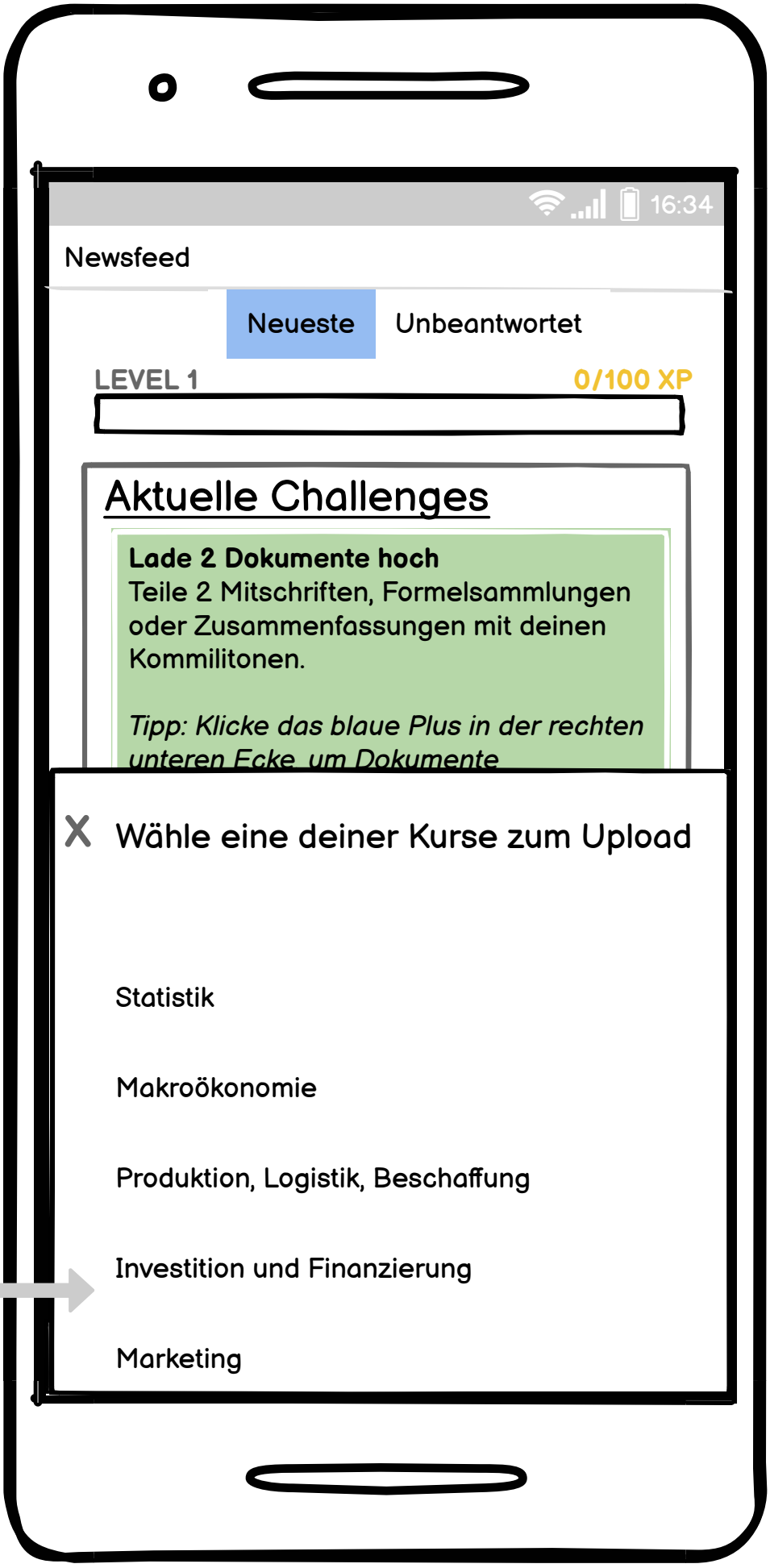
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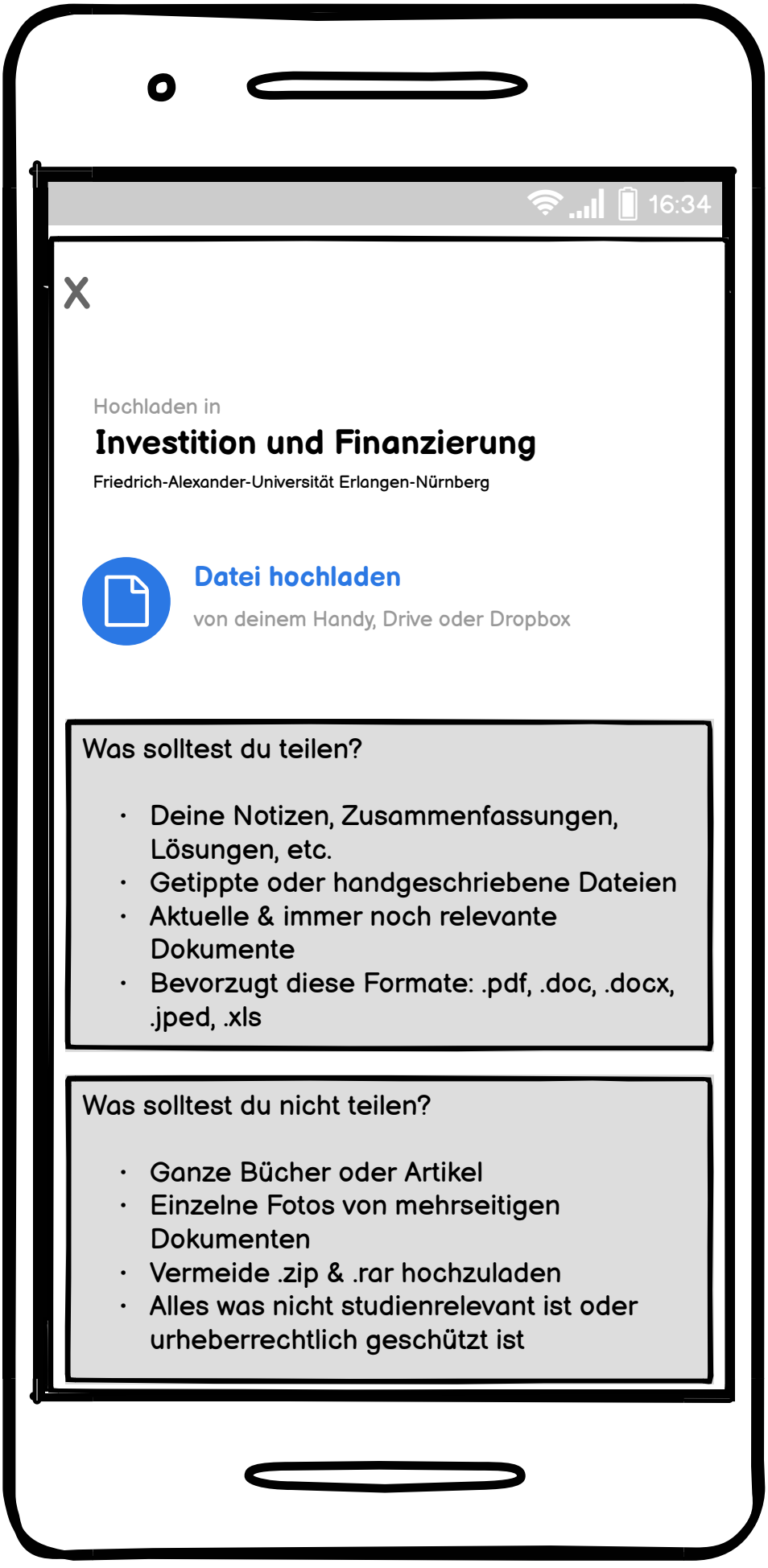

16:34
Newsfeed
Neueste
Unbeantwortet
2
Anonyme Nutzerin
vor 23 Stunden
Marketing
5
Anonyme Rakete
vor 23 Stunden
Frage stellen
Dokument hochladen
Karteikarten erstellen
7

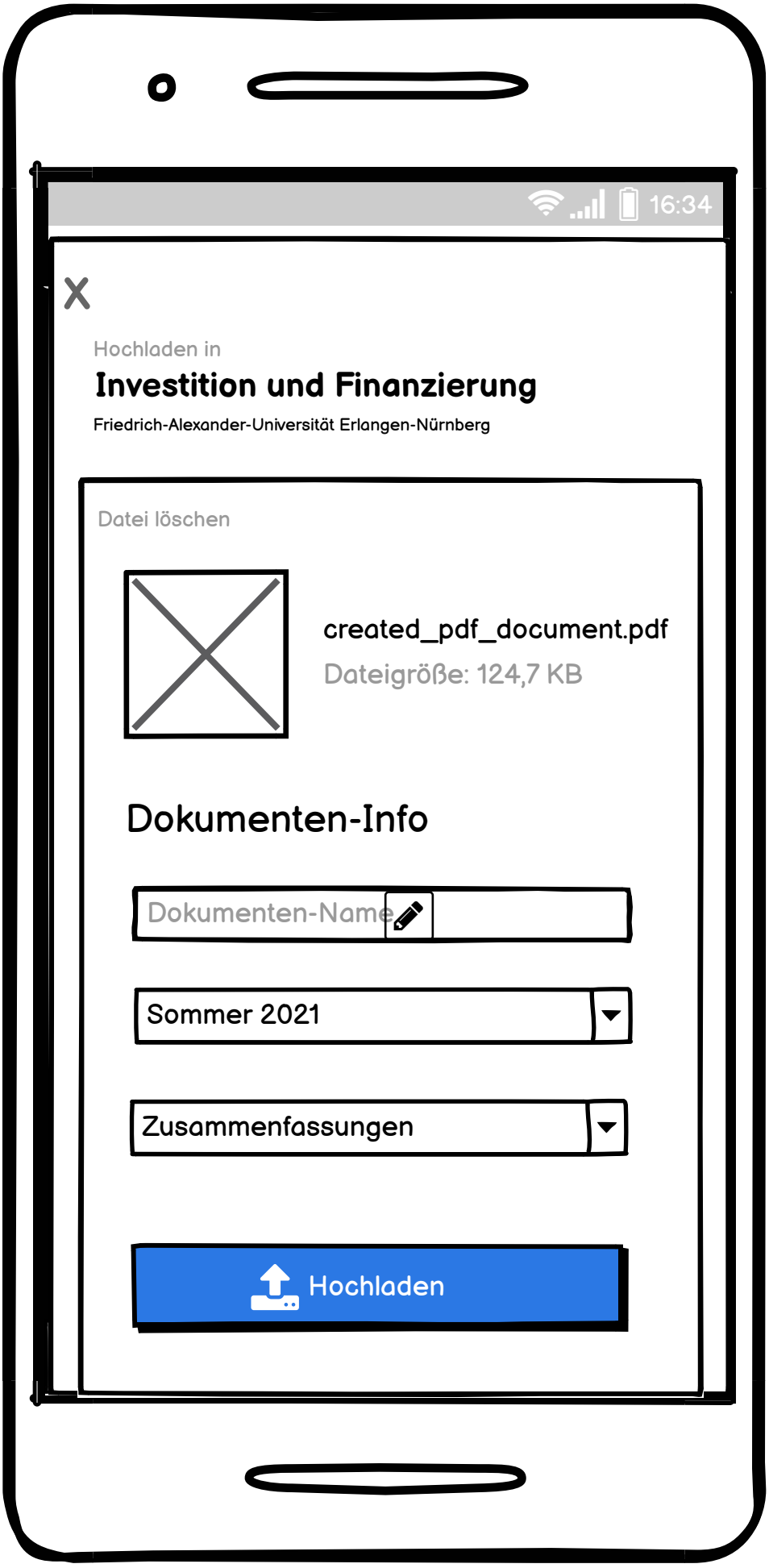


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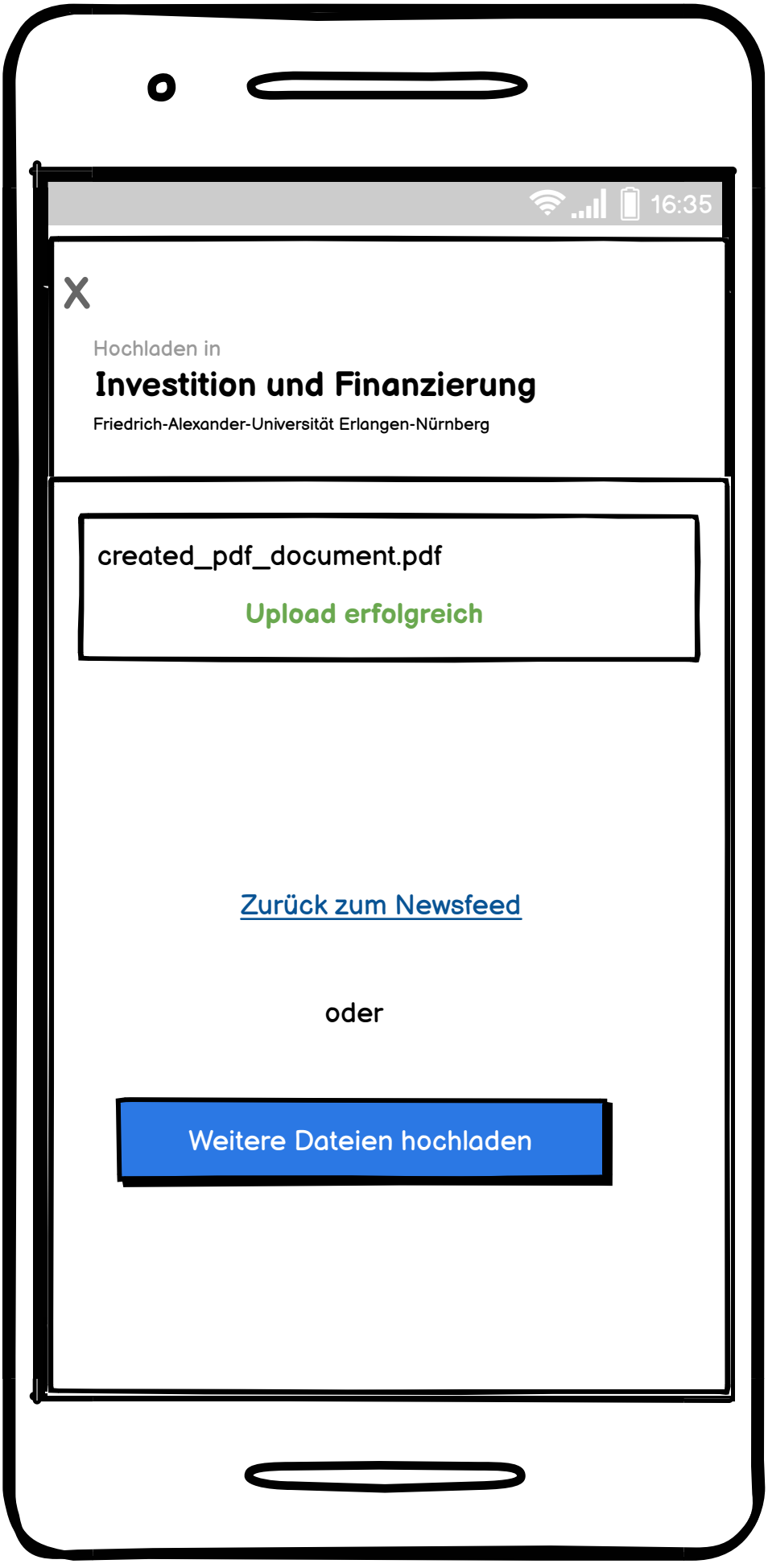

16:34
Newsfeed
Neueste
Unbeantwortet
2
vor 24 Stunden
7
vor X Stunden
Frage stellen
Dokument hochladen
Karteikarten erstellen

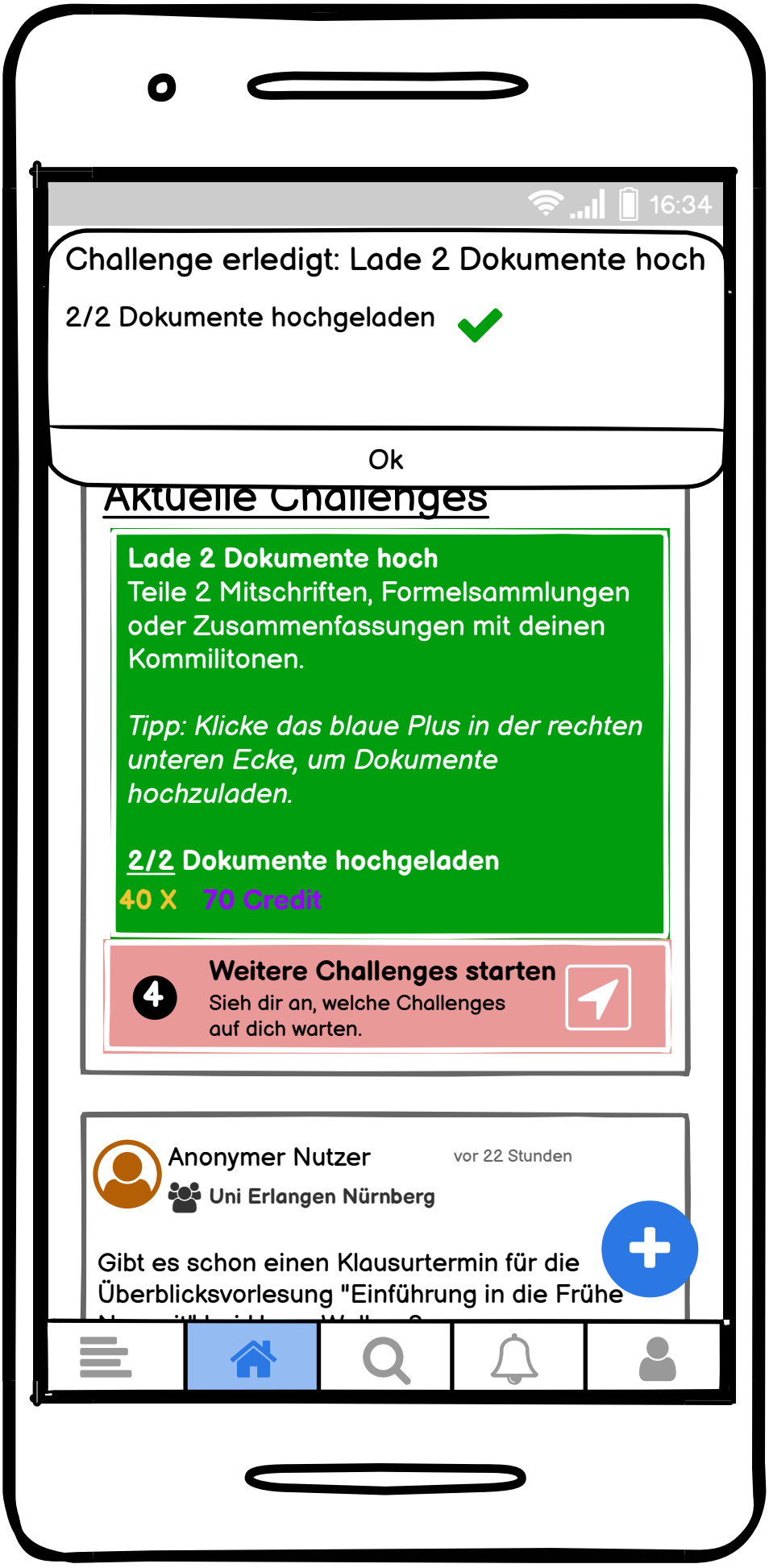

16:34
Newsfeed
Neueste
Unbeantwortet
LEVEL 1
0/100 XP
Aktuelle Challenges
Lade 2 Dokumente hoch
Teile 2 Mitschriften, Formelsammlungen oder Zusammenfassungen mit deinen Kommilitonen.
Tipp: Klicke das blaue Plus in der rechten unteren Ecke, um Dokumente
X
Wähle eine deiner Kurse zum Upload
Statistik
Makroökonomie
Produktion, Logistik, Beschaffung
Investition und Finanzierung
Marketing


16:34
X
Hochladen in
Investition und Finanzierung
Friedrich-Alexander-Universität Erlangen-Nürnberg
Datei hochladen
von deinem Handy, Drive oder Dropbox
Was solltest du teilen?
Deine Notizen, Zusammenfassungen, Lösungen, etc.
Getippte oder handgeschriebene Dateien
Aktuelle & immer noch relevante Dokumente
Bevorzugt diese Formate: .pdf, .doc, .docx, .jped, .xls
Was solltest du nicht teilen?
Ganze Bücher oder Artikel
Einzelne Fotos von mehrseitigen Dokumenten
Vermeide .zip & .rar hochzuladen
Alles was nicht studienrelevant ist oder urheberrechtlich geschützt ist


16:34
Hochladen in
Investition und Finanzierung
Friedrich-Alexander-Universität Erlangen-Nürnberg
Datei löschen
created_pdf_document.pdf
Dateigröße: 124,7 KB
Dokumenten-Info
Dokumenten-Name
Sommer 2021
Zusammenfassungen
Hochladen


16:34
Hochladen in
Investition und Finanzierung
Friedrich-Alexander-Universität Erlangen-Nürnberg
Datei löschen
created_pdf_document.pdf
Dateigröße: 124,7 KB
Dokumenten-Info
Meine IuF Zusammenfassung
Sommer 2021
Zusammenfassungen
Hochladen


16:35
X
Hochladen in
Investition und Finanzierung
Friedrich-Alexander-Universität Erlangen-Nürnberg
created_pdf_document.pdf
Upload erfolgreich
Zurück zum Newsfeed
oder
Weitere Dateien hochladen


16:34
Challenge erledigt: Lade 2 Dokumente hoch
2/2 Dokumente hochgeladen
Ok
Aktuelle Challenges
Lade 2 Dokumente hoch
Teile 2 Mitschriften, Formelsammlungen oder Zusammenfassungen mit deinen Kommilitonen.
Tipp: Klicke das blaue Plus in der rechten unteren Ecke, um Dokumente hochzuladen.
2/2 Dokumente hochgeladen
40 X 70 Credit
4
Weitere Challenges starten
Sieh dir an, welche Challenges auf dich warten.
Anonymer Nutzer
vor 22 Stunden
Uni Erlangen Nürnberg
Gibt es schon einen Klausurtermin für die Überblicksvorlesung "Einführung in die Frühe

16:34

Newsfeed 70 Credits

Neueste Unbeantwortet

LEVEL 1 40/100 XP

Aktuelle Challenges

Lade 2 Dokumente hoch
Teile 2 Mitschriften, Formelsammlungen oder Zusammenfassungen mit deinen Kommilitonen.

Tipp: Klicke das blaue Plus in der rechten unteren Ecke, um Dokumente hochzuladen.

2/2 Dokumente hochgeladen
40 X 70 Credit

4 **Weitere Challenges starten**
Sieh dir an, welche Challenges auf dich warten.

Anonymer Nutzer vor 22 Stunden
Uni Erlangen Nürnberg

Gibt es schon einen Klausurtermin für die Überblicksvorlesung "Einführung in die Frühe

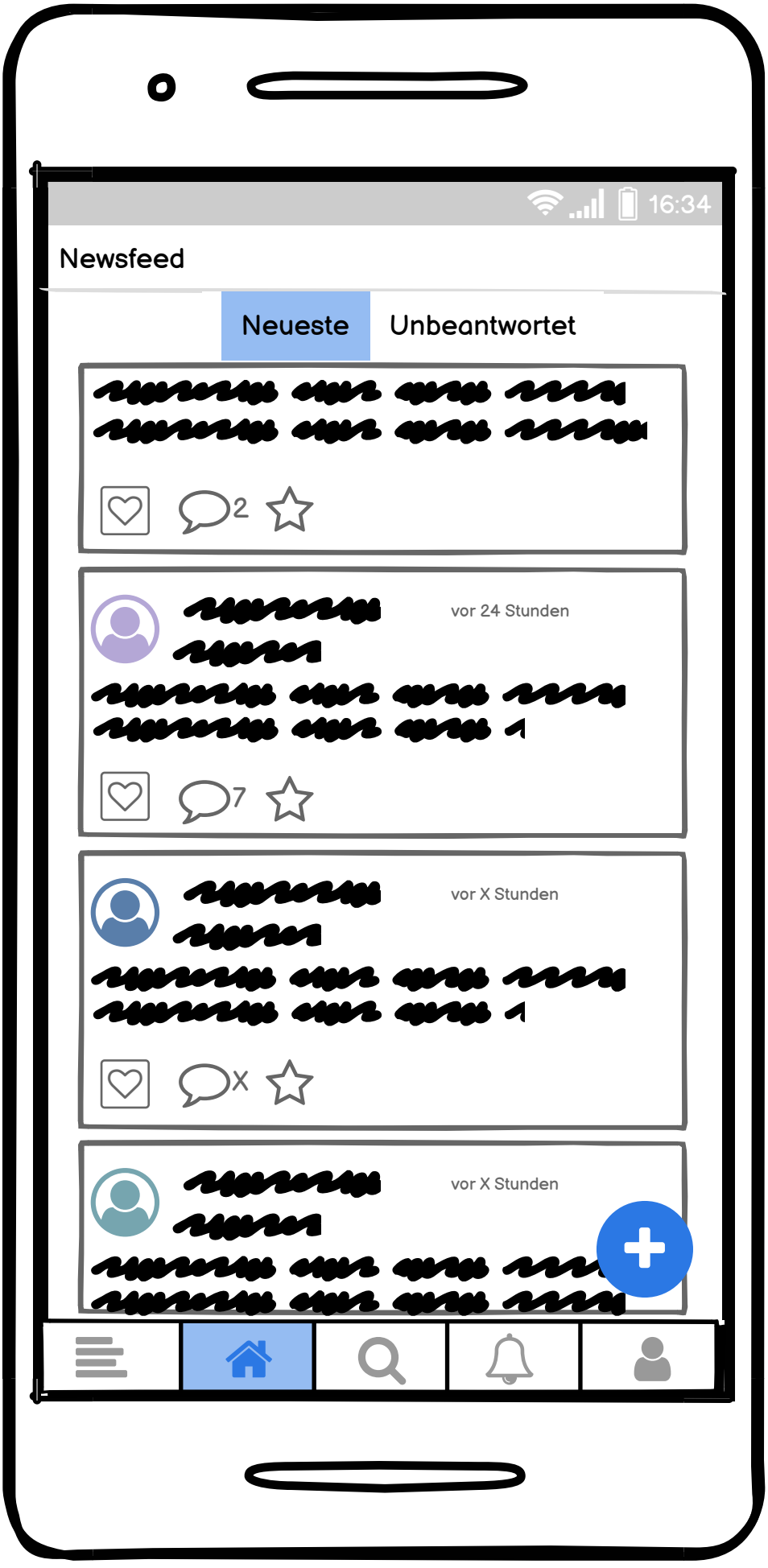
use arrows to scroll



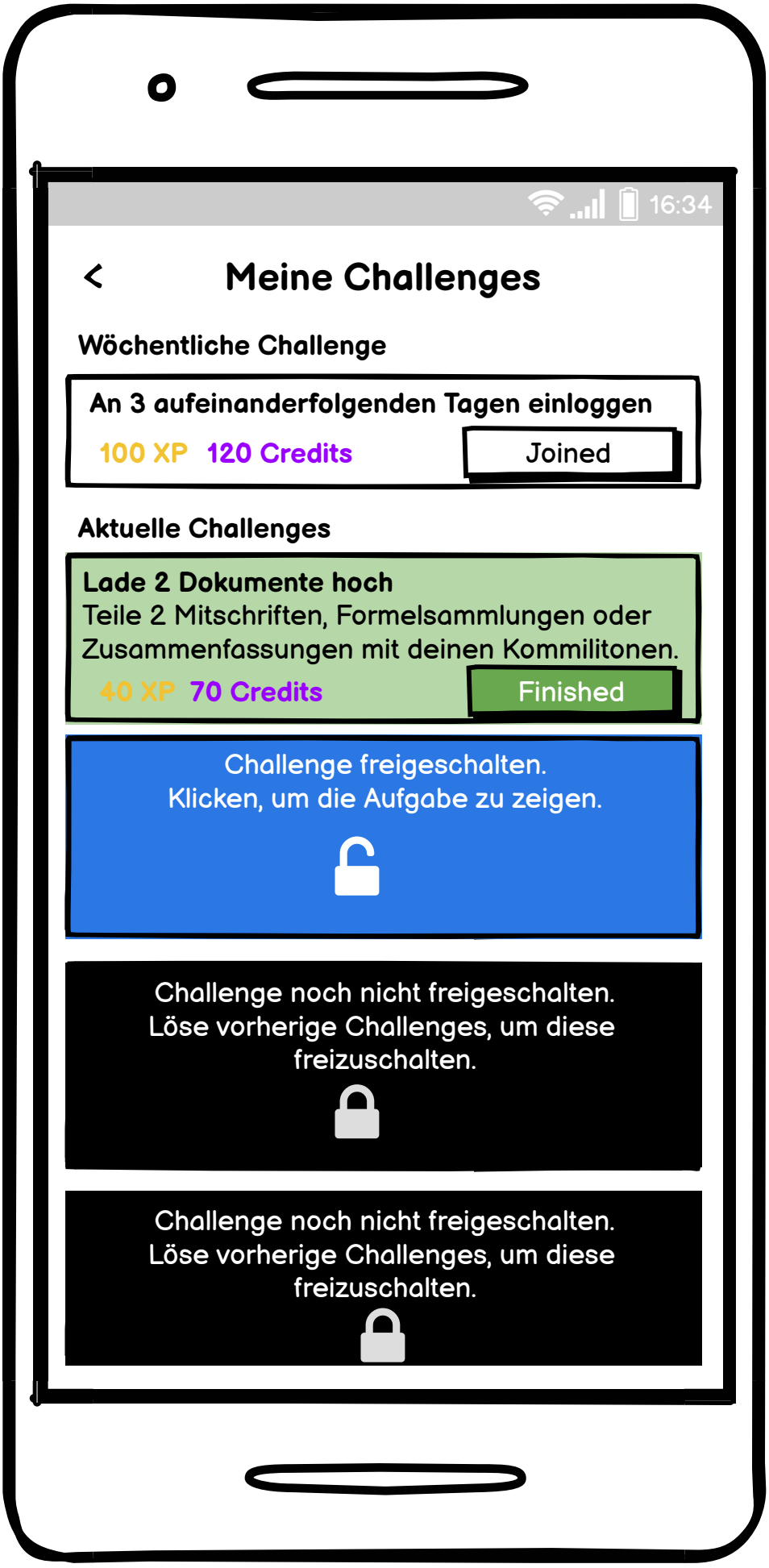
use arrows to scroll

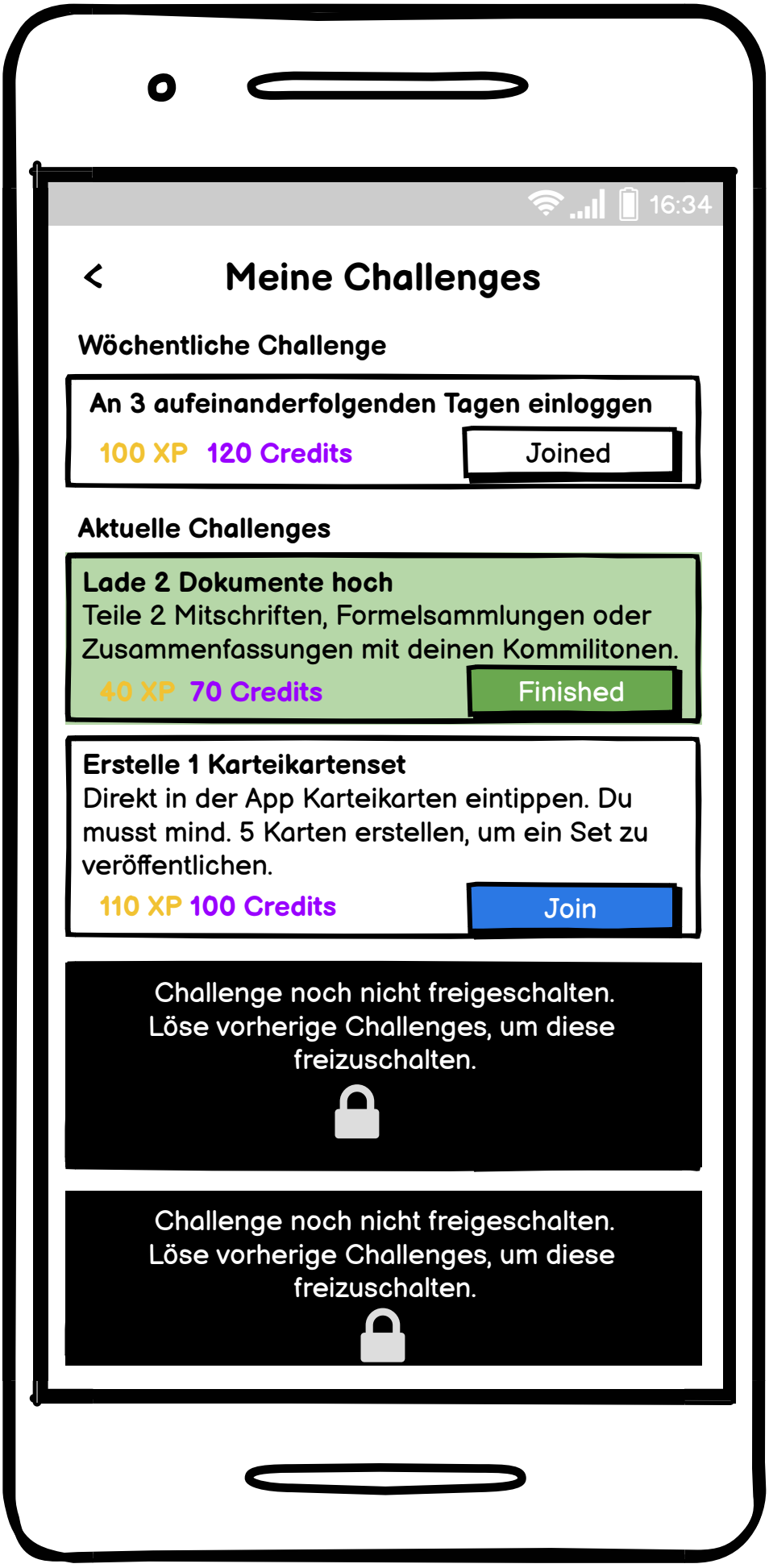


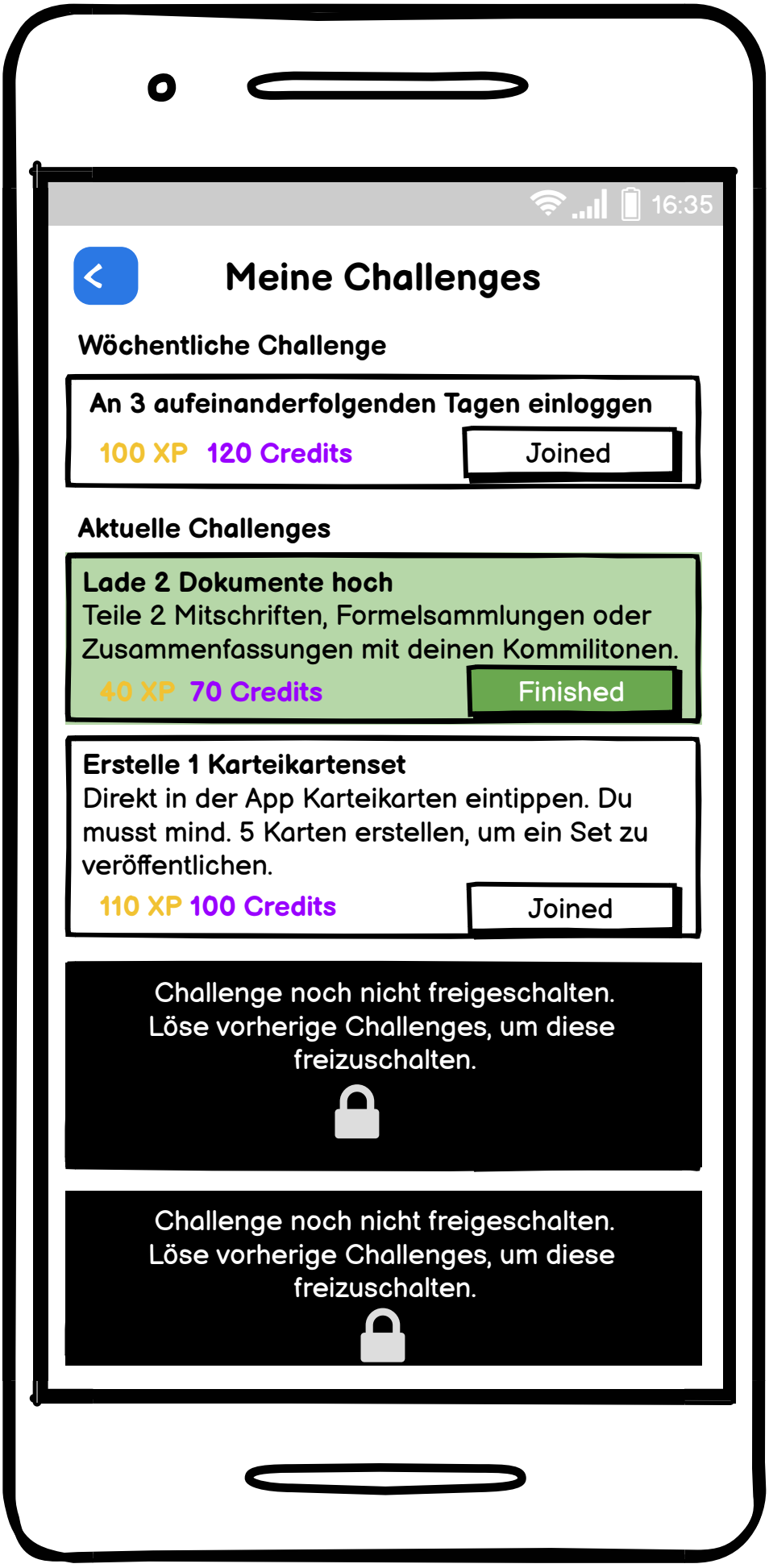
use arrows to scroll

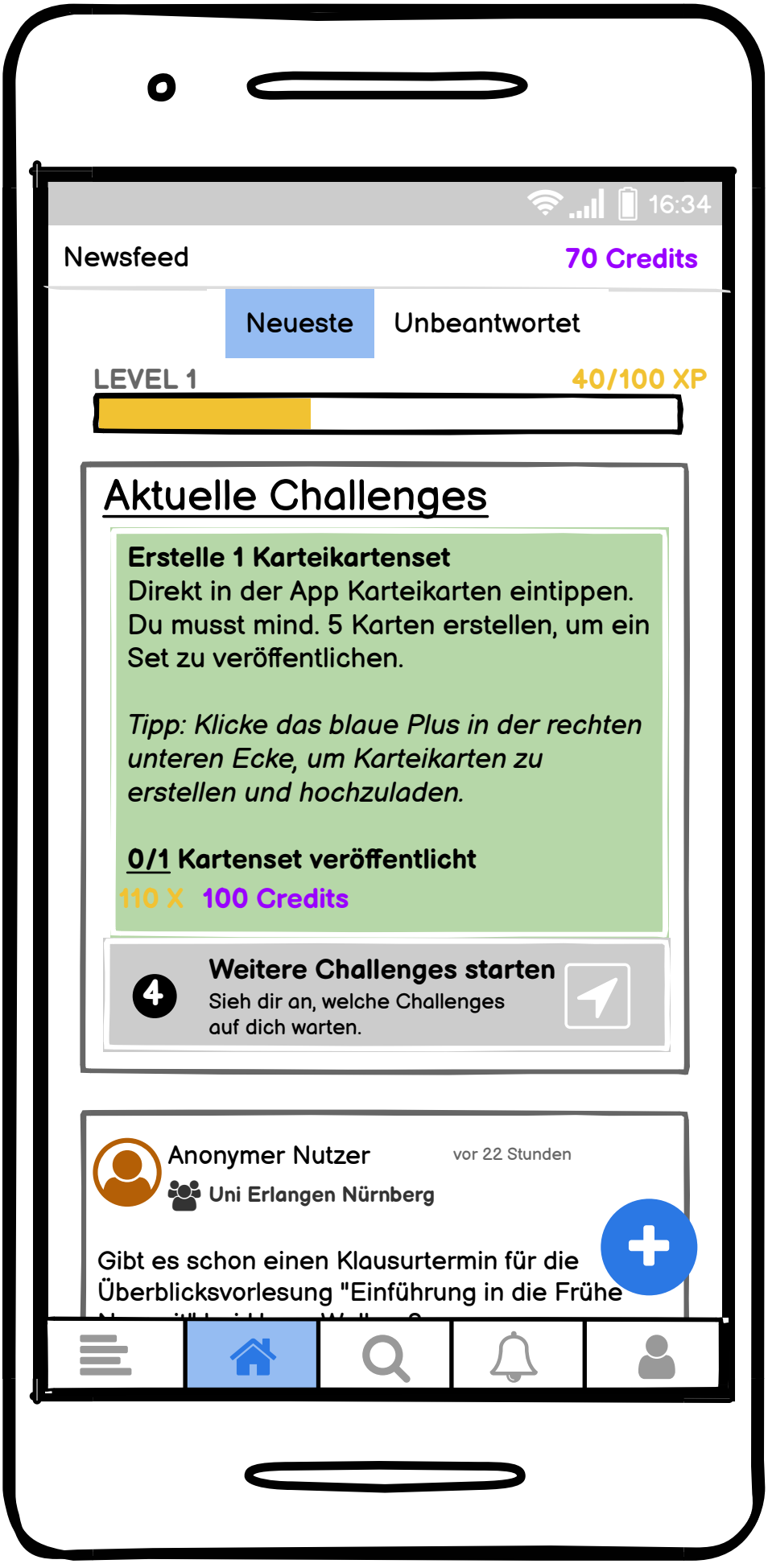


use arrows to scroll

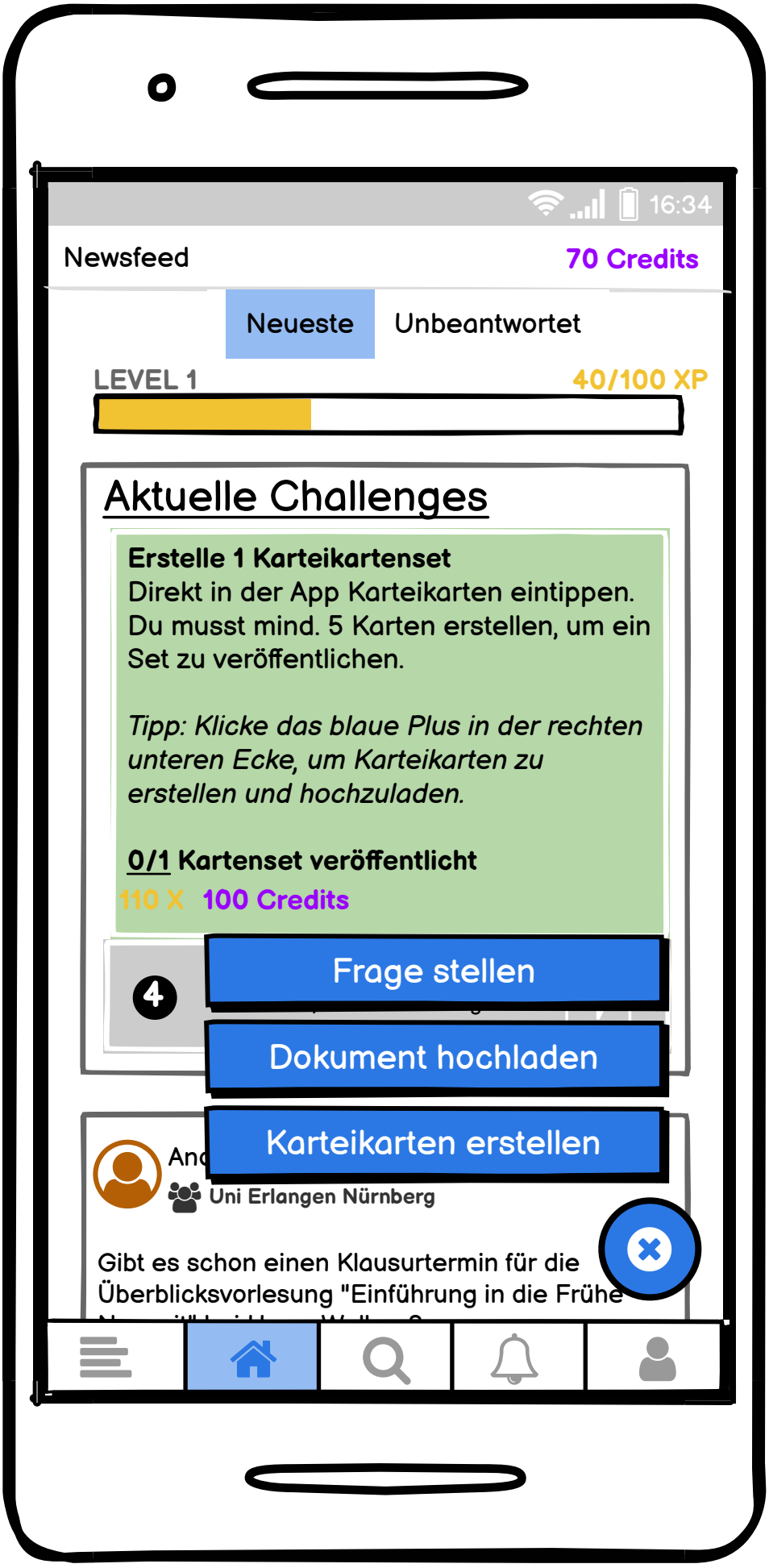

16:34
Meine Challenges
Wöchentliche Challenge
An 3 aufeinanderfolgenden Tagen einloggen
100 XP 120 Credits
Joined
Aktuelle Challenges
Lade 2 Dokumente hoch
Teile 2 Mitschriften, Formelsammlungen oder Zusammenfassungen mit deinen Kommilitonen.
40 XP 70 Credits
Finished
Challenge freigeschalten.
Klicken, um die Aufgabe zu zeigen.
Challenge noch nicht freigeschalten.
Löse vorherige Challenges, um diese freizuschalten.
Challenge noch nicht freigeschalten.
Löse vorherige Challenges, um diese freizuschalten.


16:34
Meine Challenges
Wöchentliche Challenge
An 3 aufeinanderfolgenden Tagen einloggen
100 XP 120 Credits
Joined
Aktuelle Challenges
Lade 2 Dokumente hoch
Teile 2 Mitschriften, Formelsammlungen oder Zusammenfassungen mit deinen Kommilitonen.
40 XP 70 Credits
Finished
Erstelle 1 Karteikartenset
Direkt in der App Karteikarten eintippen. Du musst mind. 5 Karten erstellen, um ein Set zu veröffentlichen.
110 XP 100 Credits
Join
Challenge noch nicht freigeschalten. Löse vorherige Challenges, um diese freizuschalten.
Challenge noch nicht freigeschalten. Löse vorherige Challenges, um diese freizuschalten.


16:35
Meine Challenges
Wöchentliche Challenge
An 3 aufeinanderfolgenden Tagen einloggen
100 XP 120 Credits
Joined
Aktuelle Challenges
Lade 2 Dokumente hoch
Teile 2 Mitschriften, Formelsammlungen oder Zusammenfassungen mit deinen Kommilitonen.
40 XP 70 Credits
Finished
Erstelle 1 Karteikartenset
Direkt in der App Karteikarten eintippen. Du musst mind. 5 Karten erstellen, um ein Set zu veröffentlichen.
110 XP 100 Credits
Joined
Challenge noch nicht freigeschalten. Löse vorherige Challenges, um diese freizuschalten.
Challenge noch nicht freigeschalten. Löse vorherige Challenges, um diese freizuschalten.



use arrows to scroll


16:34
Newsfeed
70 Credits
Neueste
Unbeantwortet
LEVEL 1
40/100 XP
Aktuelle Challenges
Erstelle 1 Karteikartenset
Direkt in der App Karteikarten eintippen. Du musst mind. 5 Karten erstellen, um ein Set zu veröffentlichen.
Tipp: Klicke das blaue Plus in der rechten unteren Ecke, um Karteikarten zu erstellen und hochzuladen.
0/1 Kartenset veröffentlicht
110 X 100 Credits
4
Frage stellen
Dokument hochladen
Karteikarten erstellen
Uni Erlangen Nürnberg
Gibt es schon einen Klausurtermin für die Überblicksvorlesung "Einführung in die Frühe



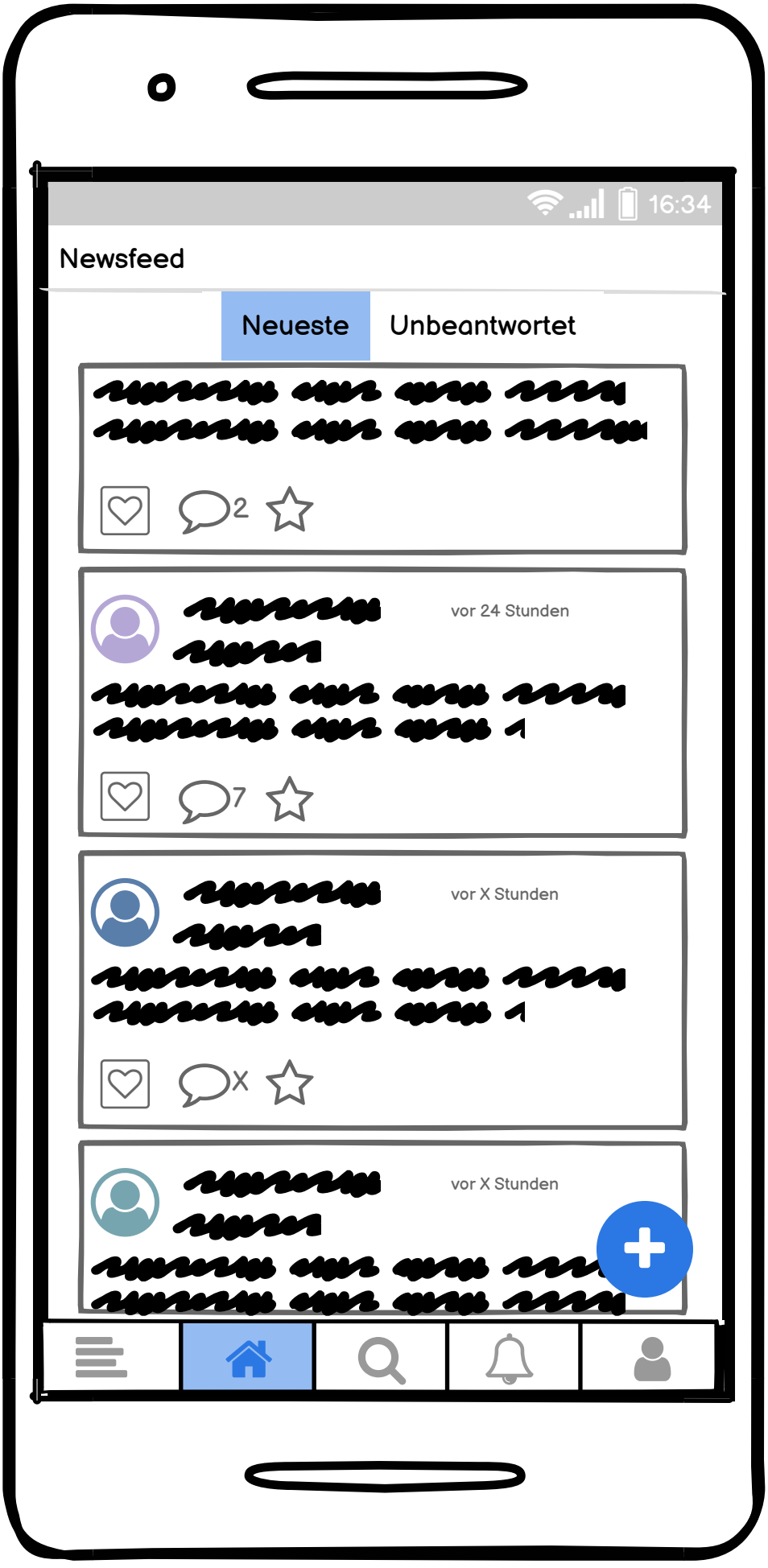
use arrows to scroll


16:35
Newsfeed
Neueste
Unbeantwortet
0/1 Kartenset veröffentlicht
110 X 100 Credits
4
Challenges starten
Sieh dir an, welche Challenges auf dich warten.
Anonymer Nutzer
vor 22 Stunden
Uni Erlangen Nürnberg
Gibt es schon einen Klausurtermin für die Überblicksvorlesung "Einführung in die Frühe Neuzeit" bei Herrn Wellner?
2
Frage stellen
Dokument hochladen
Karteikarten erstellen
5
Anonyme Rakete
vor 23 Stunden



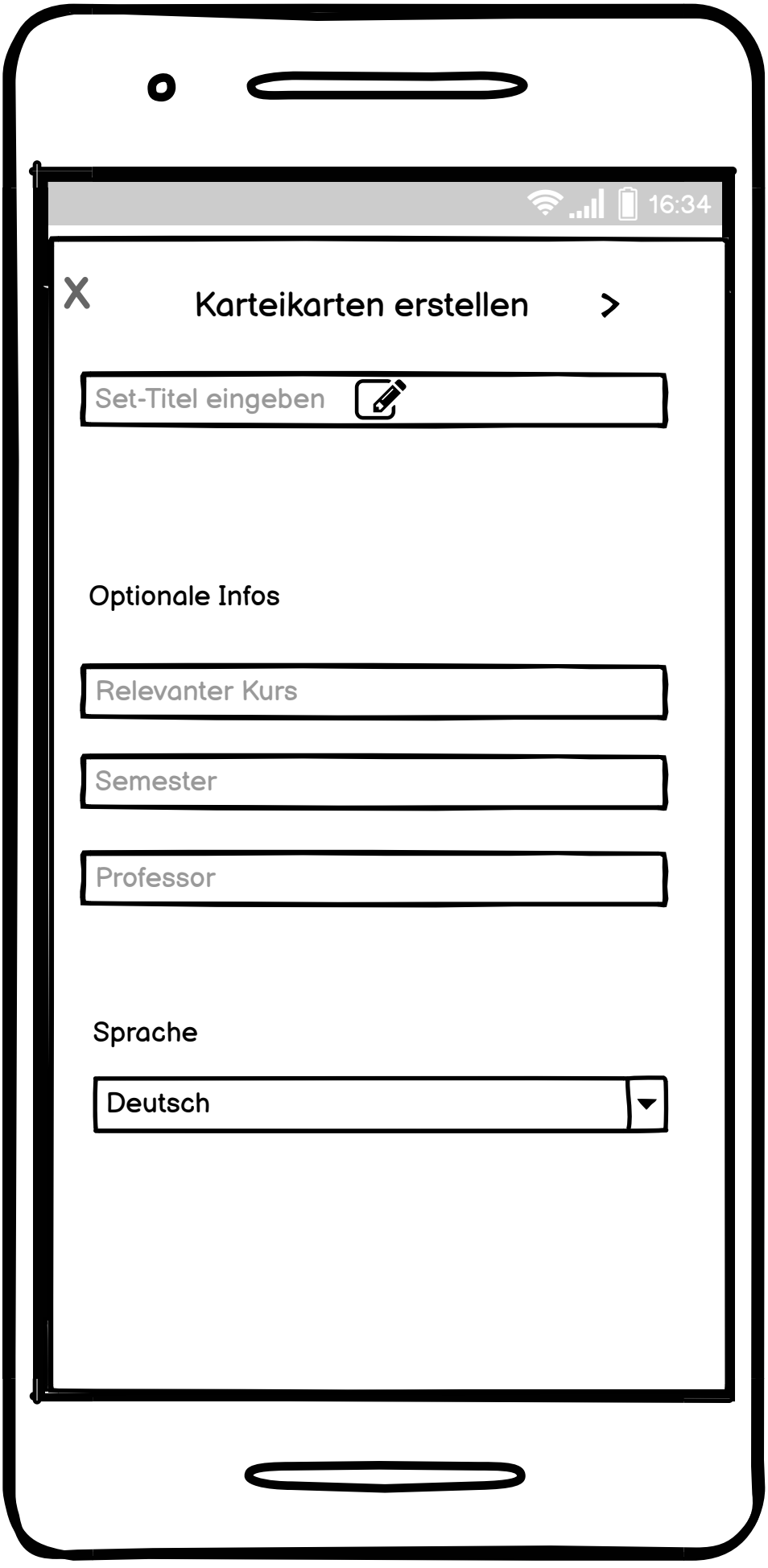
use arrows to scroll

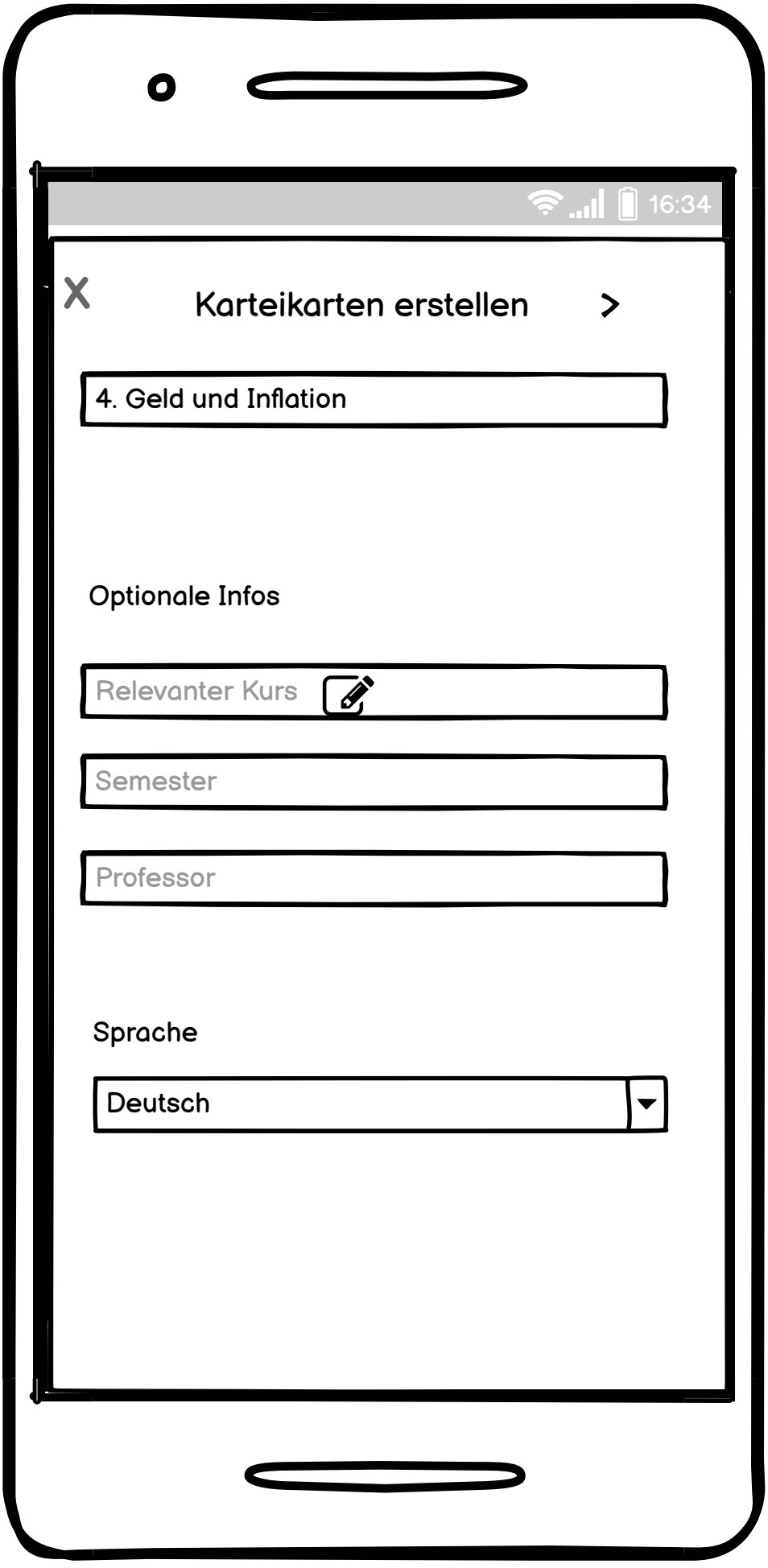

16:34
Newsfeed
Neueste
Unbeantwortet
2
Anonyme Nutzerin
vor 23 Stunden
Marketing
5
Anonyme Rakete
vor 23 Stunden
Frage stellen
Dokument hochladen
Karteikarten erstellen
7

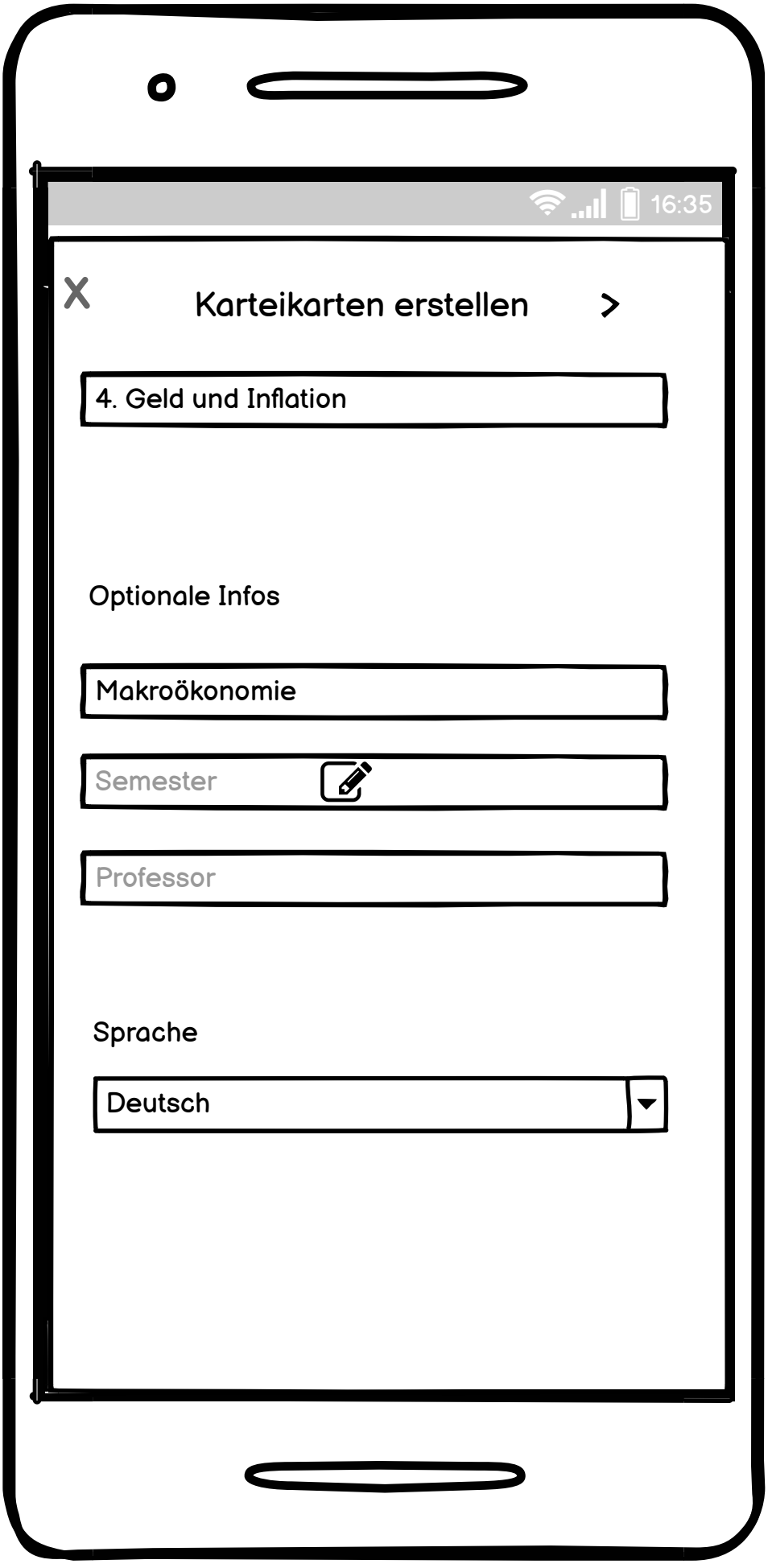


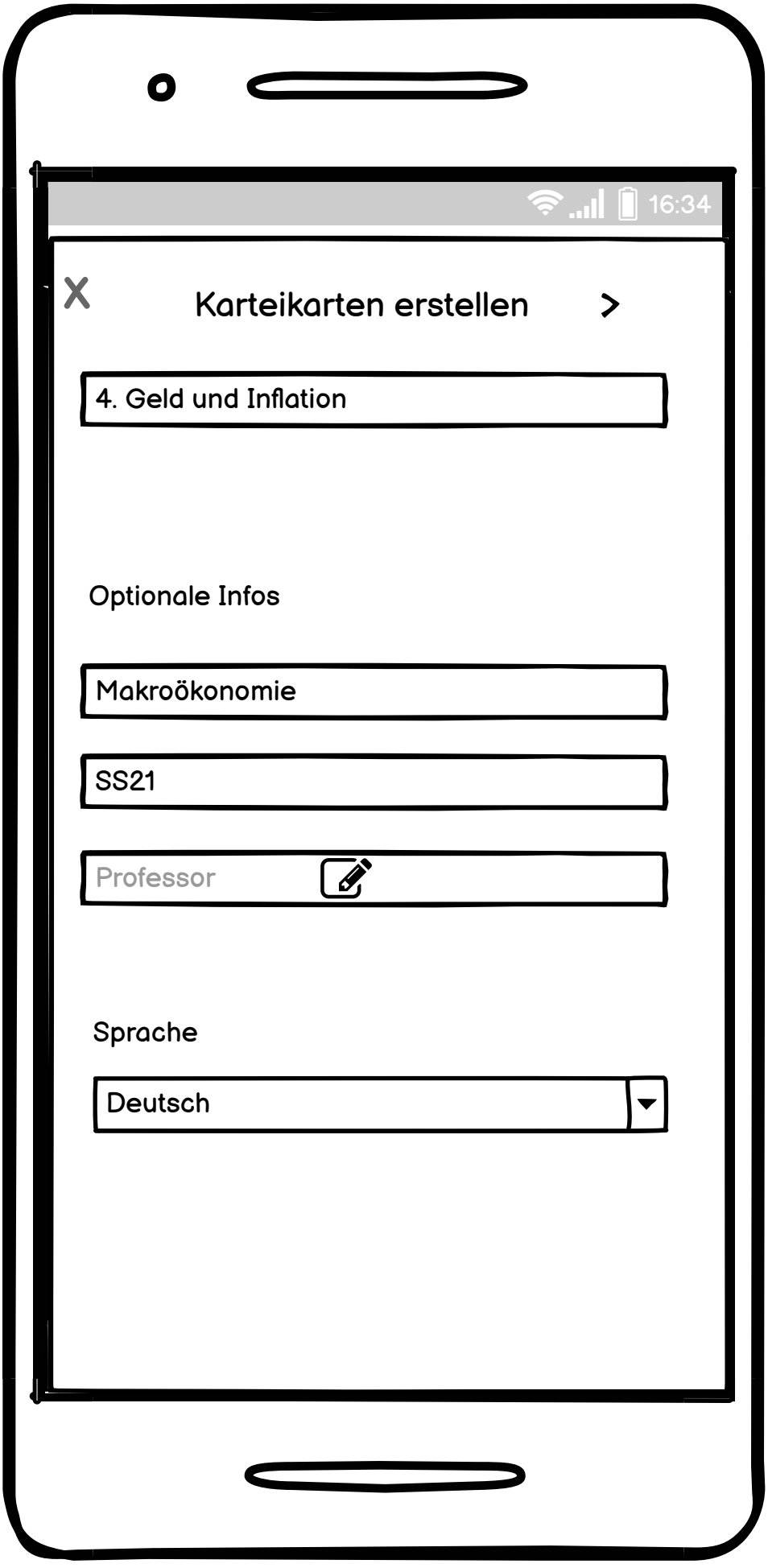
use arrows to scroll

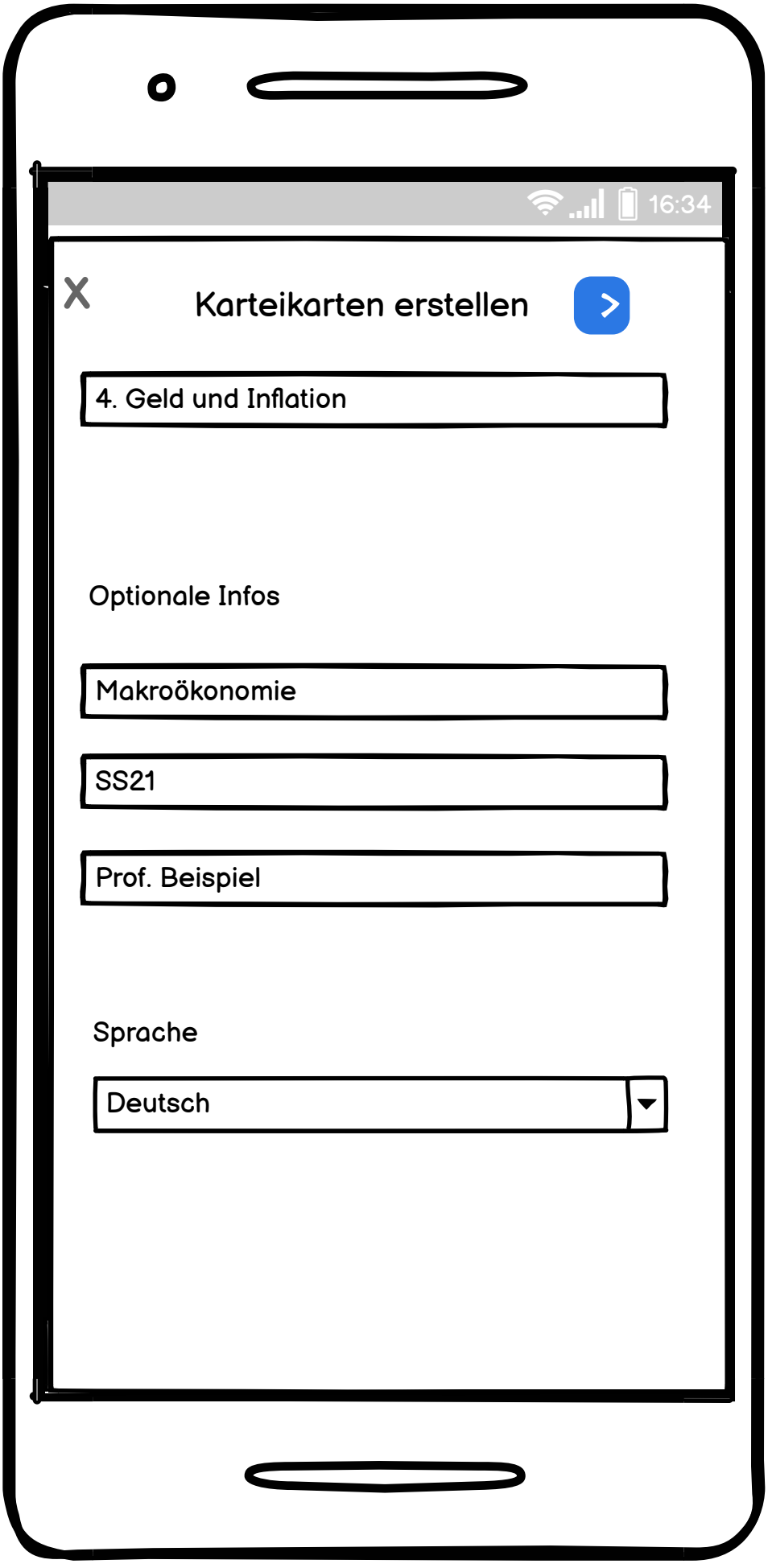

16:34
Newsfeed
Neueste
Unbeantwortet
2
vor 24 Stunden
7
vor X Stunden
Frage stellen
Dokument hochladen
Karteikarten erstellen


16:34
X
Karteikarten erstellen
>
Set-Titel eingeben
Optionale Infos
Relevanter Kurs
Semester
Professor
Sprache
Deutsch

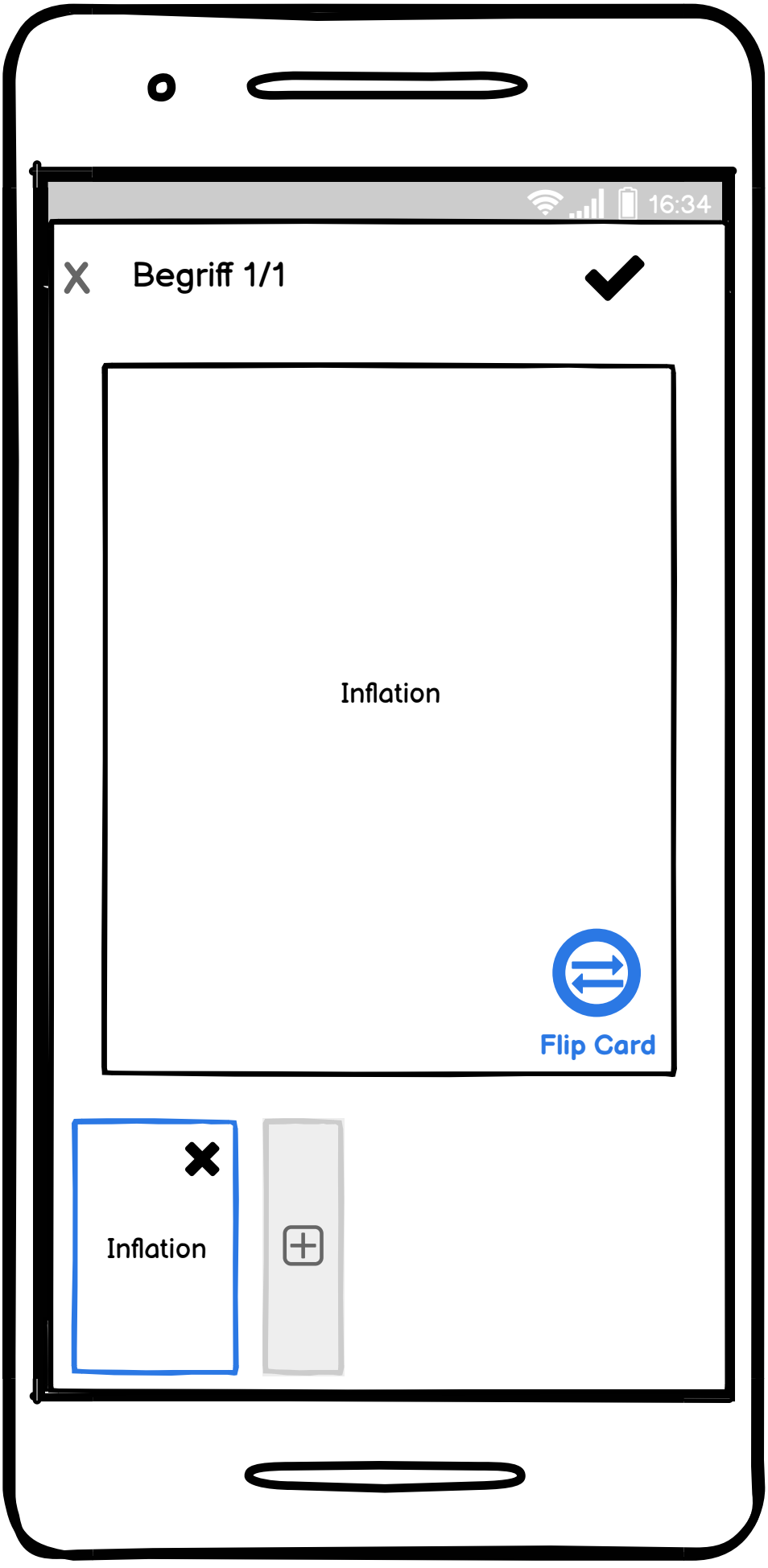

16:34
X
Karteikarten erstellen
>
4. Geld und Inflation
Optionale Infos
Relevanter Kurs
Semester
Professor
Sprache
Deutsch


16:35
X
Karteikarten erstellen
>
4. Geld und Inflation
Optionale Infos
Makroökonomie
Semester
Professor
Sprache
Deutsch

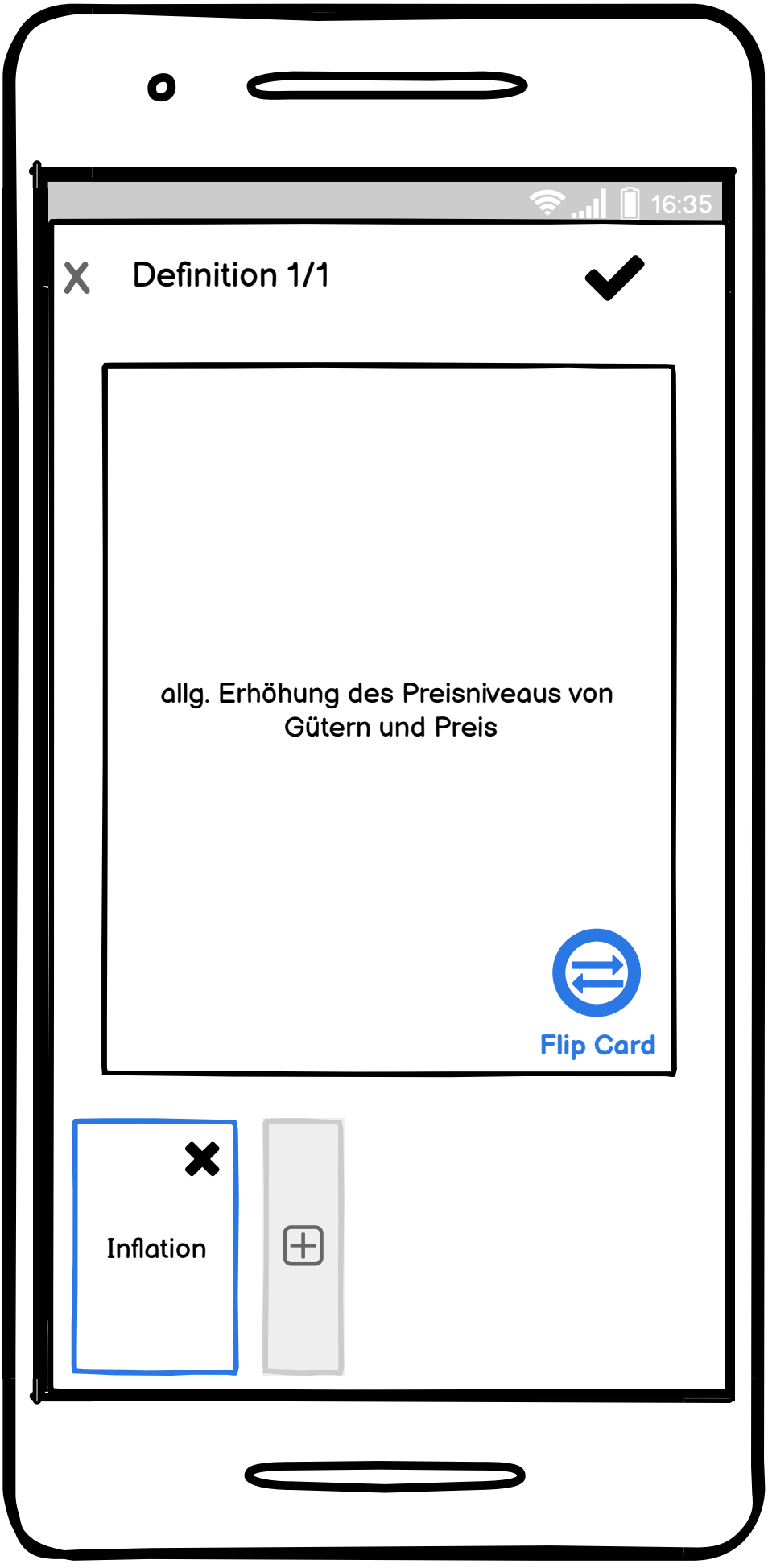

16:34
Karteikarten erstellen
4. Geld und Inflation
Optionale Infos
Makroökonomie
SS21
Professor
Sprache
Deutsch

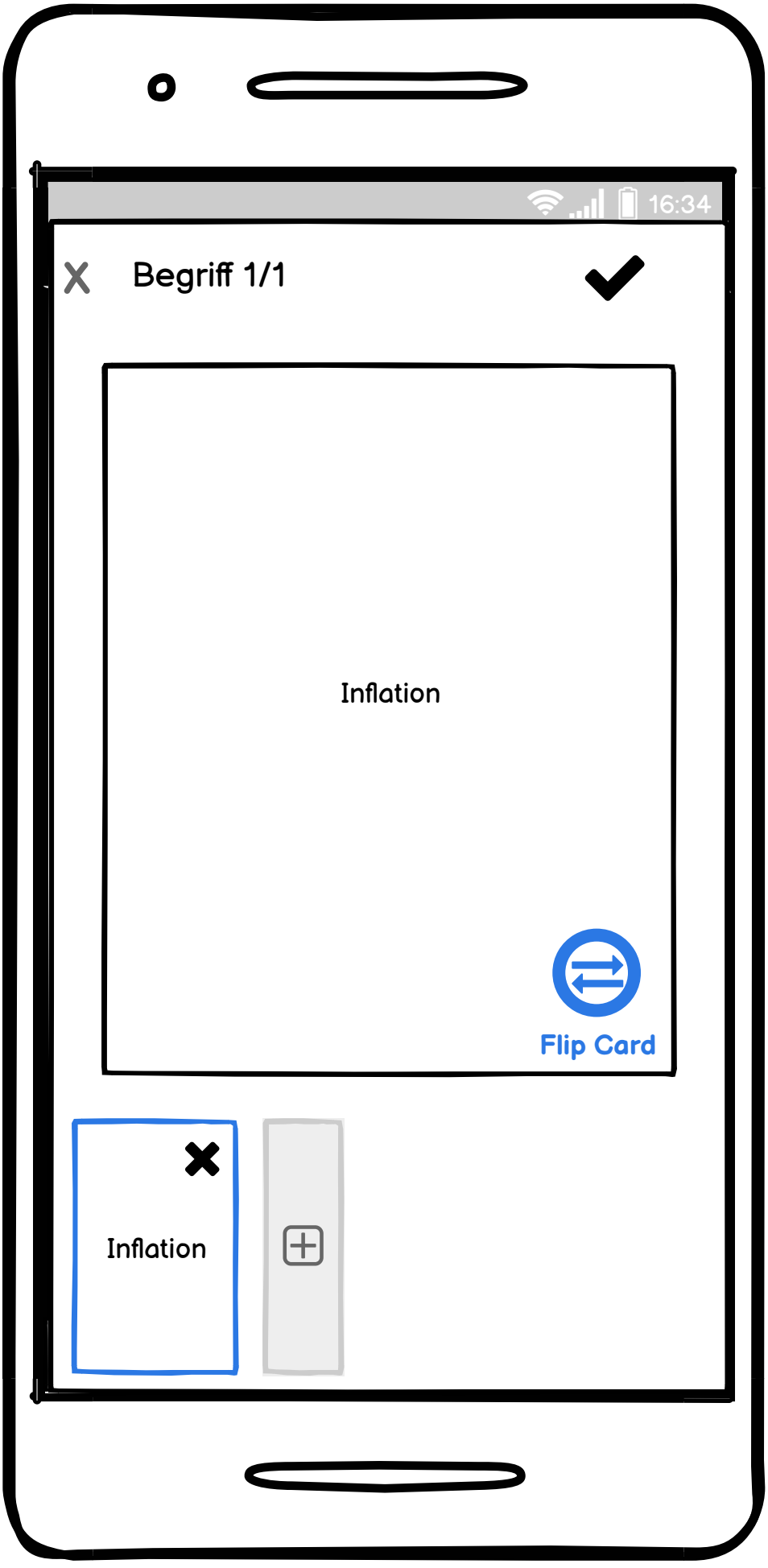

16:34
X
Karteikarten erstellen
4. Geld und Inflation
Optionale Infos
Makroökonomie
SS21
Prof. Beispiel
Sprache
Deutsch


16:34
Begriff 1/1
Tippen, um Begriff einzugeben
Flip Card

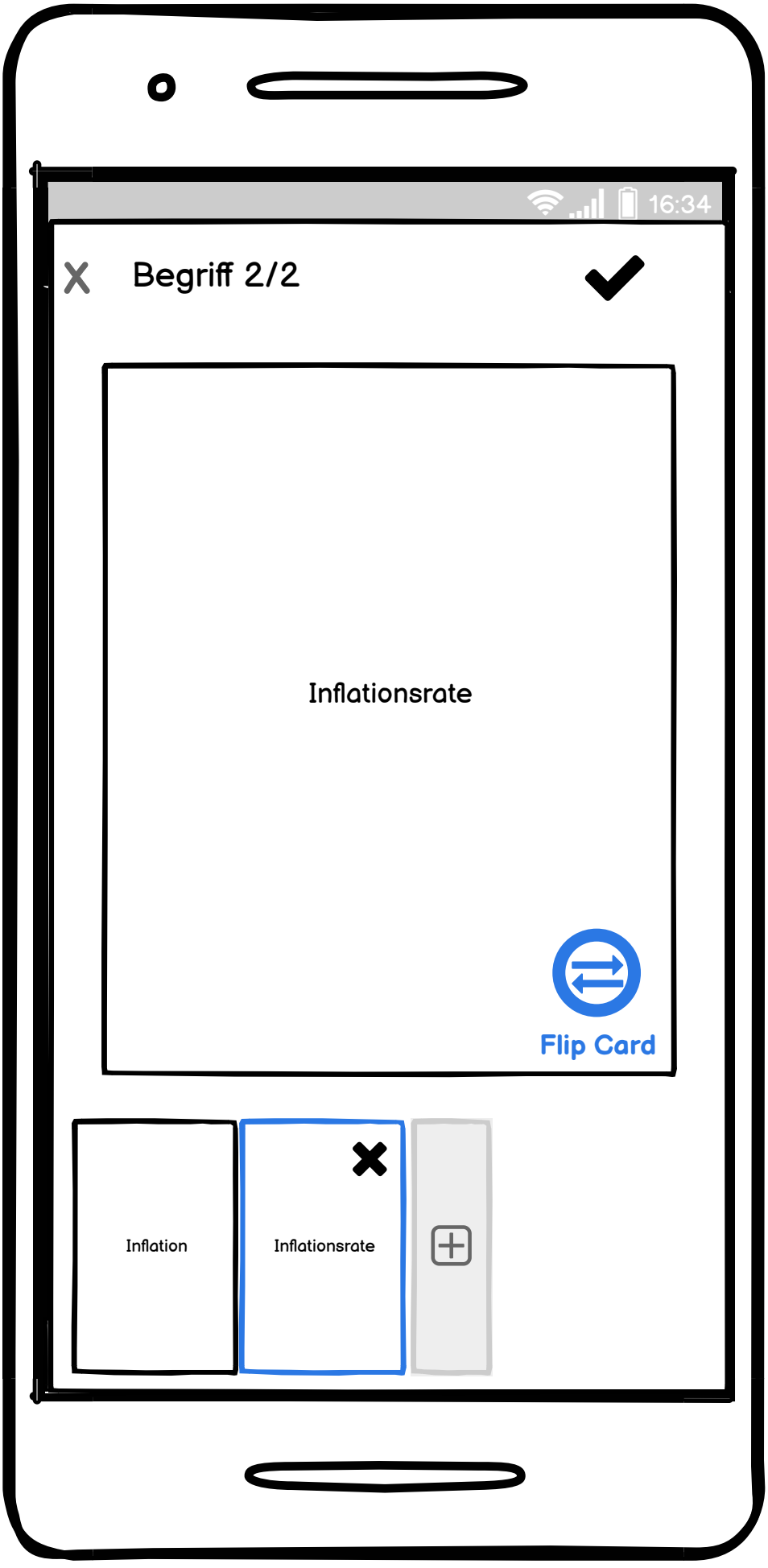

16:34
Begriff 1/1
Inflation
Flip Card
Inflation


16:34
Definition 1/1
Tippen, um Definition einzugeben
Flip Card
Inflation

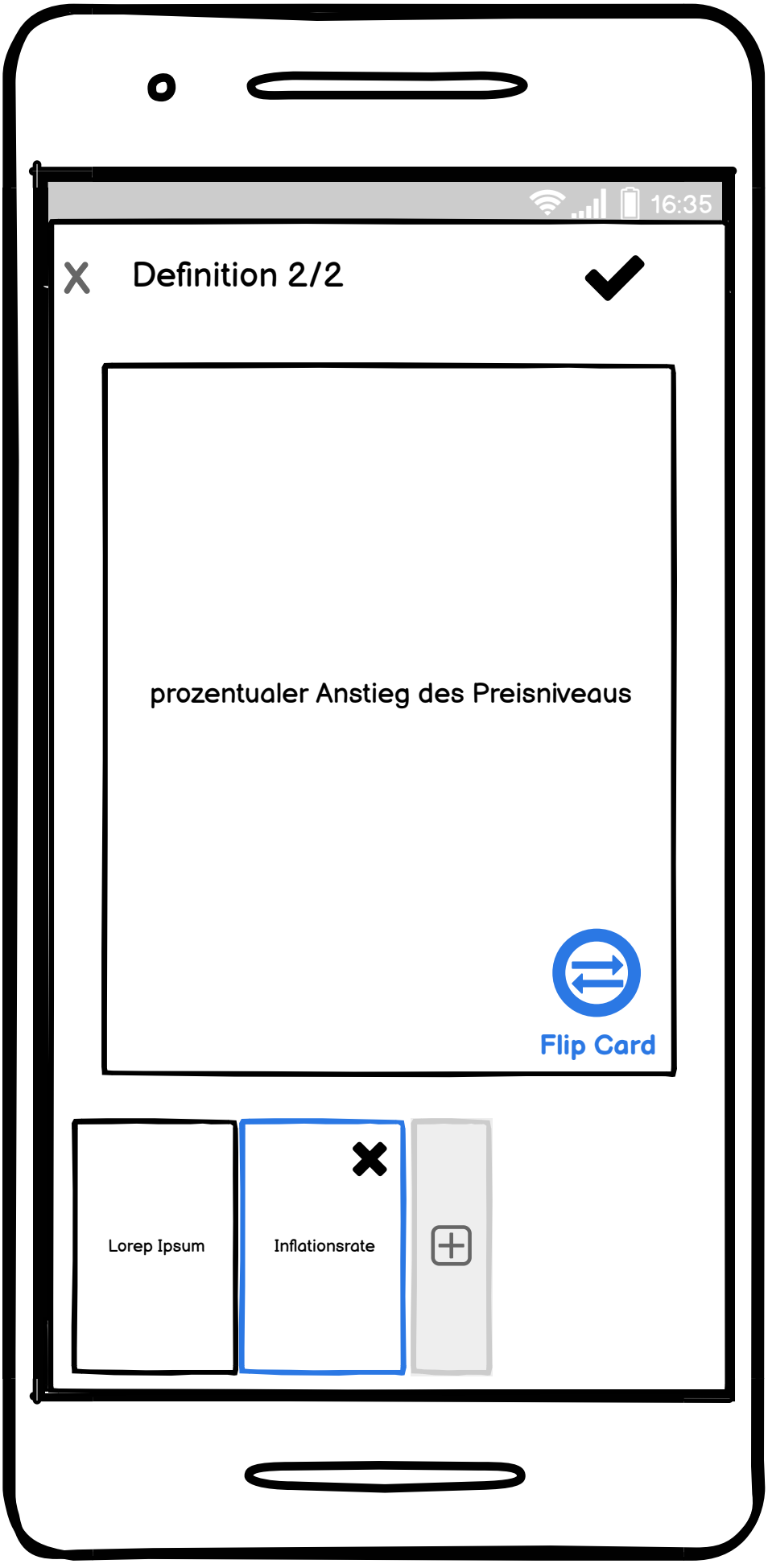

16:35
Definition 1/1
allg. Erhöhung des Preisniveaus von Gütern und Preis
Flip Card
Inflation

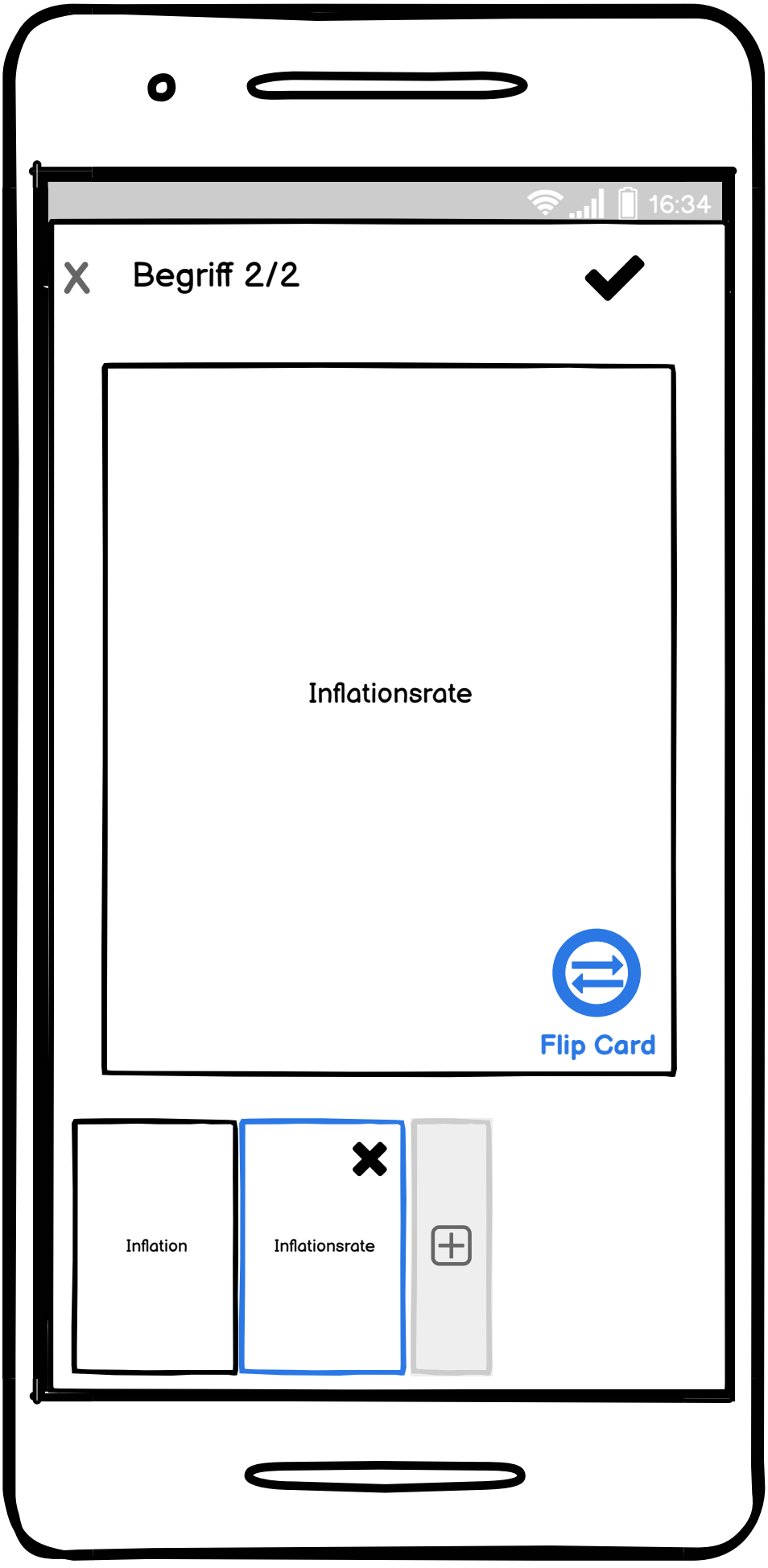

16:34
Begriff 1/1
Inflation
Flip Card
Inflation


16:34
Begriff 2/2
Tippen, um Begriff einzugeben
Flip Card
Inflation


16:34
Begriff 2/2
Inflationsrate
Flip Card
Inflation
Inflationsrate


16:34
Definition 2/2
Tippen, um Definition einzugeben
Flip Card
Lorep Ipsum
Inflationsrate


16:35
Definition 2/2
prozentualer Anstieg des Preisniveaus
Flip Card
Lorep Ipsum
Inflationsrate


16:34
Begriff 2/2
Inflationsrate
Flip Card
Inflation
Inflationsrate


16:35
Begriff 3/3
Tippen, um Begriff einzugeben
Flip Card
Lorep Ipsum
Inflationsrate


16:34
Begriff 3/3
3 Funktionen des Geldes
Flip Card
Lorep Ipsum
Inflationsrate
3 Funktionen des Geldes


16:34
Definition 3/3
Tippen, um Definition einzugeben
Flip Card
Lorep Ipsum
Inflationsrate
3 Funktionen des Geldes


16:34
Definition 3/3
Mittel zur Aufbewahrung
Recheneinheit
Tauschmittel (Transaktion)
Flip Card
Lorep Ipsum
Inflationsrate
3 Funktionen des Geldes


16:34
Begriff 3/3
3 Funktionen des Geldes
Flip Card
Lorep Ipsum
Inflationsrate
3 Funktionen des Geldes


16:34
Begriff 4/4
Tippen, um Begriff einzugeben
Flip Card
Inflationsrate
3 Funktionen des Geldes


16:34
Begriff 4/4
Vorteil des Geldes
Flip Card
Inflationsrate
3 Funktionen des Geldes
Vorteil des Geldes


16:35
Definition 4/4
Tippen, um Definition einzugeben
Flip Card
Inflationsrate
3 Funktionen des Geldes
Vorteil des Geldes

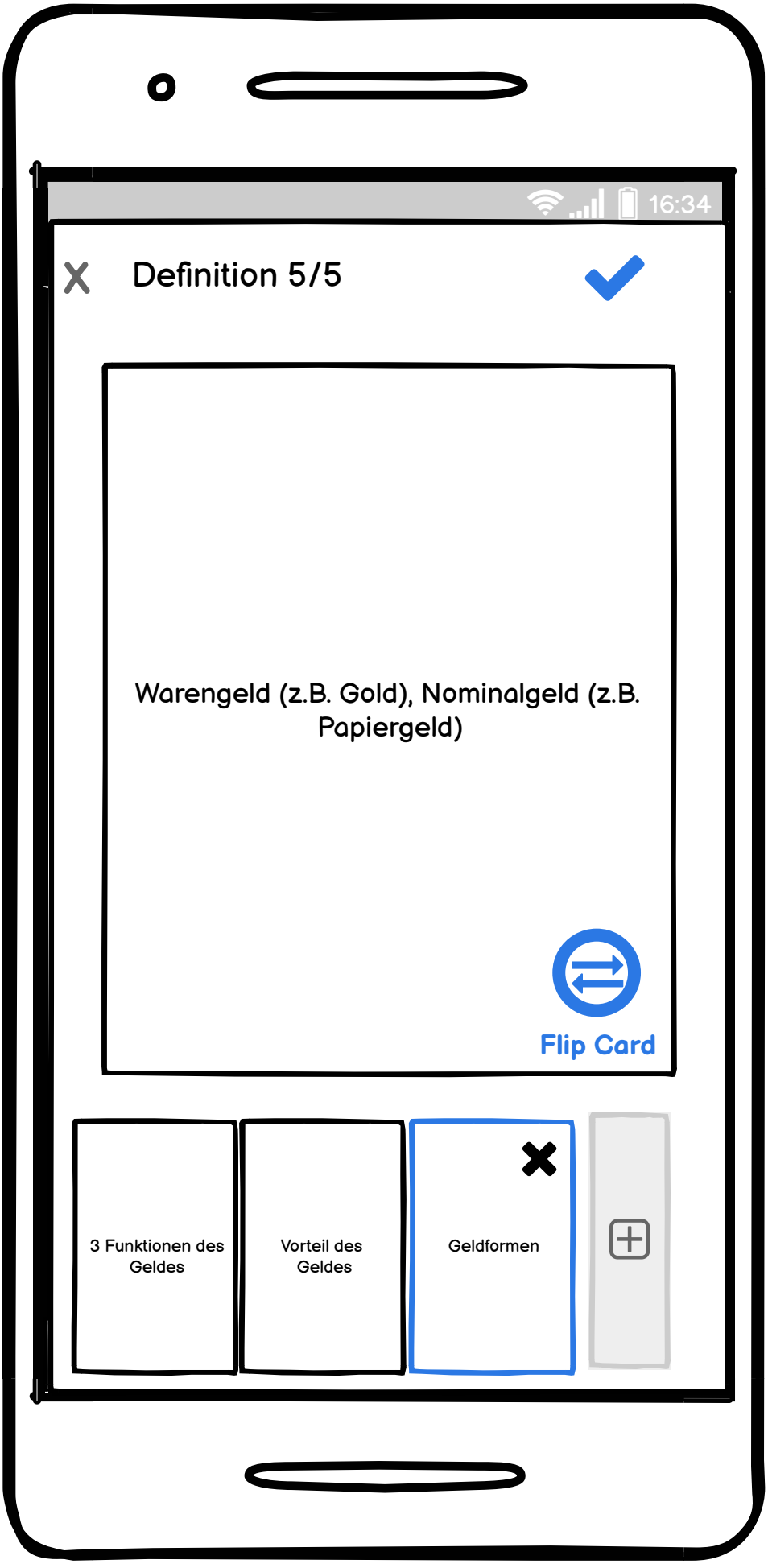

16:35
Definition 4/4
ohne Geld würde eine Naturaltauschwirtschaft herrschen: usw...
Flip Card
Inflationsrate
3 Funktionen des Geldes
Vorteil des Geldes


16:34
Begriff 4/4
Vorteil des Geldes
Flip Card
Inflationsrate
3 Funktionen des Geldes
Vorteil des Geldes


16:34
Begriff 5/5
Tippen, um Begriff einzugeben
Flip Card
3 Funktionen des Geldes
Vorteil des Geldes


16:34
Begriff 5/5
Geldformen
Flip Card
3 Funktionen des Geldes
Vorteil des Geldes
Geldformen


16:34
Definition 5/5
Tippen, um Definition einzugeben
Flip Card
3 Funktionen des Geldes
Vorteil des Geldes
Geldformen


16:34
Definition 5/5
Warengeld (z.B. Gold), Nominalgeld (z.B. Papiergeld)
Flip Card
3 Funktionen des Geldes
Vorteil des Geldes
Geldformen


16:34
Begriff 5/5
Geldformen
Flip Card
3 Funktionen des Geldes
Vorteil des Geldes
Geldformen

16:34

Challenge erledigt: Erstelle 1 Karteikarten S

1/1 Kartenset veröffentlicht

Ok

Neues Badge!

Du hast das zweite Level erreicht. Starke Leistung!

2

Ok

erstellen und hochzuladen.

1/1 Kartenset veröffentlicht

110 XP100 Credits

4 **Weitere Challenges starten**

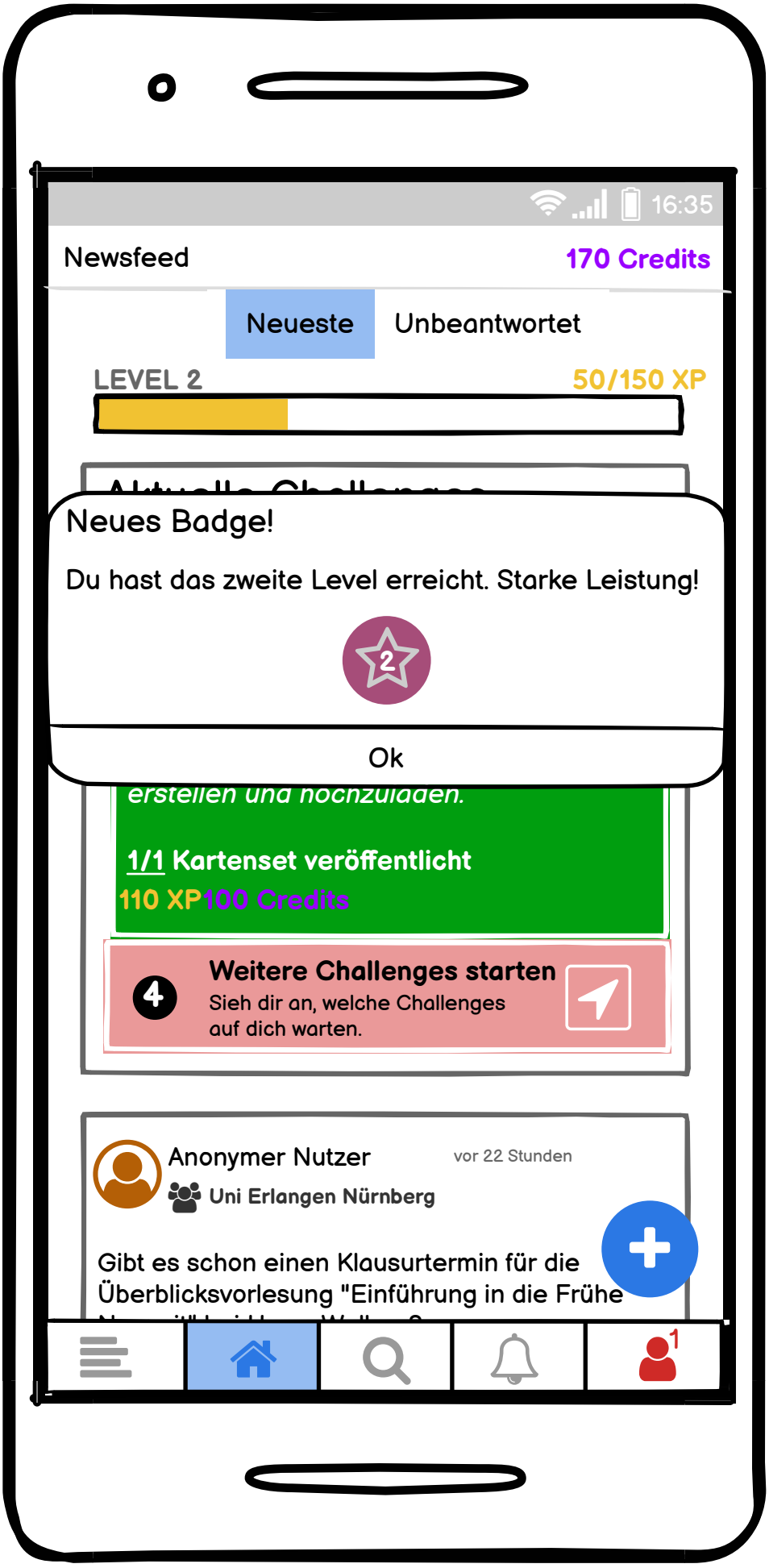
Sieh dir an, welche Challenges auf dich warten.

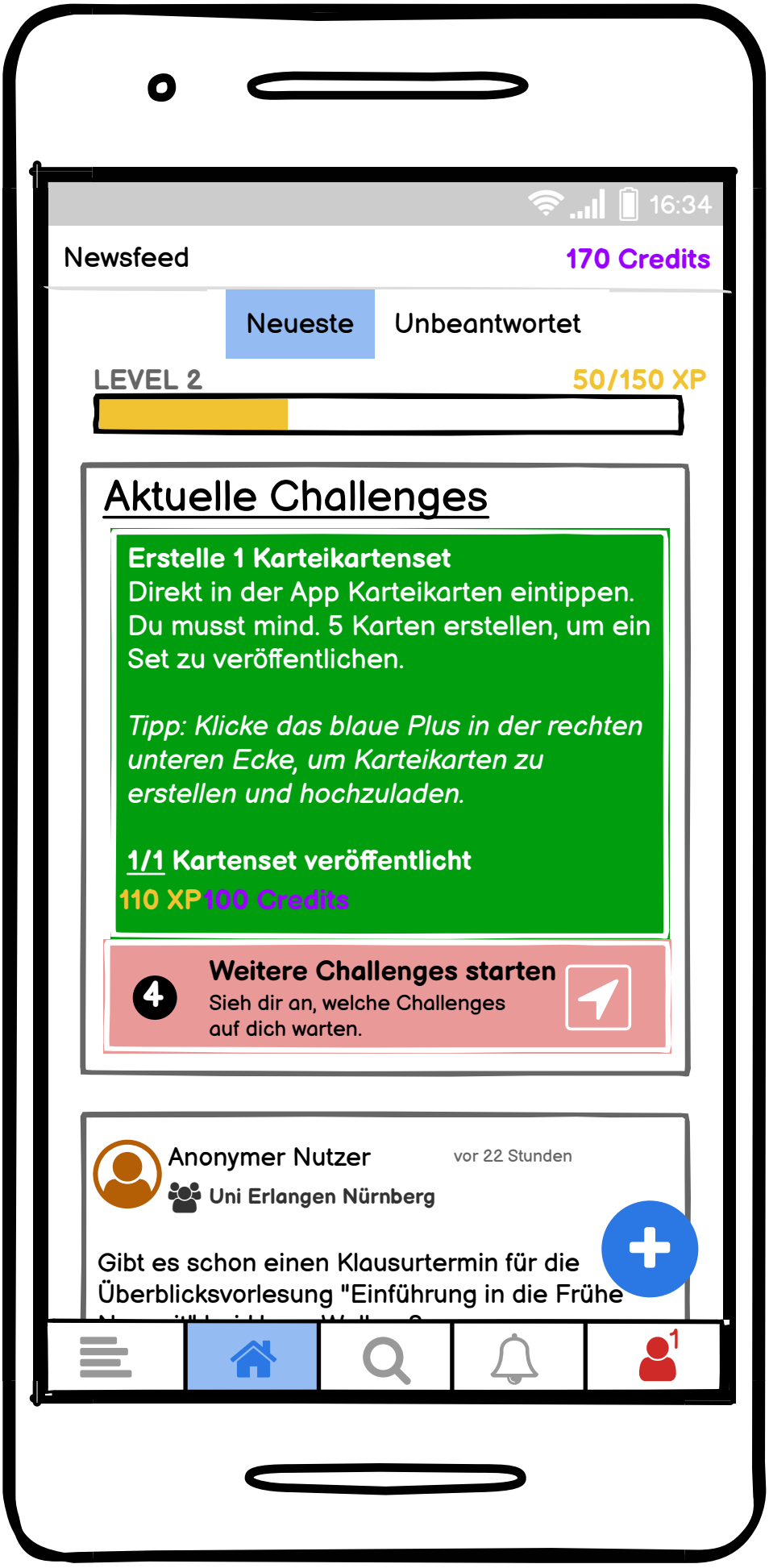
Anonymer Nutzer vor 22 Stunden

Uni Erlangen Nürnberg

Gibt es schon einen Klausurtermin für die Überblicksvorlesung "Einführung in die Frühe


16:35
Challenge erledigt: Erstelle 1 Karteikarten S
1/1 Kartenset veröffentlicht
Ok
Aktuelle Challenges
Erstelle 1 Karteikartenset
Direkt in der App Karteikarten eintippen. Du musst mind. 5 Karten erstellen, um ein Set zu veröffentlichen.
Tipp: Klicke das blaue Plus in der rechten unteren Ecke, um Karteikarten zu erstellen und hochzuladen.
1/1 Kartenset veröffentlicht
110 XP100 Credits
4
Weitere Challenges starten
Sieh dir an, welche Challenges auf dich warten.
Anonymer Nutzer
vor 22 Stunden
Uni Erlangen Nürnberg
Gibt es schon einen Klausurtermin für die Überblicksvorlesung "Einführung in die Frühe
1


16:35
Newsfeed
170 Credits
Neueste
Unbeantwortet
LEVEL 2
50/150 XP
Neues Badge!
Du hast das zweite Level erreicht. Starke Leistung!
2
Ok
erstellen und hochzuladen.
1/1 Kartenset veröffentlicht
110 XP100 Credits
4
Weitere Challenges starten
Sieh dir an, welche Challenges
auf dich warten.
Anonymer Nutzer
vor 22 Stunden
Uni Erlangen Nürnberg
Gibt es schon einen Klausurtermin für die
Überblicksvorlesung "Einführung in die Frühe
1


16:34
Newsfeed
170 Credits
Neueste
Unbeantwortet
LEVEL 2
50/150 XP
Aktuelle Challenges
Erstelle 1 Karteikartenset
Direkt in der App Karteikarten eintippen. Du musst mind. 5 Karten erstellen, um ein Set zu veröffentlichen.
Tipp: Klicke das blaue Plus in der rechten unteren Ecke, um Karteikarten zu erstellen und hochzuladen.
1/1 Kartenset veröffentlicht
110 XP100 Credits
4
Weitere Challenges starten
Sieh dir an, welche Challenges auf dich warten.
Anonymer Nutzer
vor 22 Stunden
Uni Erlangen Nürnberg
Gibt es schon einen Klausurtermin für die Überblicksvorlesung "Einführung in die Frühe
1

16:34

Profil Bearbeiten

Bild auswählen

BWLJustus

Männlich

Dein Studium

Studium hinzufügen

Friedrich-Alexander-Universität Erlangen

Betriebswirtschaftslehre

AKTUELLES LEVEL

LEVEL 2

50/150 XP

ERFOLGE

UPLOADER LVL 2

use arrows to scroll

16:35

Profil Bearbeiten

Friedrich-Alexander-Universität Erlangen

Betriebswirtschaftslehre >

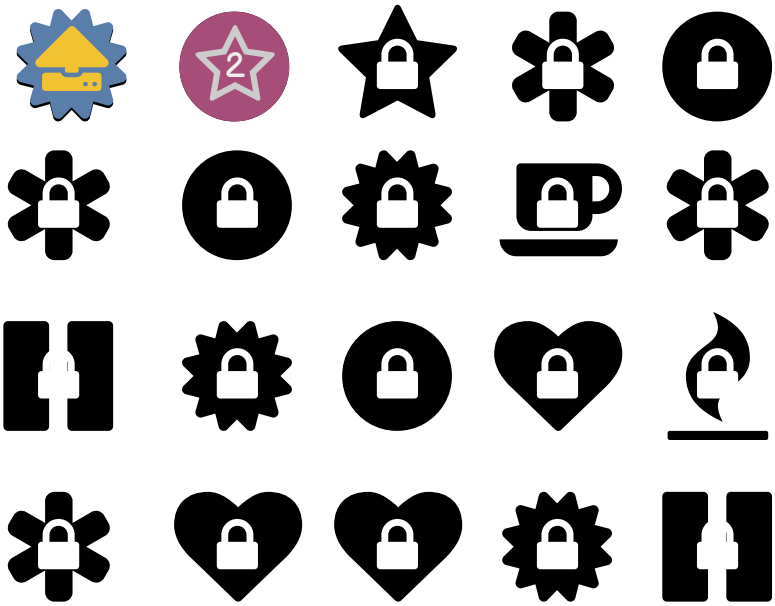
AKTUELLES LEVEL

LEVEL 2

50/150 XP

ERFOLGE

UPLOADER LVL 2





use arrows to scroll



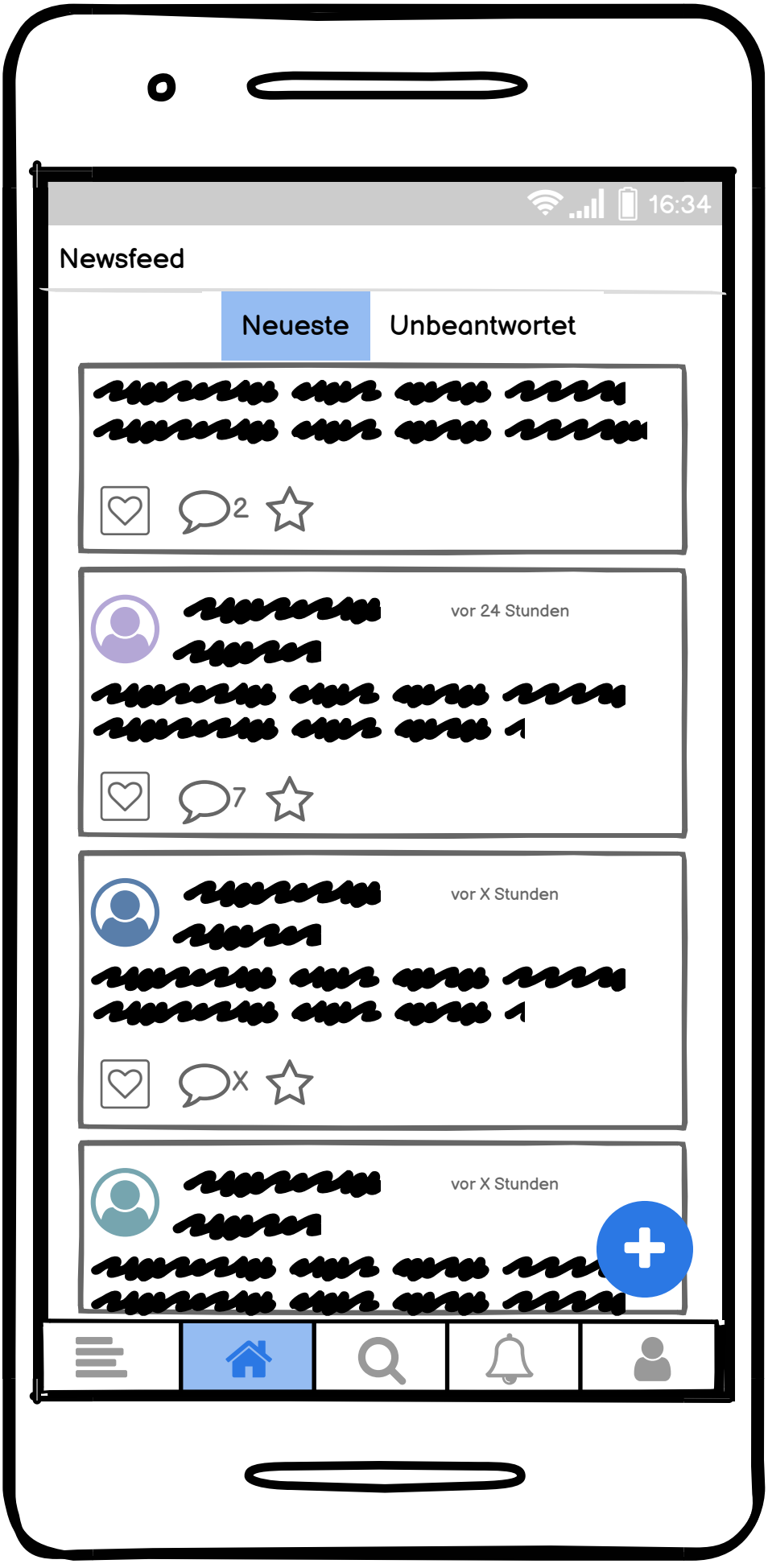
use arrows to scroll



use arrows to scroll



use arrows to scroll



use arrows to scroll


16:35
Meine Challenges
Wöchentliche Challenge
An 3 aufeinanderfolgenden Tagen einloggen
100 XP 120 Credits
Joined
Aktuelle Challenges
Lade 2 Dokumente hoch
Teile 2 Mitschriften, Formelsammlungen oder Zusammenfassungen mit deinen Kommilitonen.
40 XP 70 Credits
Finished
Erstelle 1 Karteikartenset
Direkt in der App Karteikarten eintippen. Du musst mind. 5 Karten erstellen, um ein Set zu veröffentlichen.
110 XP 100 Credits
Finished
Challenge freigeschalten.
Klicken, um die Aufgabe zu zeigen.
Challenge noch nicht freigeschalten.
Löse vorherige Challenges, um diese freizuschalten.


16:34
Meine Challenges
Wöchentliche Challenge
An 3 aufeinanderfolgenden Tagen einloggen
100 XP 120 Credits
Joined
Aktuelle Challenges
Lade 2 Dokumente hoch
Teile 2 Mitschriften, Formelsammlungen oder Zusammenfassungen mit deinen Kommilitonen.
40 XP 70 Credits
Finished
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110 XP 100 Credits
Finished
Erstelle 2 Posts
Stelle und beantworte Fragen, teile deine Erfahrungen oder gebe hilfreichen Input in Kursen oder Gruppen.
90 XP 100 Credits
Join
Challenge noch nicht freigeschalten.
Löse vorherige Challenges, um diese freizuschalten.


16:35
Meine Challenges
Wöchentliche Challenge
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100 XP 120 Credits
Joined
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90 XP 100 Credits
Joined
Challenge noch nicht freigeschalten.
Löse vorherige Challenges, um diese freizuschalten.

16:34

Newsfeed 170 Credits

Neueste Unbeantwortet

LEVEL 2 50/150 XP

Aktuelle Challenges

Erstelle 2 Posts
Stelle und beantworte Fragen, teile deine Erfahrungen oder gebe hilfreichen Input in Kursen oder Gruppen.

Tipp: Klicke das blaue Plus in der rechten unteren Ecke, um Fragen zu stellen.

0/2 Posts veröffentlicht

90 XP 100 Credits

4 **Weitere Challenges starten**
Sieh dir an, welche Challenges auf dich warten.

Anonymer Nutzer vor 22 Stunden
Uni Erlangen Nürnberg

Gibt es schon einen Klausurtermin für die Überblicksvorlesung "Einführung in die Frühe

use arrows to scroll


16:34
Newsfeed
170 Credits
Neueste
Unbeantwortet
LEVEL 2
50/150 XP
Aktuelle Challenges
Erstelle 2 Posts
Stelle und beantworte Fragen, teile deine Erfahrungen oder gebe hilfreichen Input in Kursen oder Gruppen.
Tipp: Klicke das blaue Plus in der rechten unteren Ecke, um Fragen zu stellen.
0/2 Posts veröffentlicht
90 XP 100 Credits
4
Frage stellen
Dokument hochladen
Karteikarten erstellen
Uni Erlangen Nürnberg
Gibt es schon einen Klausurtermin für die Überblicksvorlesung "Einführung in die Frühe



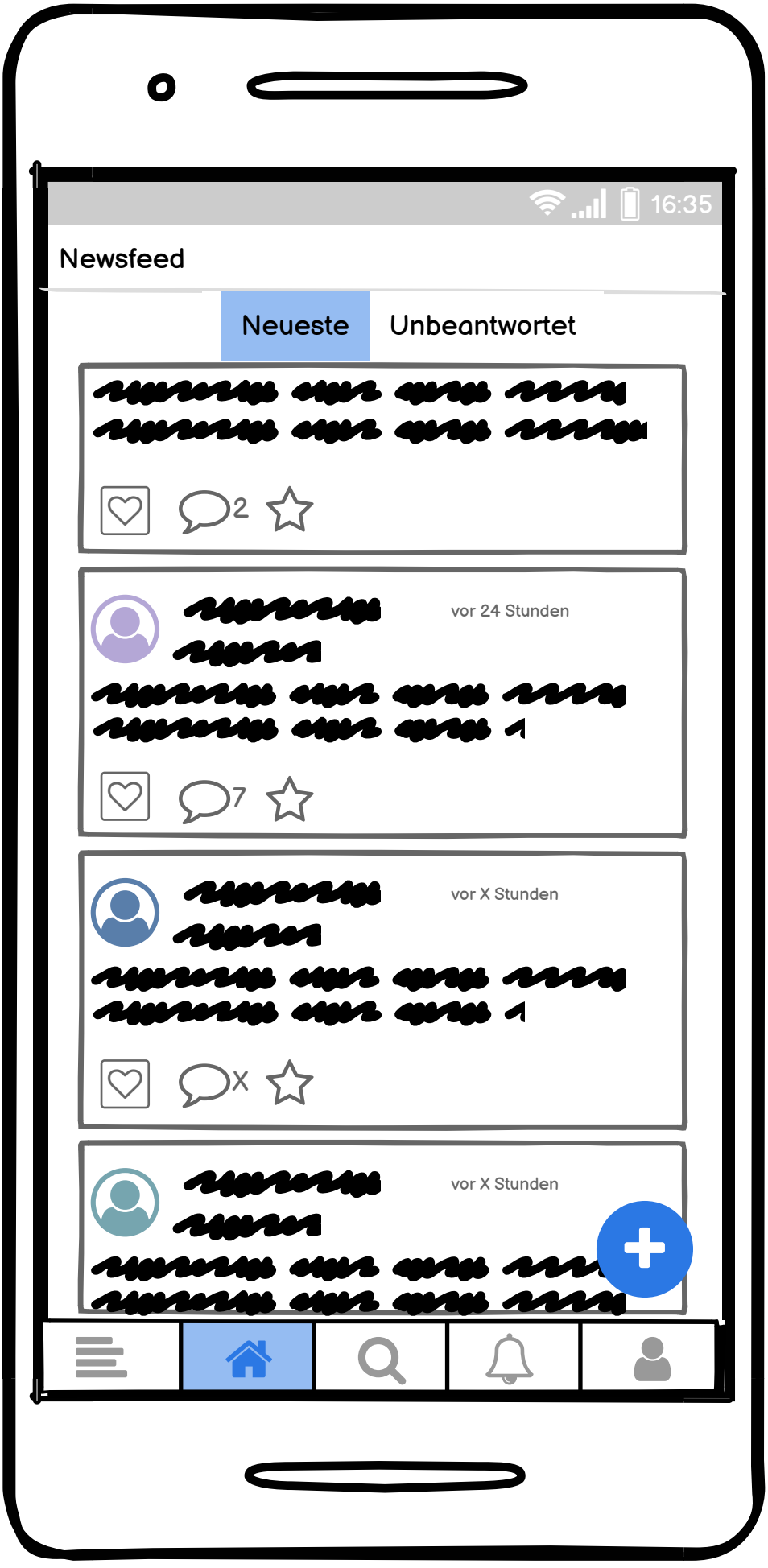
use arrows to scroll


16:34
Newsfeed
Neueste
Unbeantwortet
0/2 Posts veröffentlicht
90 XP 100 Credits
4
Weitere Challenges starten
Sieh dir an, welche Challenges
auf dich warten.
Anonymer Nutzer
vor 22 Stunden
Uni Erlangen Nürnberg
Gibt es schon einen Klausurtermin für die Überblicksvorlesung "Einführung in die Frühe Neuzeit" bei Herrn Wellner?
2
Frage stellen
Dokument hochladen
Karteikarten erstellen
5
Anonyme Rakete
vor 23 Stunden



use arrows to scroll


16:34
Newsfeed
Neueste
Unbeantwortet
2
Anonyme Nutzerin
vor 23 Stunden
Marketing
5
Anonyme Rakete
vor 23 Stunden
Frage stellen
Dokument hochladen
Karteikarten erstellen
7

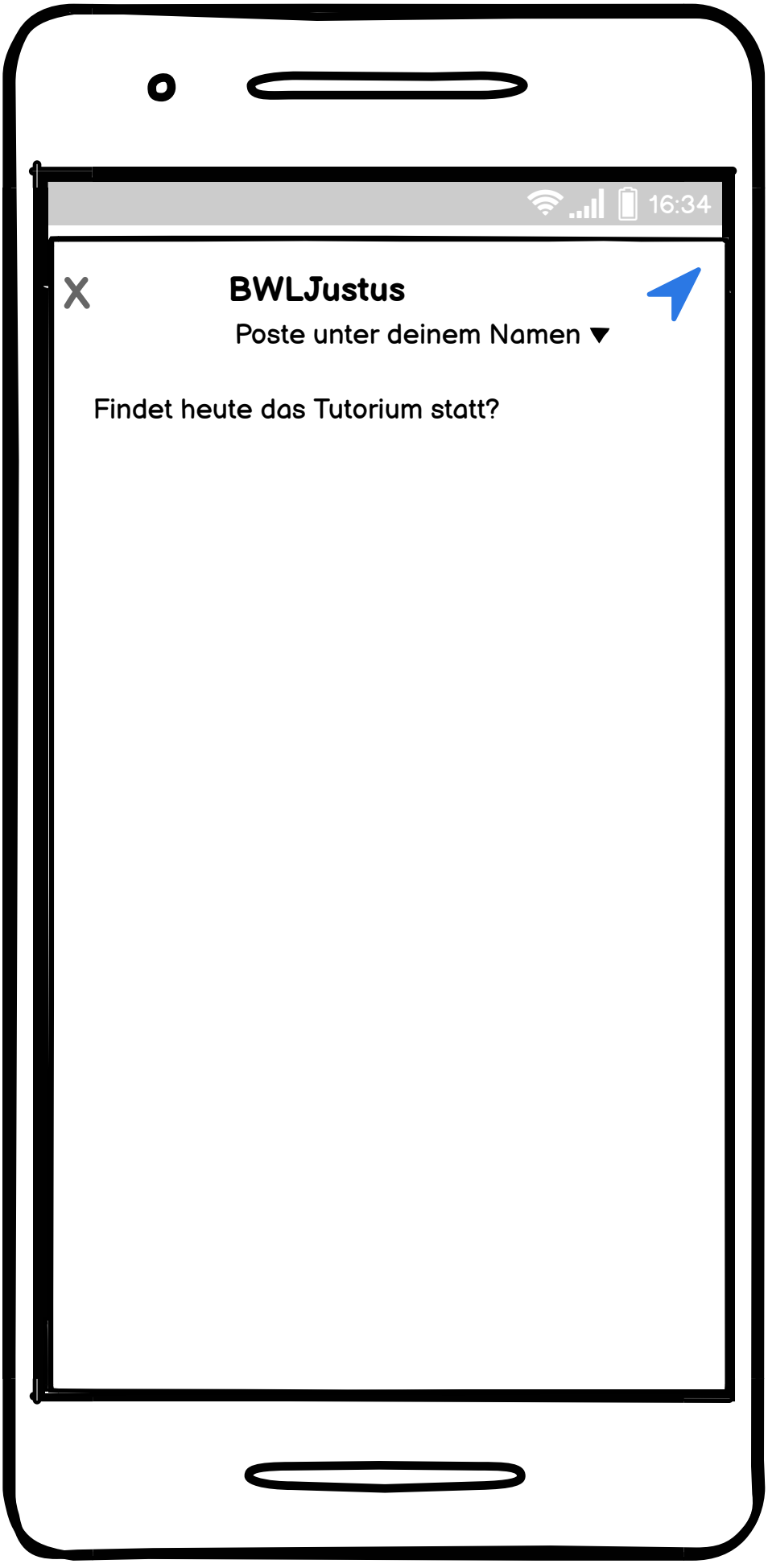


use arrows to scroll


16:34
Newsfeed
Neueste
Unbeantwortet
2
vor 24 Stunden
7
vor X Stunden
Frage stellen
Dokument hochladen
Karteikarten erstellen


16:34
Newsfeed
170 Credits
Neueste
Unbeantwortet
LEVEL 2
50/150 XP
Aktuelle Challenges
Erstelle 2 Posts
Stelle und beantworte Fragen, teile deine Erfahrungen oder gebe hilfreichen Input in Kursen oder Gruppen.
Tipp: Klicke das blaue Plus in der rechten unteren Ecke, um Fragen zu stellen.
Wo möchtest du posten?
Kurse
Gruppen
Statistik
Makroökonomie
Produktion, Logistik, Beschaffung
Investition und Finanzierung
Marketing


16:35
BWLJustus
Poste unter deinem Namen
Was möchtest du in Statistik posten?


16:34
BWLJustus
Poste unter deinem Namen
Findet heute das Tutorium statt?

16:34

Newsfeed 170 Credits

Neueste Unbeantwortet

LEVEL 2 50/150 XP

Aktuelle Challenges

Erstelle 2 Posts
Stelle und beantworte Fragen, teile deine Erfahrungen oder gebe hilfreichen Input in Kursen oder Gruppen.

Tipp: Klicke das blaue Plus in der rechten unteren Ecke, um Fragen zu stellen.

1/2 Posts veröffentlicht

90 XP 100 Credits

4 **Weitere Challenges starten**
Sieh dir an, welche Challenges auf dich warten.

BWLJustus vor 1 Minute

Statistik

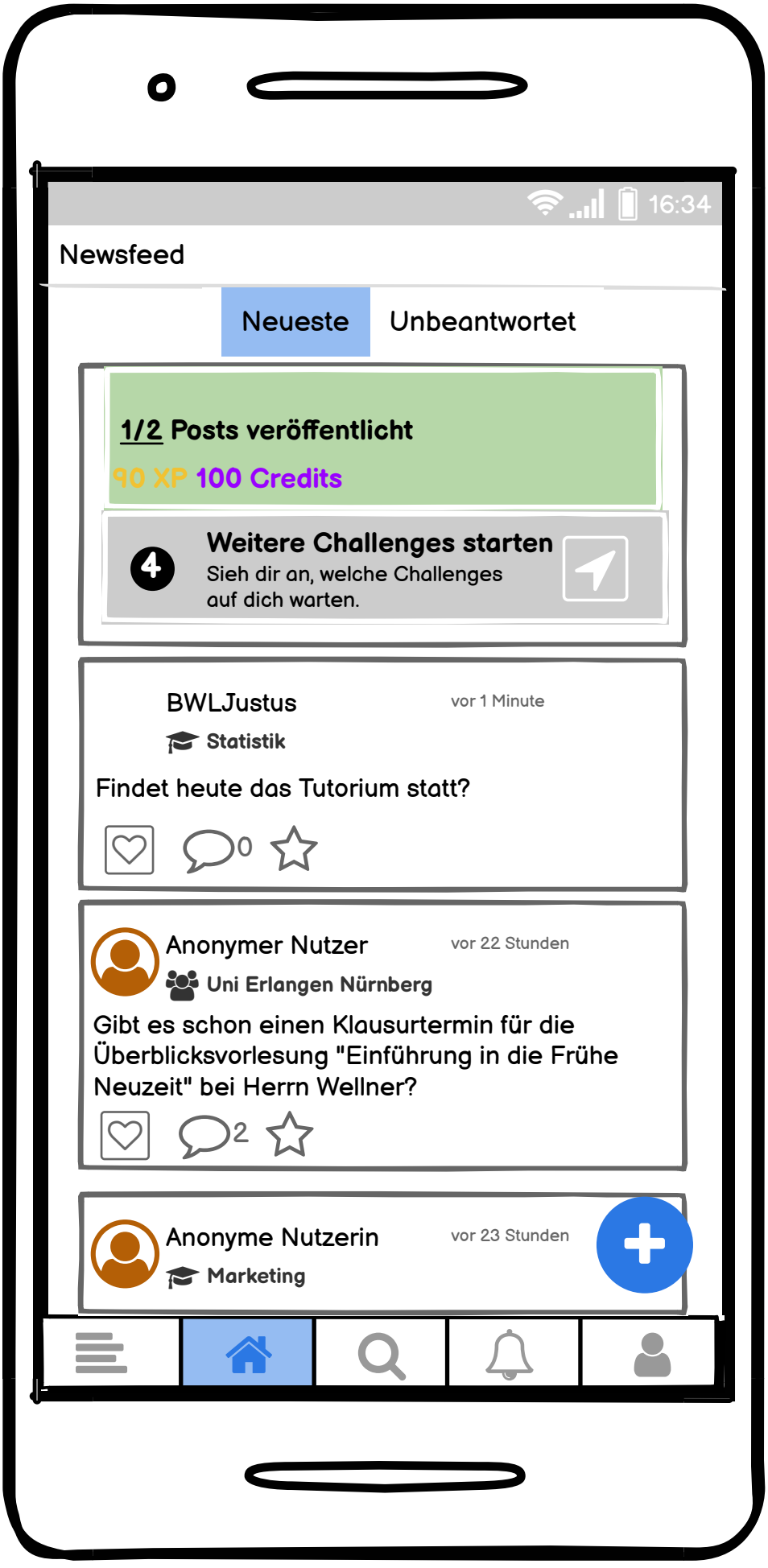
Findet heute das Tutorium statt?

0

use arrows to scroll




16:34
Newsfeed
170 Credits
Neueste
Unbeantwortet
LEVEL 2
50/150 XP
Aktuelle Challenges
Erstelle 2 Posts
Stelle und beantworte Fragen, teile deine Erfahrungen oder gebe hilfreichen Input in Kursen oder Gruppen.
Tipp: Klicke das blaue Plus in der rechten unteren Ecke, um Fragen zu stellen.
1/2 Posts veröffentlicht
90 XP 100 Credits
4
Frage stellen
Dokument hochladen
Karteikarten erstellen
BW
Findet heute das Tutorium statt?
0



use arrows to scroll


16:34
Newsfeed
Neueste
Unbeantwortet
1/2 Posts veröffentlicht
90 XP 100 Credits
4
Weitere Challenges starten
Sieh dir an, welche Challenges
auf dich warten.
BWLJustus
vor 1 Minute
Statistik
Findet heute das Tutorium statt?
0
Frage stellen
Dokument hochladen
Karteikarten erstellen
Anonyme Nutzerin
vor 23 Stunden
Marketing



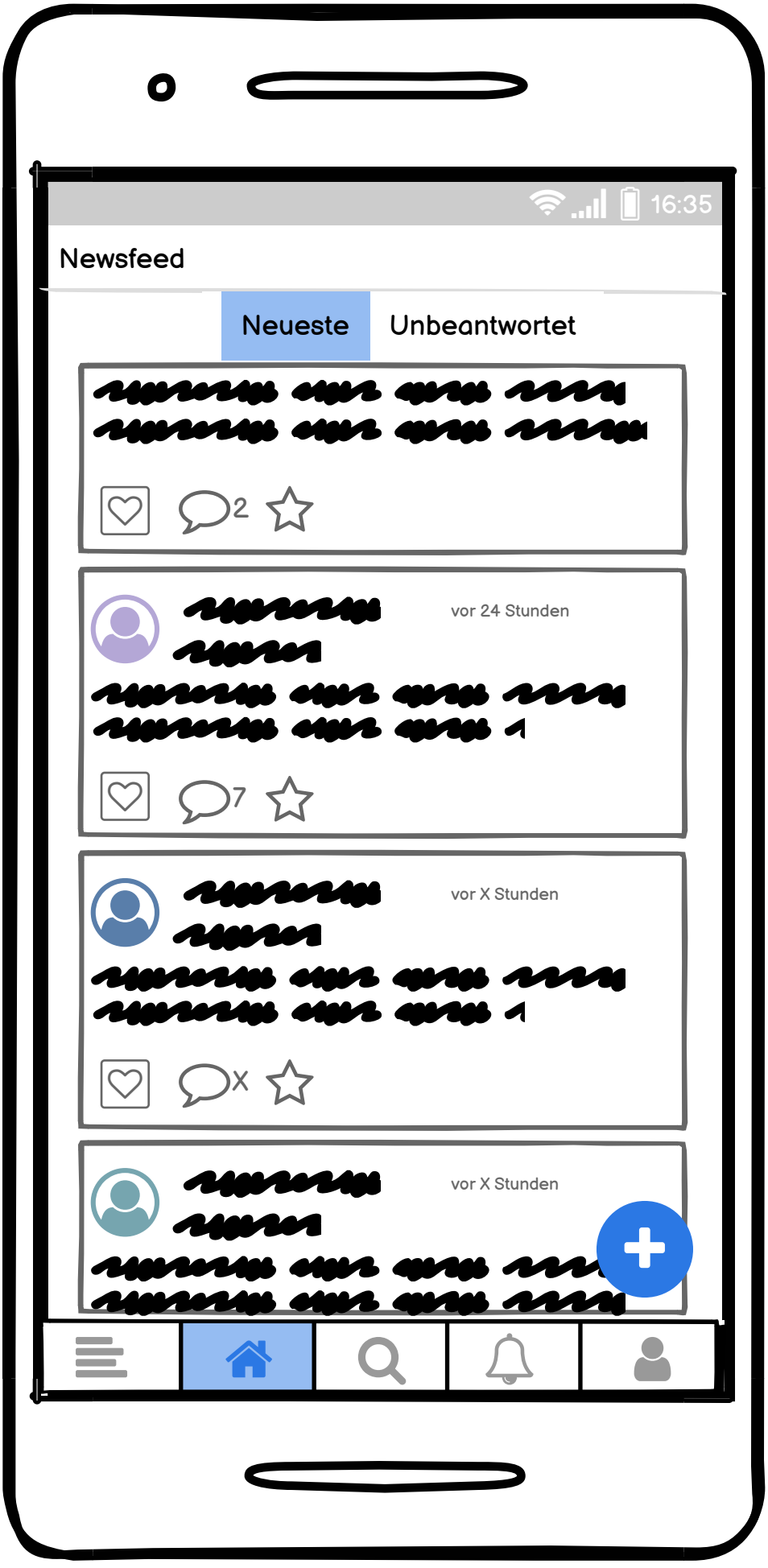
use arrows to scroll


16:35
Newsfeed
Neueste
Unbeantwortet
Findet heute das Tutorium statt?
0
Anonymer Nutzer
vor 22 Stunden
Uni Erlangen Nürnberg
Gibt es schon einen Klausurtermin für die Überblicksvorlesung "Einführung in die Frühe Neuzeit" bei Herrn Wellner?
2
Anonyme Nutzerin
vor 23 Stunden
Marketing
Frage stellen
Dokument hochladen
Karteikarten erstellen



use arrows to scroll


16:34
Newsfeed
Neueste
Unbeantwortet
2
Anonyme Nutzerin
vor 23 Stunden
Marketing
5
Anonyme Rakete
vor 23 Stunden
Frage stellen
Dokument hochladen
Karteikarten erstellen
7

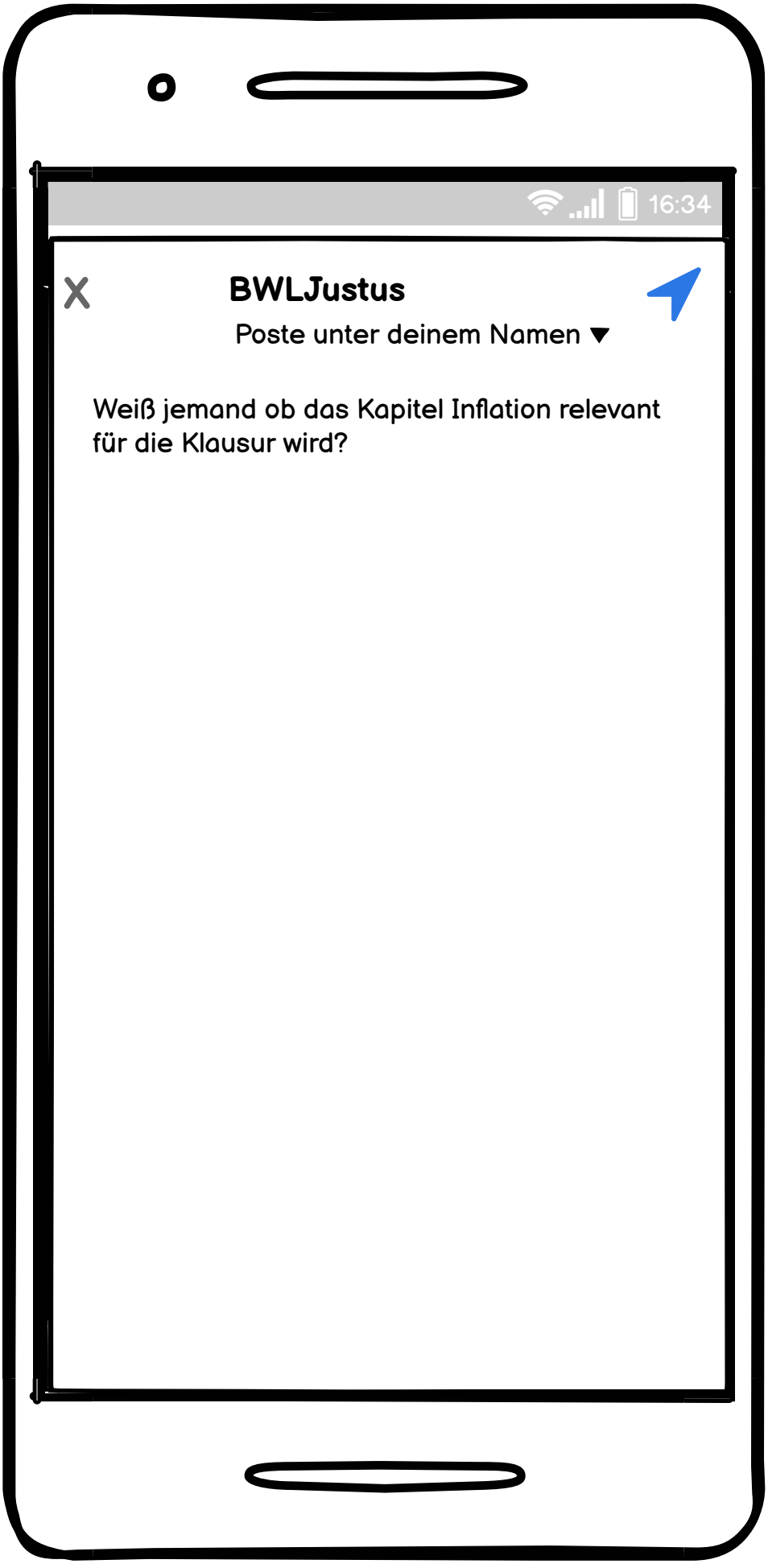


use arrows to scroll


16:34
Newsfeed
Neueste
Unbeantwortet
2
vor 24 Stunden
7
vor X Stunden
Frage stellen
Dokument hochladen
Karteikarten erstellen


16:34
Newsfeed
170 Credits
Neueste
Unbeantwortet
LEVEL 2
50/150 XP
Aktuelle Challenges
Erstelle 2 Posts
Stelle und beantworte Fragen, teile deine Erfahrungen oder gebe hilfreichen Input in Kursen oder Gruppen.
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Makroökonomie
Produktion, Logistik, Beschaffung
Investition und Finanzierung
Marketing


16:35
BWLJustus
Poste unter deinem Namen
Was möchtest du in Makroökonomie posten?


16:34
BWLJustus
Poste unter deinem Namen
Weiß jemand ob das Kapitel Inflation relevant für die Klausur wird?


16:35
Challenge erledigt: Erstelle 2 Posts
2/2 Posts erstellt
Ok
Aktuelle Challenges
Erstelle 2 Posts
Stelle und beantworte Fragen, teile deine Erfahrungen oder gebe hilfreichen Input in Kursen oder Gruppen.
Tipp: Klicke das blaue Plus in der rechten unteren Ecke, um Fragen zu stellen.
2/2 Posts veröffentlicht
90 XP 100 Credits
4
Weitere Challenges starten
Sieh dir an, welche Challenges auf dich warten.
BWLJustus
vor 1 Minute
Makroökonomie
Weiß jemand ob das Kapitel Inflation relevan
die Klausur wird?
0

16:34

Newsfeed 170 Credits

Neueste Unbeantwortet

LEVEL 2 140/150 XP

Aktuelle Challenges

Erstelle 2 Posts
Stelle und beantworte Fragen, teile deine Erfahrungen oder gebe hilfreichen Input in Kursen oder Gruppen.

Tipp: Klicke das blaue Plus in der rechten unteren Ecke, um Fragen zu stellen.

2/2 Posts veröffentlicht

90 XP 100 Credits

4 **Weitere Challenges starten**
Sieh dir an, welche Challenges auf dich warten.

BWLJustus vor 1 Minute

Makroökonomie

Weiß jemand ob das Kapitel Inflation relevan
die Klausur wird?

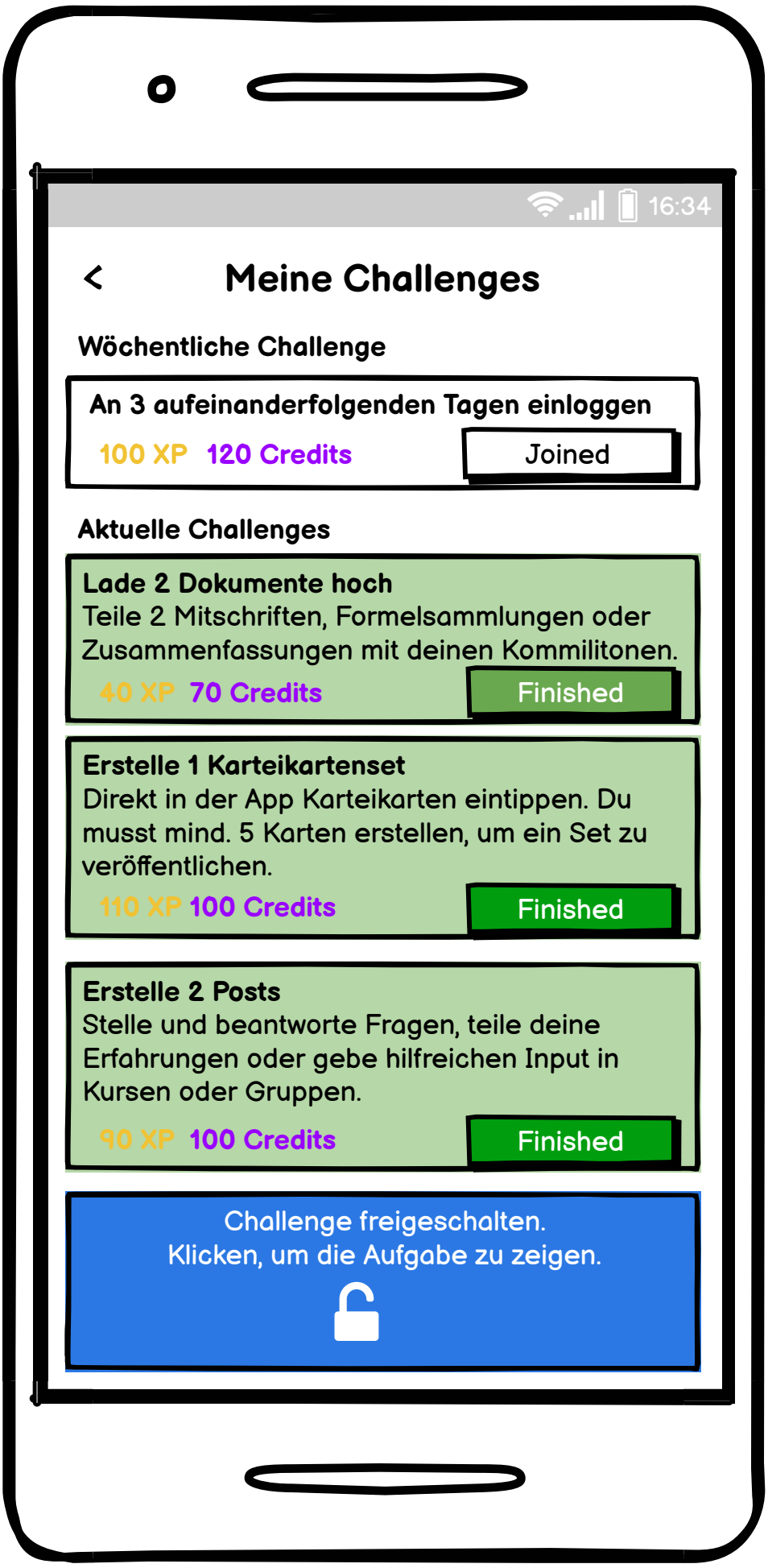
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use arrows to scroll





use arrows to scroll


16:34
Meine Challenges
Wöchentliche Challenge
An 3 aufeinanderfolgenden Tagen einloggen
100 XP 120 Credits
Joined
Aktuelle Challenges
Lade 2 Dokumente hoch
Teile 2 Mitschriften, Formelsammlungen oder Zusammenfassungen mit deinen Kommilitonen.
40 XP 70 Credits
Finished
Erstelle 1 Karteikartenset
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110 XP 100 Credits
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Erstelle 2 Posts
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Challenge freigeschalten.
Klicken, um die Aufgabe zu zeigen.


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Wöchentliche Challenge
An 3 aufeinanderfolgenden Tagen einloggen
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110 XP 100 Credits
Finished
Erstelle 2 Posts
Stelle und beantworte Fragen, teile deine Erfahrungen oder gebe hilfreichen Input in Kursen oder Gruppen.
90 XP 100 Credits
Finished
Lade 1 Freund ein
Teile den Link per Whatsapp, Email oder Facebook.
120 XP 200 Credits
Join


16:34
Meine Challenges
Wöchentliche Challenge
An 3 aufeinanderfolgenden Tagen einloggen
100 XP 120 Credits
Joined
Aktuelle Challenges
Lade 2 Dokumente hoch
Teile 2 Mitschriften, Formelsammlungen oder Zusammenfassungen mit deinen Kommilitonen.
40 XP 70 Credits
Finished
Erstelle 1 Karteikartenset
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120 XP 200 Credits
Joined

16:34

Newsfeed 170 Credits

Neueste Unbeantwortet

LEVEL 2 140/150 XP

Aktuelle Challenges

Lade 1 Freund ein
Teile den Link per Whatsapp, Email oder Facebook.

Tipp: Klicke auf das Profil-Icon im unteren Reiter, um Freunde einzuladen.

0/1 Freund eingeladen

120 X 200 Credits

4 **Weitere Challenges starten**
Sieh dir an, welche Challenges auf dich warten.

BWLJustus vor 1 Minute

Makroökonomie

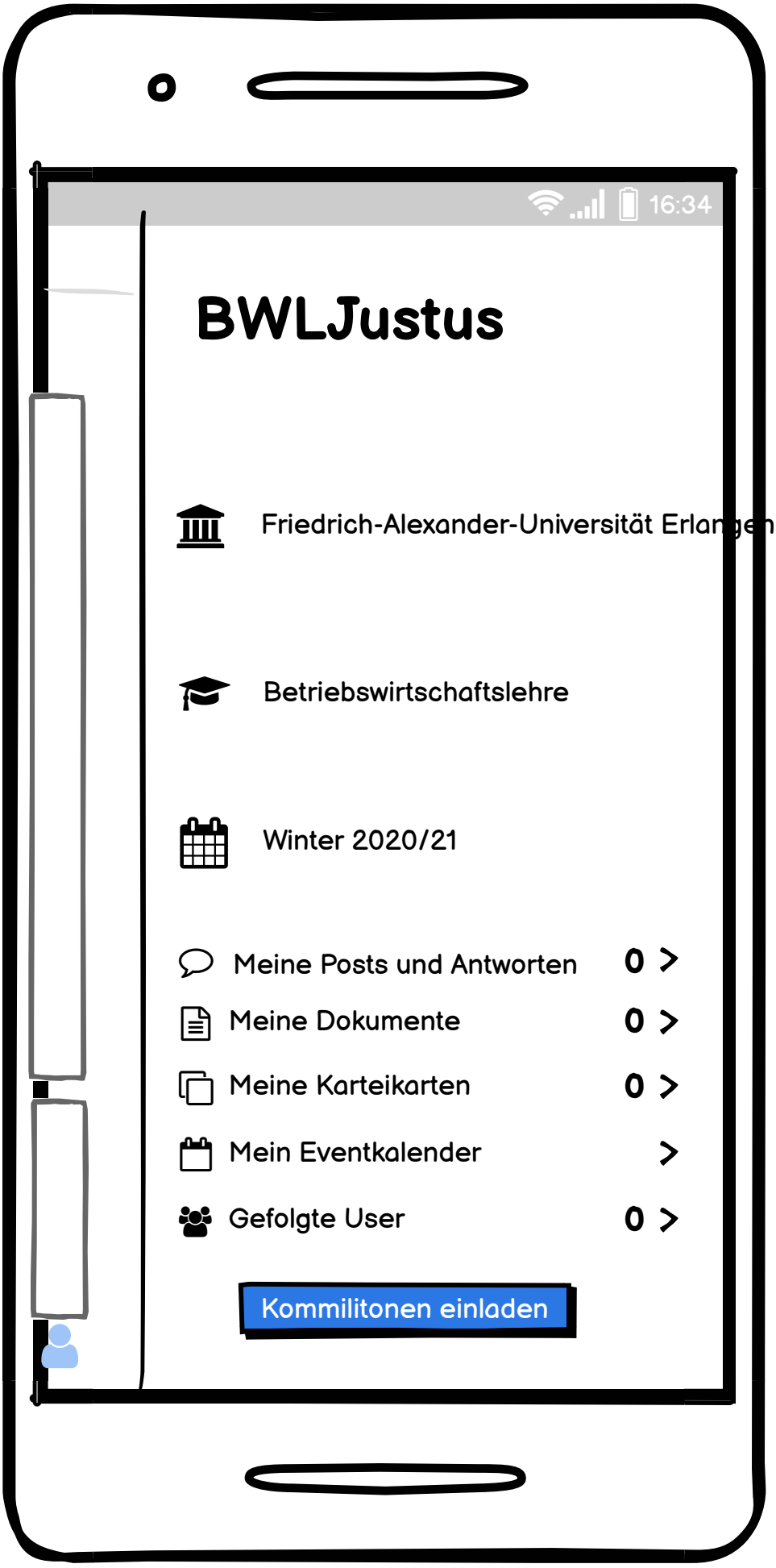
Weiß jemand ob das Kapitel Inflation relevant die Klausur wird?

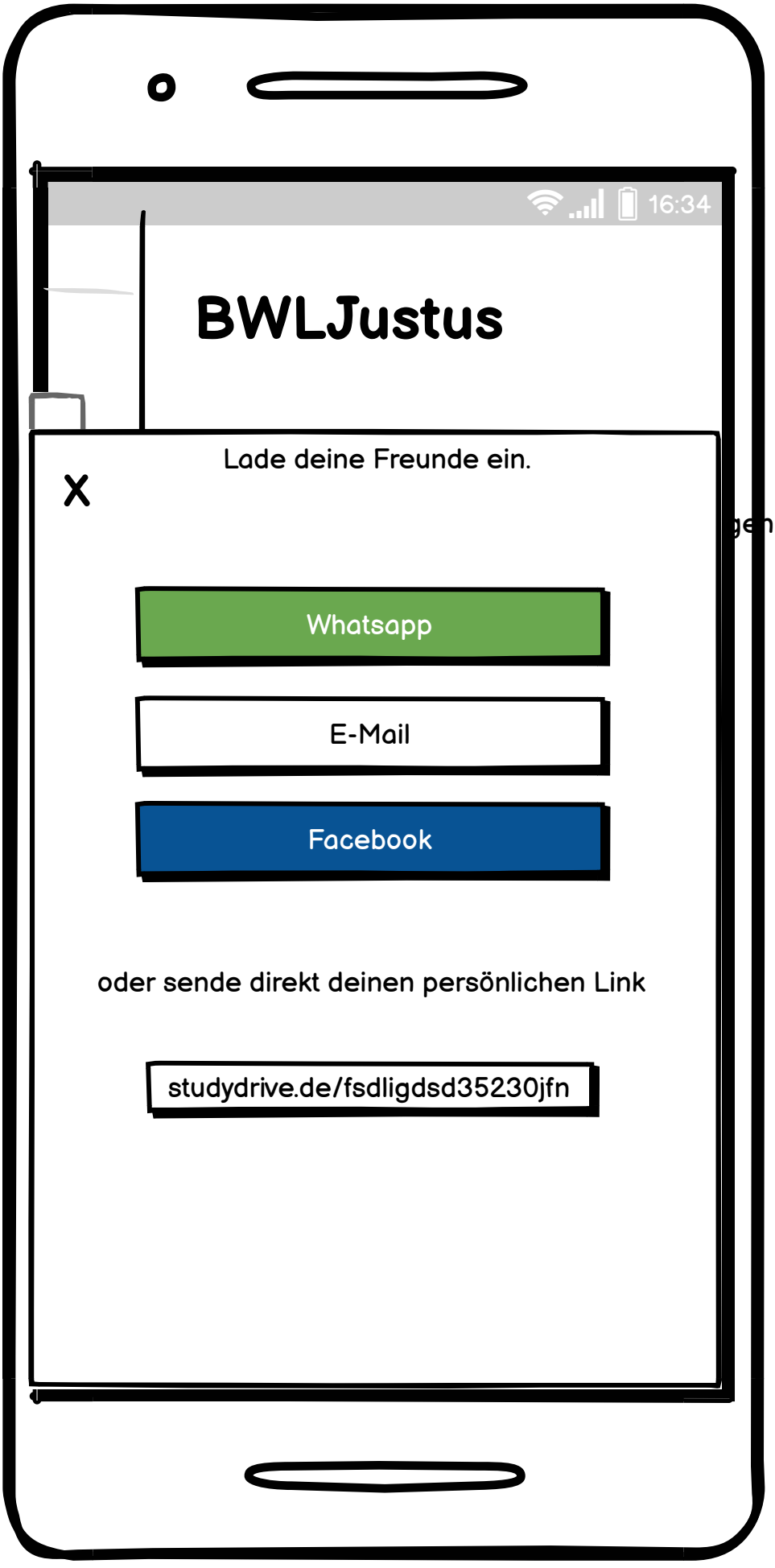
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use arrows to scroll

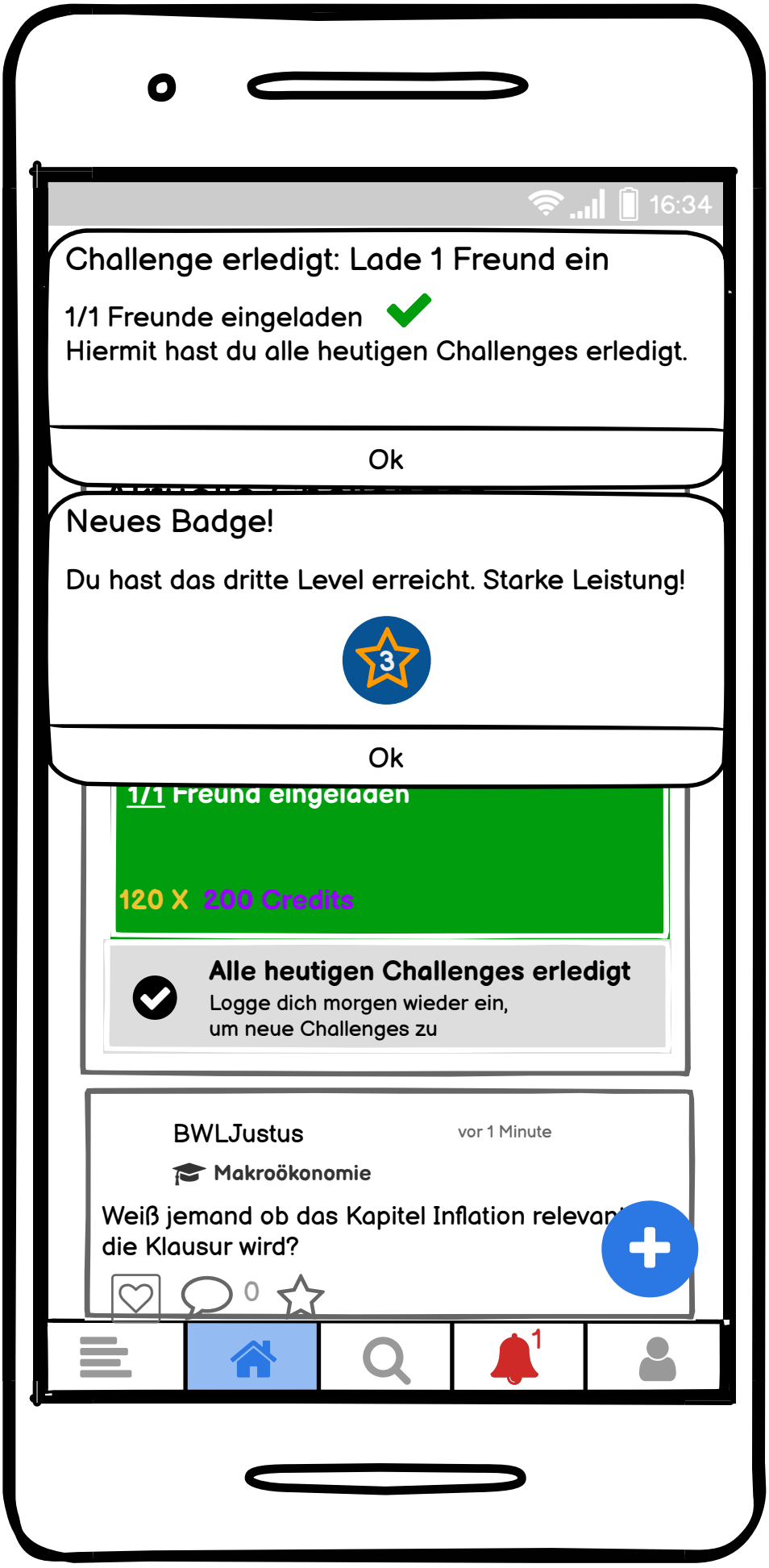


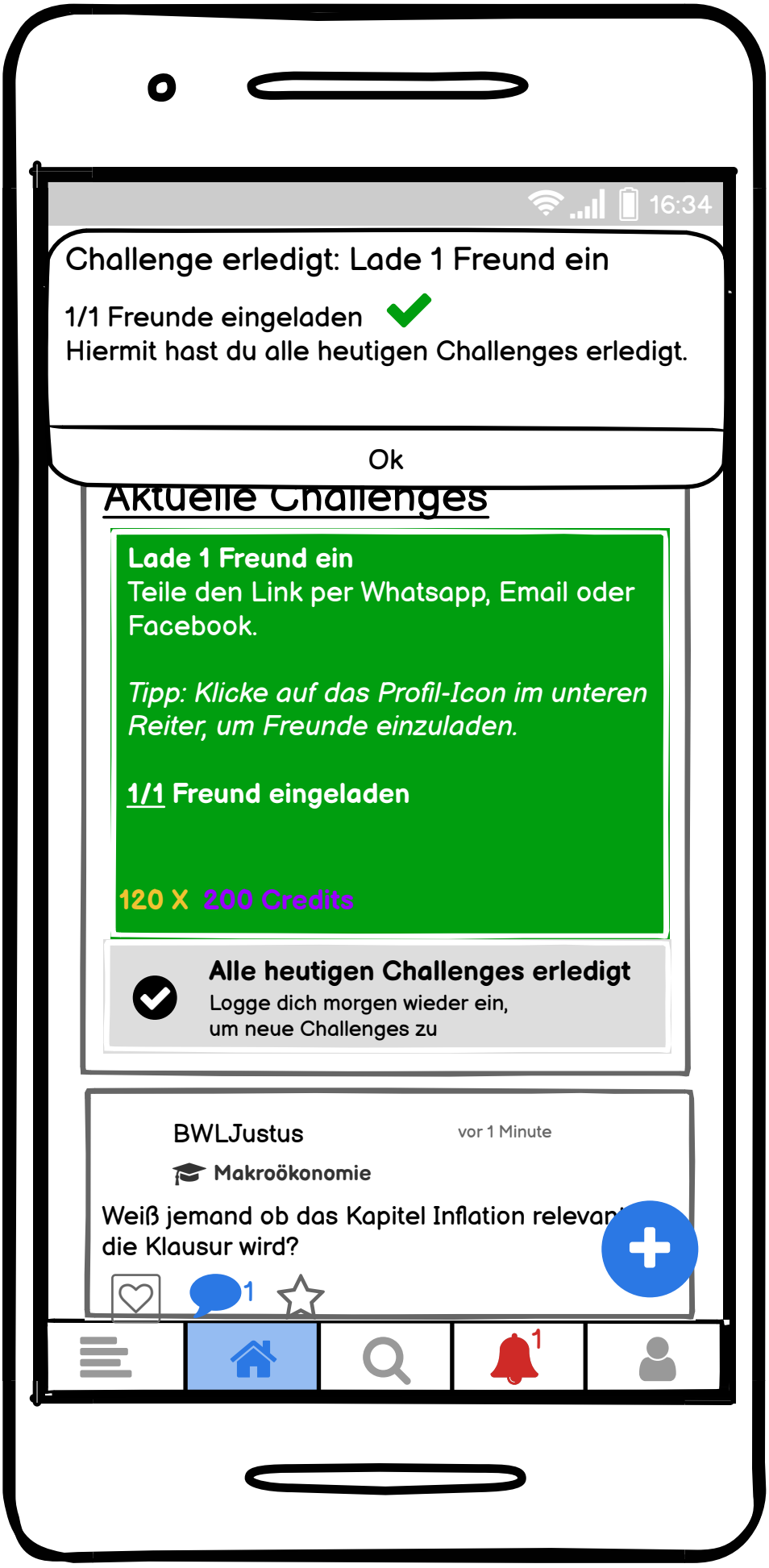
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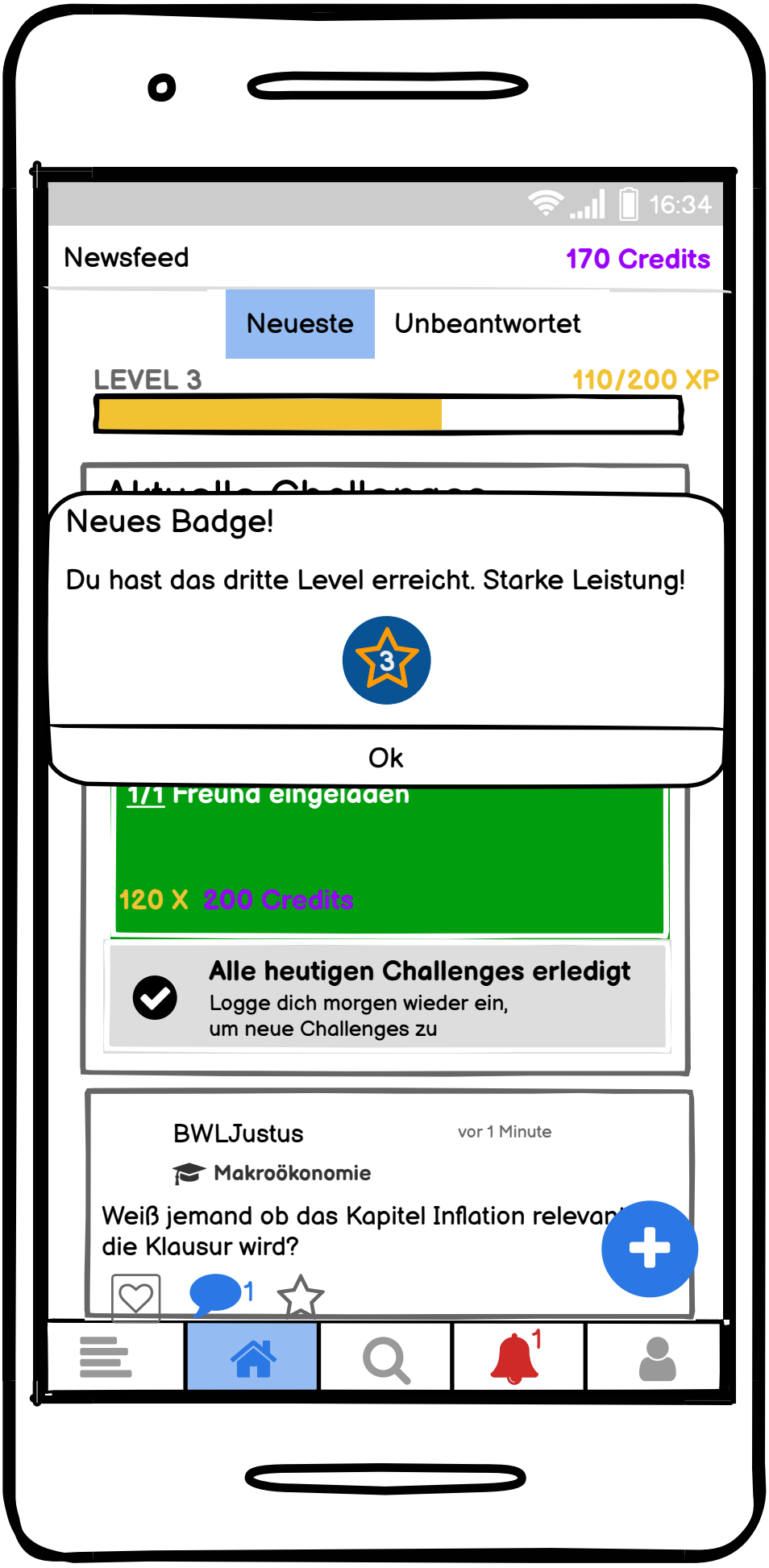

16:34
BWLJustus
Friedrich-Alexander-Universität Erlangen
Betriebswirtschaftslehre
Winter 2020/21
Meine Posts und Antworten 0 >
Meine Dokumente 0 >
Meine Karteikarten 0 >
Mein Eventkalender >
Gefolgte User 0 >
Kommilitonen einladen


16:34
BWLJustus
Lade deine Freunde ein.
X
Whatsapp
E-Mail
Facebook
oder sende direkt deinen persönlichen Link
studydrive.de/fsdligdsd35230jfn


16:34
BWLJustus
X
Erfolgreich eingeladen!
Dieses Fenster kann geschlossen werden.

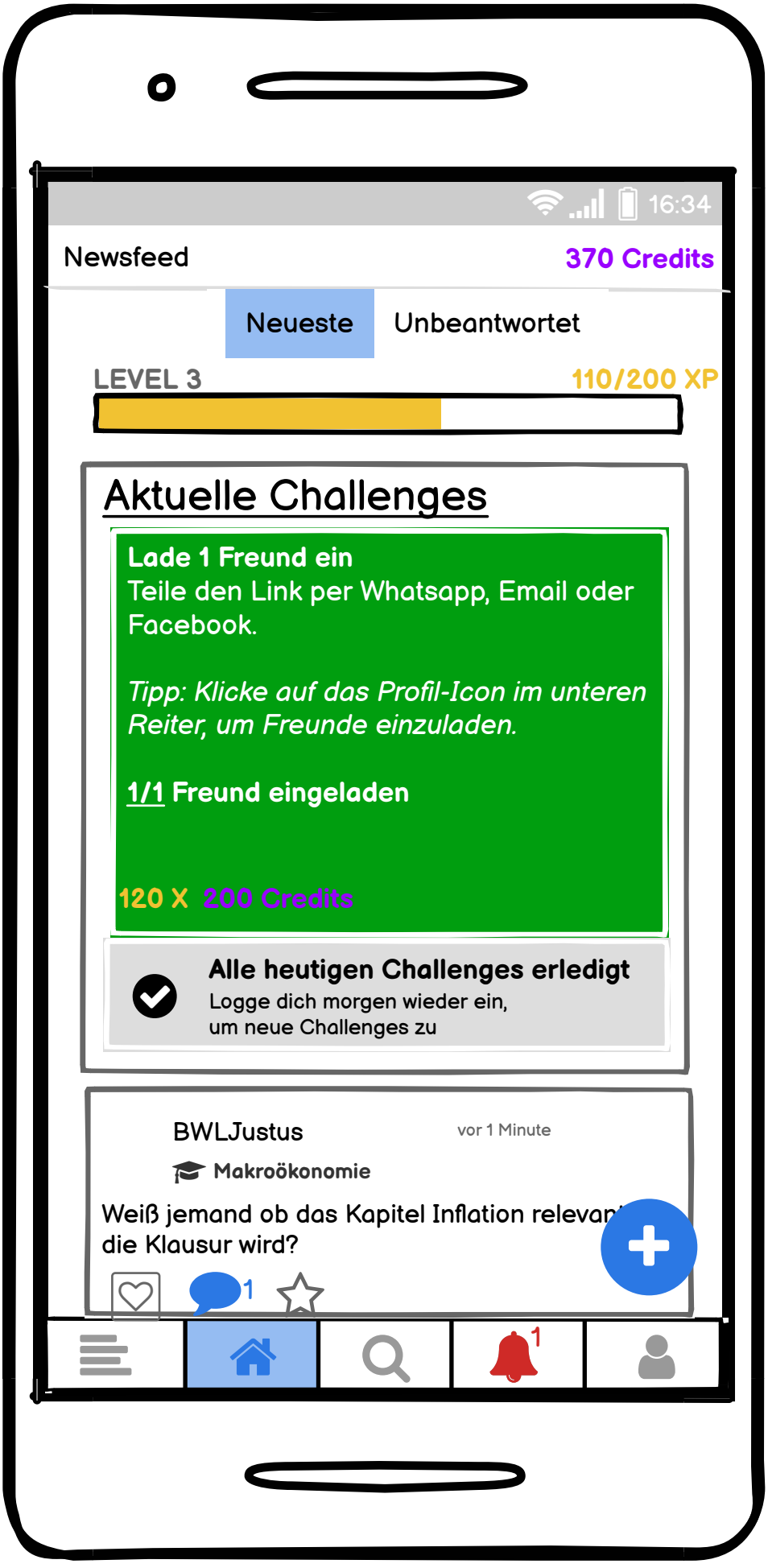

16:34
Challenge erledigt: Lade 1 Freund ein
1/1 Freunde eingeladen
Hiermit hast du alle heutigen Challenges erledigt.
Ok
Neues Badge!
Du hast das dritte Level erreicht. Starke Leistung!
3
Ok
1/1 Freund eingeladen
120 X 200 Credits
Alle heutigen Challenges erledigt
Logge dich morgen wieder ein,
um neue Challenges zu
BWLJustus
vor 1 Minute
Makroökonomie
Weiß jemand ob das Kapitel Inflation relevan
die Klausur wird?
0
1


16:34
Challenge erledigt: Lade 1 Freund ein
1/1 Freunde eingeladen
Hiermit hast du alle heutigen Challenges erledigt.
Ok
Aktuelle Challenges
Lade 1 Freund ein
Teile den Link per Whatsapp, Email oder Facebook.
Tipp: Klicke auf das Profil-Icon im unteren Reiter, um Freunde einzuladen.
1/1 Freund eingeladen
120 X 200 Credits
Alle heutigen Challenges erledigt
Logge dich morgen wieder ein, um neue Challenges zu
BWLJustus
vor 1 Minute
Makroökonomie
Weiß jemand ob das Kapitel Inflation relevan
die Klausur wird?
1
1


16:34
Newsfeed
170 Credits
Neueste
Unbeantwortet
LEVEL 3
110/200 XP
Neues Badge!
Du hast das dritte Level erreicht. Starke Leistung!
3
Ok
1/1 Freund eingeladen
120 X 200 Credits
Alle heutigen Challenges erledigt
Logge dich morgen wieder ein,
um neue Challenges zu
BWLJustus
vor 1 Minute
Makroökonomie
Weiß jemand ob das Kapitel Inflation relevan
die Klausur wird?
1
1



use arrows to scroll

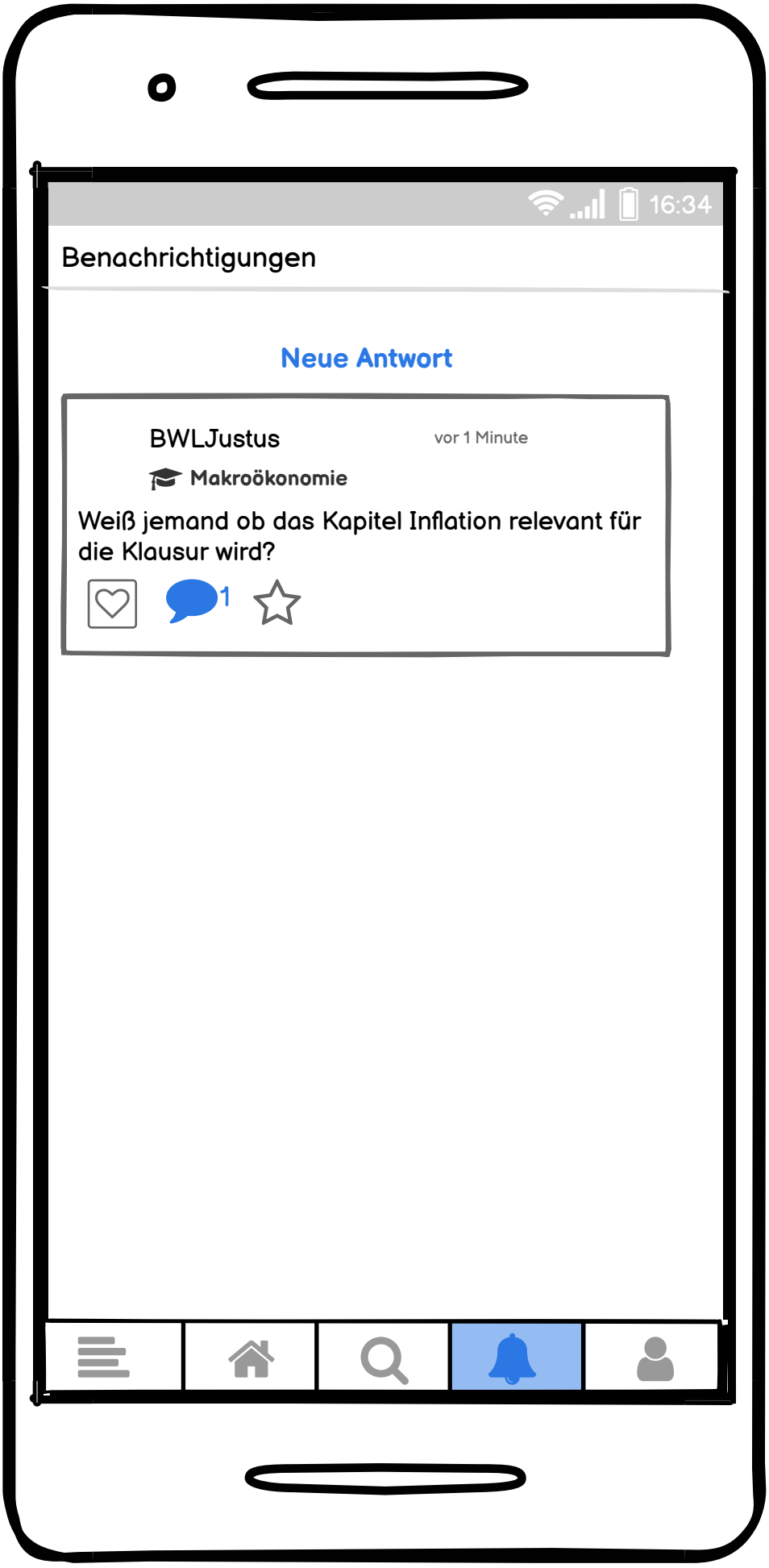


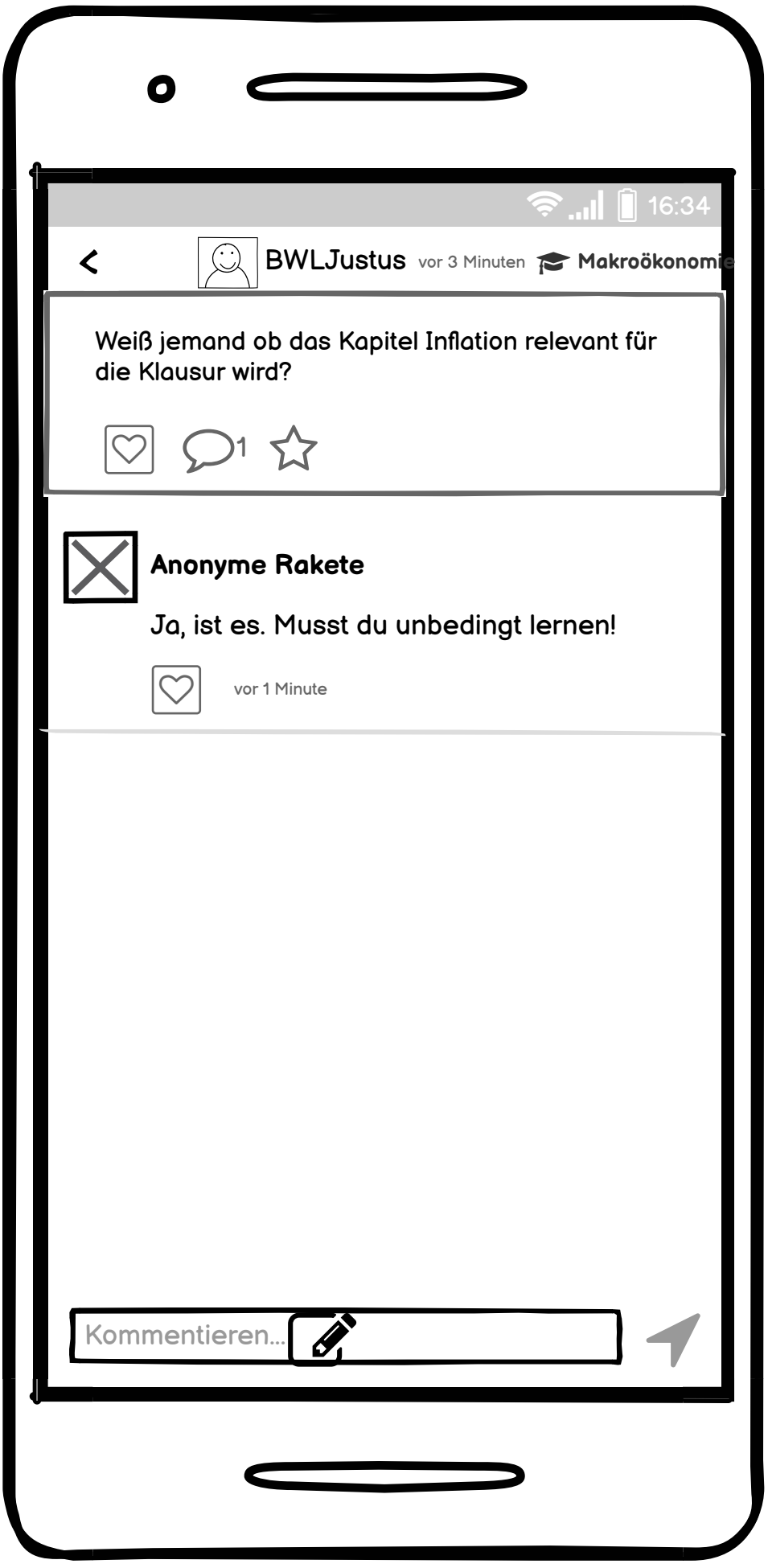
use arrows to scroll

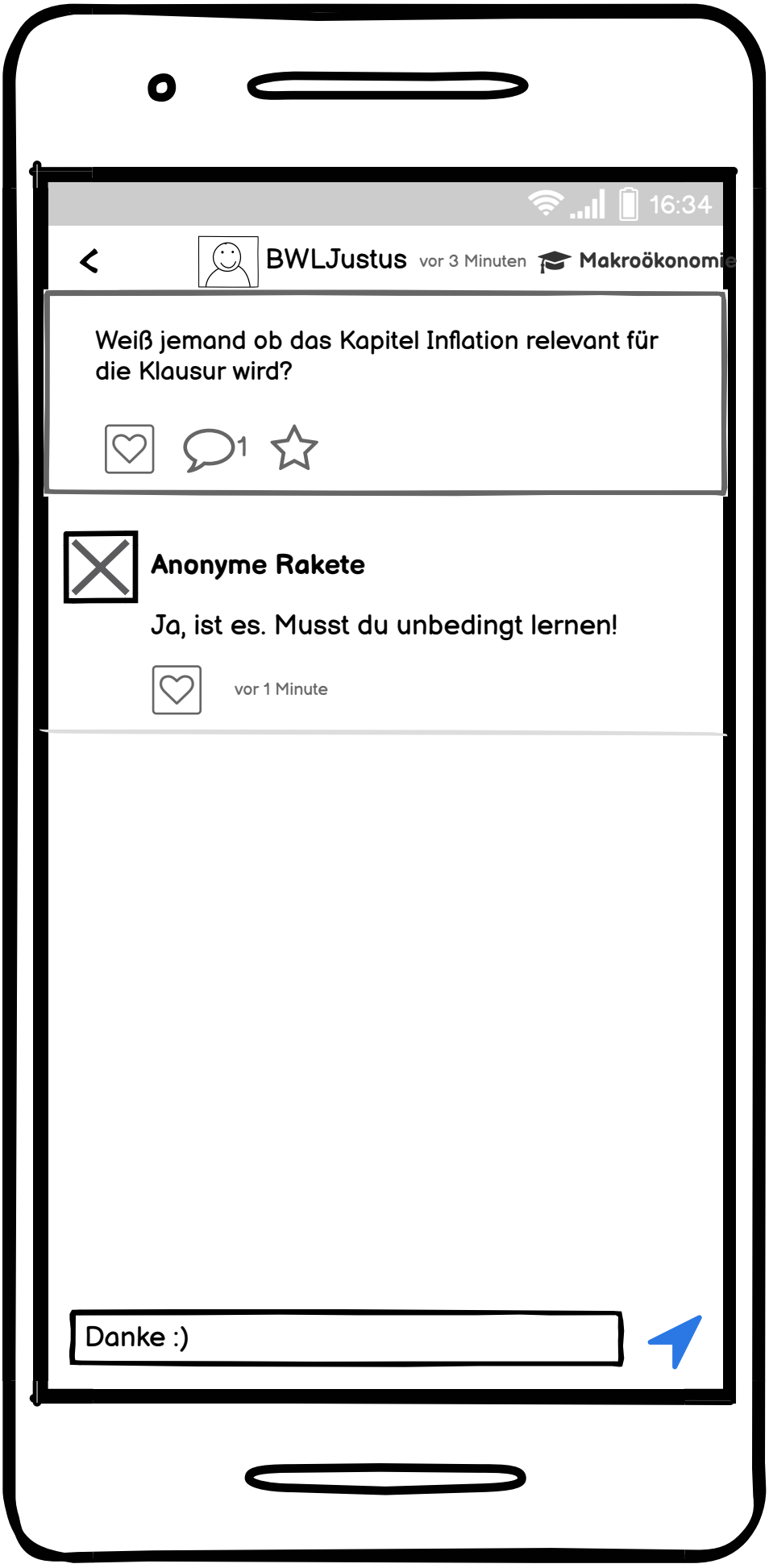




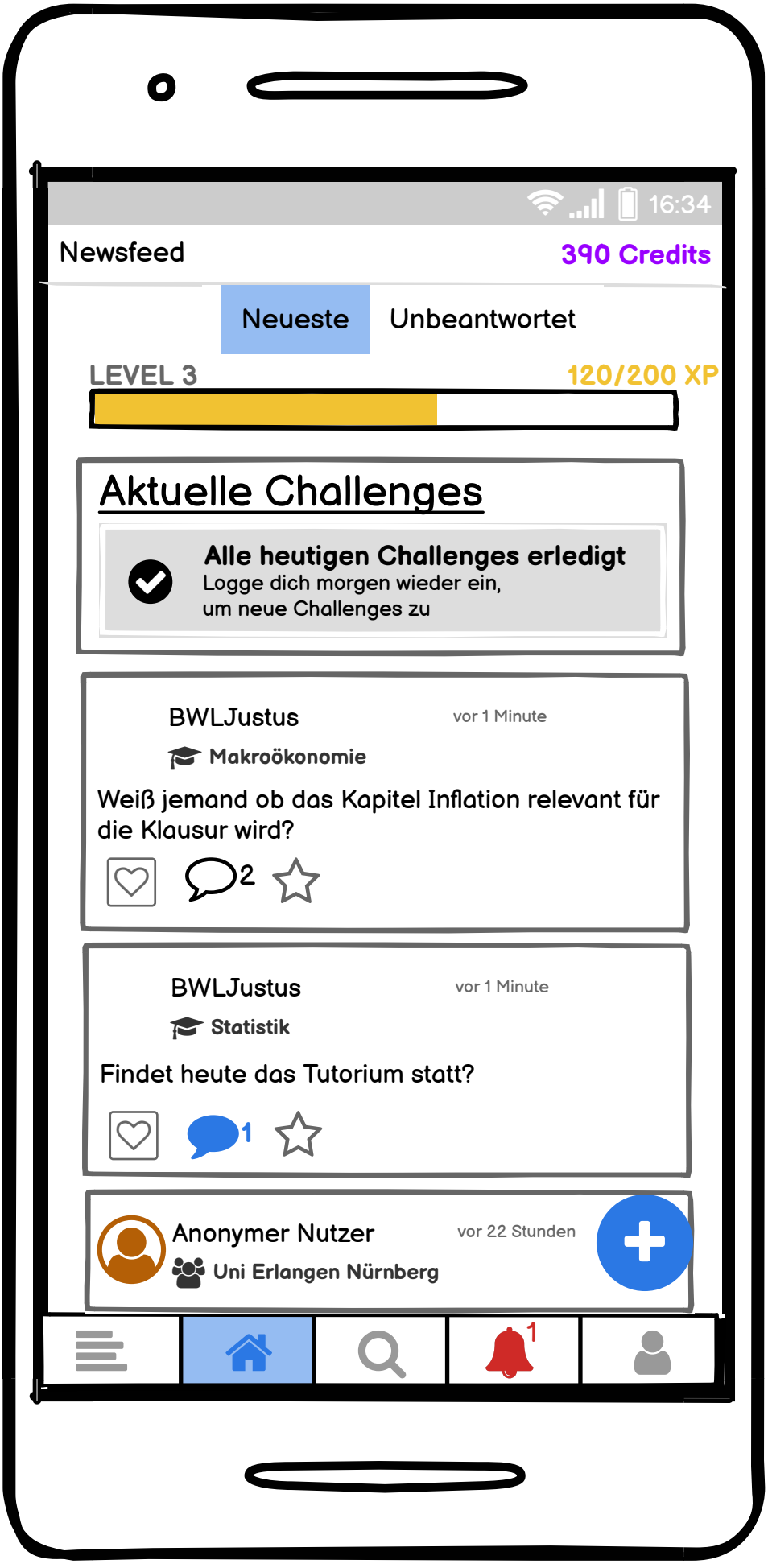
use arrows to scroll


16:34
Benachrichtigungen
Neue Antwort
BWLJustus
vor 1 Minute
Makroökonomie
Weiß jemand ob das Kapitel Inflation relevant für die Klausur wird?
1


16:34
BWLJustus
vor 3 Minuten
Makroökonomie
Weiß jemand ob das Kapitel Inflation relevant für die Klausur wird?
1
Anonyme Rakete
Ja, ist es. Musst du unbedingt lernen!
vor 1 Minute
Kommentieren...


16:34
BWLJustus
vor 3 Minuten
Makroökonomie
Weiß jemand ob das Kapitel Inflation relevant für die Klausur wird?
1
Anonyme Rakete
Ja, ist es. Musst du unbedingt lernen!
vor 1 Minute
Danke :)


16:34
BWLJustus
vor 3 Minuten
Makroökonomie
Weiß jemand ob das Kapitel Inflation relevant für die Klausur wird?
1
Anonyme Rakete
Ja, ist es. Musst du unbedingt lernen!
vor 1 Minute
BWLJustus
Danke :)
vor 1 Minute
Kommentieren...



use arrows to scroll



use arrows to scroll


16:34
Benachrichtigungen
Neue Antwort
BWLJustus
vor 1 Minute
Statistik
Findet heute das Tutorium statt?
1


16:34
BWLJustus
vor 5 Minuten
Statistik
Findet heute das Tutorium statt?
1
Anonymer Esel
Hey, ja es findet heute statt. Die Uhrzeit hat sich aber verschoben, d.h. es fängt erst um 14.00 Uhr an und geht bis 14.30 ... davor hat es ja immer schon um 11.30 Uhr angefangen.
Hast du den Zoom-Link?
vor 1 Minute
Kommentieren...


16:34
BWLJustus
vor 5 Minuten
Statistik
Findet heute das Tutorium statt?
1
Anonymer Esel
Hey, ja es findet heute statt. Die Uhrzeit hat sich aber verschoben, d.h. es fängt erst um 14.00 Uhr an und geht bis 14.30 ... davor hat es ja immer schon um 11.30 Uhr angefangen.
Hast du den Zoom-Link?
vor 1 Minute
Nein den Zugangslink hab ich nicht :/


16:34
BWLJustus
vor 5 Minuten
Statistik
Findet heute das Tutorium statt?
1
Anonymer Esel
Hey, ja es findet heute statt. Die Uhrzeit hat sich aber verschoben, d.h. es fängt erst um 14.00 Uhr an und geht bis 14.30 ... davor hat es ja immer schon um 11.30 Uhr angefangen.
Hast du den Zoom-Link?
vor 1 Minute
BWLJustus
Nein den Zugangslink hab ich nicht :/
vor 1 Minute
Antworten laden
Kommentieren...

16:34

BWLJustus vor 5 Minuten Statistik

Findet heute das Tutorium statt?

1

.. davor hat es ja immer schon um 11.30 Uhr angefangen.
Hast du den Zoom-Link?

vor 1 Minute

BWLJustus

Nein den Zugangslink hab ich nicht :/

vor 1 Minute

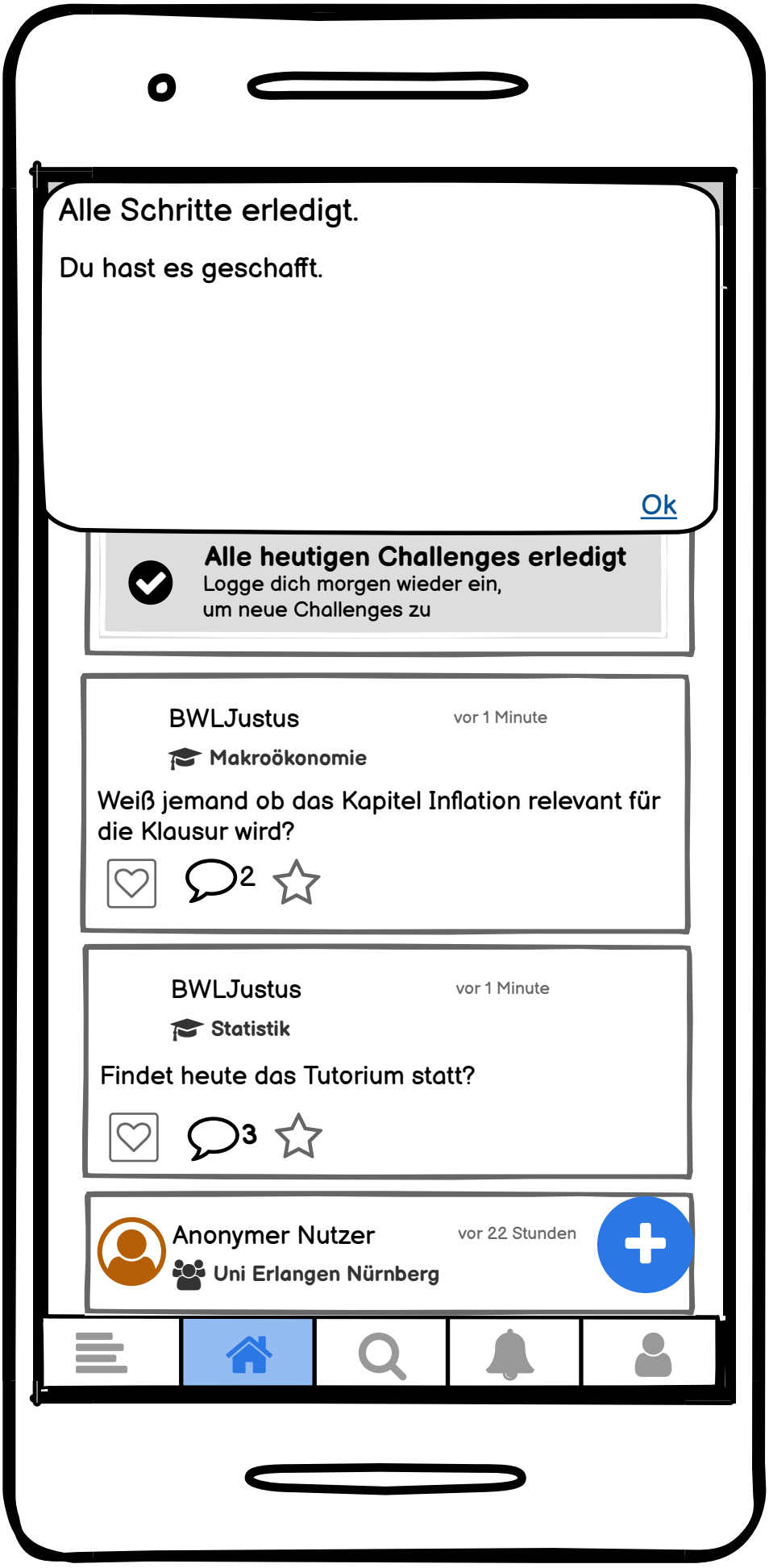
Anonymer Esel

Du findest den Link im Moodle Kurs "Statistik". Musst einfach nach unten scrollen.

Bis später dann :)

vor 1 Minute

Kommentieren...


Alle Schritte erledigt.
Du hast es geschafft.
Ok
Alle heutigen Challenges erledigt
Logge dich morgen wieder ein,
um neue Challenges zu
BWLJustus
vor 1 Minute
Makroökonomie
Weiß jemand ob das Kapitel Inflation relevant für die Klausur wird?
2
BWLJustus
vor 1 Minute
Statistik
Findet heute das Tutorium statt?
3
Anonymer Nutzer
vor 22 Stunden
Uni Erlangen Nürnberg

16:34

Danke für die Teilnahme!

Das Programm kann jetzt geschlossen werden.

Bitte fülle noch kurz den Abschluss-Fragebogen aus. Hierfür wieder die Typeform Umfrage öffnen und "I finished the recording" klicken.



Attachment 2: Non-Gamified Prototype




16:30
Studydrive
Studieren ist leichter, wenn wir einander helfen
Erhalte Hilfe & finde Lernunterlagen auf Europas größter Studierendenplattform. 100% kostenlos!
Weiter mit Facebook
Weiter mit E-Mail
Weiter mit Google
Weiter mit Apple

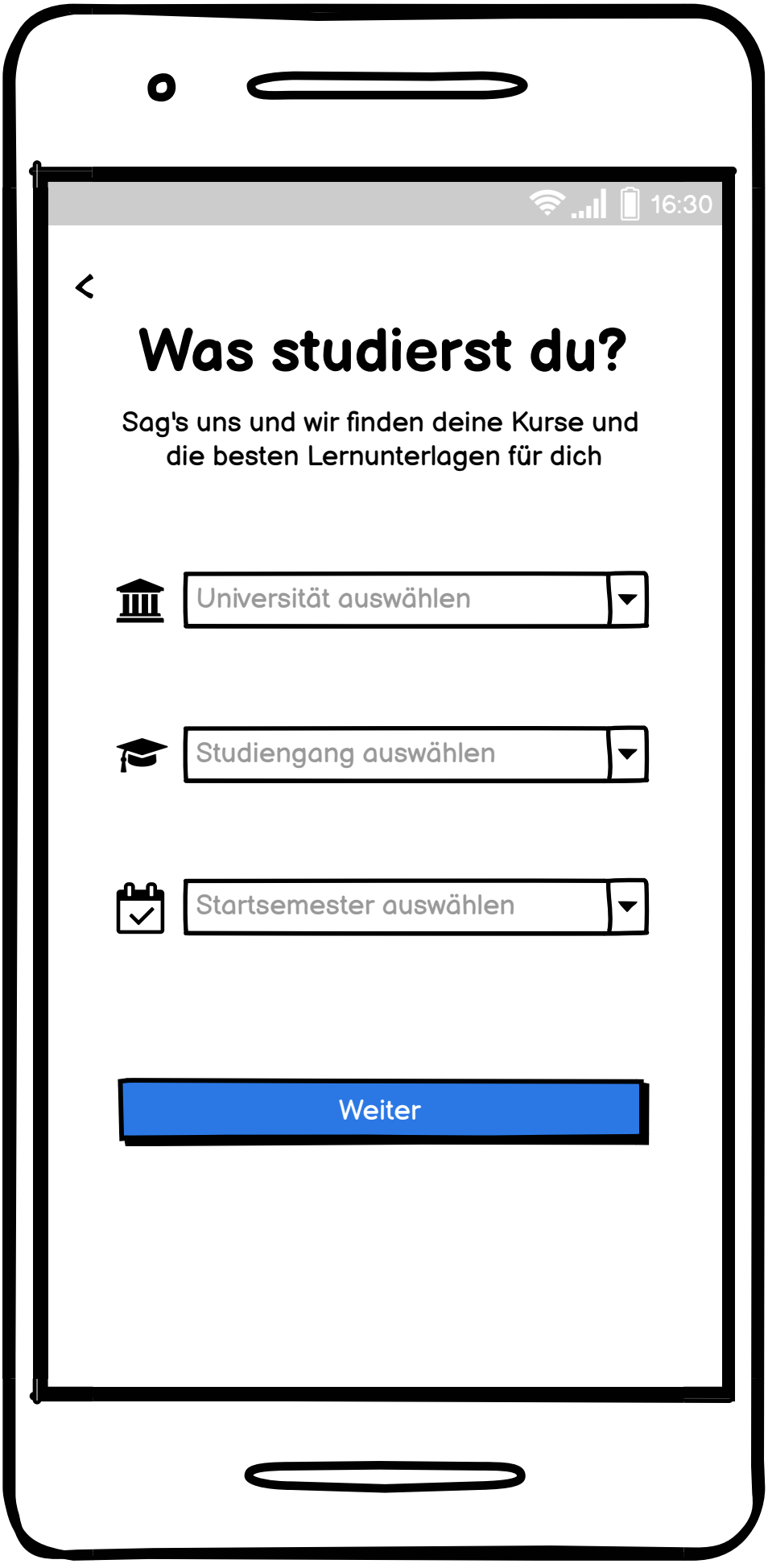

16:30
Studydrive
Mit E-Mail registrieren oder anmelden.
E-Mail eingeben
Weiter

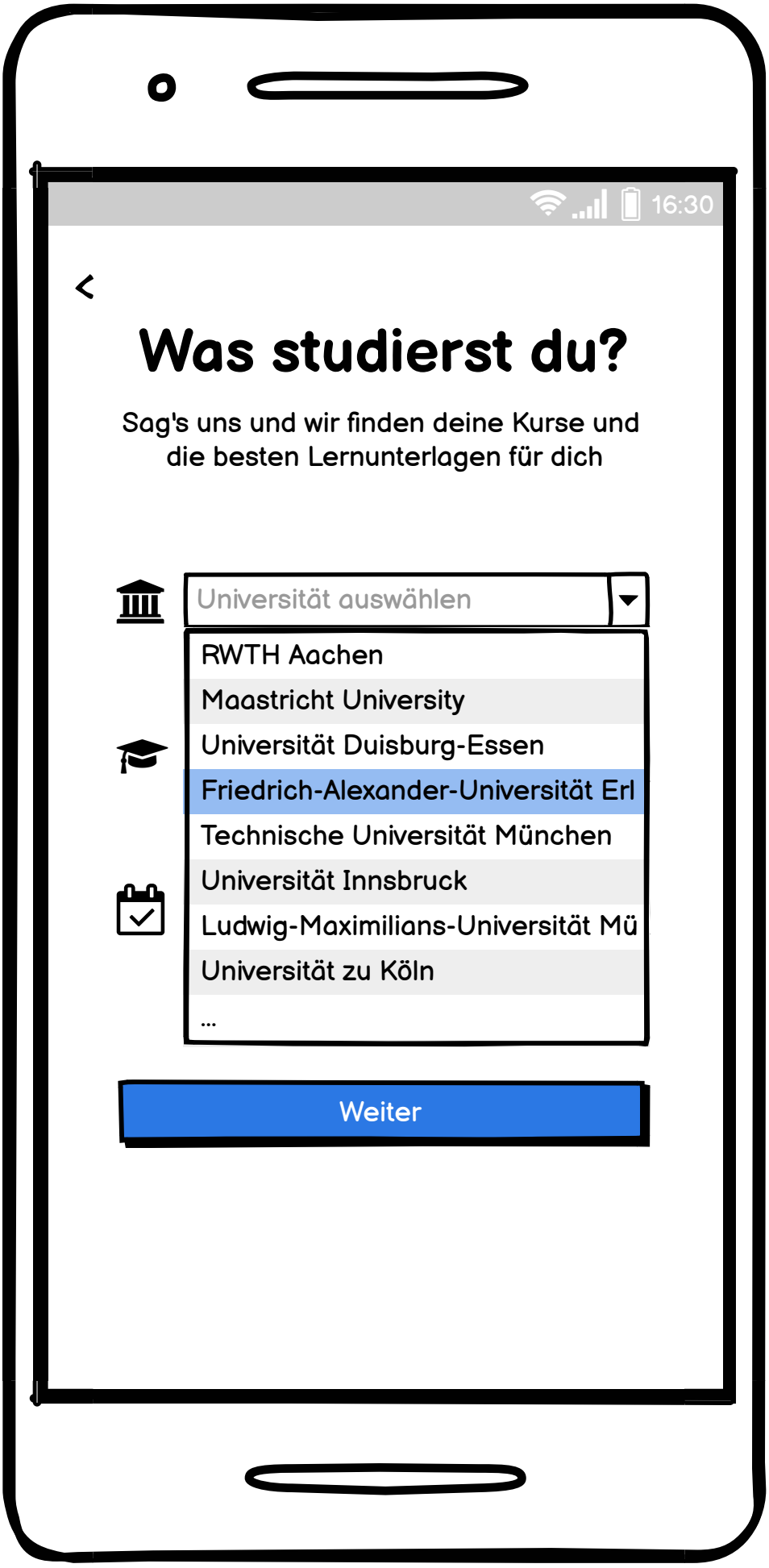

16:30
Studydrive
Mit E-Mail registrieren oder anmelden.
Beispielemail@gmail.com
Weiter

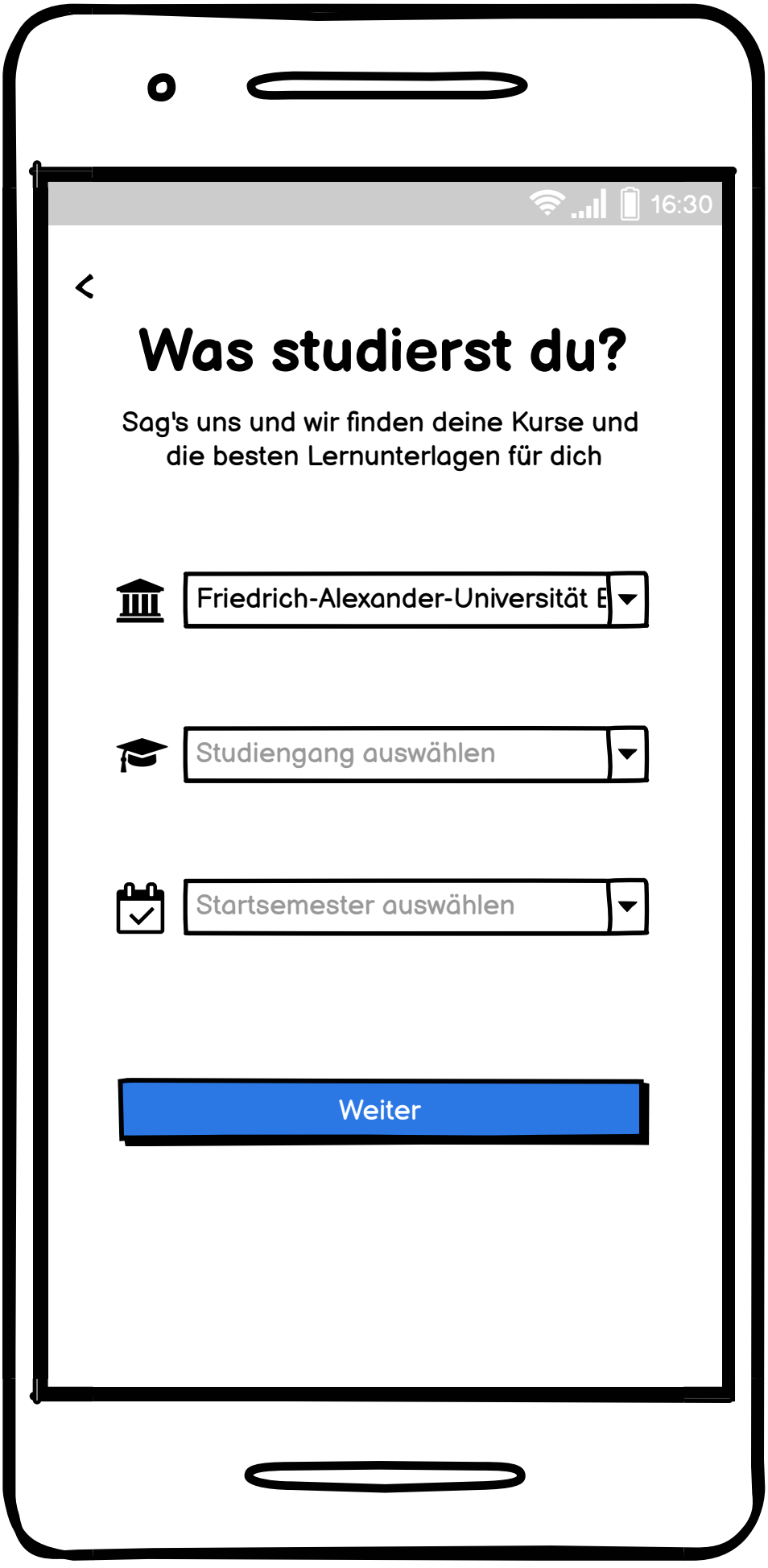

16:30
Studydrive
Willkommen bei Studydrive!
Registriere dich als neuer Nutzer
Beispielemail@gmail.com
Benutzernamen eingeben
Passwort eingeben
Registrieren
Mit der Registrierung stimmst du unseren AGB
und unserer Datenschutzerklärung zu.
Du hast bereits ein Konto? Anmelden.

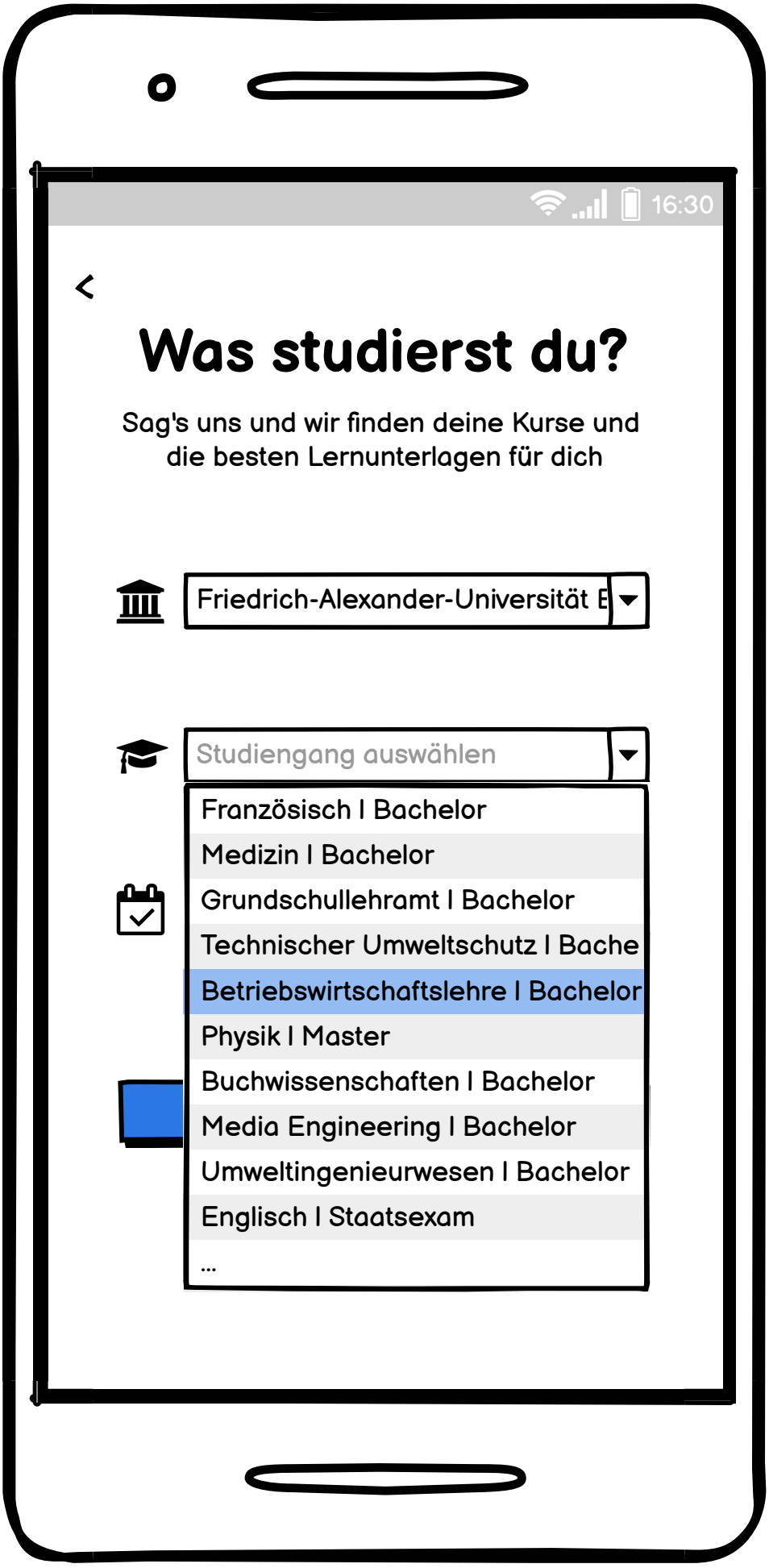

16:30
Studydrive
Willkommen bei Studydrive!
Registriere dich als neuer Nutzer
Beispielemail@gmail.com
BWLJustus
Passwort eingeben
Registrieren
Mit der Registrierung stimmst du unseren AGB
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Du hast bereits ein Konto? Anmelden.

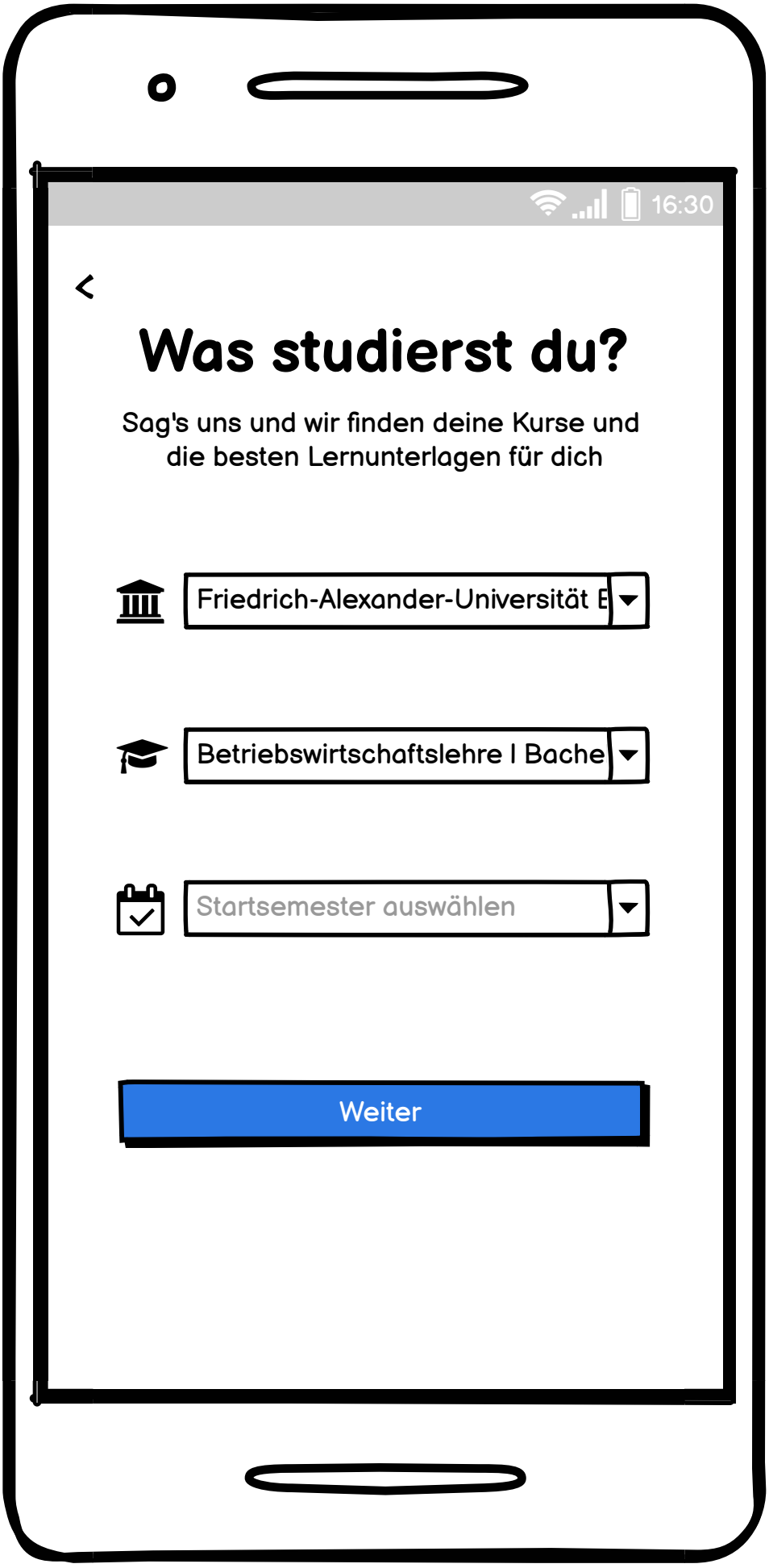

16:30
Studydrive
Willkommen bei Studydrive!
Registriere dich als neuer Nutzer
Beispielemail@gmail.com
BWLJustus
****
Registrieren
Mit der Registrierung stimmst du unseren AGB
und unserer Datenschutzerklärung zu.
Du hast bereits ein Konto? Anmelden.

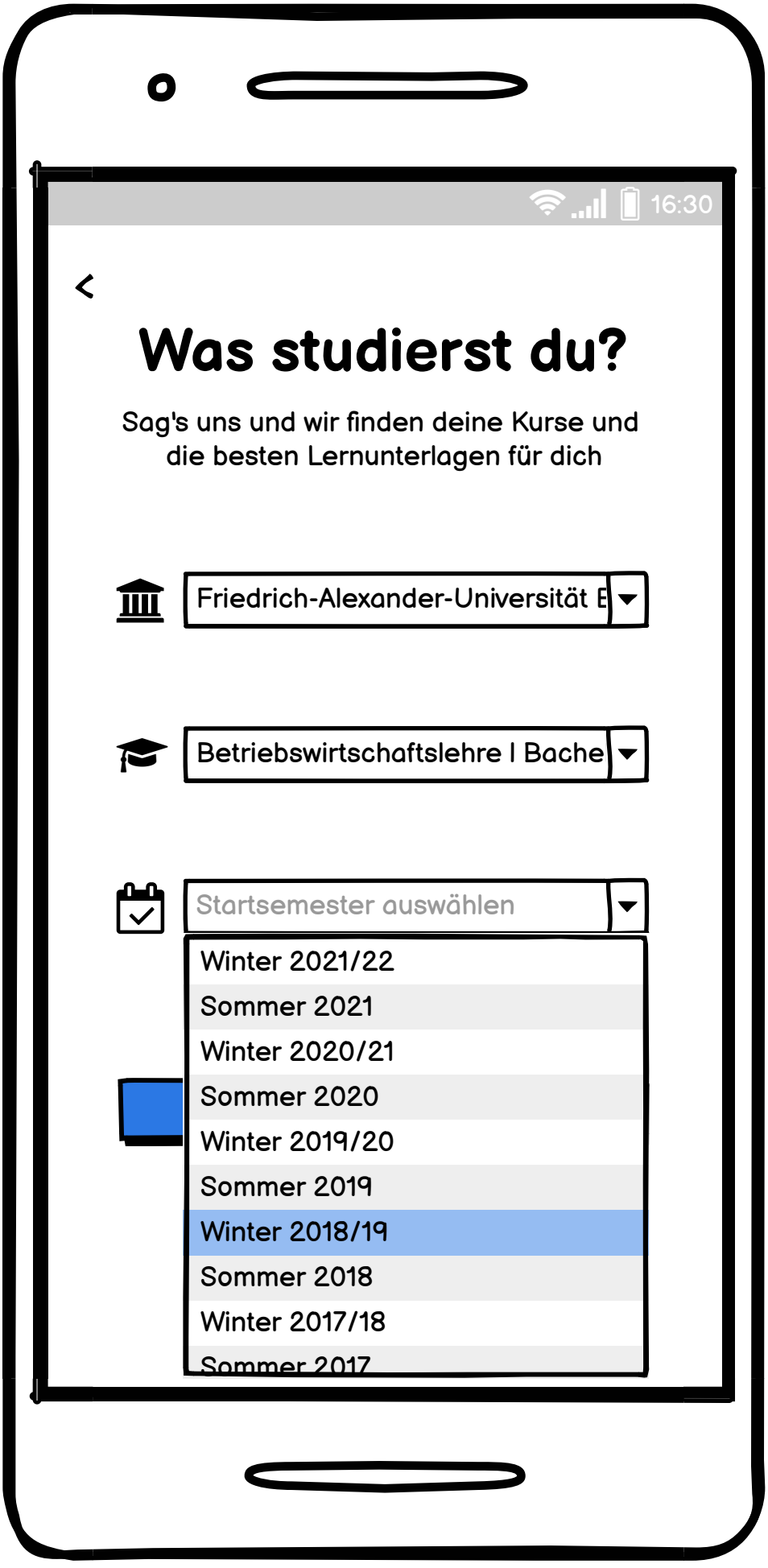

16:30
Was studierst du?
Sag's uns und wir finden deine Kurse und die besten Lernunterlagen für dich
Universität auswählen
Studiengang auswählen
Startsemester auswählen
Weiter

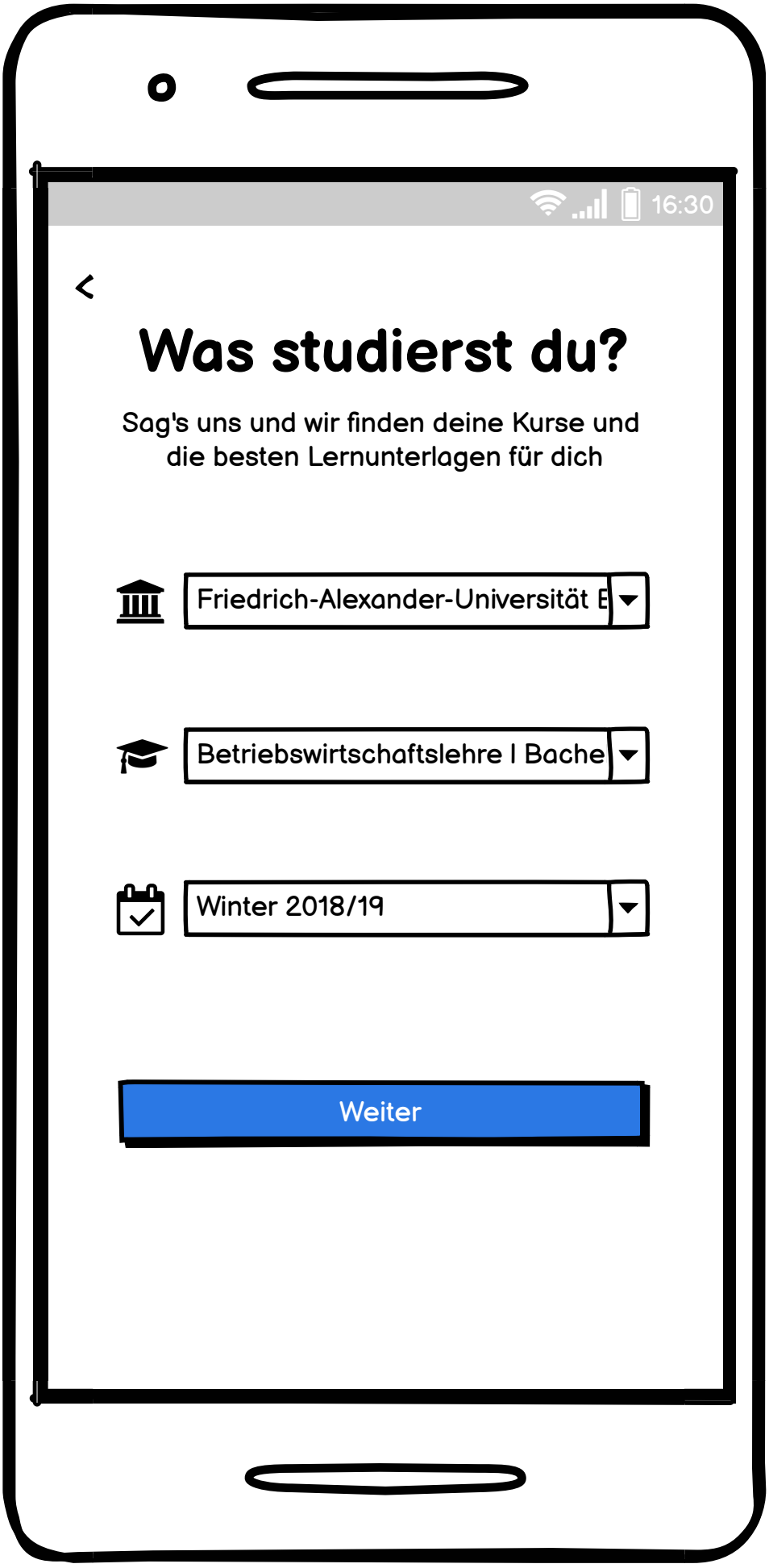

16:30
Was studierst du?
Sag's uns und wir finden deine Kurse und die besten Lernunterlagen für dich
Universität auswählen
RWTH Aachen
Maastricht University
Universität Duisburg-Essen
Friedrich-Alexander-Universität Erl
Technische Universität München
Universität Innsbruck
Ludwig-Maximilians-Universität Mü
Universität zu Köln
...
Weiter

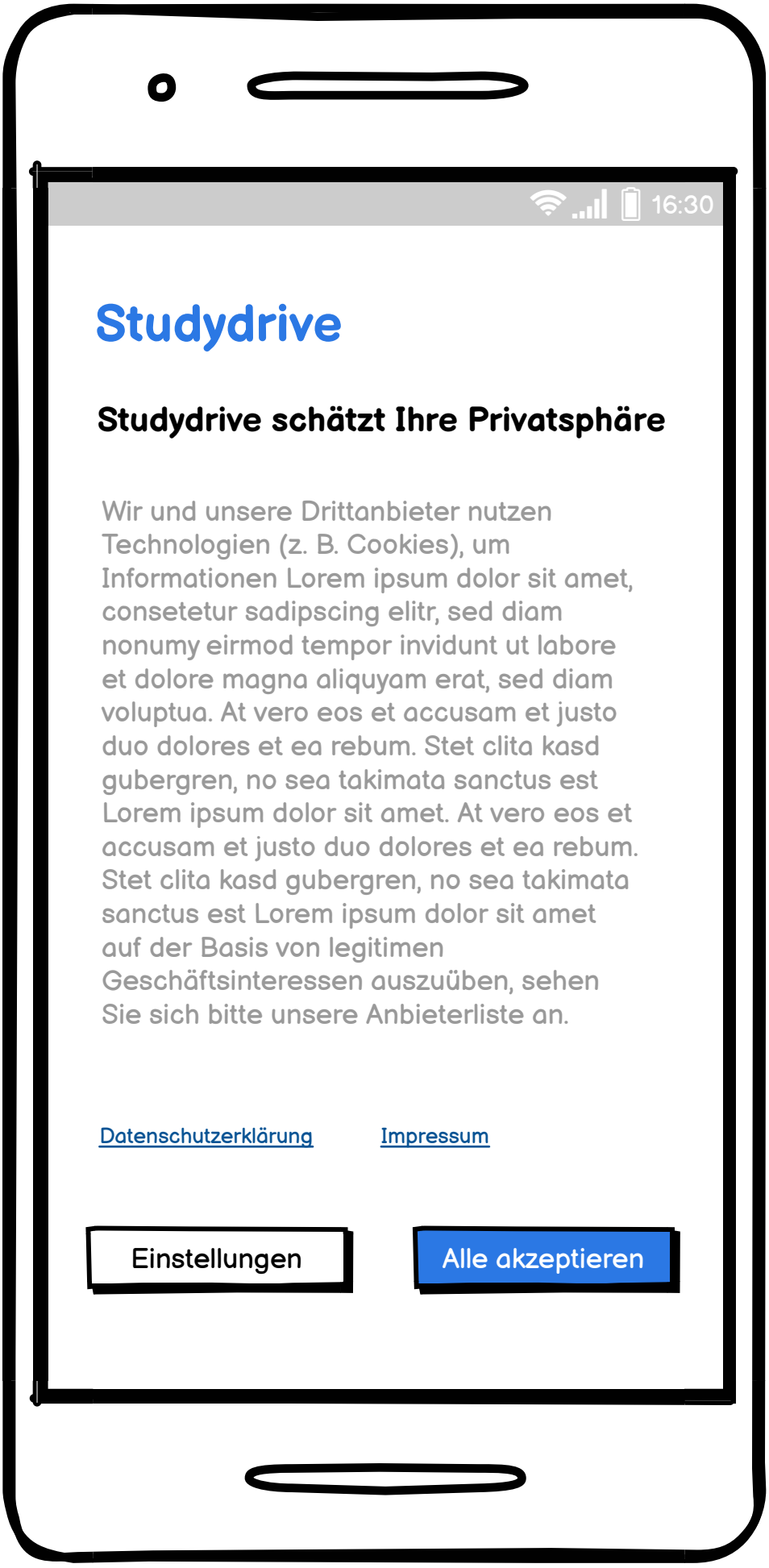

16:30
Was studierst du?
Sag's uns und wir finden deine Kurse und die besten Lernunterlagen für dich
Friedrich-Alexander-Universität E
Studiengang auswählen
Startsemester auswählen
Weiter


16:30
Was studierst du?
Sag's uns und wir finden deine Kurse und die besten Lernunterlagen für dich
Friedrich-Alexander-Universität E
Studiengang auswählen
Französisch I Bachelor
Medizin I Bachelor
Grundschullehramt I Bachelor
Technischer Umweltschutz I Bache
Betriebswirtschaftslehre I Bachelor
Physik I Master
Buchwissenschaften I Bachelor
Media Engineering I Bachelor
Umweltingenieurwesen I Bachelor
Englisch I Staatsexam
...


16:30
Was studierst du?
Sag's uns und wir finden deine Kurse und die besten Lernunterlagen für dich
Friedrich-Alexander-Universität E
Betriebswirtschaftslehre I Bache
Startsemester auswählen
Weiter

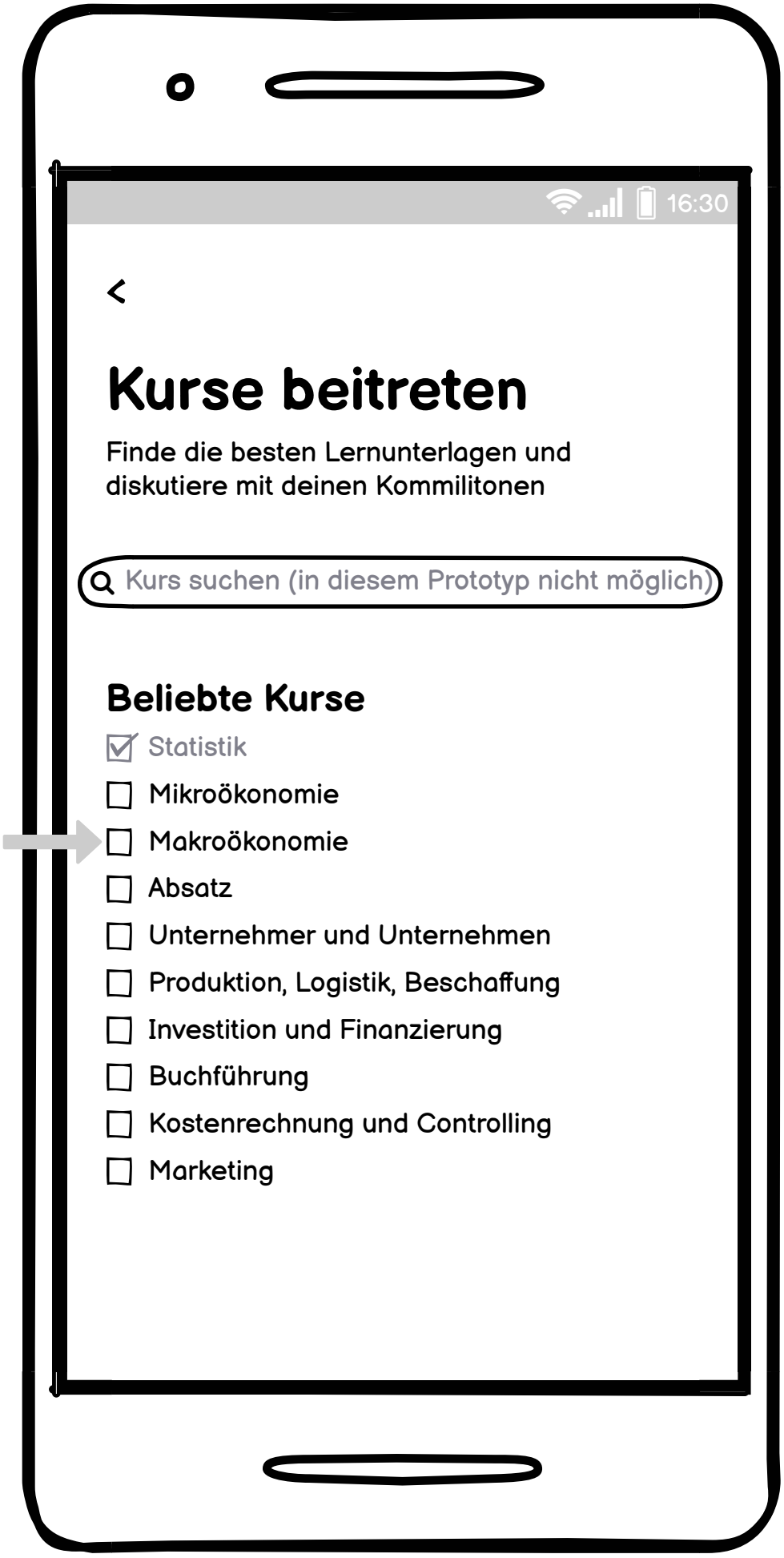

16:30
Was studierst du?
Sag's uns und wir finden deine Kurse und die besten Lernunterlagen für dich
Friedrich-Alexander-Universität E
Betriebswirtschaftslehre I Bache
Startsemester auswählen
Winter 2021/22
Sommer 2021
Winter 2020/21
Sommer 2020
Winter 2019/20
Sommer 2019
Winter 2018/19
Sommer 2018
Winter 2017/18
Sommer 2017

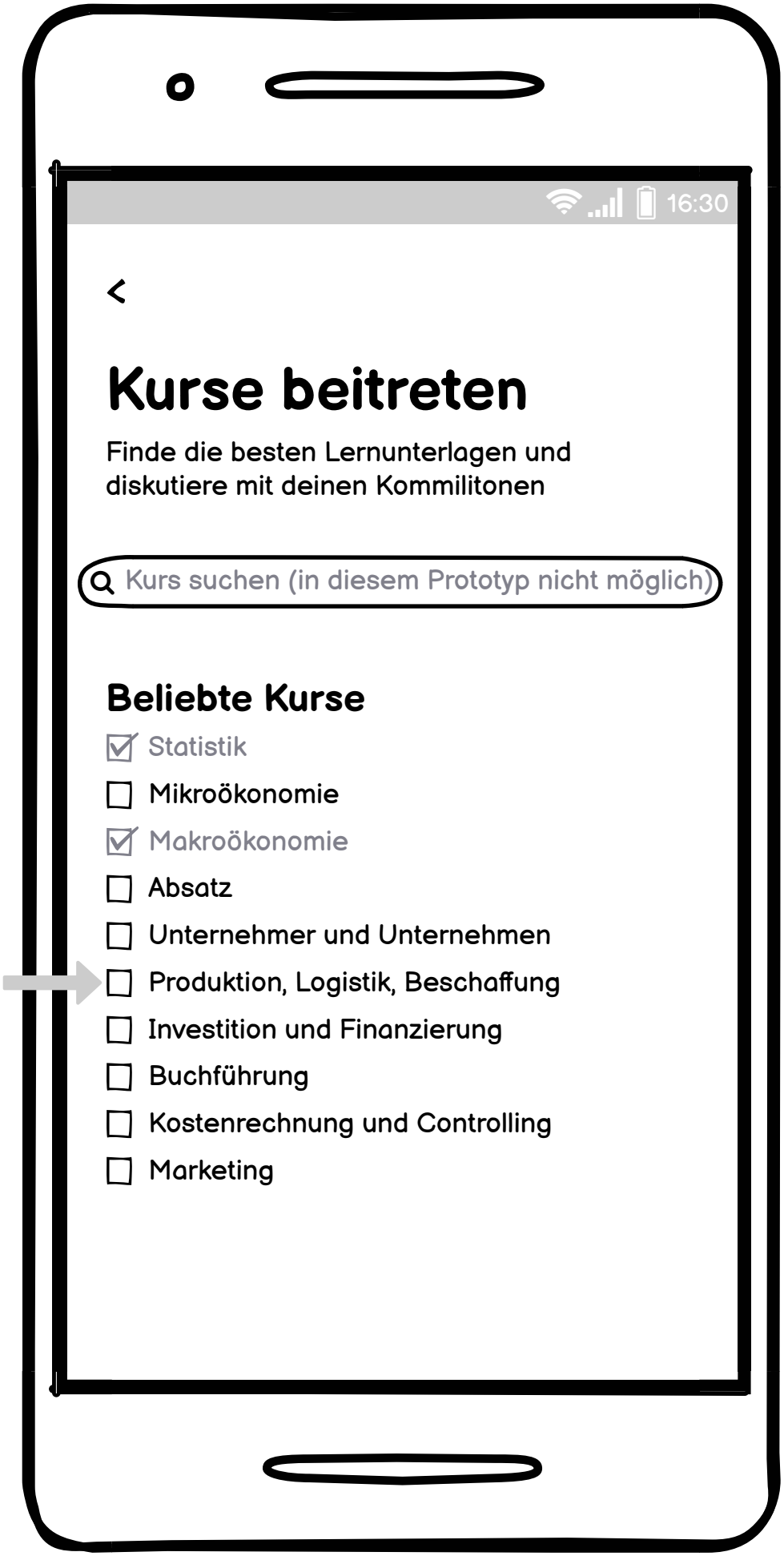

16:30
Was studierst du?
Sag's uns und wir finden deine Kurse und die besten Lernunterlagen für dich
Friedrich-Alexander-Universität E
Betriebswirtschaftslehre | Bache
Winter 2018/19
Weiter

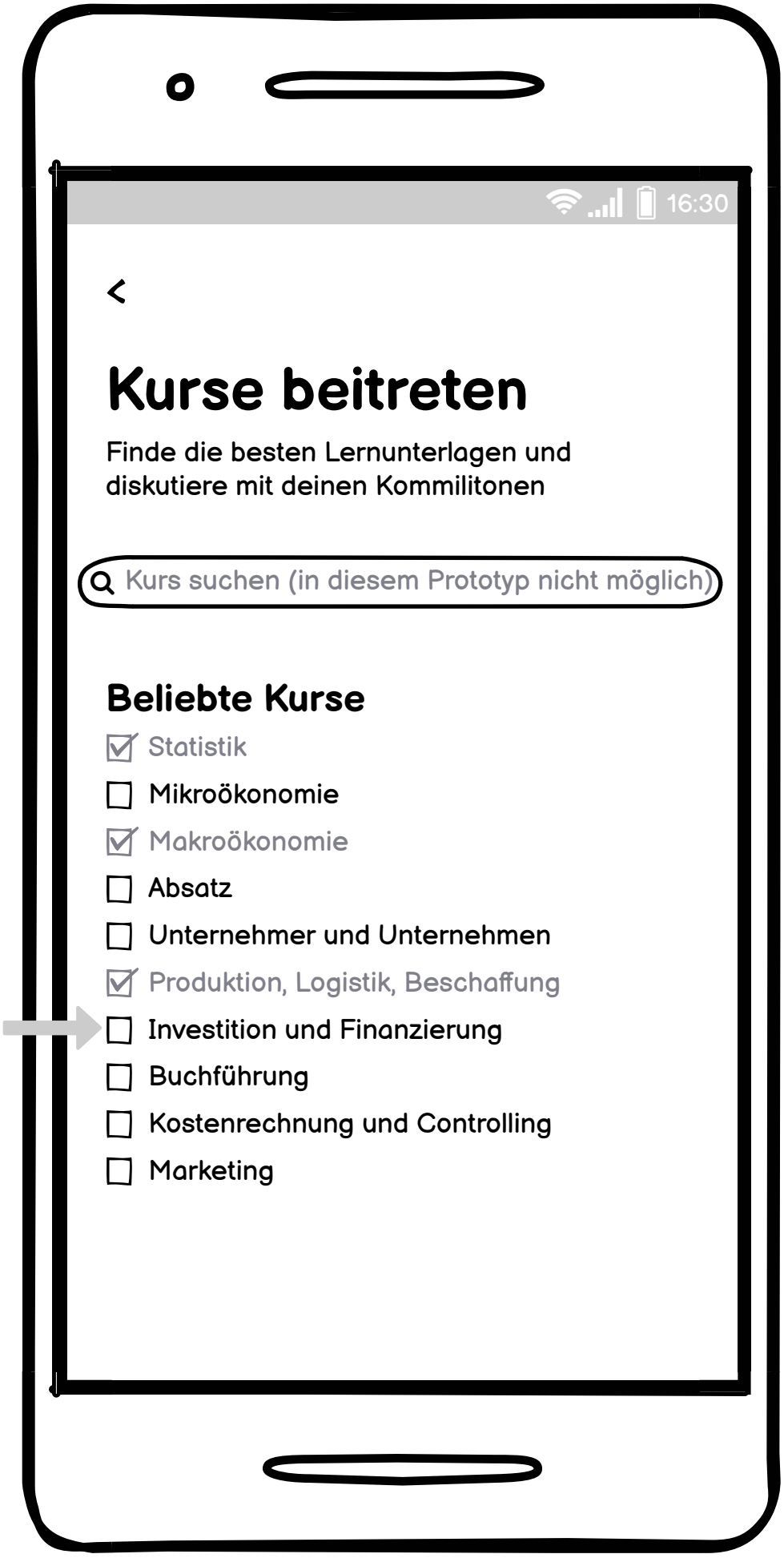

16:30
Studydrive
Studydrive schätzt Ihre Privatsphäre
Wir und unsere Drittanbieter nutzen Technologien (z. B. Cookies), um Informationen Lorem ipsum dolor sit amet, consetetur sadipscing elitr, sed diam nonumy eirmod tempor invidunt ut labore et dolore magna aliquyam erat, sed diam voluptua. At vero eos et accusam et justo duo dolores et ea rebum. Stet clita kasd gubergren, no sea takimata sanctus est Lorem ipsum dolor sit amet. At vero eos et accusam et justo duo dolores et ea rebum. Stet clita kasd gubergren, no sea takimata sanctus est Lorem ipsum dolor sit amet auf der Basis von legitimen Geschäftsinteressen auszuüben, sehen Sie sich bitte unsere Anbieterliste an.
Datenschutzerklärung
Impressum
Einstellungen
Alle akzeptieren

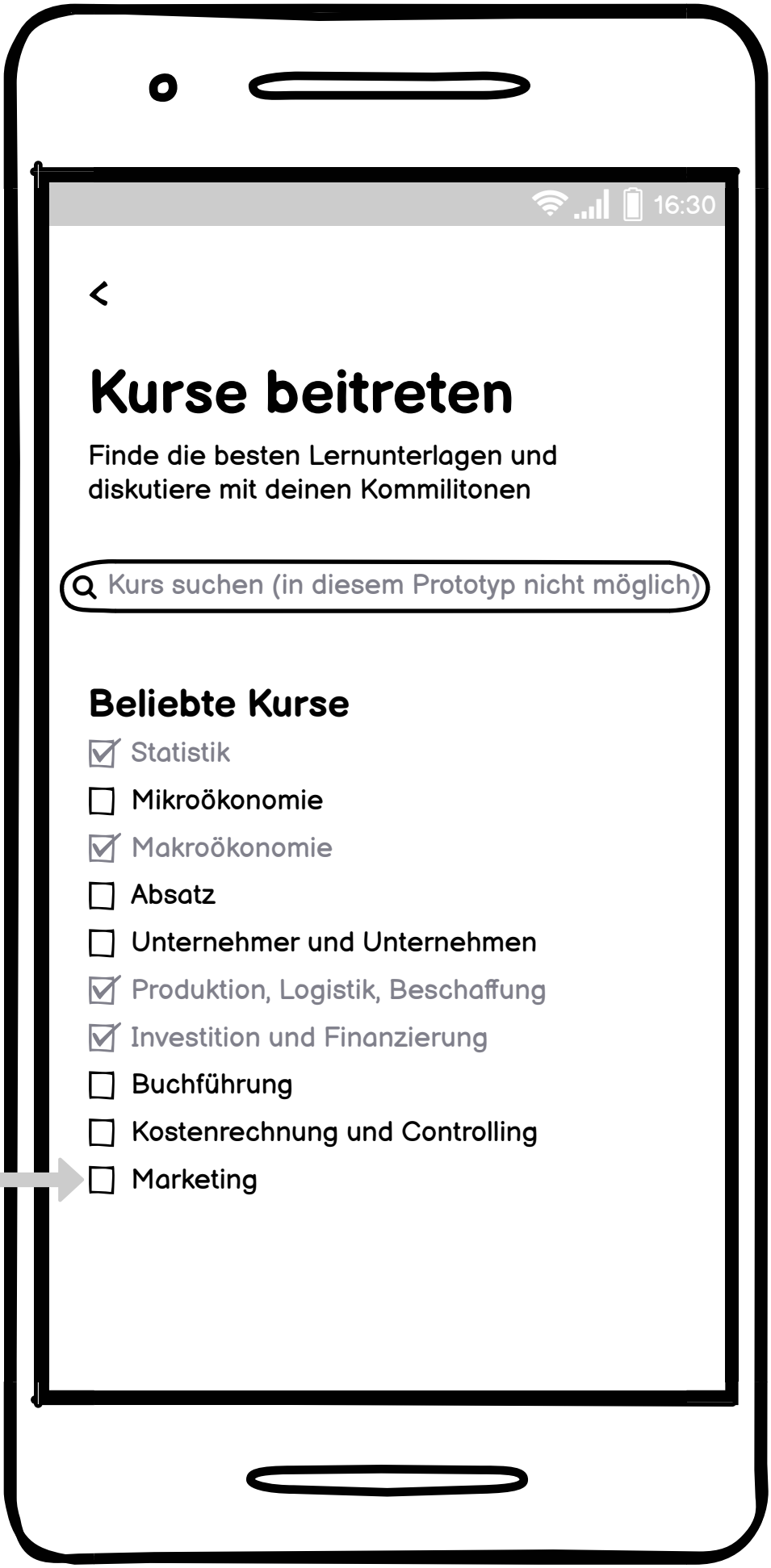

16:30
Newsfeed
Neueste
Unbeantwortet
Erste Schritte
1
Kurse hinzufügen
Finde Lernunterlagen und triff deine Kommilitonen
2
Gruppen hinzufügen
Tausche dich aus & erhalte Hilfe
3
Profil vervollständigen
Klicken um Profilinfo zu ändern
4
E-Mail bestätigen
E-Mail checken oder klicken um erneut zu senden
Anonymer Nutzer
vor 22 Stunden
Uni Erlangen Nürnberg
Gibt es schon einen Klausurtermin für die Überblicksvorlesung "Einführung in die Frühe


16:30
<
Kurse beitreten
Finde die besten Lernunterlagen und diskutiere mit deinen Kommilitonen
Kurs suchen (in diesem Prototyp nicht möglich)
Beliebte Kurse
Statistik
Mikroökonomie
Makroökonomie
Absatz
Unternehmer und Unternehmen
Produktion, Logistik, Beschaffung
Investition und Finanzierung
Buchführung
Kostenrechnung und Controlling
Marketing


16:30
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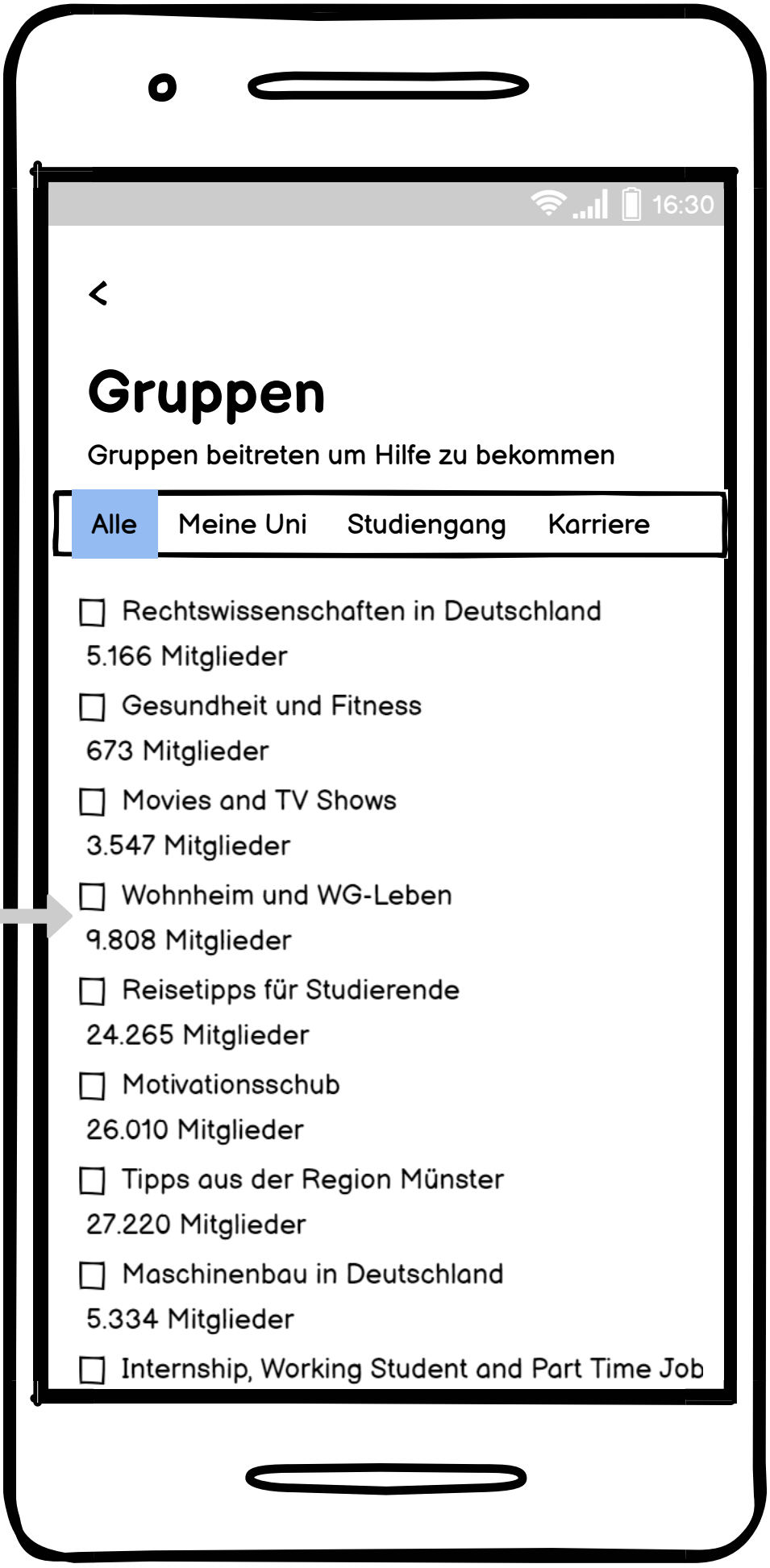

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Marketing

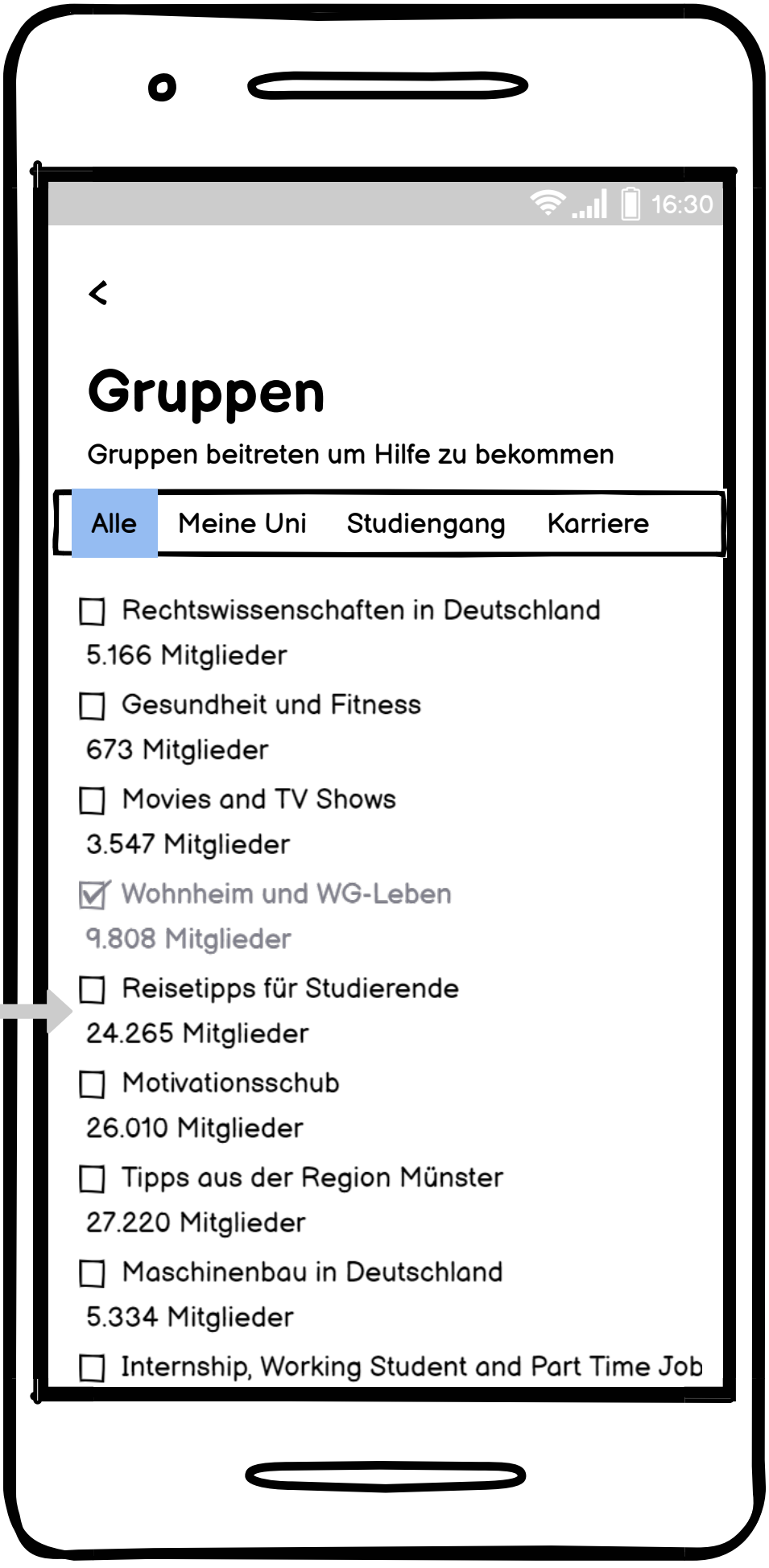

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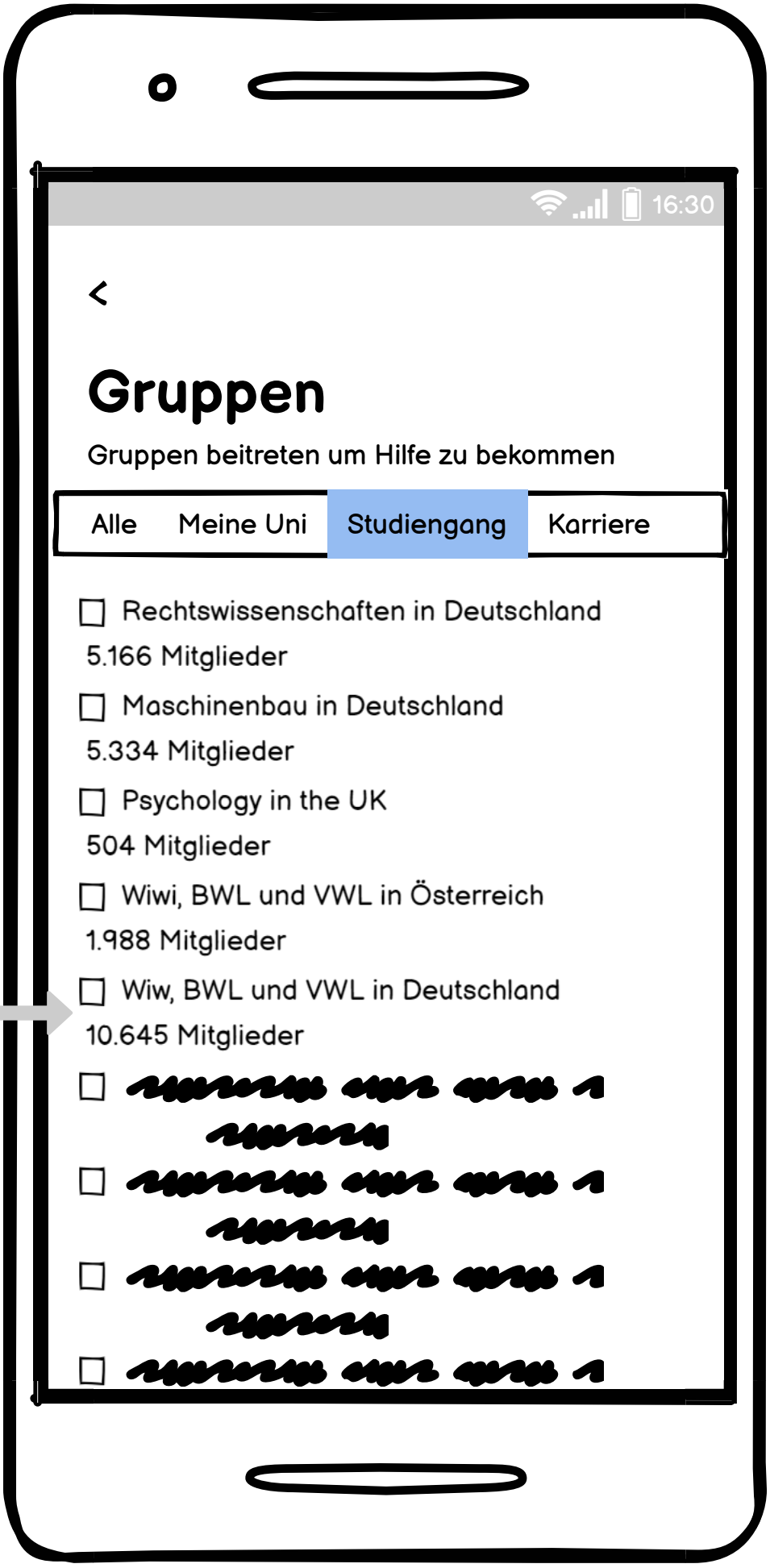

16:30
Newsfeed
Neueste
Unbeantwortet
Erste Schritte
1
Kurse hinzufügen
Finde Lernunterlagen und triff deine Kommilitonen
2
Gruppen hinzufügen
Tausche dich aus & erhalte Hilfe
3
Profil vervollständigen
Klicken um Profilinfo zu ändern
4
E-Mail bestätigen
E-Mail checken oder klicken um erneut zu senden
Anonymer Nutzer
vor 22 Stunden
Uni Erlangen Nürnberg
Gibt es schon einen Klausurtermin für die Überblicksvorlesung "Einführung in die Frühe

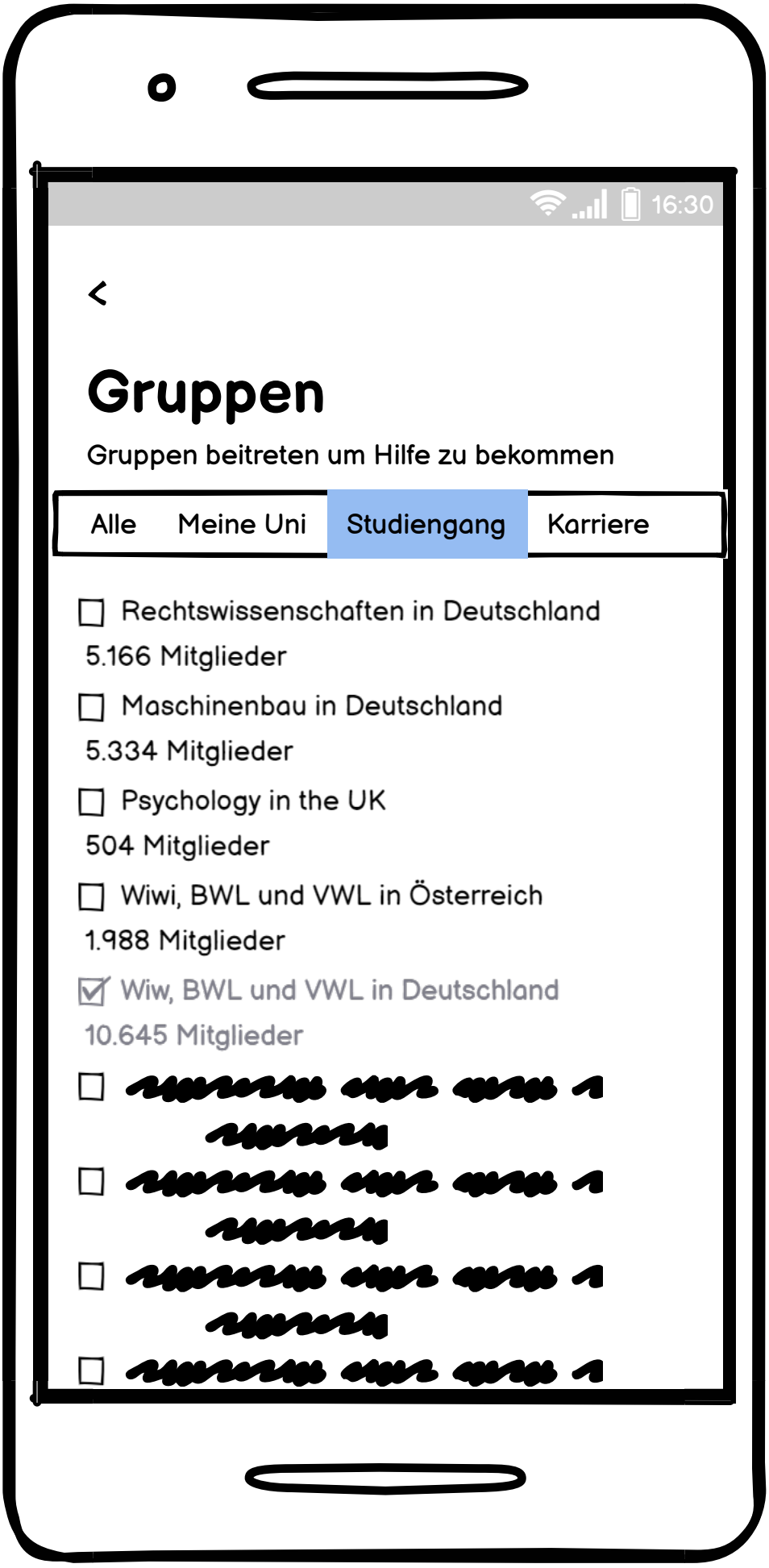

16:30
Gruppen
Gruppen beitreten um Hilfe zu bekommen
Alle
Meine Uni
Studiengang
Karriere
Rechtswissenschaften in Deutschland
5.166 Mitglieder
Gesundheit und Fitness
673 Mitglieder
Movies and TV Shows
3.547 Mitglieder
Wohnheim und WG-Leben
9.808 Mitglieder
Reisetipps für Studierende
24.265 Mitglieder
Motivationsschub
26.010 Mitglieder
Tipps aus der Region Münster
27.220 Mitglieder
Maschinenbau in Deutschland
5.334 Mitglieder
Internship, Working Student and Part Time Job


16:30
Gruppen
Gruppen beitreten um Hilfe zu bekommen
Alle
Meine Uni
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Karriere
Rechtswissenschaften in Deutschland
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Maschinenbau in Deutschland
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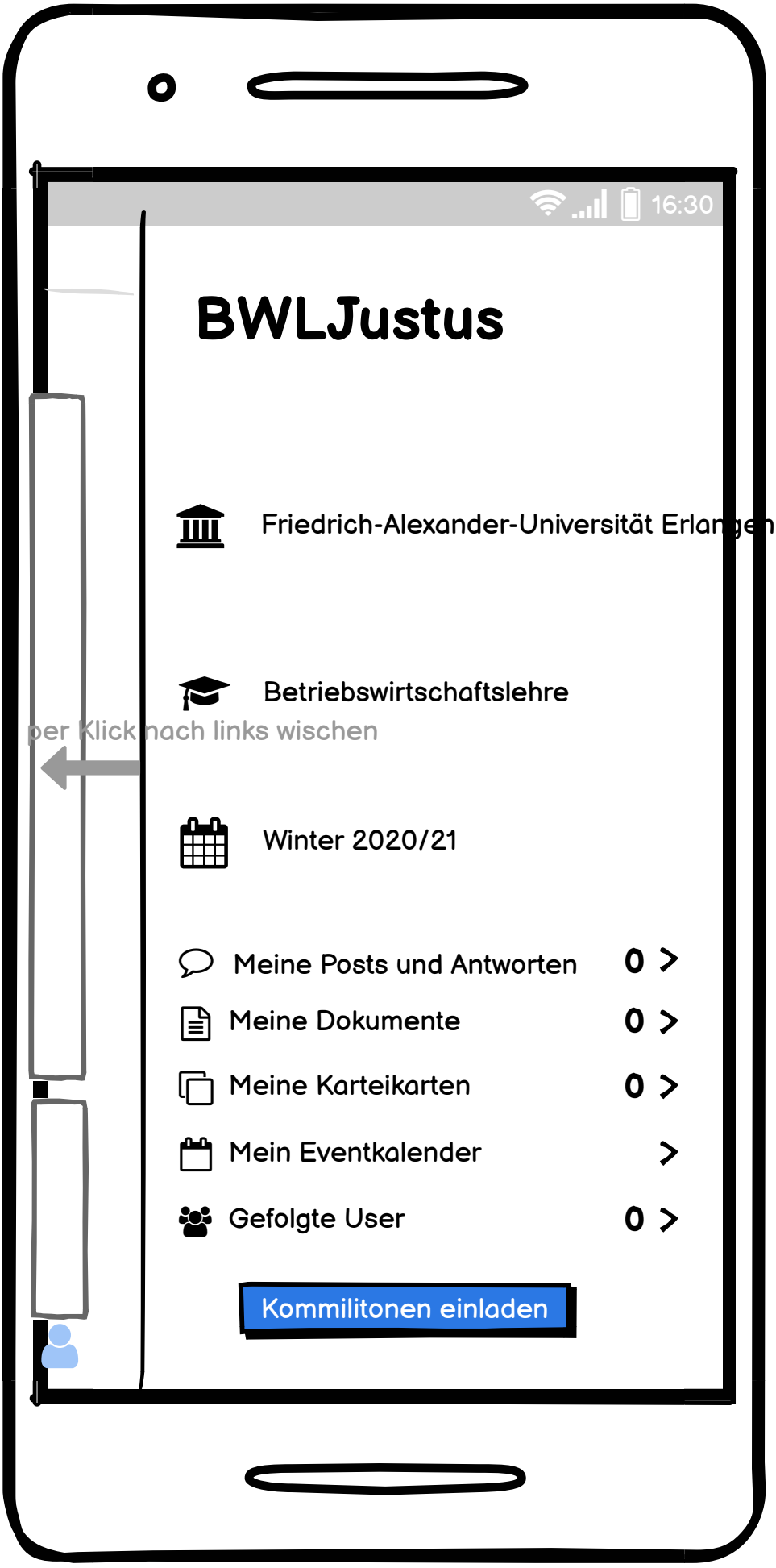

16:30
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Meine Uni
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Karriere
Rechtswissenschaften in Deutschland
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Maschinenbau in Deutschland
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16:30
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Meine Uni
Studiengang
Karriere
Rechtswissenschaften in Deutschland
5.166 Mitglieder
Maschinenbau in Deutschland
5.334 Mitglieder
Psychology in the UK
504 Mitglieder
Wiwi, BWL und VWL in Österreich
1.988 Mitglieder
Wiw, BWL und VWL in Deutschland
10.645 Mitglieder


16:30
Gruppen
Gruppen beitreten um Hilfe zu bekommen
Alle
Meine Uni
Studiengang
Karriere
Rechtswissenschaften in Deutschland
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Maschinenbau in Deutschland
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Psychology in the UK
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Wiwi, BWL und VWL in Österreich
1.988 Mitglieder
Wiw, BWL und VWL in Deutschland
10.645 Mitglieder


16:30
Newsfeed
Neueste
Unbeantwortet
Erste Schritte
1
Kurse hinzufügen
Finde Lernunterlagen und triff deine Kommilitonen
2
Gruppen hinzufügen
Tausche dich aus & erhalte Hilfe
3
Profil ansehen u. vervollständigen
Klicken um Profilinfo zu ändern
4
E-Mail bestätigen
E-Mail checken oder klicken um erneut zu senden
Anonymer Nutzer
vor 22 Stunden
Uni Erlangen Nürnberg
Gibt es schon einen Klausurtermin für die Überblicksvorlesung "Einführung in die Frühe


16:30
< Profil Bearbeiten
BWLJustus
Männlich
Dein Studium
Studium hinzufügen
Friedrich-Alexander-Universität Erlangen
Betriebswirtschaftslehre
>


16:30
BWLJustus
Friedrich-Alexander-Universität Erlangen
Betriebswirtschaftslehre
per Klick nach links wischen
Winter 2020/21
Meine Posts und Antworten 0 >
Meine Dokumente 0 >
Meine Karteikarten 0 >
Mein Eventkalender >
Gefolgte User 0 >
Kommilitonen einladen

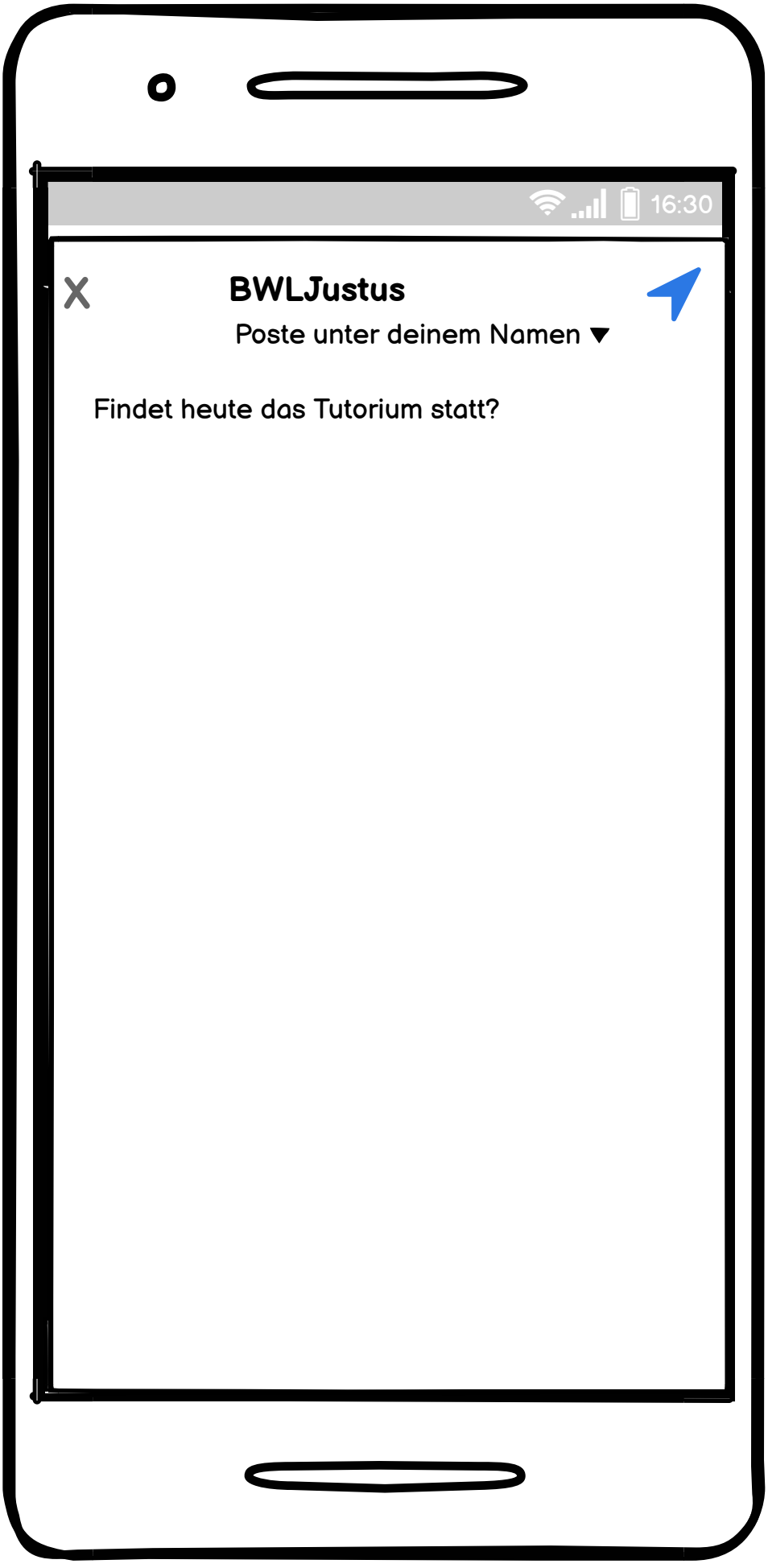

16:30
Newsfeed
Neueste
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Erste Schritte
1
Kurse hinzufügen
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Profil vervollständigen
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E-Mail bestätigen
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Anonymer Nutzer
vor 22 Stunden
Uni Erlangen Nürnberg
Gibt es schon einen Klausurtermin für die Überblicksvorlesung "Einführung in die Frühe


16:30
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Anonymer Nutzer
vor 22 Stunden
Uni Erlangen Nürnberg
Gibt es schon einen Klausurtermin für die Überblicksvorlesung "Einführung in die Frühe


16:30
Newsfeed
Neueste
Unbeantwortet
Erste Schritte
1
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Finde Lernunterlagen und triff deine Kommilitonen
2
Gruppen hinzufügen
Tausche dich aus & erhalte Hilfe
3
Profil vervollständigen
Klicken um Profilinfo zu ändern
4
Frage stellen
Dokument hochladen
Karteikarten erstellen
Uni Erlangen Nürnberg
Gibt es schon einen Klausurtermin für die Überblicksvorlesung "Einführung in die Frühe


16:30
Newsfeed
Neueste
Unbeantwortet
Erste Schritte
1
Kurse hinzufügen
Finde Lernunterlagen und triff deine Kommilitonen
2
Gruppen hinzufügen
Tausche dich aus & erhalte Hilfe
X
Wo möchtest du posten?
Kurse
Gruppen
Statistik
Makroökonomie
Produktion, Logistik, Beschaffung
Investition und Finanzierung
Marketing


16:30
BWLJustus
Poste unter deinem Namen
Was möchtest du in Statistik posten?

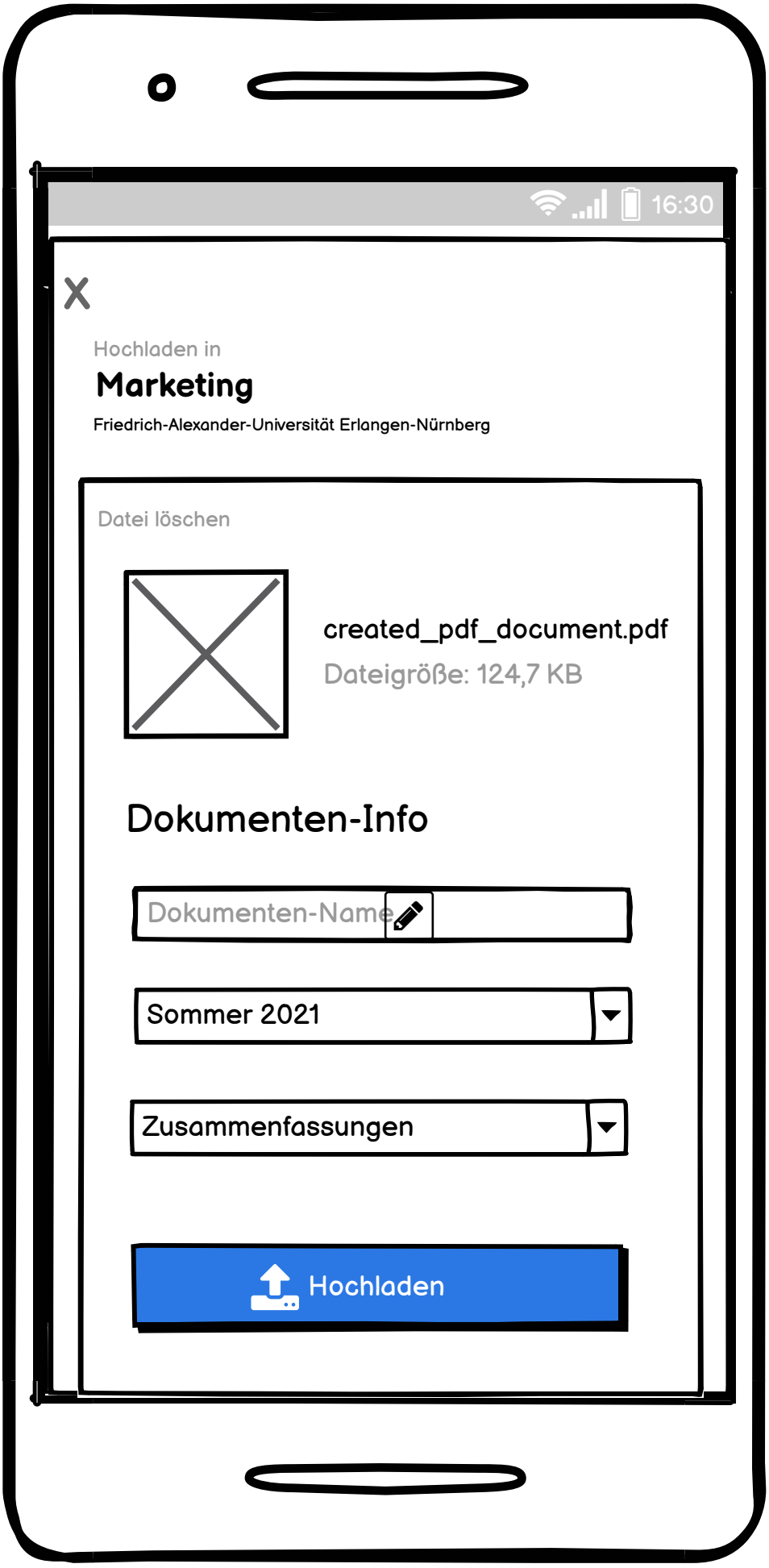

16:30
BWLJustus
Poste unter deinem Namen
Findet heute das Tutorium statt?


16:30
Newsfeed
Neueste
Unbeantwortet
BWLJustus
vor 1 Minute
Statistik
Findet heute das Tutorium statt?
0
Anonymer Nutzer
vor 22 Stunden
Uni Erlangen Nürnberg
Gibt es schon einen Klausurtermin für die Überblicksvorlesung "Einführung in die Frühe Neuzeit" bei Herrn Wellner?
0
Anonyme Nutzerin
vor 23 Stunden
Marketing
5

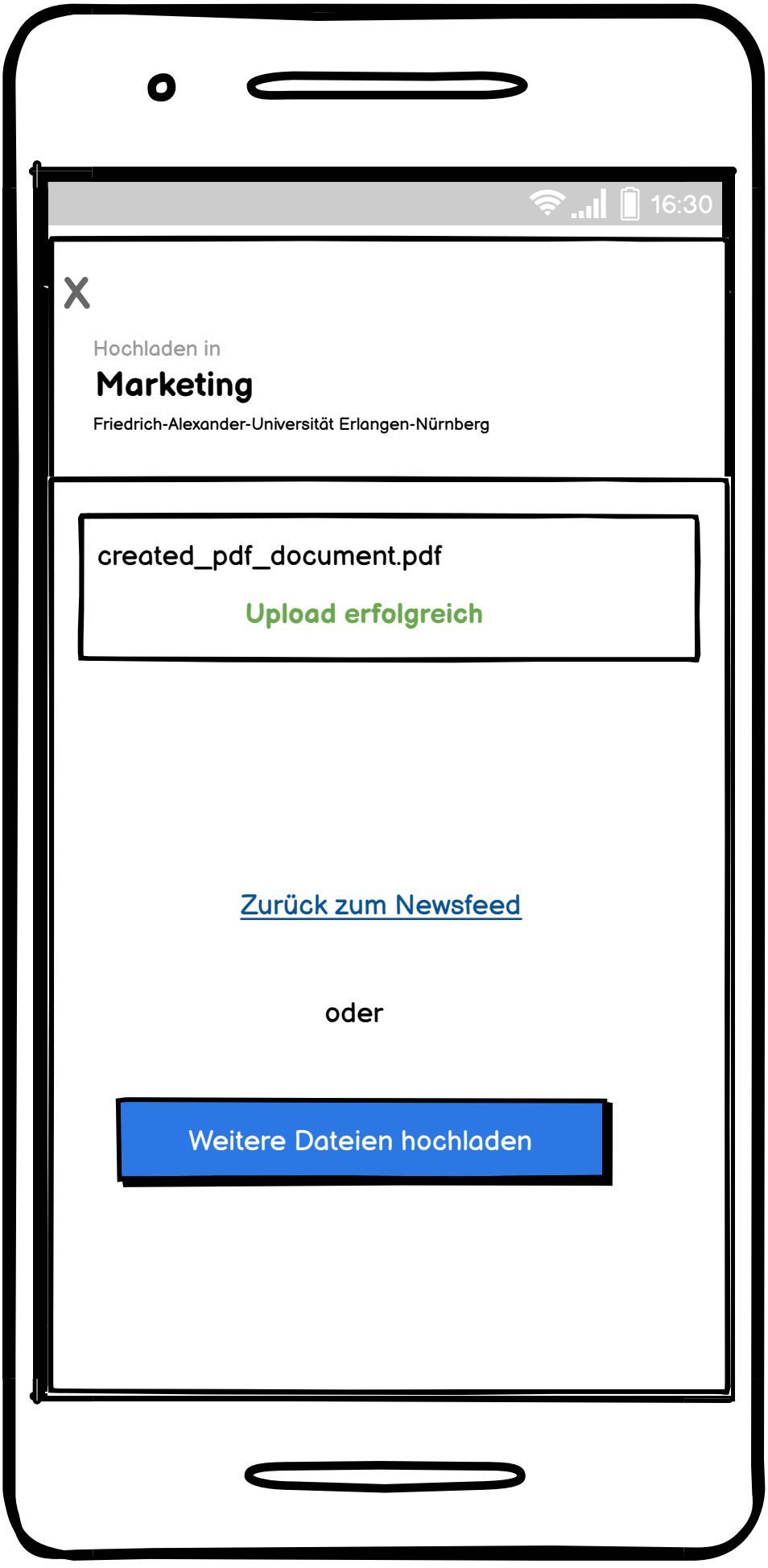

16:30
Newsfeed
Neueste
Unbeantwortet
BWLJustus
vor 1 Minute
Statistik
Findet heute das Tutorium statt?
0
Anonymer Nutzer
vor 22 Stunden
Uni Erlangen Nürnberg
Gibt es schon einen Klausurtermin für die Überblicksvorlesung "Einführung in die Frühe Neuzeit" bei Herrn Wellner?
Frage stellen
Dokument hochladen
Karteikarten erstellen
5


16:30
Newsfeed
Neueste
Unbeantwortet
BWLJustus
vor 1 Minute
Statistik
Findet heute das Tutorium statt?
0
Anonymer Nutzer
vor 22 Stunden
Wähle eine deiner Kurse zum Upload
Statistik
Makroökonomie
Produktion, Logistik, Beschaffung
Investition und Finanzierung
Marketing


16:30
X
Hochladen in
Marketing
Friedrich-Alexander-Universität Erlangen-Nürnberg
Datei hochladen
von deinem Handy, Drive oder Dropbox
Was solltest du teilen?
Deine Notizen, Zusammenfassungen, Lösungen, etc.
Getippte oder handgeschriebene Dateien
Aktuelle & immer noch relevante Dokumente
Bevorzugt diese Formate: .pdf, .doc, .docx, .jped, .xls
Was solltest du nicht teilen?
Ganze Bücher oder Artikel
Einzelne Fotos von mehrseitigen Dokumenten
Vermeide .zip & .rar hochzuladen
Alles was nicht studienrelevant ist oder urheberrechtlich geschützt ist

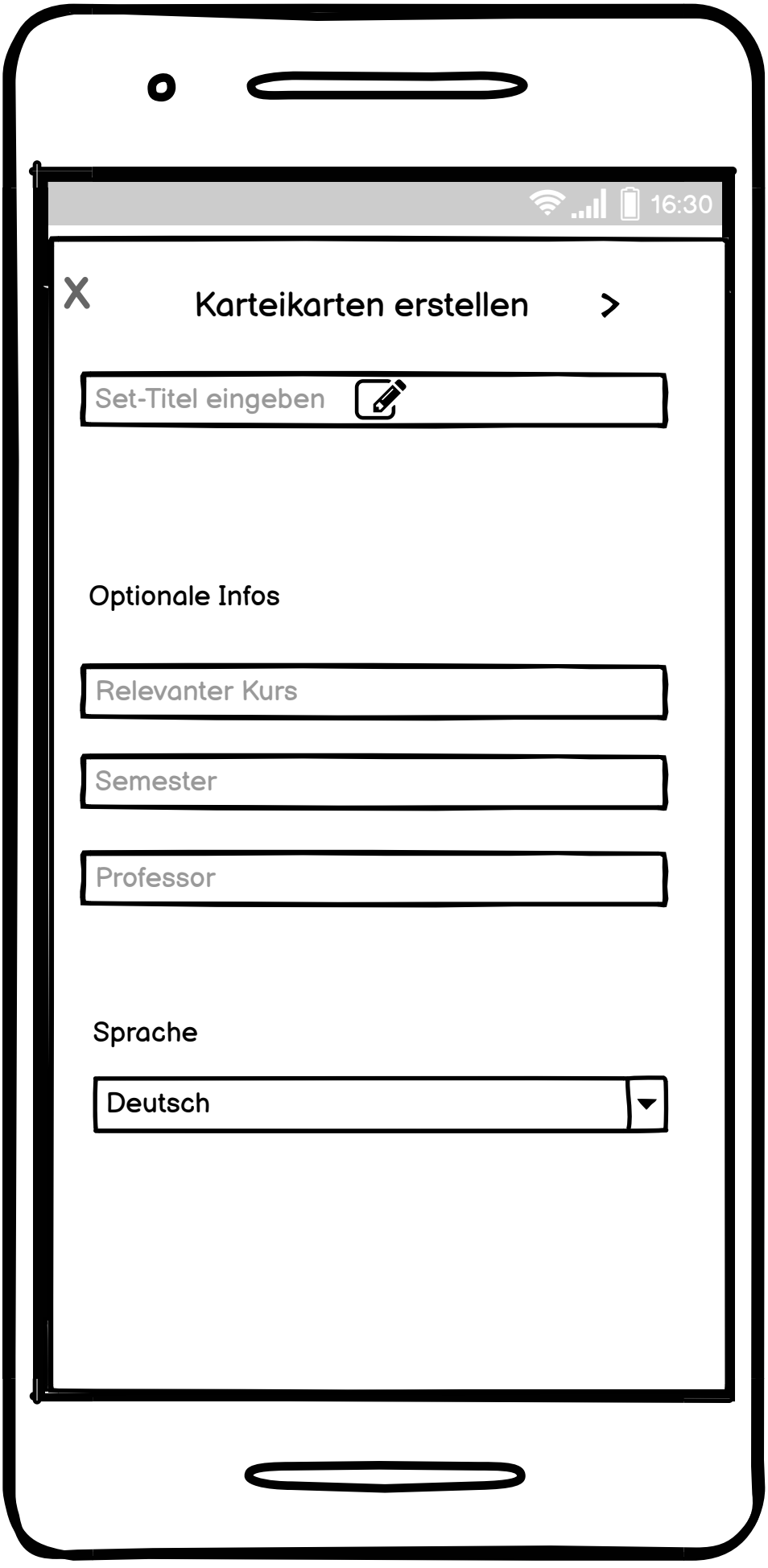

16:30
Hochladen in
Marketing
Friedrich-Alexander-Universität Erlangen-Nürnberg
Datei löschen
created_pdf_document.pdf
Dateigröße: 124,7 KB
Dokumenten-Info
Dokumenten-Name
Sommer 2021
Zusammenfassungen
Hochladen

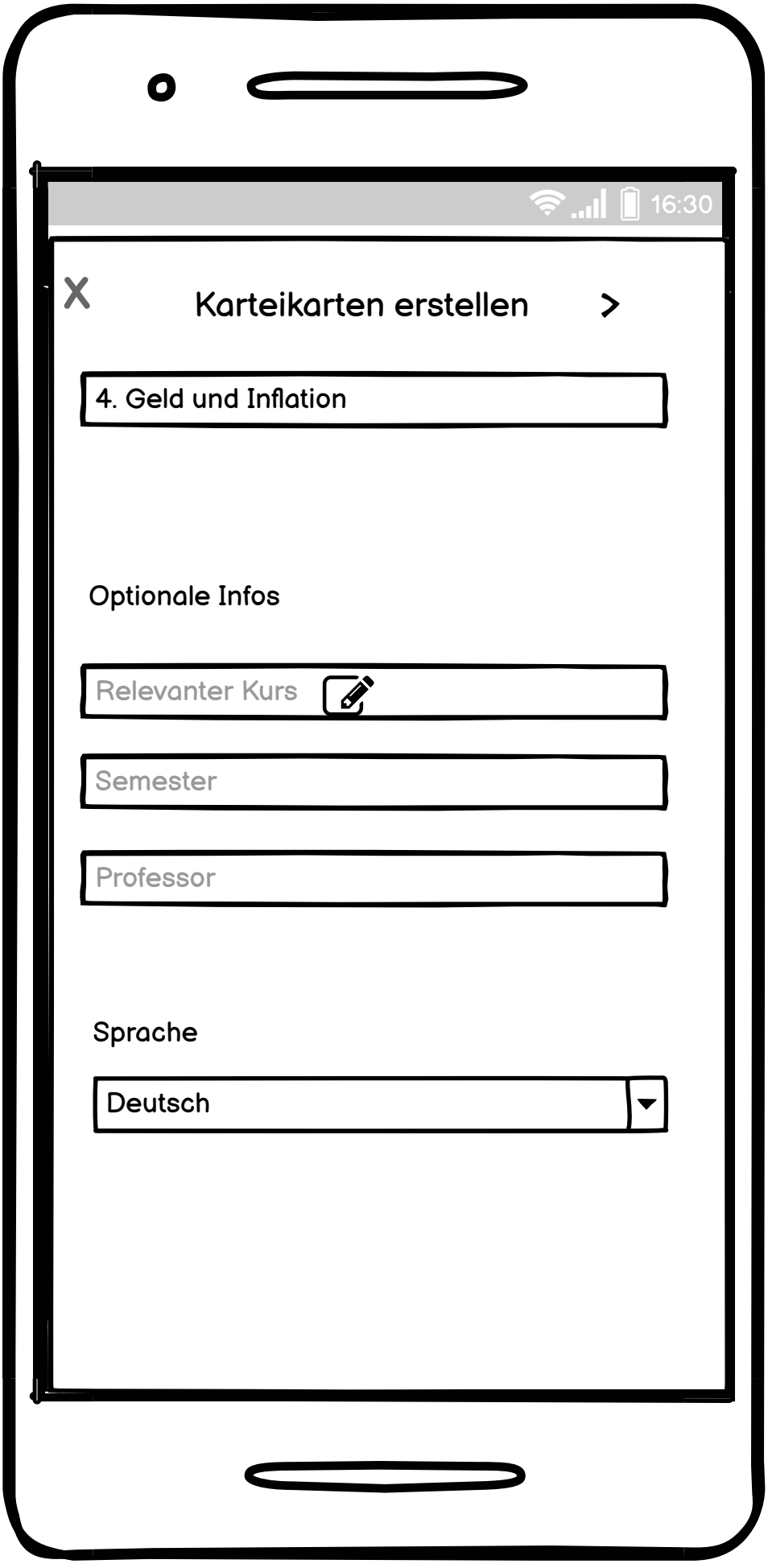

16:30
Hochladen in
Marketing
Friedrich-Alexander-Universität Erlangen-Nürnberg
Datei löschen
created_pdf_document.pdf
Dateigröße: 124,7 KB
Dokumenten-Info
Meine Marketing Zusammenfassung
Sommer 2021
Zusammenfassungen
Hochladen

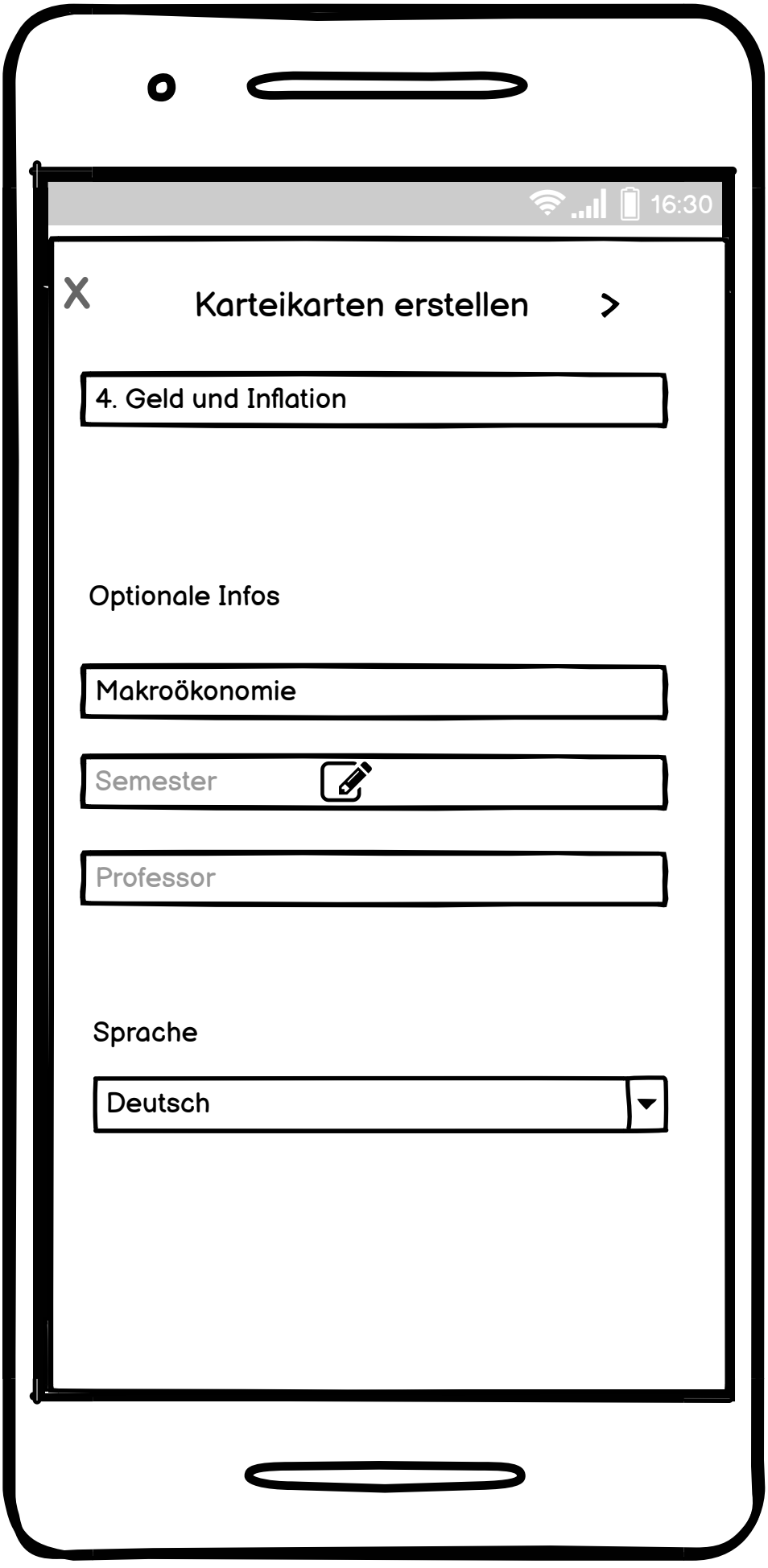

16:30
X
Hochladen in
Marketing
Friedrich-Alexander-Universität Erlangen-Nürnberg
created_pdf_document.pdf
Upload erfolgreich
Zurück zum Newsfeed
oder
Weitere Dateien hochladen

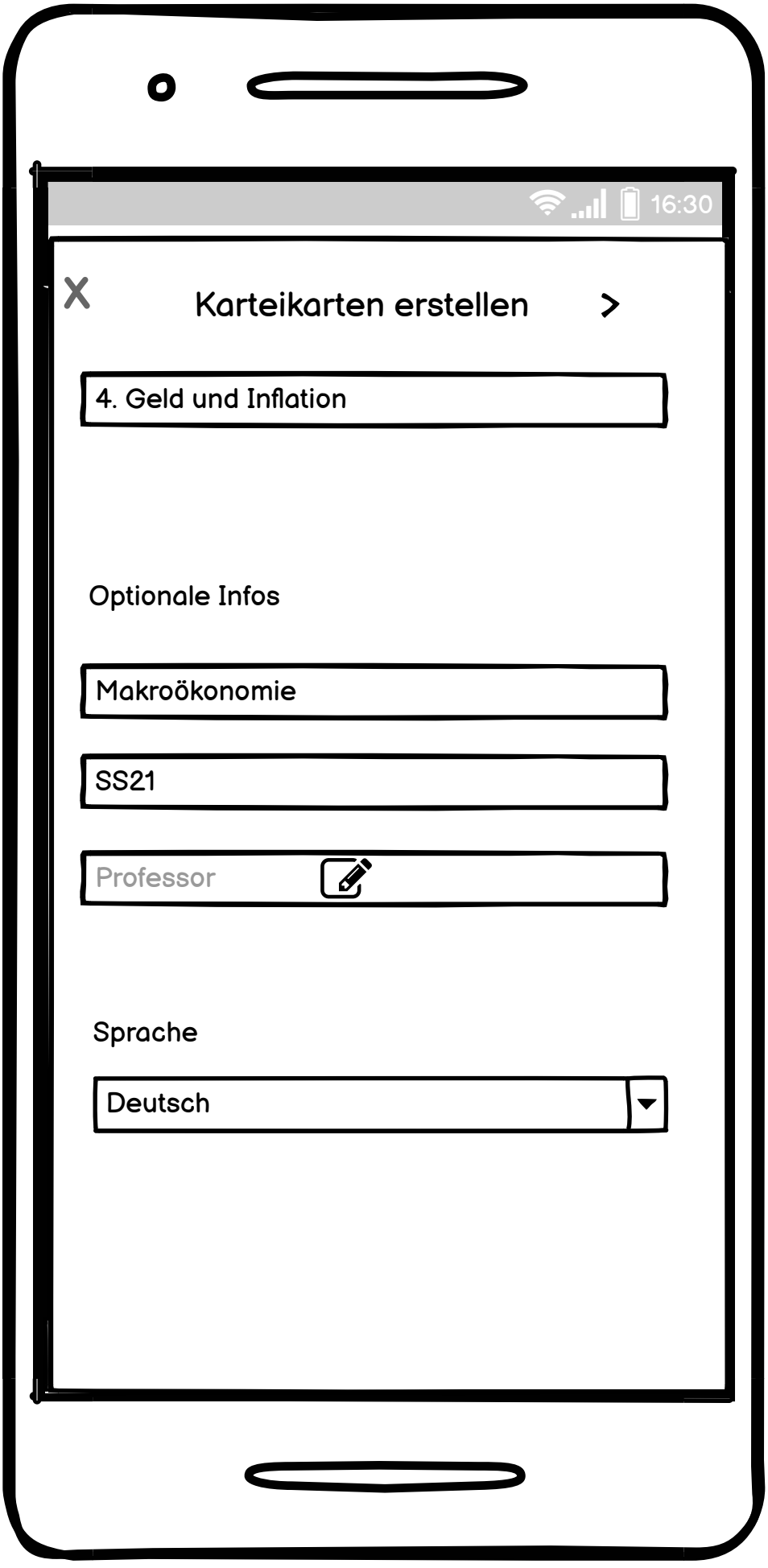

16:30
Newsfeed
Neueste
Unbeantwortet
BWLJustus
vor 1 Minute
Statistik
Findet heute das Tutorium statt?
0
Anonymer Nutzer
vor 22 Stunden
Uni Erlangen Nürnberg
Gibt es schon einen Klausurtermin für die Überblicksvorlesung "Einführung in die Frühe Neuzeit" bei Herrn Wellner?
0
Anonyme Nutzerin
vor 23 Stunden
Marketing
5

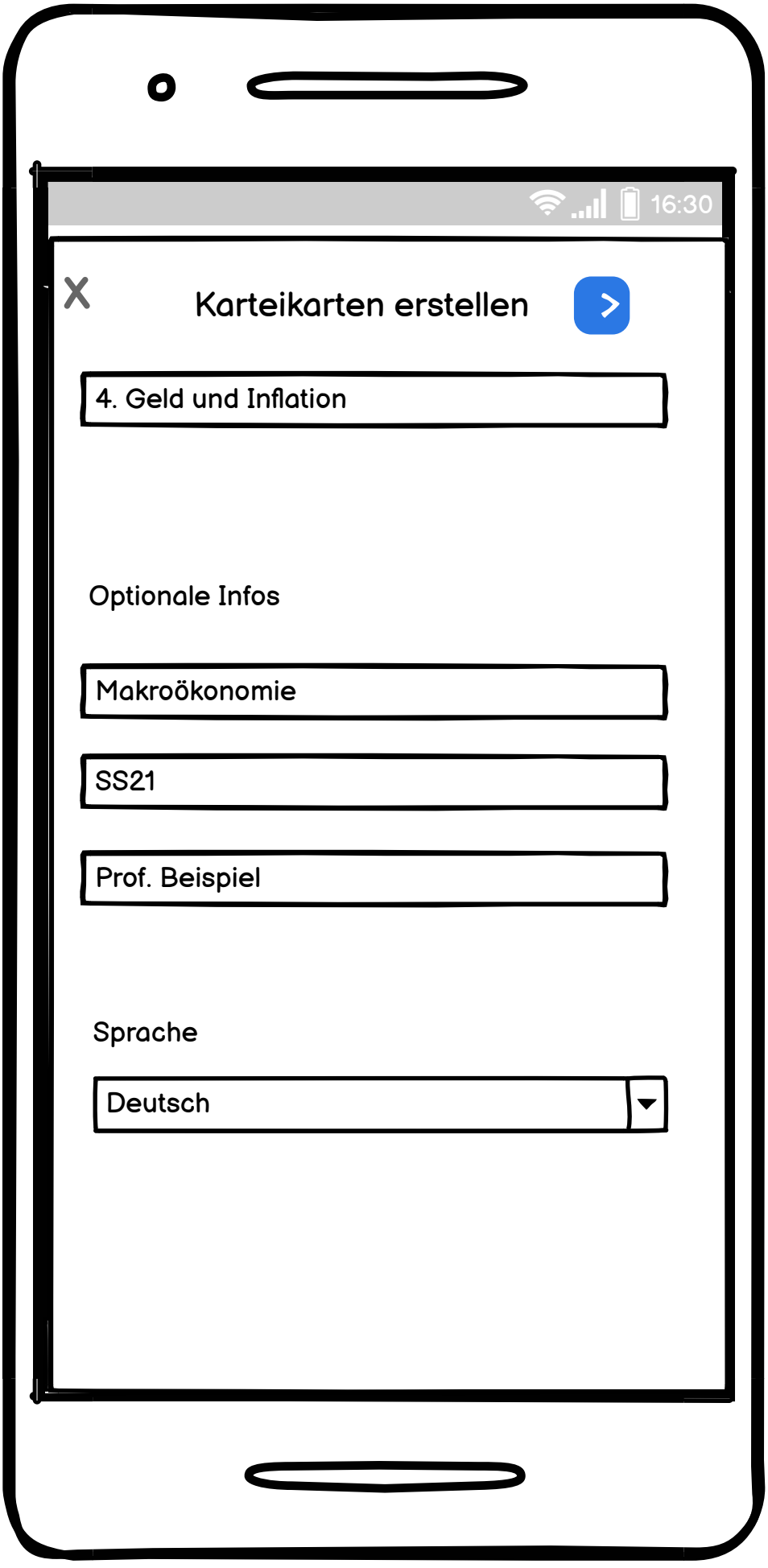

16:30
Newsfeed
Neueste
Unbeantwortet
BWLJustus
vor 2 Minuten
Statistik
Findet heute das Tutorium statt?
0
Anonymer Nutzer
vor 22 Stunden
Uni Erlangen Nürnberg
Gibt es schon einen Klausurtermin für die Überblicksvorlesung "Einführung in die Frühe Neuzeit" bei Herrn Wellner?
Frage stellen
Dokument hochladen
Karteikarten erstellen
5


16:30
X
Karteikarten erstellen
>
Set-Titel eingeben
Optionale Infos
Relevanter Kurs
Semester
Professor
Sprache
Deutsch

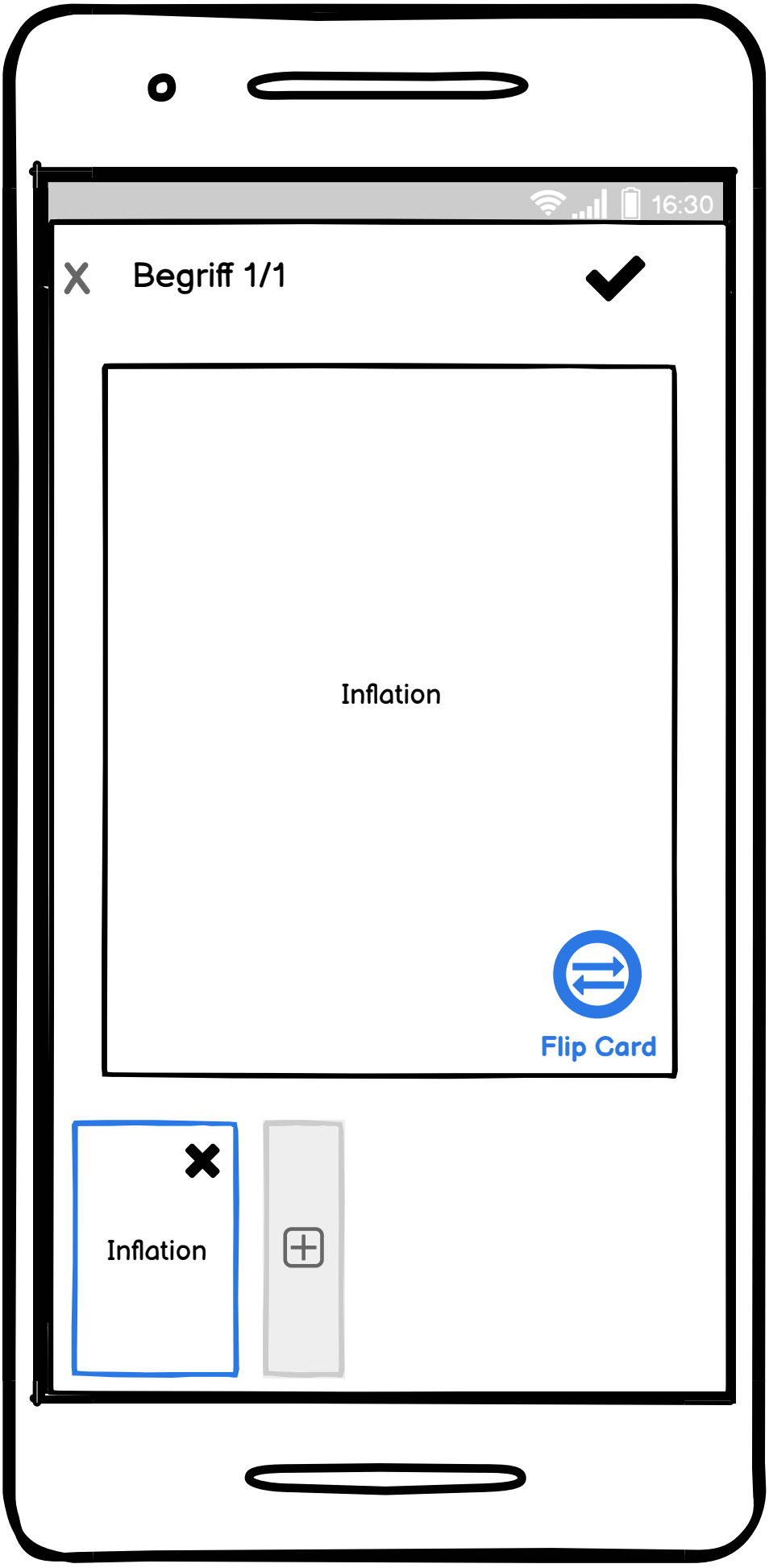

16:30
X
Karteikarten erstellen
>
4. Geld und Inflation
Optionale Infos
Relevanter Kurs
Semester
Professor
Sprache
Deutsch


16:30
X
Karteikarten erstellen
>
4. Geld und Inflation
Optionale Infos
Makroökonomie
Semester
Professor
Sprache
Deutsch

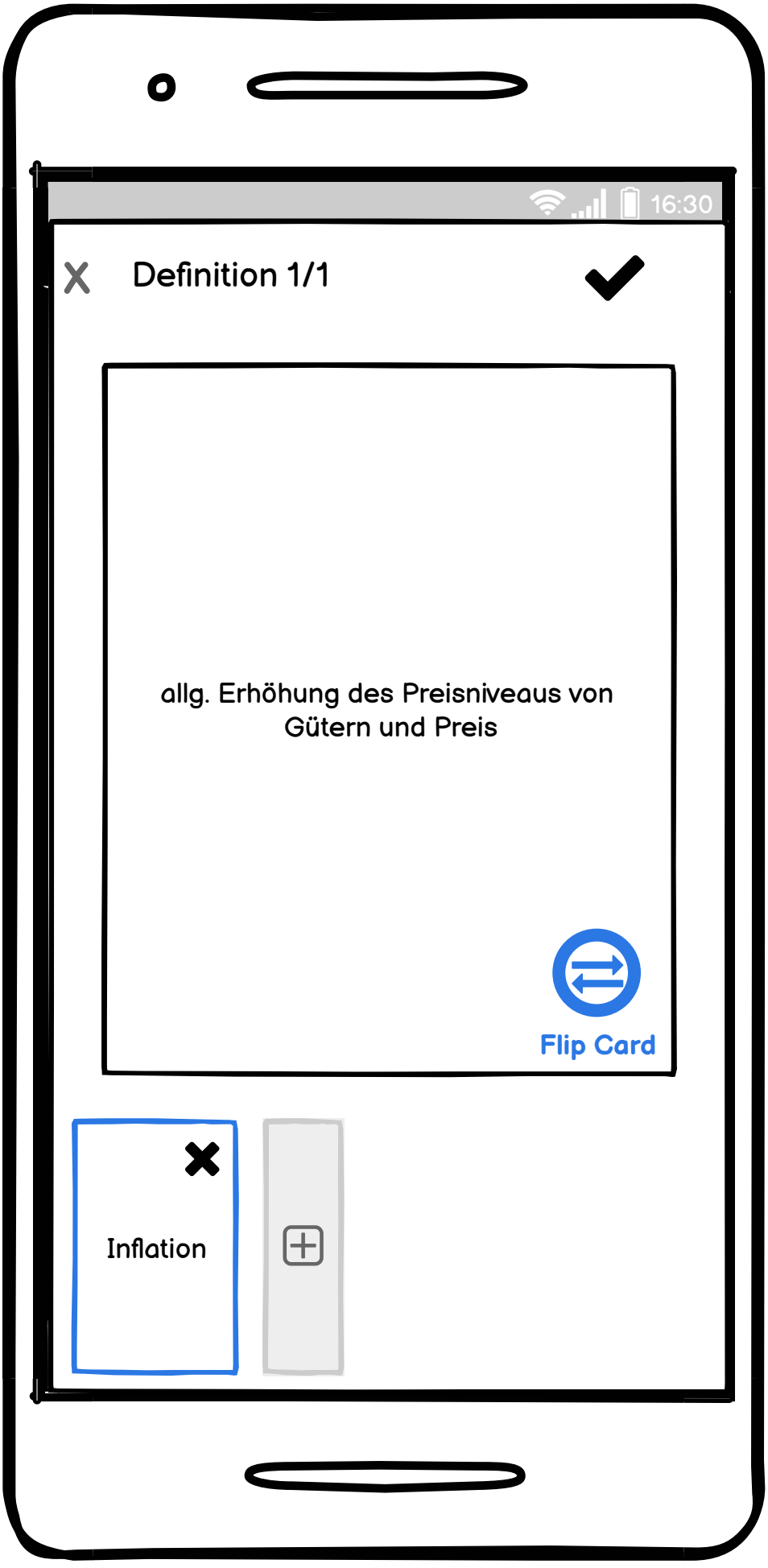

16:30
Karteikarten erstellen
4. Geld und Inflation
Optionale Infos
Makroökonomie
SS21
Professor
Sprache
Deutsch


16:30
X
Karteikarten erstellen
4. Geld und Inflation
Optionale Infos
Makroökonomie
SS21
Prof. Beispiel
Sprache
Deutsch

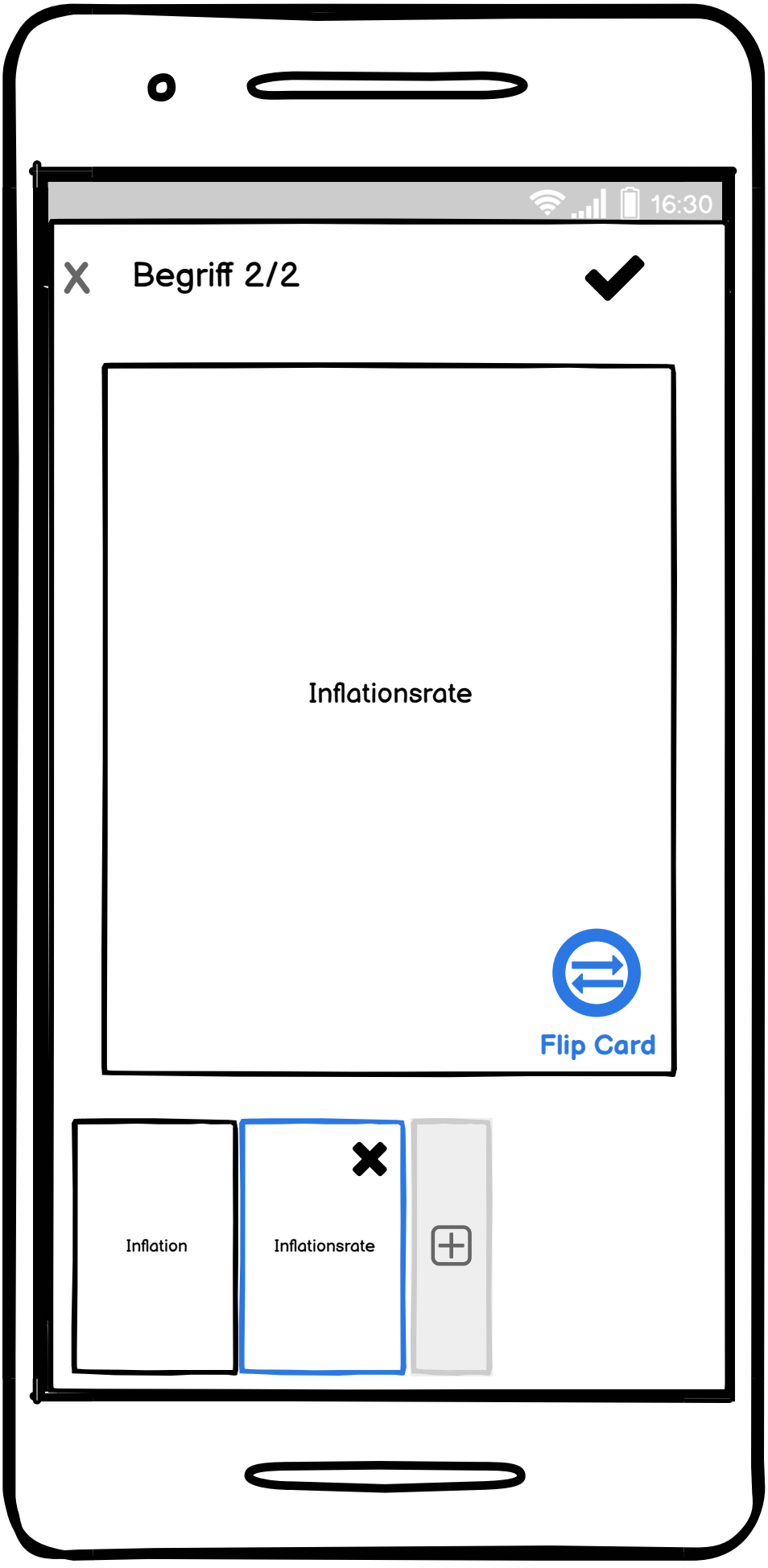

16:30
Begriff 1/1
Tippen, um Begriff einzugeben
Flip Card


16:30
Begriff 1/1
Inflation
Flip Card
Inflation

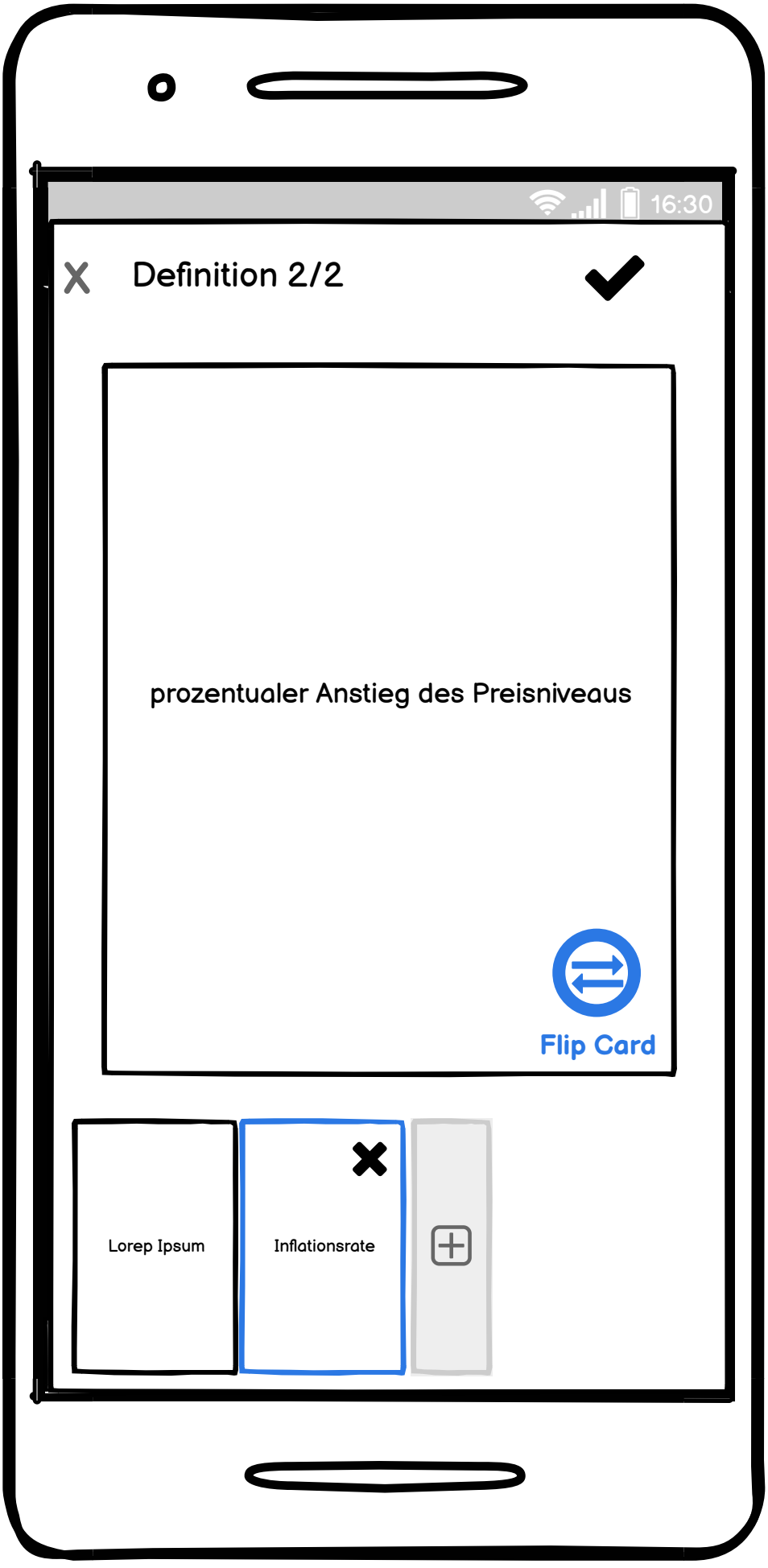

16:30
Definition 1/1
Tippen, um Definition einzugeben
Flip Card
Inflation


16:30
Definition 1/1
allg. Erhöhung des Preisniveaus von Gütern und Preis
Flip Card
Inflation


16:30
Begriff 2/2
Tippen, um Begriff einzugeben
Flip Card
Inflation


16:30
Begriff 2/2
Inflationsrate
Flip Card
Inflation
Inflationsrate


16:30
Definition 2/2
Tippen, um Definition einzugeben
Flip Card
Lorep Ipsum
Inflationsrate


16:30
Definition 2/2
prozentualer Anstieg des Preisniveaus
Flip Card
Lorep Ipsum
Inflationsrate


16:30
Begriff 3/3
Tippen, um Begriff einzugeben
Flip Card
Lorep Ipsum
Inflationsrate


16:30
Begriff 3/3
3 Funktionen des Geldes
Flip Card
Lorep Ipsum
Inflationsrate
3 Funktionen des Geldes


16:30
Definition 3/3
Tippen, um Definition einzugeben
Flip Card
Lorep Ipsum
Inflationsrate
3 Funktionen des Geldes


16:30
Definition 3/3
Mittel zur Aufbewahrung
Recheneinheit
Tauschmittel (Transaktion)
Flip Card
Lorep Ipsum
Inflationsrate
3 Funktionen des Geldes


16:30
Begriff 4/4
Tippen, um Begriff einzugeben
Flip Card
Inflationsrate
3 Funktionen des Geldes


16:30
Begriff 4/4
Vorteil des Geldes
Flip Card
Inflationsrate
3 Funktionen des Geldes
Vorteil des Geldes

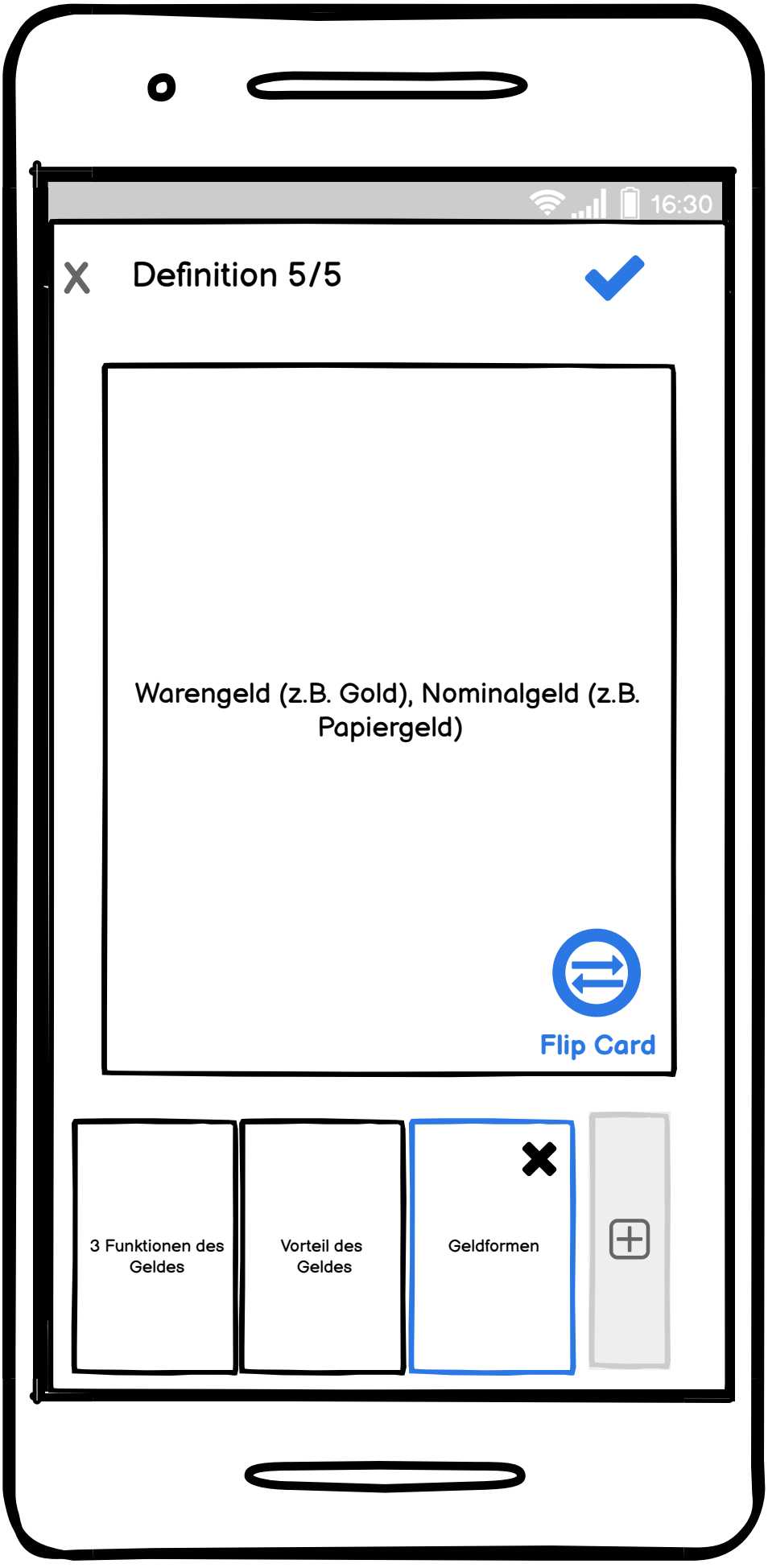

16:30
Definition 4/4
Tippen, um Definition einzugeben
Flip Card
Inflationsrate
3 Funktionen des Geldes
Vorteil des Geldes


16:30
Definition 4/4
ohne Geld würde eine Naturaltauschwirtschaft herrschen: usw...
Flip Card
Inflationsrate
3 Funktionen des Geldes
Vorteil des Geldes


16:30
Begriff 5/5
Tippen, um Begriff einzugeben
Flip Card
3 Funktionen des Geldes
Vorteil des Geldes

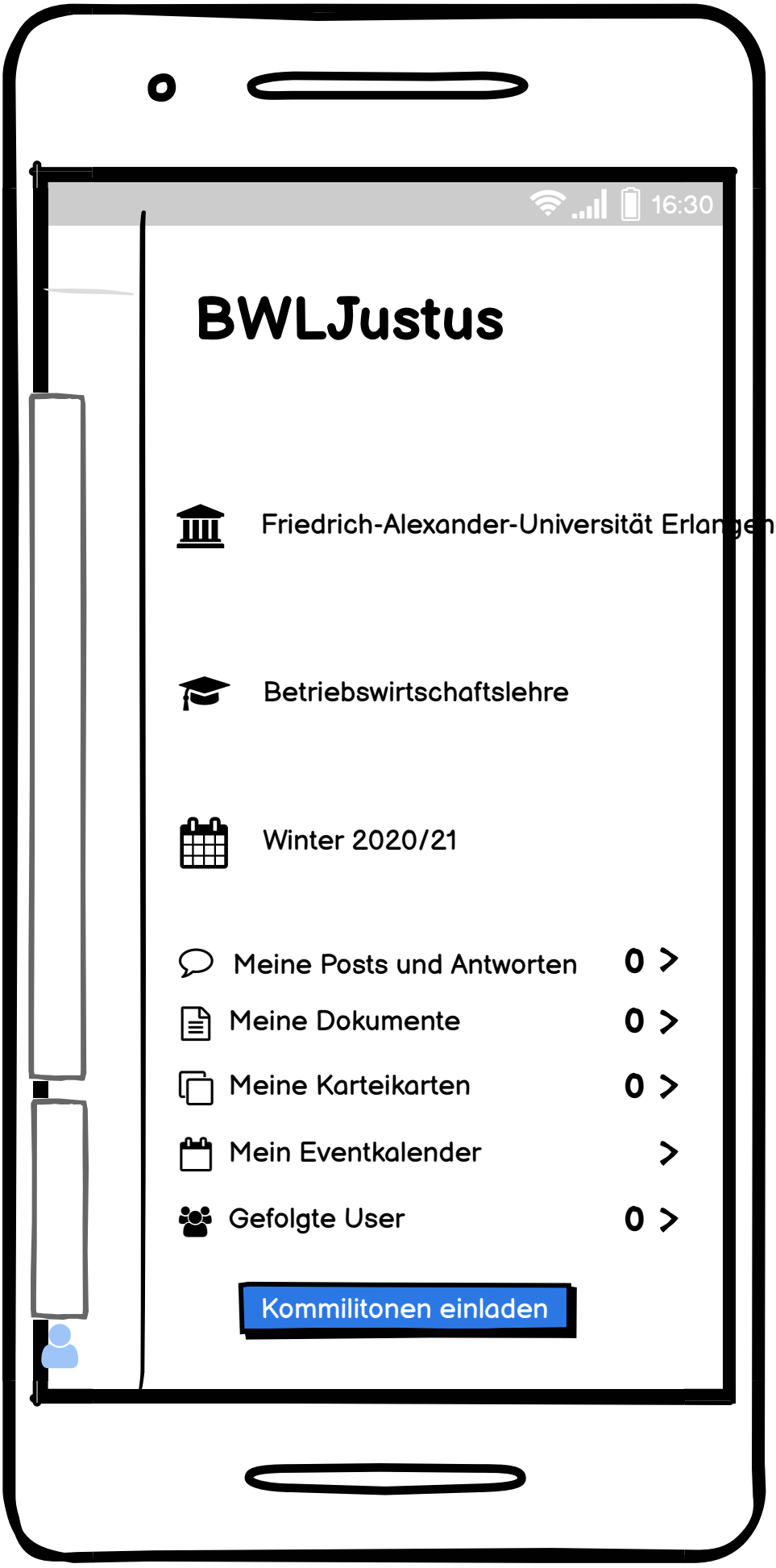

16:30
Begriff 5/5
Geldformen
Flip Card
3 Funktionen des Geldes
Vorteil des Geldes
Geldformen

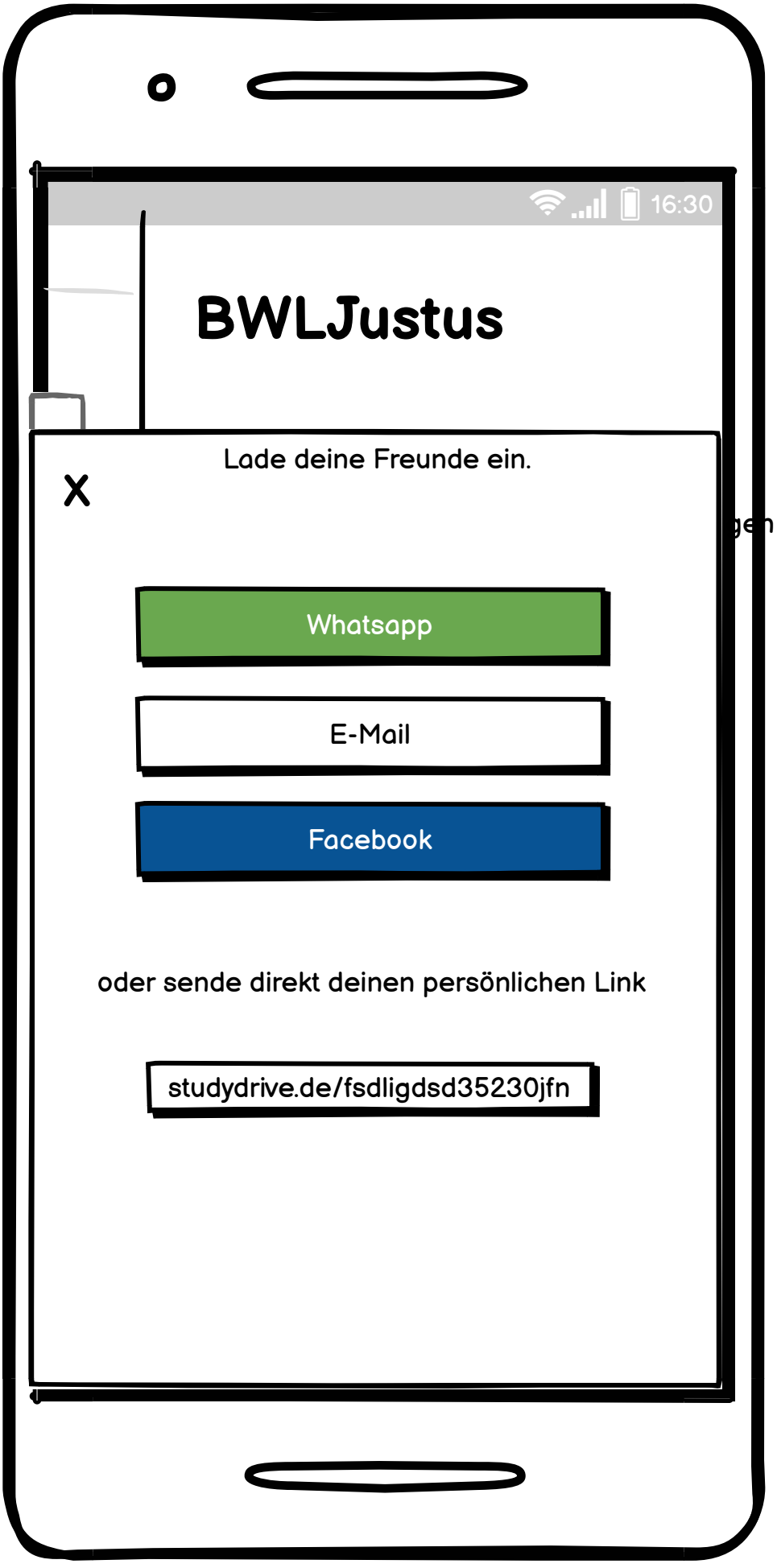

16:30
Definition 5/5
Tippen, um Definition einzugeben
Flip Card
3 Funktionen des Geldes
Vorteil des Geldes
Geldformen


16:30
Definition 5/5
Warengeld (z.B. Gold), Nominalgeld (z.B. Papiergeld)
Flip Card
3 Funktionen des Geldes
Vorteil des Geldes
Geldformen


16:30
X
Lorep Ipsum x/x
Nicht genug Karteikarten für ein öffentliches Lernset
Erstelle mindestens 5 Karteikarten um dein Lernset fertig zu stellen.
Weitere Karteikarten erstellen


16:30
Newsfeed
Neueste
Unbeantwortet
BWLJustus
vor 4 Minuten
Statistik
Findet heute das Tutorium statt?
0
Anonymer Nutzer
vor 22 Stunden
Uni Erlangen Nürnberg
Gibt es schon einen Klausurtermin für die Überblicksvorlesung "Einführung in die Frühe Neuzeit" bei Herrn Wellner?
0
Anonyme Nutzerin
vor 23 Stunden
Marketing
5


16:30
BWLJustus
Friedrich-Alexander-Universität Erlangen
Betriebswirtschaftslehre
Winter 2020/21
Meine Posts und Antworten 0 >
Meine Dokumente 0 >
Meine Karteikarten 0 >
Mein Eventkalender >
Gefolgte User 0 >
Kommilitonen einladen


16:30
BWLJustus
Lade deine Freunde ein.
X
Whatsapp
E-Mail
Facebook
oder sende direkt deinen persönlichen Link
studydrive.de/fsdligdsd35230jfn

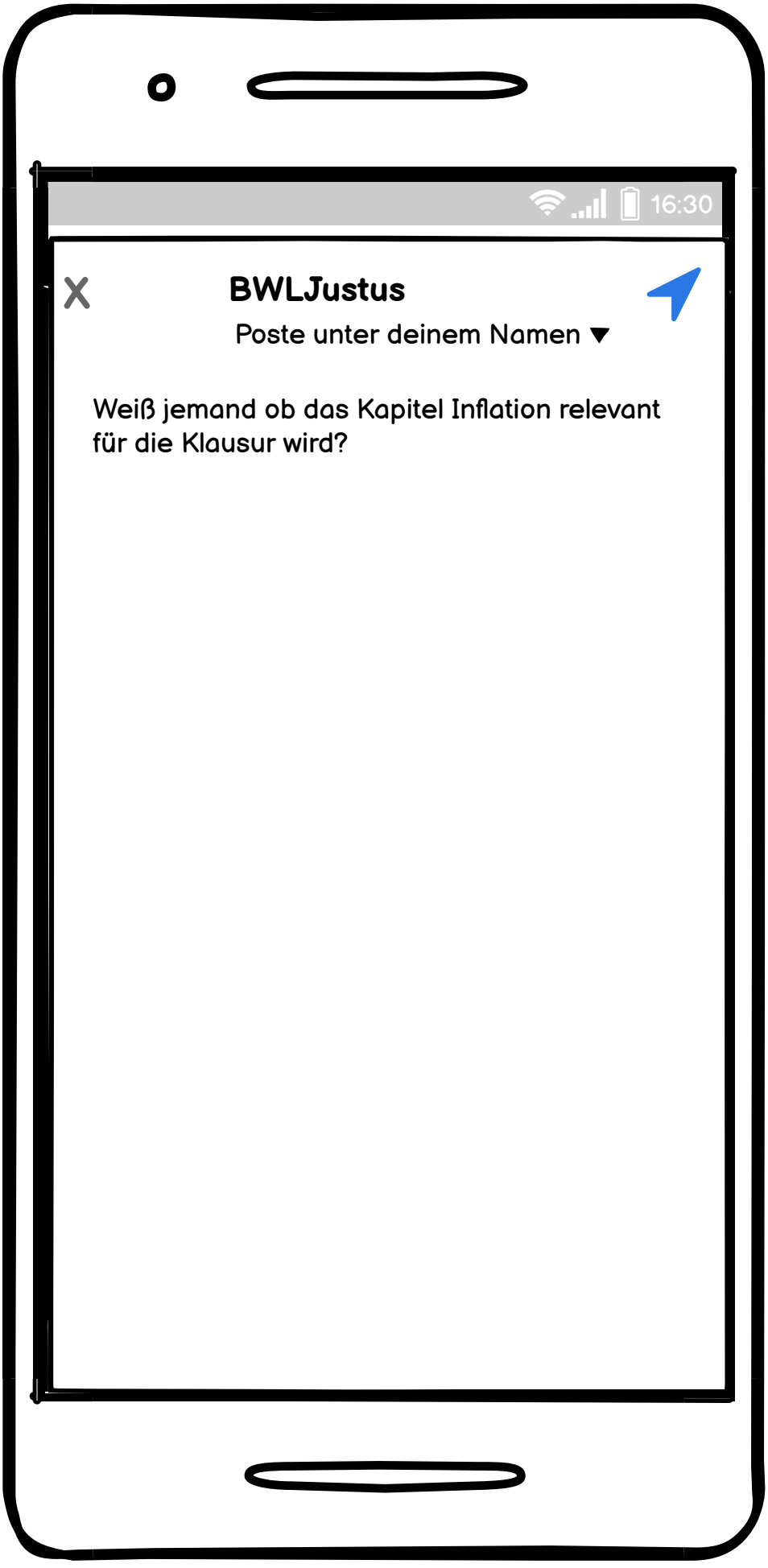

16:30
BWLJustus
X
Erfolgreich eingeladen!
Dieses Fenster kann geschlossen werden.


16:30
Newsfeed
Neueste
Unbeantwortet
BWLJustus
vor 5 Minuten
Statistik
Findet heute das Tutorium statt?
0
Anonymer Nutzer
vor 22 Stunden
Uni Erlangen Nürnberg
Gibt es schon einen Klausurtermin für die Überblicksvorlesung "Einführung in die Frühe Neuzeit" bei Herrn Wellner?
0
Anonyme Nutzerin
vor 23 Stunden
Marketing
5


16:30
Newsfeed
Neueste
Unbeantwortet
BWLJustus
vor 5 Minuten
Statistik
Findet heute das Tutorium statt?
0
Anonymer Nutzer
vor 22 Stunden
Uni Erlangen Nürnberg
Gibt es schon einen Klausurtermin für die Überblicksvorlesung "Einführung in die Frühe Neuzeit" bei Herrn Wellner?
Frage stellen
Dokument hochladen
Karteikarten erstellen
5


16:30
Newsfeed
Neueste
Unbeantwortet
BWLJustus
vor 5 Minuten
Statistik
Findet heute das Tutorium statt?
0
Wo möchtest du posten?
Kurse
Gruppen
Statistik
Makroökonomie
Produktion, Logistik, Beschaffung
Investition und Finanzierung
Marketing

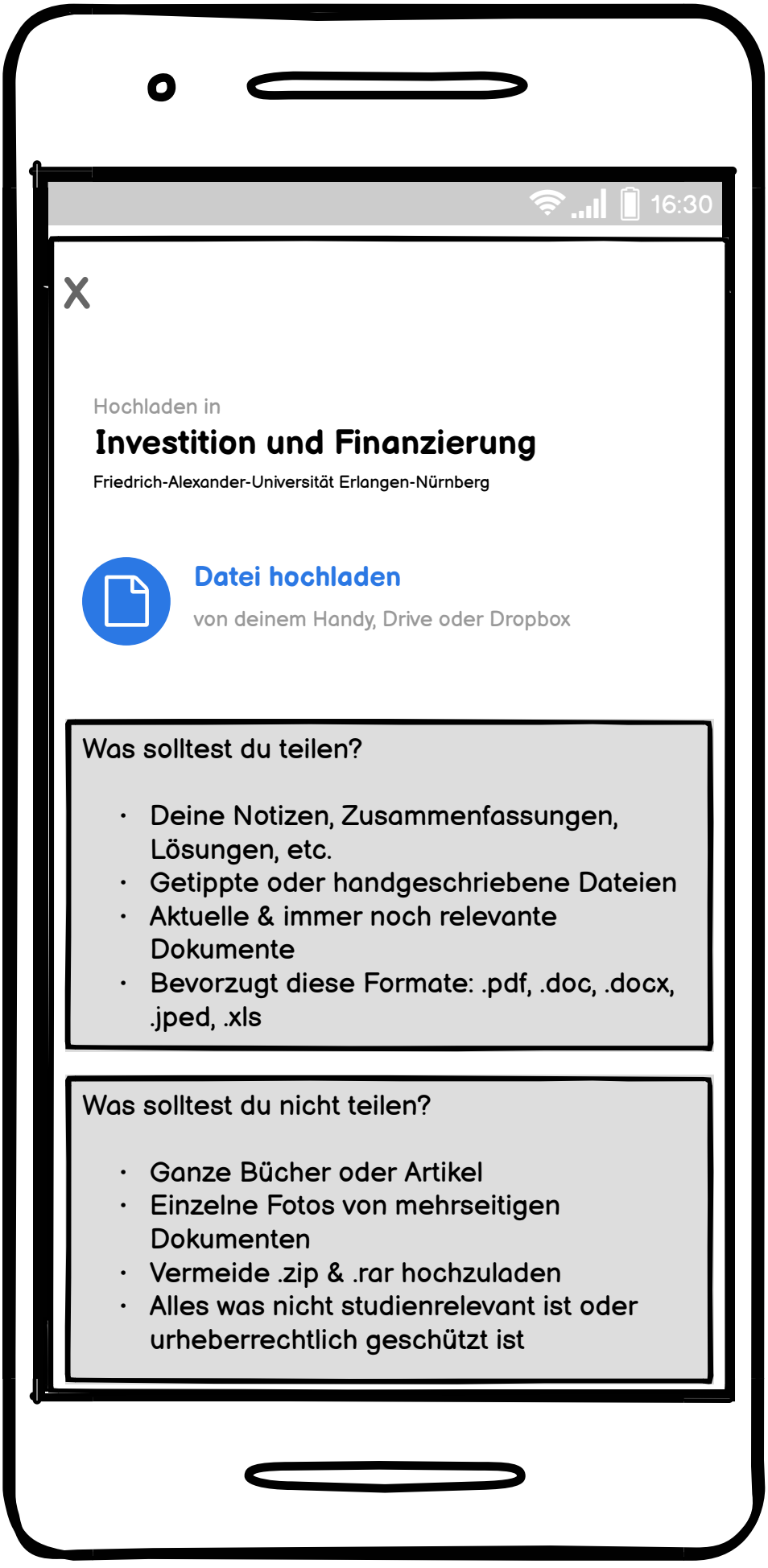

16:30
BWLJustus
Poste unter deinem Namen
Was möchtest du in Makroökonomie posten?

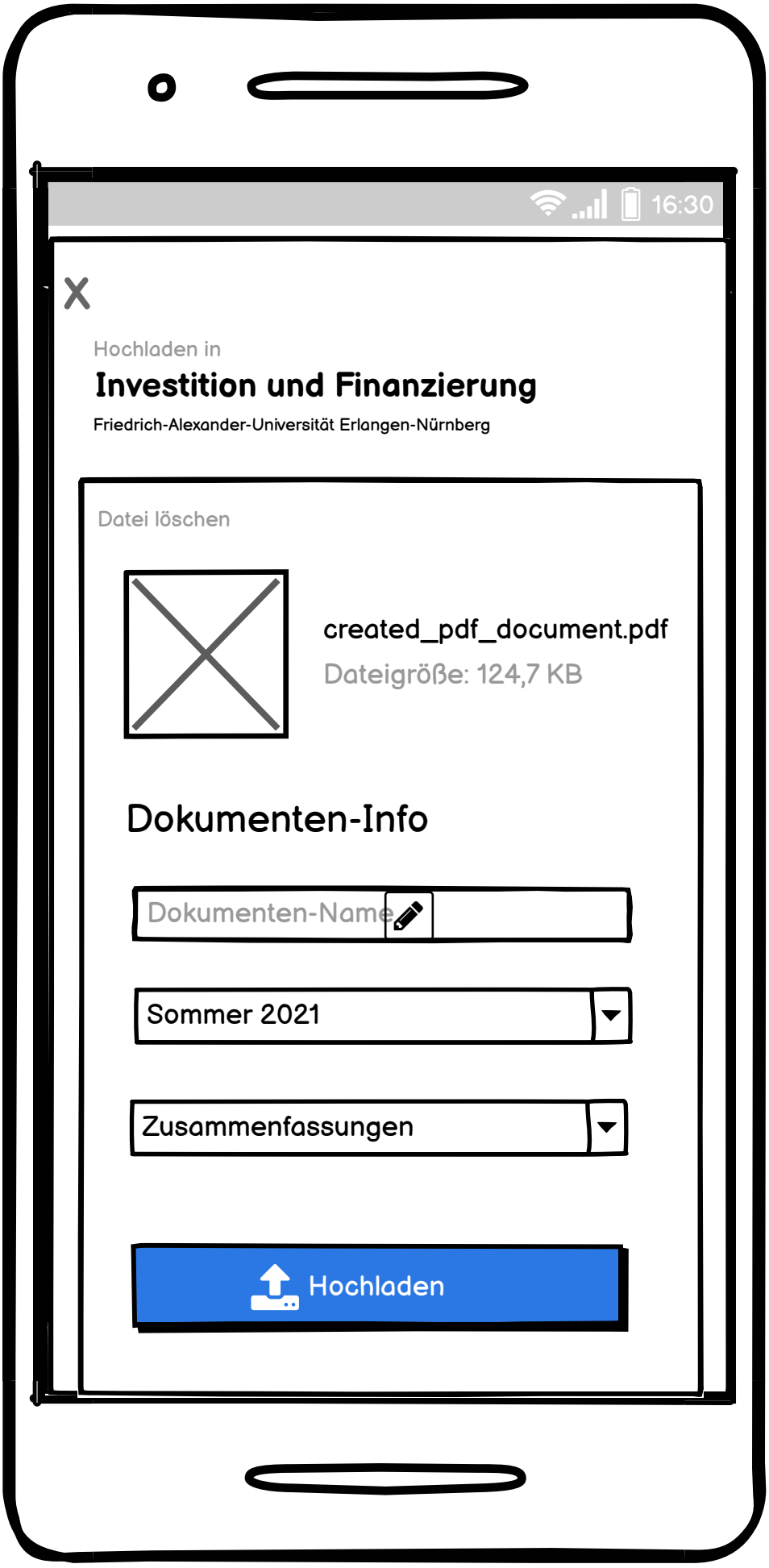

16:30
BWLJustus
Poste unter deinem Namen
Weiß jemand ob das Kapitel Inflation relevant für die Klausur wird?


16:30
Newsfeed
Neueste
Unbeantwortet
BWLJustus
vor 1 Minute
Makroökonomie
Weiß jemand ob das Kapitel Inflation relevant für die Klausur wird?
0
BWLJustus
vor 5 Minuten
Statistik
Findet heute das Tutorium statt?
0
Anonymer Nutzer
vor 22 Stunden
Uni Erlangen Nürnberg
Gibt es schon einen Klausurtermin für die Überblicksvorlesung "Einführung in die Früh Neuzeit" bei Herrn Wellner?

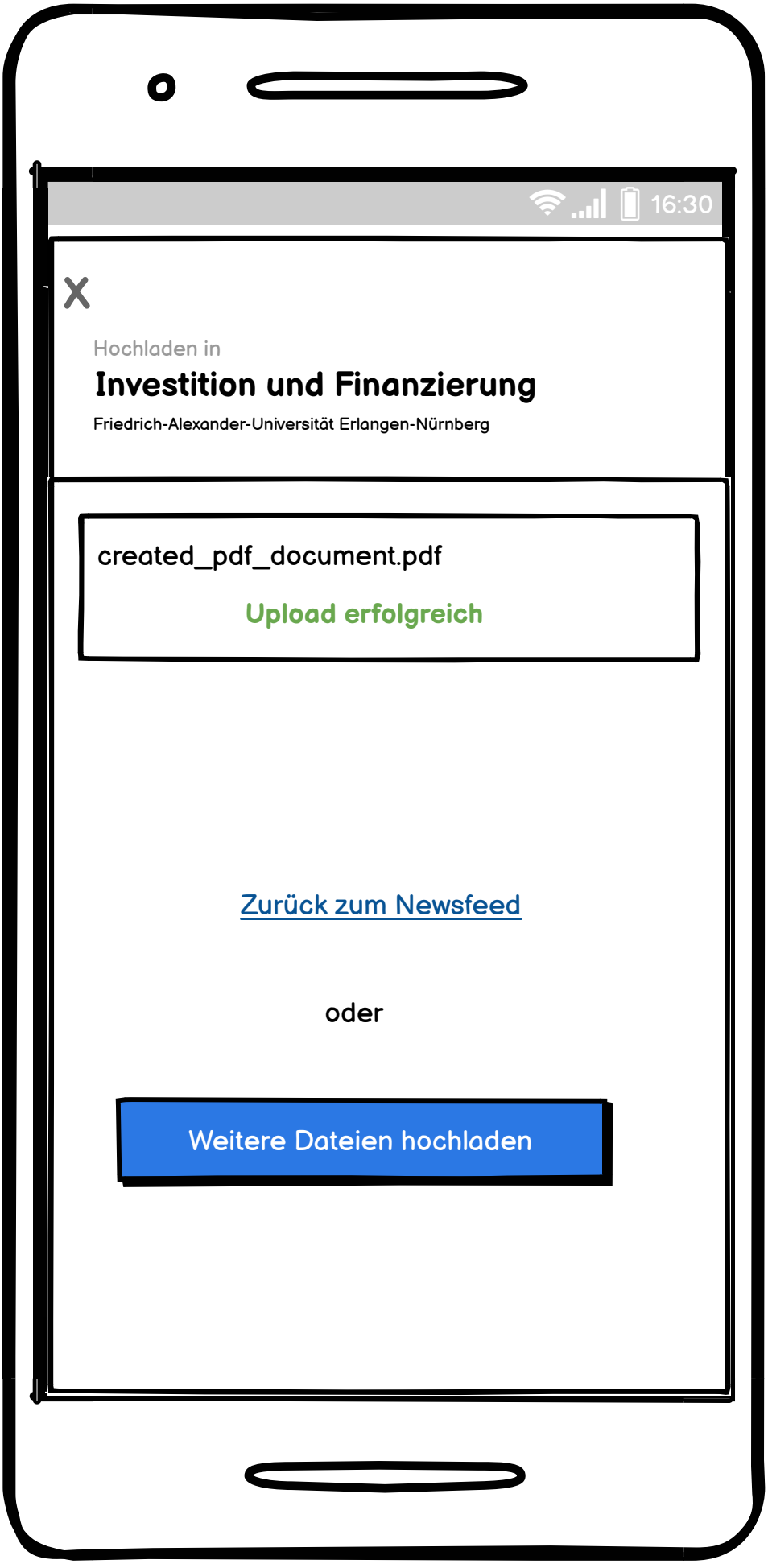

16:30
Newsfeed
Neueste
Unbeantwortet
BWLJustus
vor 1 Minute
Makroökonomie
Weiß jemand ob das Kapitel Inflation relevant für die Klausur wird?
0
BWLJustus
vor 5 Minuten
Statistik
Findet heute das Tutorium statt?
Frage stellen
Dokument hochladen
Karteikarten erstellen
Überblicksvorlesung "Einführung in die Früh
Neuzeit" bei Herrn Wellner?


16:30
Newsfeed
Neueste
Unbeantwortet
BWLJustus
vor 1 Minute
Statistik
Findet heute das Tutorium statt?
0
Anonymer Nutzer
vor 22 Stunden
Wähle eine deiner Kurse zum Upload
Statistik
Makroökonomie
Produktion, Logistik, Beschaffung
Investition und Finanzierung
Marketing


16:30
X
Hochladen in
Investition und Finanzierung
Friedrich-Alexander-Universität Erlangen-Nürnberg
Datei hochladen
von deinem Handy, Drive oder Dropbox
Was solltest du teilen?
Deine Notizen, Zusammenfassungen, Lösungen, etc.
Getippte oder handgeschriebene Dateien
Aktuelle & immer noch relevante Dokumente
Bevorzugt diese Formate: .pdf, .doc, .docx, .jped, .xls
Was solltest du nicht teilen?
Ganze Bücher oder Artikel
Einzelne Fotos von mehrseitigen Dokumenten
Vermeide .zip & .rar hochzuladen
Alles was nicht studienrelevant ist oder urheberrechtlich geschützt ist


16:30
Hochladen in
Investition und Finanzierung
Friedrich-Alexander-Universität Erlangen-Nürnberg
Datei löschen
created_pdf_document.pdf
Dateigröße: 124,7 KB
Dokumenten-Info
Dokumenten-Name
Sommer 2021
Zusammenfassungen
Hochladen


16:30
Hochladen in
Investition und Finanzierung
Friedrich-Alexander-Universität Erlangen-Nürnberg
Datei löschen
created_pdf_document.pdf
Dateigröße: 124,7 KB
Dokumenten-Info
Meine IuF Zusammenfassung
Sommer 2021
Zusammenfassungen
Hochladen

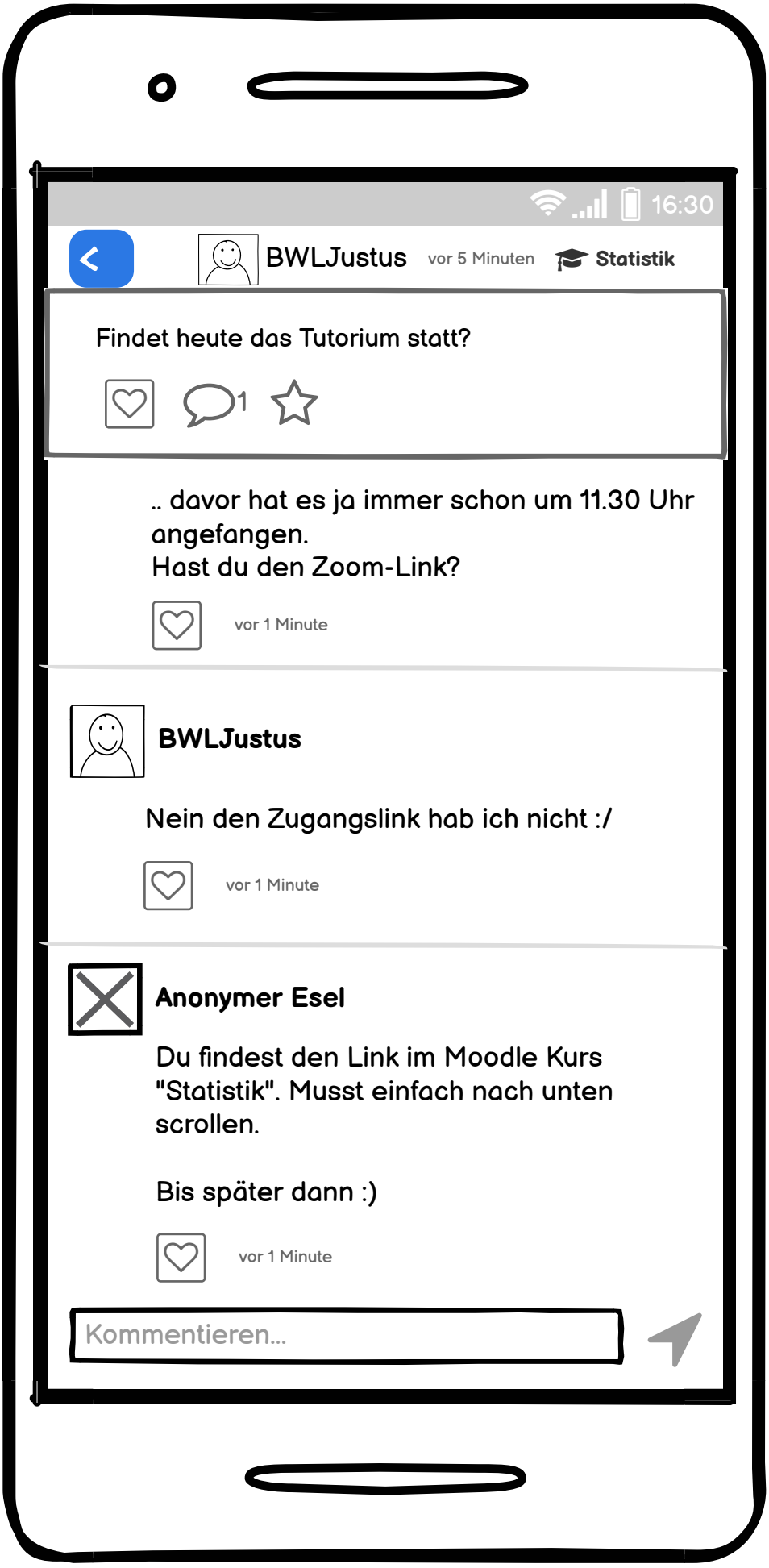

16:30
X
Hochladen in
Investition und Finanzierung
Friedrich-Alexander-Universität Erlangen-Nürnberg
created_pdf_document.pdf
Upload erfolgreich
Zurück zum Newsfeed
oder
Weitere Dateien hochladen


16:30
Newsfeed
Neueste
Unbeantwortet
BWLJustus
vor 1 Minute
Makroökonomie
Weiß jemand ob das Kapitel Inflation relevant für die Klausur wird?
0
BWLJustus
vor 5 Minuten
Statistik
Findet heute das Tutorium statt?
1
Anonymer Nutzer
vor 22 Stunden
Uni Erlangen Nürnberg
Gibt es schon einen Klausurtermin für die Überblicksvorlesung "Einführung in die Früh Neuzeit" bei Herrn Wellner?
1

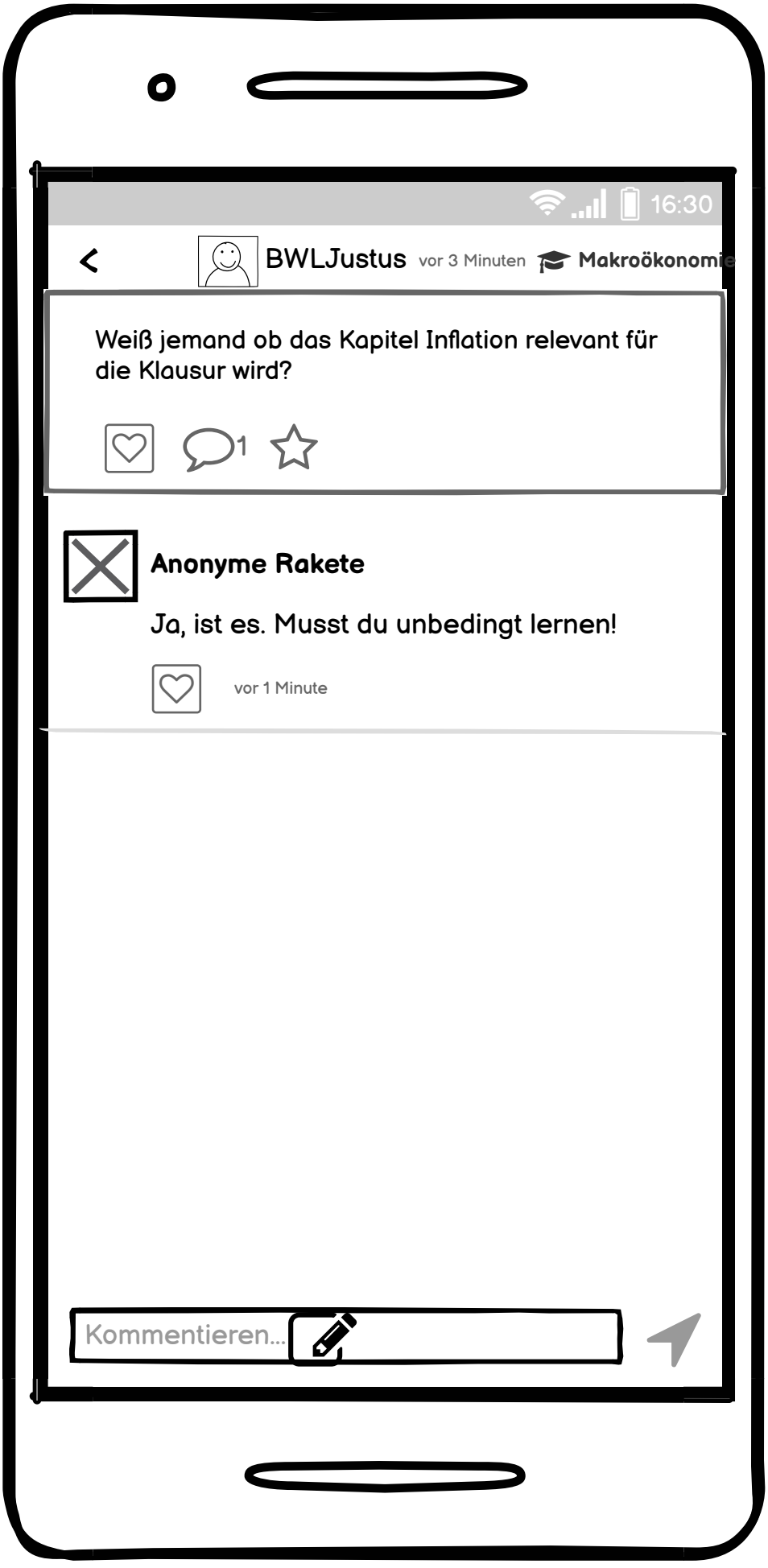

16:30
BWLJustus
vor 5 Minuten
Statistik
Findet heute das Tutorium statt?
1
Anonymer Esel
Hey, ja es findet heute statt. Die Uhrzeit hat sich aber verschoben, d.h. es fängt erst um 14.00 Uhr an und geht bis 14.30 ... davor hat es ja immer schon um 11.30 Uhr angefangen.
Hast du den Zoom-Link?
vor 1 Minute
Kommentieren...


16:30
BWLJustus
vor 5 Minuten
Statistik
Findet heute das Tutorium statt?
1
Anonymer Esel
Hey, ja es findet heute statt. Die Uhrzeit hat sich aber verschoben, d.h. es fängt erst um 14.00 Uhr an und geht bis 14.30 ... davor hat es ja immer schon um 11.30 Uhr angefangen.
Hast du den Zoom-Link?
vor 1 Minute
Nein den Zugangslink hab ich nicht :/


16:30
BWLJustus
vor 5 Minuten
Statistik
Findet heute das Tutorium statt?
1
Anonymer Esel
Hey, ja es findet heute statt. Die Uhrzeit hat sich aber verschoben, d.h. es fängt erst um 14.00 Uhr an und geht bis 14.30 ... davor hat es ja immer schon um 11.30 Uhr angefangen.
Hast du den Zoom-Link?
vor 1 Minute
BWLJustus
Nein den Zugangslink hab ich nicht :/
vor 1 Minute
Antworten laden
Kommentieren...


16:30
BWLJustus
vor 5 Minuten
Statistik
Findet heute das Tutorium statt?
1
.. davor hat es ja immer schon um 11.30 Uhr angefangen.
Hast du den Zoom-Link?
vor 1 Minute
BWLJustus
Nein den Zugangslink hab ich nicht :/
vor 1 Minute
Anonymer Esel
Du findest den Link im Moodle Kurs "Statistik". Musst einfach nach unten scrollen.
Bis später dann :)
vor 1 Minute
Kommentieren...


16:30
Newsfeed
Neueste
Unbeantwortet
BWLJustus
vor 1 Minute
Makroökonomie
Weiß jemand ob das Kapitel Inflation relevant für die Klausur wird?
1
BWLJustus
vor 5 Minuten
Statistik
Findet heute das Tutorium statt?
3
Anonymer Nutzer
vor 22 Stunden
Uni Erlangen Nürnberg
Gibt es schon einen Klausurtermin für die Überblicksvorlesung "Einführung in die Früh
Neuzeit" bei Herrn Wellner?


16:30
BWLJustus
vor 3 Minuten
Makroökonomie
Weiß jemand ob das Kapitel Inflation relevant für die Klausur wird?
1
Anonyme Rakete
Ja, ist es. Musst du unbedingt lernen!
vor 1 Minute
Kommentieren...


16:30
BWLJustus
vor 3 Minuten
Makroökonomie
Weiß jemand ob das Kapitel Inflation relevant für die Klausur wird?
1
Anonyme Rakete
Ja, ist es. Musst du unbedingt lernen!
vor 1 Minute
BWLJustus
Danke :)
vor 1 Minute
Kommentieren...

16:30

Danke für die Teilnahme!

Das Programm kann jetzt geschlossen werden.

Bitte fülle noch kurz den Abschluss-Fragebogen aus. Hierfür wieder die Typeform Umfrage öffnen und "I finished the recording" klicken.



Plagiarism Declaration in Accordance with Examination Rules

I herewith declare that I worked on this thesis independently. Furthermore, it was not submitted to any other examining committee. All sources and aids used in this thesis, including literal and analogous citations, have been identified.



Sebastian Nikol